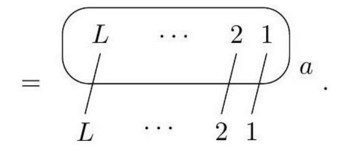
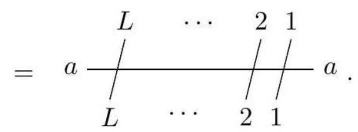
cardona-qft-methods — formula report

3760 inline formulas (section omitted; all rendered), 1012 display equations, 8 tables, 40 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0717 **[conf 0.003]**	275	■ 0.003 ● C|-4	`\begin{aligned} & \hat{T}=\operatorname{tr}_{a} M_{a}(u) \\ &=\left.\right|_{L} ^{L} \cdots \\ & \cdots \cdots \\ & \cdots 2 \\ & L \end{aligned} a .`	$\hat{T}=\operatorname{tr}_a M_a(u)$ $=\|_L^L\cdots$ $\cdots\cdots \quad a.$ $\cdots 2$ L	$\hat{T}=\operatorname{tr}_a M_a(u)$ 
cardona-qft-methods_EQ0889 **[conf 0.077]**	345	■ 0.077 ● W|-2	`\begin{array}{rll} \operatorname{Map}(8, M)=L M \times_{M} L M & \longrightarrow L M \times L M \\ \downarrow & \Delta & \\ M & \downarrow & \downarrow \\ \downarrow^{e v \times e v} \end{array} c M \times M \text { }`	$\operatorname{Map}(8,M)=LM\times_M LM \longrightarrow LM\times LM$ $\downarrow \quad \Delta$ $M \quad \downarrow \qquad \downarrow \quad cM\times M$ $\downarrow^{ev\times ev}$	$\begin{array}{ccc} Map(8,M)=LM\times_M LM & \longrightarrow & LM\times LM \\ \downarrow & & \downarrow{\scriptstyle ev\times ev} \\ M & \xrightarrow{\Delta} & M\times M \end{array}$
cardona-qft-methods_EQ0908 **[conf 0.080]**	350	■ 0.080 ● C|+5	`\begin{array}{rlc} P_{g} M:{ }_{1} \times{ }_{0} P_{h} M & \longrightarrow & P_{g} M \times P_{h} M \\ \downarrow & & \downarrow \\ \downarrow & e_{0} \circ \pi_{1} & \\ M & & \downarrow \\ \downarrow & & \downarrow \times M \end{array}`	$P_gM:{}_1\times{}_0P_hM \longrightarrow P_gM\times P_hM$ $\downarrow \qquad \downarrow$ $\downarrow\ e_0\circ\pi_1$ $M \qquad \downarrow$ $\downarrow \qquad \downarrow\times M$	$\begin{array}{ccc} P_gM:{}_1\times_0 P_hM & \longrightarrow & P_gM\times P_hM \\ \downarrow{\scriptstyle e_0\circ\pi_1} & & \downarrow{\scriptstyle e_0\times e_0} \\ M & \xrightarrow{\Delta_g} & M\times M \end{array}$
cardona-qft-methods_EQ0077	046	■ 0.150 ● U|+44	`\left.\begin{array}{ccccccc} 1 & \rightarrow & \operatorname{Inn}(\mathcal{A}) & \longrightarrow & \operatorname{Aut}(\mathcal{A}) & \longrightarrow & \operatorname{Aut}(\mathcal{A}) / \operatorname{Inn}(\mathcal{A}) \end{array} \right\rvert\, \begin{gathered} 1, \\ 1 \\ \hline \end{gathered} \quad \mathcal{G} \quad \longrightarrow \quad \mathcal{G} \rtimes \operatorname{Diff}(M) \quad \longrightarrow \quad \operatorname{Diff}(M) \quad \rightarrow \quad 1 .`	*(not rendered)*	$\begin{array}{ccccccccc} 1 & \to & Inn(\mathcal{A}) & \longrightarrow & Aut(\mathcal{A}) & \longrightarrow & Aut(\mathcal{A})/Inn(\mathcal{A}) & \to & 1, \\ 1 & \to & \mathcal{G} & \longrightarrow & \mathcal{G}\rtimes\mathrm{Diff}(M) & \longrightarrow & \mathrm{Diff}(M) & \to & 1. \end{array}$
cardona-qft-methods_EQ0716	275	■ 0.155 ● W|+2	`\begin{aligned} & M_{a}(u)=L_{a, L}(u) L_{a, L-1}(u) \cdots L_{a, 2}(u) L_{a, 1}(u) \\ &=\left.a\right|_{L} ^{L} \cdots \\ & \cdots \cdots \\ & L \end{aligned}`	$M_a(u)=L_{a,L}(u)L_{a,L-1}(u)\cdots L_{a,2}(u)L_{a,1}(u)$ $=a\|_L^L\cdots$ $\cdots\cdots$ L	$M_a(u)=L_{a,L}(u)L_{a,L-1}(u)\cdots L_{a,2}(u)L_{a,1}(u)$ 

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0654 (4)	261	■ 0.167 ● U|+55	\phi_{\theta}(x) \phi_{\theta}(y)=e^{-\frac{1}{2} \frac{ \partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{ \partial}{\partial y^{\nu}}} \phi_{0}(x) \phi_{0}(y) e^{ \frac{1}{2}\left(\frac{\overleftarrow{\delta}}{\partial x^{\mu}}+\frac{\overleftarrow{□}}{\partial y^{\mu}} \right) \theta^{\mu \nu} P_{\nu}} \neq \phi_{0}(x) \phi_{0}(y) .	*(not rendered)*	$\phi_\theta(x)\phi_\theta(y)=e^{-\frac{1}{2}\frac{\partial}{\partial x^\mu}\theta^{\mu\nu}\frac{\partial}{\partial y^\nu}}\phi_0(x)\phi_0(y)e^{\frac{1}{2}\left(\frac{\overleftarrow{\partial}}{\partial x^\mu}+\frac{\overleftarrow{\partial}}{\partial y^\mu}\right)\theta^{\mu\nu}P_\nu}\neq\phi_0(x)\phi_0(y).$
cardona-qft-methods_ EQ0325 (11)	161	■ 0.219 ● N|+0	\begin{aligned} d \eta_{a b}=D \eta_{a b}=0, & \\ d \epsilon_{a_{1} a_{2} \cdots s_{D}}=D \epsilon_{a_{1} a_{2} \cdots a_{D}}= & 0 . \end{aligned}	$\begin{aligned} d\eta_{ab}=D\eta_{ab}=0,& \\ d\epsilon_{a_1a_2\cdots s_D}=D\epsilon_{a_1a_2\cdots a_D}=&0.\end{aligned}$	$\begin{aligned} d\eta_{ab}&=D\eta_{ab}=0, \\ d\epsilon_{a_1a_2\cdots s_D}&=D\epsilon_{a_1a_2\cdots a_D}=0.\end{aligned}$
cardona-qft-methods_ EQ0066 (19)	041	■ 0.286 ● C|+4	\begin{aligned} & a_{0}(f, P)=(4 \pi)^{-d / 2} \int_{M} d v o l_{g} \operatorname{tr}_{V}(f), \\ & a_{2}(f, P)=\frac{(4 \pi)^{-d / 2}}{6} \int_{M} d v o l_{g} \operatorname{tr}_{V}[(f(6 E+s)], \\ & \begin{aligned} a_{4}(f, P)=\frac{(4 \pi)^{-d / 2}}{360} \int_{M} d v o l_{g} \operatorname{tr}_{V}\left[f \left(60 E_{; k k}+60 E s+180 E^{2}+12 R_{; k k}\right.\right. \\ \left.\left.+5 s^{2}-2 R_{i j} R_{i j}+2 R_{i j k l} R_{i j k l}+30 \Omega_{i j} \Omega_{i j}\right)\right] \end{aligned} \end{aligned}	$\begin{aligned} & a_0(f,P)=(4\pi)^{-d/2}\int_M dvol_g\,\mathrm{tr}_V(f), \\ & a_2(f,P)=\frac{(4\pi)^{-d/2}}{6}\int_M dvol_g\,\mathrm{tr}_V[(f(6E+s)], \\ & \begin{aligned} a_4(f,P)=\frac{(4\pi)^{-d/2}}{360}\int_M dvol_g\,\mathrm{tr}_V\left[f\left(60E_{;kk}+60Es+180E^2+12R_{;kk}\right.\right. \\ \left.\left.+5s^2-2R_{ij}R_{ij}+2R_{ijkl}R_{ijkl}+30\Omega_{ij}\Omega_{ij}\right)\right]\end{aligned}\end{aligned}$	$\begin{aligned} a_0(f,P)&=(4\pi)^{-d/2}\int_M dvol_g\,\mathrm{tr}_V(f)\,, \\ a_2(f,P)&=\tfrac{(4\pi)^{-d/2}}{6}\int_M dvol_g\,\mathrm{tr}_V\big[(f(6E+s)\big], \qquad (1 \\ a_4(f,P)&=\tfrac{(4\pi)^{-d/2}}{360}\int_M dvol_g\,\mathrm{tr}_V\big[f(60E_{;kk}+60Es+180E^2+12R_{;kk} \\ &\quad +5s^2-2R_{ij}R_{ij}+2R_{ijkl}R_{ijkl}+30\Omega_{ij}\Omega_{ij})\big].\end{aligned}$
cardona-qft-methods_ EQ0824 (3)	313	■ 0.289 ● N|+2	\begin{gathered} \delta \psi_{M}=\nabla_{M} \epsilon+\frac{1}{4} H_{M} \mathcal{P} \epsilon+\frac{1}{16} e^{\phi} \sum_{n} \not_{n}^{(10)} \Gamma_{M} \mathcal{P}_{n} \epsilon, \\ \delta \lambda=\left(\not \partial \phi+\frac{1}{2} \not H \mathcal{P}\right) \epsilon+\frac{1}{8} e^{\phi} \sum_{n}(-1)^{n}(5-n) \nvdash_{n}^{(10)} \mathcal{P}_{n} \epsilon . \end{gathered}	$\begin{gathered}\delta\psi_M=\nabla_M\epsilon+\frac{1}{4}H_M\mathcal{P}\epsilon+\frac{1}{16}e^\phi\sum_n \not{n}^{(10)}\,\Gamma_M\mathcal{P}_n\epsilon, \\ \delta\lambda=\left(\not\partial\phi+\frac{1}{2}\not H\mathcal{P}\right)\epsilon+\frac{1}{8}e^\phi\sum_n(-1)^n(5-n)\nvdash_n^{(10)}\mathcal{P}_n\epsilon.\end{gathered}$	$\begin{aligned}\delta\psi_M&=\nabla_M\epsilon+\frac{1}{4}\not H_M\mathcal{P}\epsilon+\frac{1}{16}e^\phi\sum_n \not F_n^{(10)}\,\Gamma_M\mathcal{P}_n\,\epsilon\,, \\ \delta\lambda&=\left(\not\partial\phi+\frac{1}{2}\not H\mathcal{P}\right)\epsilon+\frac{1}{8}e^\phi\sum_n(-1)^n(5-n)\not F_n^{(10)}\,\mathcal{P}_n\epsilon\,.\end{aligned}$
cardona-qft-methods_ EQ0791 (32)	300	■ 0.292 ● W|+0	\begin{array}{rc} \mathrm{R}-\mathrm{R}: \xi_{M N} \psi_{0}^{M} \tilde{\psi}_{0}^{N}|0\rangle, & \mathrm{NS}-\mathrm{R}: \xi_{M N} \psi_{1 / 2}^{M} \tilde{\psi}_{0}^{N}|0\rangle, \\ \mathrm{NS}-\mathrm{NS}: \xi_{M N} \psi_{1 / 2}^{M} \tilde{\psi}_{1 / 2}^{N}|0\rangle, & \mathrm{R}-\mathrm{NS}: \xi_{M N} \psi_{0}^{M} \tilde{\psi}_{1 / 2}^{N}|0\rangle . \end{array}	$\begin{array}{rc}\mathrm{R}-\mathrm{R}:\xi_{MN}\psi_0^M\tilde\psi_0^N\vert 0\rangle, & \mathrm{NS}-\mathrm{R}:\xi_{MN}\psi_{1/2}^M\tilde\psi_0^N\vert 0\rangle, \\ \mathrm{NS}-\mathrm{NS}:\xi_{MN}\psi_{1/2}^M\tilde\psi_{1/2}^N\vert 0\rangle, & \mathrm{R}-\mathrm{NS}:\xi_{MN}\psi_0^M\tilde\psi_{1/2}^N\vert 0\rangle.\end{array}$	$\begin{array}{ll}\mathrm{R}-\mathrm{R}:\ \xi_{MN}\psi_0^M\tilde\psi_0^N\vert 0\rangle\,, & \mathrm{NS}-\mathrm{R}:\xi_{MN}\psi_{1/2}^M\tilde\psi_0^N\vert 0\rangle\,, \\ \mathrm{NS}-\mathrm{NS}:\xi_{MN}\psi_{1/2}^M\tilde\psi_{1/2}^N\vert 0\rangle\,, & \mathrm{R}-\mathrm{NS}:\xi_{MN}\psi_0^M\tilde\psi_{1/2}^N\vert 0\rangle\,.\end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_ EQ1012</td><td>377</td><td>■ 0.300 ● W|+0</td><td><code>\begin{aligned} &\left.\omega\right|_{x=0} \cong \chi_{0}(\omega)=\omega(1)=\langle\nu, \nu\rangle=1, \\ & \left.\partial_{x} \omega\right|_{x=0} \cong \chi_{0}(X \omega)=\left\langle\nu, X^{\dagger} \nu\right\rangle= \left\langle\binom{ 1}{0},\left(\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array}\right)\binom{1}{0}\right\rangle=0, \\ &\left.\partial_{x}^{2} \omega\right|_{x=0} \cong \chi_{0}\left(X^{2} \omega\right)=\left\langle\nu, \left(X^{\dagger}\right)^{2} \nu\right\rangle=\left \langle\binom{ 1}{0},\left(\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array}\right)\binom{1}{0}\right\rangle=-1, \end{aligned}</code></td><td>$$\begin{aligned} &\left.\omega\right|_{x=0} \cong \chi_{0}(\omega)=\omega(1)=\langle\nu, \nu\rangle=1, \\ & \left.\partial_{x} \omega\right|_{x=0} \cong \chi_{0}(X \omega)=\left\langle\nu, X^{\dagger} \nu\right\rangle= \left\langle\binom{ 1}{0},\left(\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array}\right)\binom{1}{0}\right\rangle=0, \\ &\left.\partial_{x}^{2} \omega\right|_{x=0} \cong \chi_{0}\left(X^{2} \omega\right)=\left\langle\nu, \left(X^{\dagger}\right)^{2} \nu\right\rangle=\left \langle\binom{ 1}{0},\left(\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array}\right)\binom{1}{0}\right\rangle=-1, \end{aligned}$$</td><td>$$\begin{aligned} \omega|_{x=0} &\cong \chi_0(\omega) = \omega(1) = \langle\nu,\nu\rangle = 1, \\ \partial_x\omega|_{x=0} &\cong \chi_0(X\omega) = \langle\nu, X^\dagger\nu\rangle = \langle (\begin{smallmatrix}1\\0\end{smallmatrix}), (\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}) (\begin{smallmatrix}1\\0\end{smallmatrix})\rangle = 0, \\ \partial_x^2\omega|_{x=0} &\cong \chi_0(X^2\omega) = \langle\nu, (X^\dagger)^2\nu\rangle = \langle (\begin{smallmatrix}1\\0\end{smallmatrix}), (\begin{smallmatrix}-1&0\\0&-1\end{smallmatrix}) (\begin{smallmatrix}1\\0\end{smallmatrix})\rangle = -1, \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0681</td><td>269</td><td>■ 0.304 ● N|+1</td><td><code>\begin{gathered} " \text { Group" } \approx e^{i \text { "Algebra" }}, \\ S U(2) \approx e^{i \mathfrak{s u}(2)} \end{gathered}</code></td><td>$$\begin{gathered} " \text { Group" } \approx e^{i \text { "Algebra" }}, \\ S U(2) \approx e^{i \mathfrak{s u}(2)} \end{gathered}$$</td><td>$$\begin{gathered} \text{“}Group\text{”} \approx e^{i \text{“}Algebra\text{”}}, \\ SU(2) \approx e^{i\mathfrak{su}(2)} . \end{gathered}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0158 (4)</td><td>087</td><td>■ 0.311 ● N|+0</td><td><code>\mathrm{t}-\operatorname{Ind}=i_{!}^{-1} j_{!}: K \left(T^{*} M\right) \rightarrow K(\{p t\}) \simeq \mathbb{Z} .</code></td><td>$$\mathrm{t}-\operatorname{Ind}=i_{!}^{-1} j_{!}: K \left(T^{*} M\right) \rightarrow K(\{p t\}) \simeq \mathbb{Z} .$$</td><td>$$\text{t-Ind} \;=\; i_!^{-1} j_! : K(T^*M) \to K(\{pt\}) \;\simeq\; \mathbb{Z}.$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0501</td><td>226</td><td>■ 0.333 ● N|+0</td><td><code>\left.\sum_{\text {generations }}\left(m_{\nu}^{2}+m_{e}^{2}+3 m_{u}^{2}+3 m_{d}^{2}\right) \right|_{\Lambda=\Lambda_{u n i f}}=\left.8 M_{W}^{2} \right|_{\Lambda=\Lambda_{u n i f}}</code></td><td>$$\left.\sum_{\text {generations }}\left(m_{\nu}^{2}+m_{e}^{2}+3 m_{u}^{2}+3 m_{d}^{2}\right) \right|_{\Lambda=\Lambda_{u n i f}}=\left.8 M_{W}^{2} \right|_{\Lambda=\Lambda_{u n i f}}$$</td><td>$$\sum_{generations} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2)|_{\Lambda=\Lambda_{unif}} = 8M_W^2|_{\Lambda=\Lambda_{unif}}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0049 (14)</td><td>035</td><td>■ 0.362 ● N|+0</td><td><code>\left.\mathcal{H}=L^{2}((M, g), S)\right)=\left\{\psi \in \Gamma^{\infty}(M, S) \mid \int_{M}\langle\psi, \psi \rangle_{x} d \operatorname{vol}_{g}(x)<\infty\right\}</code></td><td>$$\left.\mathcal{H}=L^{2}((M, g), S)\right)=\left\{\psi \in \Gamma^{\infty}(M, S) \mid \int_{M}\langle\psi, \psi \rangle_{x} d \operatorname{vol}_{g}(x)<\infty\right\}$$</td><td>$$\mathcal{H} = L^2\big((M,g),S\big) = \{\, \psi \in \Gamma^\infty(M,S) \mid \int_M \langle\psi,\psi\rangle_x \, dvol_g(x) < \infty \,\}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0795 (36)</td><td>301</td><td>■ 0.396 ● W|+0</td><td><code>\begin{aligned} & \operatorname{type} \text { IIB } \mathrm{NS}-\mathrm{R}: \mathbf{8}_{v} \otimes \mathbf{8}_{s}=\mathbf{8}_{s} \oplus \mathbf{5 6}_{s}= \lambda^{1} \oplus \Psi^{1 M}, \\ & \mathrm{R}- \mathrm{NS}: \mathbf{8}_{s} \otimes \mathbf{8}_{v}= \mathbf{8}_{s} \oplus \mathbf{5 6}_{s}=\lambda^{2} \oplus \Psi^{2 M}, \\ & \text { type IIA } \mathrm{NS}- \mathrm{R}: \mathbf{8}_{v} \otimes \mathbf{8}_{c}= \mathbf{8}_{c} \oplus \mathbf{5 6}_{c}=\lambda^{1} \oplus \Psi^{1 M}, \\ & \mathrm{R}-\mathrm{NS}: \mathbf{8}_{s} \otimes \mathbf{8}_{v}=\mathbf{8}_{s} \oplus \mathbf{5 6}_{s}=\lambda^{2} \oplus \Psi^{2 M} . \end{aligned}</code></td><td>$$\begin{aligned} & \operatorname{type} \text { IIB } \mathrm{NS}-\mathrm{R}: \mathbf{8}_{v} \otimes \mathbf{8}_{s}=\mathbf{8}_{s} \oplus \mathbf{5 6}_{s}= \lambda^{1} \oplus \Psi^{1 M}, \\ & \mathrm{R}- \mathrm{NS}: \mathbf{8}_{s} \otimes \mathbf{8}_{v}= \mathbf{8}_{s} \oplus \mathbf{5 6}_{s}=\lambda^{2} \oplus \Psi^{2 M}, \\ & \text { type IIA } \mathrm{NS}- \mathrm{R}: \mathbf{8}_{v} \otimes \mathbf{8}_{c}= \mathbf{8}_{c} \oplus \mathbf{5 6}_{c}=\lambda^{1} \oplus \Psi^{1 M}, \\ & \mathrm{R}-\mathrm{NS}: \mathbf{8}_{s} \otimes \mathbf{8}_{v}=\mathbf{8}_{s} \oplus \mathbf{5 6}_{s}=\lambda^{2} \oplus \Psi^{2 M} . \end{aligned}$$</td><td>$$\begin{aligned} \text{type IIB NS} - \mathrm{R}: &\;\; \mathbf{8}_v \otimes \mathbf{8}_s = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^1 \oplus \Psi^{1\,M}, \\ \mathrm{R} - \mathrm{NS}: &\;\; \mathbf{8}_s \otimes \mathbf{8}_v = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^2 \oplus \Psi^{2\,M}, \\ \text{type IIA NS} - \mathrm{R}: &\;\; \mathbf{8}_v \otimes \mathbf{8}_c = \mathbf{8}_c \oplus \mathbf{56}_c = \lambda^1 \oplus \Psi^{1\,M}, \\ \mathrm{R} - \mathrm{NS}: &\;\; \mathbf{8}_s \otimes \mathbf{8}_v = \mathbf{8}_s \oplus \mathbf{56}_s = \lambda^2 \oplus \Psi^{2\,M}. \end{aligned}$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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<tr><td>cardona-qft-methods_EQ0028</td><td>027</td><td>0.431 W|+1</td><td><code>\begin{aligned} W R e s\left((1+\Delta)^{-d / 2}\right) & =\int_{M}|d x| \int_{\mathbb{S}^{d-1}}\|\xi\|_{x}^{-d} d \xi=\int_{M}|d x| \sqrt{\operatorname{det} g_{x}} \operatorname{Vol}\left(\mathbb{S}^{\mathrm{d}-1}\right) \\ & =\operatorname{Vol}\left(\mathbb{S}^{\mathrm{d}-1}\right) \int_{\mathrm{M}}\left|\operatorname{dvolg}_{\mathrm{g}}\right|=\operatorname{Vol}\left(\mathbb{S}^{\mathrm{d}-1}\right) \end{aligned}</code></td><td>$\begin{aligned} W R e s\left((1+\Delta)^{-d / 2}\right) & =\int_{M}|d x| \int_{\mathbb{S}^{d-1}}\|\xi\|_{x}^{-d} d \xi=\int_{M}|d x| \sqrt{\operatorname{det} g_{x}} \operatorname{Vol}\left(\mathbb{S}^{\mathrm{d}-1}\right) \\ & =\operatorname{Vol}\left(\mathbb{S}^{\mathrm{d}-1}\right) \int_{\mathrm{M}}\left|\operatorname{dvolg}_{\mathrm{g}}\right|=\operatorname{Vol}\left(\mathbb{S}^{\mathrm{d}-1}\right) \end{aligned}$</td><td>$\begin{aligned} WRes\big((1+\Delta)^{-d/2}\big) & = \int_M |dx| \int_{\mathbb{S}^{d-1}} \|\xi\|_x^{-d}\, d\xi = \int_M |dx| \sqrt{det\, g_x}\, \mathrm{Vol}(\mathbb{S}^{\mathrm{d}-1}) \\ & = \mathrm{Vol}(\mathbb{S}^{\mathrm{d}-1}) \int_{\mathrm{M}} |\mathrm{dvol}_{\mathrm{g}}| = \mathrm{Vol}(\mathbb{S}^{\mathrm{d}-1}). \end{aligned}$ □</td></tr>
<tr><td>cardona-qft-methods_EQ0148 (1)</td><td>083</td><td>0.450 N|+1</td><td><code>\left.\mathcal{D} \rightsquigarrow \sigma_{L}(\mathcal{D}) \rightsquigarrow \text { an element of } K \text {-theory } \rightsquigarrow \mathrm{t} \text { - } \operatorname{Ind}\left(\sigma_{L}(\mathcal{D})\right)\right) \in \mathbb{Z} \text {. }</code></td><td>$\mathcal{D} \rightsquigarrow \sigma_{L}(\mathcal{D}) \rightsquigarrow \text { an element of } K \text {-theory } \rightsquigarrow \mathrm{t} \text { - } \operatorname{Ind}\left(\sigma_{L}(\mathcal{D})\right)) \in \mathbb{Z} \text {. }$</td><td>$\mathcal{D} \rightsquigarrow \sigma_L(\mathcal{D}) \rightsquigarrow \text{ an element of } K\text{-theory} \rightsquigarrow \text{t-Ind}(\sigma_L(\mathcal{D}))) \in \mathbb{Z}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0515 (1)</td><td>236</td><td>0.478 N|-1</td><td><code>\lambda_{C}=\frac{\hbar}{M c} \leq L \text { or } M \geq \frac{\hbar}{L c} \simeq 10^{19} \mathrm{GeV} \text { (Planck mass) } \simeq 10^{-5} \text { grams } .</code></td><td>$\lambda_{C}=\frac{\hbar}{M c} \leq L \text { or } M \geq \frac{\hbar}{L c} \simeq 10^{19} \mathrm{GeV} \text { (Planck mass) } \simeq 10^{-5} \text { grams } .$</td><td>$\lambda_C = \frac{\hbar}{Mc} \leq L \text{ or } M \geq \frac{\hbar}{Lc} \simeq 10^{19}\text{GeV (Planck mass)} \simeq 10^{-5}\text{grams.}$</td></tr>
<tr><td>cardona-qft-methods_EQ0019 (7)</td><td>024</td><td>0.490 N|-1</td><td><code>W \operatorname{Res} P=\underset{s=0}{\operatorname{Res}} \operatorname{Tr}\left(P|\mathcal{D}|^{-s}\right) .</code></td><td>$W \operatorname{Res} P=\underset{s=0}{\operatorname{Res}} \operatorname{Tr}\left(P|\mathcal{D}|^{-s}\right) .$</td><td>$WResP = \underset{s=0}{\mathrm{Res}}\, \mathrm{Tr}\big(P|\mathcal{D}|^{-s}\big).$</td></tr>
<tr><td>cardona-qft-methods_EQ0324 (10)</td><td>161</td><td>0.495 W|+2</td><td><code>\begin{aligned} \phi_{\|}^{a}(x)-\phi^{a}(x) & =d \phi^{a}(x)+\omega_{b}^{a}(x) \phi^{b}(x)=D \phi^{a}(x) \\ \omega^{\prime a}(x) & =\Lambda_{c}^{a}(x) \Lambda_{b}^{d}(x) \omega_{d}^{c}(x)+\Lambda_{c}^{a}(x) d \Lambda_{b}^{c}(x) \end{aligned}</code></td><td>$\begin{aligned} \phi_{\|}^{a}(x)-\phi^{a}(x) & =d \phi^{a}(x)+\omega_{b}^{a}(x) \phi^{b}(x)=D \phi^{a}(x) \\ \omega^{\prime a}(x) & =\Lambda_{c}^{a}(x) \Lambda_{b}^{d}(x) \omega_{d}^{c}(x)+\Lambda_{c}^{a}(x) d \Lambda_{b}^{c}(x) \end{aligned}$</td><td>$\begin{aligned} \phi^a_{\|}(x) - \phi^a(x) & = d\phi^a(x) + \omega^a_b(x)\phi^b(x) = D\phi^a(x) \\ \omega'^a_{\ b}(x) & = \Lambda^a_c(x)\Lambda^d_b(x)\omega^c_{\ d}(x) + \Lambda^a_c(x) d\Lambda^c_b(x), \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_EQ0481</td><td>219</td><td>0.498 K|+0</td><td><code>P f(\mathfrak{A})=\int e^{-\frac{1}{2} \mathfrak{A}(\tilde{\xi})} D[\tilde{\xi}],</code></td><td>$P f(\mathfrak{A})=\int e^{-\frac{1}{2} \mathfrak{A}(\tilde{\xi})} D[\tilde{\xi}],$</td><td>$Pf(\mathfrak{A}) = \int e^{-\frac{1}{2}\mathfrak{A}(\tilde{\xi})} D[\tilde{\xi}],$</td></tr>
<tr><td>cardona-qft-methods_EQ1001</td><td>375</td><td>0.508 W|+0</td><td><code>\begin{aligned} X(a b) & =(a \otimes b) \circ\left(X^{\dagger}(\cdot) \otimes \operatorname{Id}+\operatorname{Id} \otimes X^{\dagger}(\cdot)\right) \circ \Delta \\ & =(X a \otimes b+a \otimes X b) \circ \Delta=(X a) b+a(X b) . \end{aligned}</code></td><td>$\begin{aligned} X(a b) & =(a \otimes b) \circ\left(X^{\dagger}(\cdot) \otimes \operatorname{Id}+\operatorname{Id} \otimes X^{\dagger}(\cdot)\right) \circ \Delta \\ & =(X a \otimes b+a \otimes X b) \circ \Delta=(X a) b+a(X b) . \end{aligned}$</td><td>$\begin{aligned} X(ab) & = (a \otimes b) \circ \big(X^\dagger(\cdot) \otimes \mathrm{Id} + \mathrm{Id} \otimes X^\dagger(\cdot)\big) \circ \Delta \\ & = (Xa \otimes b + a \otimes Xb) \circ \Delta = (Xa)b + a(Xb). \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_EQ0712</td><td>274</td><td>0.516 N|+0</td><td><code>R_{a, b}(u)=u 1_{a, b}+i P_{a, b}=\left(\begin{array}{cccc} u+i & 0 & 0 & 0 \\ 0 & u & i & 0 \\ 0 & i & u & 0 \\ 0 & 0 & 0 & u+i \end{array}\right),</code></td><td>$R_{a, b}(u)=u 1_{a, b}+i P_{a, b}=\left(\begin{array}{cccc} u+i & 0 & 0 & 0 \\ 0 & u & i & 0 \\ 0 & i & u & 0 \\ 0 & 0 & 0 & u+i \end{array}\right),$</td><td>$R_{a,b}(u) = u1_{a,b} + iP_{a,b} = \begin{pmatrix} u+i & 0 & 0 & 0 \\ 0 & u & i & 0 \\ 0 & i & u & 0 \\ 0 & 0 & 0 & u+i \end{pmatrix},$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0890	345	0.517 C|+5	`\begin{aligned} & H_{*}(L M) \otimes H_{*}(L M) \rightarrow H_{*}(L M \times L M) \\ & H_{*} \left(\operatorname{Map}(8, M)^{e v^{*}(T M)} \right) \xrightarrow{\tau_{\Delta *}} H_{*-n}( \operatorname{Map}(8, M)) \\ & \downarrow \\ & H_{*-n}(L M) \end{aligned}`	$\begin{aligned} & H_*(LM)\otimes H_*(LM)\rightarrow H_*(LM\times LM)\\ & H_*\left(\operatorname{Map}(8,M)^{ev^*(TM)}\right)\xrightarrow{\tau_{\Delta *}} H_{*-n}(\operatorname{Map}(8,M))\\ & \downarrow\\ & H_{*-n}(LM)\end{aligned}$	$\begin{tikzcd} H_*(LM)\otimes H_*(LM) \arrow[r] & H_*(LM\times LM) \arrow[d, "\tau_{\Delta *}"'] & \\ & H_*(Map(8,M)^{ev^*(TM)}) \arrow[r, "Thom"] & H_{*-n}(Map(8,M)) \arrow[d] \\ & & H_{*-n}(LM) \end{tikzcd}$
cardona-qft-methods_ EQ0012	020	0.522 N|+0	`\sigma_{m}^{P}(x, \xi)=\lim _{t \rightarrow \infty} t^{-m} e^{-i \operatorname{th}(x)} P e^{i \operatorname{th}(x)},`	$\sigma_m^P(x,\xi)=\lim_{t\rightarrow\infty} t^{-m}e^{-i\operatorname{th}(x)}Pe^{i\operatorname{th}(x)},$	$\sigma_m^P(x,\xi)=\lim_{t\to\infty} t^{-m}\, e^{-ith(x)}\, P\, e^{ith(x)},$
cardona-qft-methods_ EQ0201	096	0.522 N|+0	`\langle d \alpha, \beta\rangle=\left\langle\alpha, d^{*} \beta\right\rangle, \quad \text { for all } \quad \alpha, \beta \in \Omega^{*}(M, E) .`	$\langle d\alpha,\beta\rangle=\left\langle\alpha,d^*\beta\right\rangle, \quad \text{for all} \quad \alpha,\beta\in\Omega^*(M,E).$	$\langle d\alpha,\beta\rangle \;=\; \langle\alpha,d^*\beta\rangle, \qquad \text{for all} \quad \alpha,\beta\in\Omega^*(M,E).$
cardona-qft-methods_ EQ0895	346	0.523 K|+0	`L M^{e v^{*}(-T M)} \wedge L M^{e v^{*}(-T M)} \rightarrow \operatorname{Map}(8, M)^{e v^{*}(-T M)} .`	$LM^{ev^*(-TM)}\wedge LM^{ev^*(-TM)}\rightarrow\operatorname{Map}(8,M)^{ev^*(-TM)}.$	$LM^{ev^*(-TM)}\wedge LM^{ev^*(-TM)}\to Map(8,M)^{ev^*(-TM)}.$
cardona-qft-methods_ EQ0099	057	0.523 W|+1	`\begin{aligned} \operatorname{Tr}\left(f\left(\mathcal{D}^{2} / \Lambda^{2}\right)\right) \sim \frac{1}{(4 \pi)^{2}} & {\left[\operatorname{rk}(\mathrm{E}) \int_{0}^{\infty} x f(x) d x \Lambda^{4}+b_{2}\left(\mathcal{D}^{2}\right) \int_{0}^{ \infty} f(x) d x \Lambda^{2}\right.} \\ & \left.+ \sum_{m=0}^{\infty}(-1)^{m} f^{(m)}(0) b_{2 m+4}\left(\mathcal{D}^{2}\right) \Lambda^{-2 m}\right], \quad \Lambda \rightarrow \infty \end{aligned}`	$\begin{aligned}\operatorname{Tr}\left(f\left(\mathcal{D}^2/\Lambda^2\right)\right)\sim\frac{1}{(4\pi)^2} & \left[\operatorname{rk}(\mathrm{E})\int_0^\infty xf(x)dx\Lambda^4+b_2\left(\mathcal{D}^2\right)\int_0^\infty f(x)dx\Lambda^2\right.\\ & \left.+\sum_{m=0}^\infty(-1)^m f^{(m)}(0)b_{2m+4}\left(\mathcal{D}^2\right)\Lambda^{-2m}\right], \quad \Lambda\rightarrow\infty\end{aligned}$	$\begin{aligned}\mathrm{Tr}\big(f(\mathcal{D}^2/\Lambda^2)\big)\sim\tfrac{1}{(4\pi)^2}\Big[\mathrm{rk(E)}\int_0^\infty xf(x)\,dx\,\Lambda^4+b_2(\mathcal{D}^2)\int_0^\infty f(x)\,dx\,\Lambda^2\\ +\sum_{m=0}^\infty(-1)^m f^{(m)}(0)\,b_{2m+4}(\mathcal{D}^2)\,\Lambda^{-2m}\Big], \quad \Lambda\to\infty\end{aligned}$
cardona-qft-methods_ EQ0154 (3)	085	0.523 K|+0	`\sigma_{L}(\mathcal{D}): \pi^{\star} E \rightarrow \pi^{\star} F .`	$\sigma_L(\mathcal{D}):\pi^\star E\rightarrow\pi^\star F.$	$\sigma_L(\mathcal{D}):\pi^\star E \;\;\to\;\; \pi^\star F.$
cardona-qft-methods_ EQ0919	352	0.527 K|+0	`C_{*}\left(P_{G} M^{-T M}\right) \stackrel{\simeq}{ \rightarrow} \mathcal{R} \mathcal{H o m}_{C^{*}(M)^{e}} \left(C^{*}(M), C^{*}(M) \# G\right)`	$C_*\left(P_GM^{-TM}\right)\stackrel{\simeq}{\rightarrow}\mathcal{R}\mathcal{H}\Updownarrow_{C^*(M)^e}\left(C^*(M),C^*(M)\#G\right)$	$C_*(P_GM^{-TM})\stackrel{\simeq}{\to}\mathcal{RH}om_{C^*(M)^e}(C^*(M),C^*(M)\#G)$
cardona-qft-methods_ EQ0356 (25)	171	0.527 S|+0	`\begin{aligned} \mathbf{P}_{8} & =R_{a_{2}}^{a_{1}} R_{a_{3}}^{a_{2}} \cdots R_{a_{1}}^{a_{4}}, \\ \left( \mathbf{P}_{4}\right)^{2} & =\left(R_{b}^{a} R_{a}^{b} \right)^{2}, \end{aligned}`	$\begin{aligned}\mathbf{P}_8 & =R_{a_2}^{a_1}R_{a_3}^{a_2}\cdots R_{a_1}^{a_4},\\ \left(\mathbf{P}_4\right)^2 & =\left(R_b^aR_a^b\right)^2,\end{aligned}$	$\begin{aligned}\mathbf{P}_8 &= R^{a_1}_{a_2}R^{a_2}_{a_3}\cdots R^{a_4}_{a_1},\\ (\mathbf{P}_4)^2 &= (R^a_bR^b_a)^2,\end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_ EQ0424</td><td>202</td><td>0.529 N|+0</td><td><code>\begin{aligned} C_{\lambda \mu \nu \kappa}=R_{\lambda \mu \nu \kappa} & -\frac{1}{2}\left(g_{\lambda \nu} R_{\mu \kappa}-g_{\lambda \kappa} R_{\mu \nu}-g_{\mu \nu} R_{ \lambda \kappa}+g_{\mu \kappa} R_{\lambda \nu}\right) \ \ & +\frac{1}{6} R_{\alpha}^{\alpha}\left(g_{\lambda \nu} g_{\mu \kappa}-g_{\lambda \kappa} g_{\mu \nu}\right) . \end{aligned}</code></td><td>$$\begin{aligned} C_{\lambda \mu \nu \kappa}=R_{\lambda \mu \nu \kappa} & -\frac{1}{2}\left(g_{\lambda \nu} R_{\mu \kappa}-g_{\lambda \kappa} R_{\mu \nu}-g_{\mu \nu} R_{\lambda \kappa}+g_{\mu \kappa} R_{\lambda \nu}\right) \\ & +\frac{1}{6} R_{\alpha}^{\alpha}\left(g_{\lambda \nu} g_{\mu \kappa}-g_{\lambda \kappa} g_{\mu \nu}\right) . \end{aligned}$$</td><td>$$C_{\lambda\mu\nu\kappa} = R_{\lambda\mu\nu\kappa} - \frac{1}{2}(g_{\lambda\nu}R_{\mu\kappa} - g_{\lambda\kappa}R_{\mu\nu} - g_{\mu\nu}R_{\lambda\kappa} + g_{\mu\kappa}R_{\lambda\nu}) + \frac{1}{6}R^{\alpha}_{\alpha}(g_{\lambda\nu}g_{\mu\kappa} - g_{\lambda\kappa}g_{\mu\nu}).$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0026</td><td>027</td><td>0.531 N|+1</td><td><code>T(P)=\mu\left(c_{P}(x)\right)=c \int_{M} c_{P}(x)|d x|=c W R e s(P)</code></td><td>$T(P)=\mu\left(c_{P}(x)\right)=c \int_{M} c_{P}(x)|d x|=c W R e s(P)$</td><td>$T(P) = \mu(c_P(x)) = c \int_M c_P(x)\,|dx| = c\, WRes(P).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0747</td><td>283</td><td>0.535 N|+0</td><td><code>\mathcal{D}=\mathcal{D}_{0}+\frac{g_{\mathrm{YM}}^{2}}{16 \pi^{2}} \hat{H}_{X X X}+O\left(g_{\mathrm{YM}}^{4} \right) .</code></td><td>$\mathcal{D}=\mathcal{D}_{0}+\frac{g_{\mathrm{YM}}^{2}}{16 \pi^{2}} \hat{H}_{X X X}+O\left(g_{\mathrm{YM}}^{4}\right) .$</td><td>$\mathcal{D} = \mathcal{D}_0 + \frac{g^2_{\mathrm{YM}}}{16\pi^2}\hat{H}_{XXX} + O(g^4_{\mathrm{YM}})\,.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0831 (10)</td><td>314</td><td>0.539 C|+3</td><td><code>\begin{aligned} & \epsilon_{\mathrm{IIA}}^{1}= \xi_{+}^{1} \otimes \eta_{+}^{1}+\xi_{-}^{1} \otimes \eta_{-}^{1} \\ & \epsilon_{\mathrm{IIA}}^{2}=\xi_{+}^{2} \otimes \eta_{-}^{2}+\xi_{-}^{2} \otimes \eta_{+}^{2} \end{aligned}</code></td><td>$$\begin{aligned} & \epsilon_{\mathrm{IIA}}^{1}=\xi_{+}^{1} \otimes \eta_{+}^{1}+\xi_{-}^{1} \otimes \eta_{-}^{1} \\ & \epsilon_{\mathrm{IIA}}^{2}=\xi_{+}^{2} \otimes \eta_{-}^{2}+\xi_{-}^{2} \otimes \eta_{+}^{2} \end{aligned}$$</td><td>$$\begin{aligned} &\epsilon^1_{\mathrm{IIA}} = \xi^1_+ \otimes \eta^1_+ + \xi^1_- \otimes \eta^1_-\ , \\ &\epsilon^2_{\mathrm{IIA}} = \xi^2_+ \otimes \eta^2_- + \xi^2_- \otimes \eta^2_+\ , \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0834 (13)</td><td>317</td><td>0.540 N|+1</td><td><code>\begin{aligned} & e^{-2 A+\phi}(\mathrm{d}+H \wedge) \left(e^{2 A-\phi} \Phi_{+}\right)=0 \\ & e^{-2 A+\phi}(\mathrm{d}+H \wedge)\left(e^{2 A-\phi} \Phi_{-}\right)= \mathrm{d} A \wedge \bar{\Phi}_{-}+\frac{\mathrm{i}}{16} e^{\phi+A} * F_{\text {IIA }}+ \end{aligned}</code></td><td>$$\begin{aligned} & e^{-2 A+\phi}(\mathrm{d}+H \wedge)\left(e^{2 A-\phi} \Phi_{+}\right)=0 \\ & e^{-2 A+\phi}(\mathrm{d}+H \wedge)\left(e^{2 A-\phi} \Phi_{-}\right)=\mathrm{d} A \wedge \bar{\Phi}_{-}+\frac{\mathrm{i}}{16} e^{\phi+A} * F_{\text {IIA }}+ \end{aligned}$$</td><td>$$\begin{aligned} &e^{-2A+\phi}(\mathrm{d}+H\wedge)(e^{2A-\phi}\Phi_+) = 0\,, \\ &e^{-2A+\phi}(\mathrm{d}+H\wedge)(e^{2A-\phi}\Phi_-) = \mathrm{d}A\wedge\bar{\Phi}_- + \frac{\mathrm{i}}{16}e^{\phi+A} * F_{\mathrm{IIA}} + \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0031</td><td>029</td><td>0.541 W|+1</td><td><code>\begin{aligned} & \sigma_{N}\left(T_{1}+T_{2}\right) \leq \sigma_{N}\left(T_{1}\right)+\sigma_{N}\left(T_{2}\right) \\ & \sigma_{N_{1}}\left(T_{1}\right)+\sigma_{N_{2}} \left(T_{2}\right) \leq \sigma_{N_{1}+N_{2}} \left(T_{1}+T_{2}\right) \text { when } T_{1}, T_{2} \geq 0 . \end{aligned}</code></td><td>$$\begin{aligned} & \sigma_{N}\left(T_{1}+T_{2}\right) \leq \sigma_{N}\left(T_{1}\right)+\sigma_{N}\left(T_{2}\right) \\ & \sigma_{N_{1}}\left(T_{1}\right)+\sigma_{N_{2}}\left(T_{2}\right) \leq \sigma_{N_{1}+N_{2}}\left(T_{1}+T_{2}\right) \text { when } T_{1}, T_{2} \geq 0 . \end{aligned}$$</td><td>$$\begin{aligned} &\sigma_N(T_1+T_2) \le \sigma_N(T_1) + \sigma_N(T_2), \\ &\sigma_{N_1}(T_1) + \sigma_{N_2}(T_2) \le \sigma_{N_1+N_2}(T_1+T_2) \text{ when } T_1, T_2 \ge 0. \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0748</td><td>284</td><td>0.547 C|-27</td><td><code>\begin{aligned} A d S_{5} & : & -R^{2} & =-Z_{0}^{2}+Z_{1}^{2}+Z_{2}^{2}+Z_{3}^{2}+Z_{4}^{2}-Z_{5}^{2} \\ S^{5} & : & R^{2} & =Y_{1}^{2}+Y_{2}^{2}+Y_{3}^{2}+Y_{4}^{2}+Y_{5}^{2}+Y_{6}^{2} \end{aligned}</code></td><td>$$\begin{aligned} A d S_{5} & : & -R^{2} & =-Z_{0}^{2}+Z_{1}^{2}+Z_{2}^{2}+Z_{3}^{2}+Z_{4}^{2}-Z_{5}^{2} \\ S^{5} & : & R^{2} & =Y_{1}^{2}+Y_{2}^{2}+Y_{3}^{2}+Y_{4}^{2}+Y_{5}^{2}+Y_{6}^{2} \end{aligned}$$</td><td>$$\begin{aligned} AdS_5 :&\quad -R^2 = -Z_0^2 + Z_1^2 + Z_2^2 + Z_3^2 + Z_4^2 - Z_5^2\,, \\ S^5 :&\quad R^2 = Y_1^2 + Y_2^2 + Y_3^2 + Y_4^2 + Y_5^2 + Y_6^2\,. \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0936</td><td>354</td><td>0.552 N|+1</td><td><code>H H^{*}(\mathbb{Z} G, \mathbb{Z} G) \cong H_{*}^{ \mathrm{pro}}\left(L B G^{-T B G}\right) .</code></td><td>$H H^{*}(\mathbb{Z} G, \mathbb{Z} G) \cong H_{*}^{\mathrm{pro}}\left(L B G^{-T B G}\right) .$</td><td>$HH^*(\mathbb{Z}G, \mathbb{Z}G) \cong H^{\mathrm{pro}}_*(LBG^{-TBG}).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0778 (19)</td><td>297</td><td>0.557 W|+0</td><td><code>T_{\alpha}^{\alpha}{ }_{\alpha}=-\frac{1}{2 \alpha^{ \prime}} \beta_{M N}^{G} g^{\alpha \beta} \partial_{ \alpha} X^{M} \partial_{\beta} X^{N}-\frac{i}{2 \alpha^{\prime}} \beta_{M N}^{B} \epsilon^{\alpha \beta} \partial_{\alpha} X^{M} \partial_{\beta} X^{N}- \frac{1}{2} \beta^{\Phi} R .</code></td><td>$T_{\alpha}^{\alpha}{ }_{\alpha}=-\frac{1}{2 \alpha^{\prime}} \beta_{M N}^{G} g^{\alpha \beta} \partial_{\alpha} X^{M} \partial_{\beta} X^{N}-\frac{i}{2 \alpha^{\prime}} \beta_{M N}^{B} \epsilon^{\alpha \beta} \partial_{\alpha} X^{M} \partial_{\beta} X^{N}-\frac{1}{2} \beta^{\Phi} R .$</td><td>$T^{\alpha}{}_{\alpha} = -\frac{1}{2\alpha'}\beta^G_{MN}g^{\alpha\beta}\partial_\alpha X^M\partial_\beta X^N - \frac{i}{2\alpha'}\beta^B_{MN}\epsilon^{\alpha\beta}\partial_\alpha X^M\partial_\beta X^N - \frac{1}{2}\beta^\Phi R\,.$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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EQ0149</td><td>083</td><td>■ 0.565
● K|+0</td><td><code>\text { elliptic operators } \rightarrow \mathbb{Z}, \quad \mathcal{D} \mapsto \text { t-Ind }\left(\sigma_{L}(\mathcal{D})\right)</code></td><td>$\text{ elliptic operators } \rightarrow \mathbb{Z}, \quad \mathcal{D} \mapsto \text{ t-Ind }\left(\sigma_{L}(\mathcal{D})\right)$</td><td>$\text{elliptic operators } \rightarrow \mathbb{Z}, \qquad \mathcal{D} \mapsto \text{ t-Ind}(\sigma_{L}(\mathcal{D}))$</td></tr>
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EQ0978 (18)</td><td>364</td><td>■ 0.570
● N|+0</td><td><code>\left[\mathcal{S}_{n}\right]=\left[\Sigma_{n}\right]-n \mathbb{T}^{n-2}=\sum_{i=2}^{n-1}\binom{n}{i} \mathbb{T}^{n-1-i} .</code></td><td>$\left[\mathcal{S}_{n}\right]=\left[\Sigma_{n}\right]-n \mathbb{T}^{n-2}=\sum_{i=2}^{n-1}\binom{n}{i} \mathbb{T}^{n-1-i}.$</td><td>$[\mathcal{S}_{n}]=[\Sigma_{n}]-n \mathbb{T}^{n-2}=\sum_{i=2}^{n-1}\binom{n}{i} \mathbb{T}^{n-1-i}.$</td></tr>
<tr><td>cardona-qft-methods_
EQ0558</td><td>246</td><td>■ 0.574
● N|+0</td><td><code>\begin{aligned} \Delta(\eta)=\sum_{\alpha} \eta_{ \alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} & \equiv \eta_{ \alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} \\ & \equiv \eta^{(1)} \otimes \eta^{(2)} . \end{aligned}</code></td><td>$\begin{aligned} \Delta(\eta)=\sum_{\alpha} \eta_{\alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} & \equiv \eta_{\alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} \\ & \equiv \eta^{(1)} \otimes \eta^{(2)}. \end{aligned}$</td><td>$\begin{aligned} \Delta(\eta)=\sum_{\alpha} \eta_{\alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} & \equiv \eta_{\alpha}^{(1)} \otimes \eta_{\alpha}^{(2)} \\ & \equiv \eta^{(1)} \otimes \eta^{(2)}. \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_
EQ0548</td><td>244</td><td>■ 0.577
● W|+1</td><td><code>\begin{aligned} & m_{r}(\xi \otimes \eta \otimes \rho)= \eta \xi \otimes \rho \\ & m_{r}^{\prime}(\xi \otimes \eta \otimes \rho)=\rho \otimes \eta \xi \end{aligned}</code></td><td>$\begin{aligned} & m_{r}(\xi \otimes \eta \otimes \rho)=\eta \xi \otimes \rho \\ & m_{r}^{\prime}(\xi \otimes \eta \otimes \rho)=\rho \otimes \eta \xi \end{aligned}$</td><td>$\begin{aligned} & m_{r}(\xi \otimes \eta \otimes \rho)=\eta \xi \otimes \rho \\ & m_{r}^{\prime}(\xi \otimes \eta \otimes \rho)=\rho \otimes \eta \xi. \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_
EQ0035 (10)</td><td>029</td><td>■ 0.578
● K|+0</td><td><code>\mathcal{L}^{1}(\mathcal{H}) \subset \mathcal{L}^{1, \infty} \subset \mathcal{L}^{p}(\mathcal{H}) \quad \text { for } p>1, \quad\|T\| \leq\|T\|_{1, \infty} \leq\|T \|_{1} .</code></td><td>$\mathcal{L}^{1}(\mathcal{H}) \subset \mathcal{L}^{1, \infty} \subset \mathcal{L}^{p}(\mathcal{H}) \quad \text{ for } p>1, \quad\|T\| \leq\|T\|_{1, \infty} \leq\|T\|_{1}.$</td><td>$\mathcal{L}^{1}(\mathcal{H}) \subset \mathcal{L}^{1, \infty} \subset \mathcal{L}^{p}(\mathcal{H}) \quad \text{for } p>1, \qquad \|T\| \leq \|T\|_{1, \infty} \leq \|T\|_{1}.$</td></tr>
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EQ0674</td><td>267</td><td>■ 0.578
● N|+0</td><td><code>Z_{s t r}\left[\left.\phi\right|_{\partial A d S}=J \right]=Z_{C F T}[J]</code></td><td>$Z_{s t r}\left[\left.\phi\right|_{\partial A d S}=J\right]=Z_{C F T}[J]$</td><td>$Z_{str}[\phi\,|_{\partial\, AdS}= J]=Z_{CFT}[J]$</td></tr>
<tr><td>cardona-qft-methods_
EQ0735 (3)</td><td>280</td><td>■ 0.578
● W|+0</td><td><code>\begin{aligned} {\left[M_{\mu \nu}, M_{\rho \sigma}\right] } & =i\left(\eta_{\mu \rho} M_{\nu \sigma}+\eta_{ \nu \sigma} M_{\mu \rho}-\eta_{\mu \sigma} M_{\nu \rho}- \eta_{\nu \rho} M_{\mu \sigma}\right), \\ {\left[M_{\mu \nu}, P_{\rho}\right] } & =i\left(\eta_{\rho \nu} P_{ \mu}-\eta_{\mu \rho} P_{\nu}\right), \\ {\left[P_{\mu}, P_{\nu}\right] } & =0 . \end{aligned}</code></td><td>$\begin{aligned} {\left[M_{\mu \nu}, M_{\rho \sigma}\right]} & =i\left(\eta_{\mu \rho} M_{\nu \sigma}+\eta_{\nu \sigma} M_{\mu \rho}-\eta_{\mu \sigma} M_{\nu \rho}-\eta_{\nu \rho} M_{\mu \sigma}\right), \\ {\left[M_{\mu \nu}, P_{\rho}\right]} & =i\left(\eta_{\rho \nu} P_{\mu}-\eta_{\mu \rho} P_{\nu}\right), \\ {\left[P_{\mu}, P_{\nu}\right]} & =0. \end{aligned}$</td><td>$\begin{aligned} [M_{\mu \nu}, M_{\rho \sigma}] & =i(\eta_{\mu \rho} M_{\nu \sigma}+\eta_{\nu \sigma} M_{\mu \rho}-\eta_{\mu \sigma} M_{\nu \rho}-\eta_{\nu \rho} M_{\mu \sigma}), \\ [M_{\mu \nu}, P_{\rho}] & =i(\eta_{\rho \nu} P_{\mu}-\eta_{\mu \rho} P_{\nu}), \\ [P_{\mu}, P_{\nu}] & =0. \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_
EQ0773 (14)</td><td>295</td><td>■ 0.590
● W|+1</td><td><code>\begin{aligned} \xi_{t} & \equiv \frac{1}{D} \eta^{M N} \xi_{M N} \\ \xi_{M N}^{s} & =\frac{1}{2}\left(\xi_{M N}+\xi_{N M}-2 \xi_{t} \eta_{M N}\right), \\ \xi_{M N}^{a} & =\frac{1}{2}\left(\xi_{M N}-\xi_{N M}\right) . \end{aligned}</code></td><td>$\begin{aligned} \xi_{t} & \equiv \frac{1}{D} \eta^{M N} \xi_{M N} \\ \xi_{M N}^{s} & =\frac{1}{2}\left(\xi_{M N}+\xi_{N M}-2 \xi_{t} \eta_{M N}\right), \\ \xi_{M N}^{a} & =\frac{1}{2}\left(\xi_{M N}-\xi_{N M}\right). \end{aligned}$</td><td>$\begin{aligned} \xi_{t} & \equiv \frac{1}{D} \eta^{M N} \xi_{M N}, \\ \xi_{M N}^{s} & =\frac{1}{2}(\xi_{M N}+\xi_{N M}-2 \xi_{t} \eta_{M N}), \\ \xi_{M N}^{a} & =\frac{1}{2}(\xi_{M N}-\xi_{N M}). \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_
EQ0399</td><td>182</td><td>■ 0.599
● S|+1</td><td><code>\begin{aligned} & \mathbf{N}_{4} \mathbf{P}_{4}^{\text {Lor }}= \\ & \left(T^{a} T_{a}-e^{a} e^{b} R_{a b}\right) R^{c}{ }_{d} R^{d}{ }_{c} \end{aligned}</code></td><td>$\begin{aligned} & \mathbf{N}_{4} \mathbf{P}_{4}^{\text{Lor}}= \\ & \left(T^{a} T_{a}-e^{a} e^{b} R_{a b}\right) R^{c}{}_{d} R^{d}{}_{c} \end{aligned}$</td><td>$\begin{aligned} & \mathbf{N}_{4} \mathbf{P}_{4}^{Lor}= \\ & (T^{a} T_{a}-e^{a} e^{b} R_{a b}) R^{c}{}_{d} R^{d}{}_{c} \end{aligned}$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0611 (19)	254	■ 0.600 • S|+1	`\begin{array}{r} P_{\mu} c_{p_{1}}^{\dagger} c_{p_{2}}^{\dagger} \ldots c_{p_{N}}^{\dagger}|0\rangle e_{p_{1}} \otimes e_{p_{2}} \otimes \ldots e_{p_{N}}+ \ \ c_{p_{1}}^{\dagger} c_{p_{2}}^{\dagger} \ldots c_{p_{N}}^{\dagger}|0\rangle\left(-\sum_{i} P_{i \mu}\right) e_{p_{1}} \otimes e_{p_{2}} \otimes \ldots e_{p_{N}}=0 \end{array}`	$\begin{array}{r} P_{\mu} c_{p_{1}}^{\dagger} c_{p_{2}}^{\dagger} \ldots c_{p_{N}}^{\dagger}\vert 0\rangle e_{p_{1}} \otimes e_{p_{2}} \otimes \ldots e_{p_{N}}+ \\ c_{p_{1}}^{\dagger} c_{p_{2}}^{\dagger} \ldots c_{p_{N}}^{\dagger}\vert 0\rangle\left(-\sum_{i} P_{i \mu}\right) e_{p_{1}} \otimes e_{p_{2}} \otimes \ldots e_{p_{N}}=0 \end{array}$	$P_{\mu} c_{p_{1}}^{\dagger} c_{p_{2}}^{\dagger} \ldots c_{p_{N}}^{\dagger}\vert 0\rangle e_{p_{1}} \otimes e_{p_{2}} \otimes \ldots e_{p_{N}} + c_{p_{1}}^{\dagger} c_{p_{2}}^{\dagger} \ldots c_{p_{N}}^{\dagger}\vert 0\rangle(-\sum_{i} P_{i \mu}) e_{p_{1}} \otimes e_{p_{2}} \otimes \ldots e_{p_{N}}=0,$
cardona-qft-methods_ EQ0478	219	■ 0.600 • N|+0	`S=\operatorname{Tr}\left(f\left(D_{A} / \Lambda\right) \right)+\frac{1}{2}\left\langle J \tilde{\xi}, D_{A} \tilde{\xi}\right\rangle, \quad \tilde{\xi} \in \mathcal{H}_{c l}^{+},`	$S=\operatorname{Tr}\left(f\left(D_{A} / \Lambda\right) \right)+\frac{1}{2}\left\langle J \tilde{\xi}, D_{A} \tilde{\xi}\right\rangle, \quad \tilde{\xi} \in \mathcal{H}_{c l}^{+},$	$S=\operatorname{Tr}(f(D_{A} / \Lambda))+\frac{1}{2}\langle J \tilde{\xi}, D_{A} \tilde{\xi}\rangle, \quad \tilde{\xi} \in \mathcal{H}_{cl}^{+},$
cardona-qft-methods_ EQ0809 (9)	308	■ 0.604 • N|+0	`\mathcal{H}=\left(\begin{array}{cc} g-B g^{-1} B & B g^{-1} \\ -g^{-1} B & g^{-1} \end{array}\right) .`	$\mathcal{H}=\left(\begin{array}{cc} g-B g^{-1} B & B g^{-1} \\ -g^{-1} B & g^{-1} \end{array}\right) .$	$\mathcal{H}=\left(\begin{array}{cc} g-B g^{-1} B & B g^{-1} \\ -g^{-1} B & g^{-1} \end{array}\right) .$
cardona-qft-methods_ EQ0465	217	■ 0.612 • W|+0	`\begin{gathered} A_{q}^{(0,1)}=\left(\begin{array}{cc} 0 & X \\ X^{\prime} & 0 \end{array}\right) \otimes 1_{3} \quad A_{1}^{(0,1)}=\left(\begin{array}{cc} 0 & Y \\ Y^{\prime} & 0 \end{array}\right) \\ X=\left(\begin{array}{cc} \Upsilon_{u}^{*} \varphi_{1} & \Upsilon_{u}^{*} \varphi_{2} \\ -\Upsilon_{d}^{*} \bar{\varphi}_{2} & \Upsilon_{d}^{*} \bar{\varphi}_{1} \end{array}\right) \quad \text { and } \quad X^{\prime}= \left(\begin{array}{cc} \Upsilon_{u} \varphi_{1}^{\prime} & \Upsilon_{d} \varphi_{2}^{\prime} \\ -\Upsilon_{u} \bar{\varphi}_{2}^{\prime} & \Upsilon_{d} \bar{\varphi}_{1}^{\prime} \end{array}\right) \\ Y= \left(\begin{array}{cc} \Upsilon_{\nu}^{*} \varphi_{1} & \Upsilon_{\nu}^{*} \varphi_{2} \\ -\Upsilon_{e}^{*} \bar{\varphi}_{2} & \Upsilon_{e}^{*} \bar{\varphi}_{1} \end{array}\right) \quad \text { and } \quad Y^{\prime}=\left(\begin{array}{cc} \Upsilon_{\nu} \varphi_{1}^{\prime} & \Upsilon_{e} \varphi_{2}^{\prime} \\ -\Upsilon_{\nu} \bar{\varphi}_{2}^{\prime} & \Upsilon_{e} \bar{\varphi}_{1}^{\prime} \end{array} \right) \end{gathered}`	$\begin{gathered} A_{q}^{(0,1)}=\left(\begin{array}{cc} 0 & X \\ X^{\prime} & 0 \end{array}\right) \otimes 1_{3} \quad A_{1}^{(0,1)}=\left(\begin{array}{cc} 0 & Y \\ Y^{\prime} & 0 \end{array}\right) \\ X=\left(\begin{array}{cc} \Upsilon_{u}^{*} \varphi_{1} & \Upsilon_{u}^{*} \varphi_{2} \\ -\Upsilon_{d}^{*} \bar{\varphi}_{2} & \Upsilon_{d}^{*} \bar{\varphi}_{1} \end{array}\right) \quad \text { and } \quad X^{\prime}= \left(\begin{array}{cc} \Upsilon_{u} \varphi_{1}^{\prime} & \Upsilon_{d} \varphi_{2}^{\prime} \\ -\Upsilon_{u} \bar{\varphi}_{2}^{\prime} & \Upsilon_{d} \bar{\varphi}_{1}^{\prime} \end{array}\right) \\ Y= \left(\begin{array}{cc} \Upsilon_{\nu}^{*} \varphi_{1} & \Upsilon_{\nu}^{*} \varphi_{2} \\ -\Upsilon_{e}^{*} \bar{\varphi}_{2} & \Upsilon_{e}^{*} \bar{\varphi}_{1} \end{array}\right) \quad \text { and } \quad Y^{\prime}=\left(\begin{array}{cc} \Upsilon_{\nu} \varphi_{1}^{\prime} & \Upsilon_{e} \varphi_{2}^{\prime} \\ -\Upsilon_{\nu} \bar{\varphi}_{2}^{\prime} & \Upsilon_{e} \bar{\varphi}_{1}^{\prime} \end{array} \right) \end{gathered}$	$\begin{gathered} A_{q}^{(0,1)}=\left(\begin{array}{cc} 0 & X \\ X^{\prime} & 0 \end{array}\right) \otimes 1_{3} \quad A_{1}^{(0,1)}=\left(\begin{array}{cc} 0 & Y \\ Y^{\prime} & 0 \end{array}\right) \\ X=\left(\begin{array}{cc} \Upsilon_{u}^{*} \varphi_{1} & \Upsilon_{u}^{*} \varphi_{2} \\ -\Upsilon_{d}^{*} \bar{\varphi}_{2} & \Upsilon_{d}^{*} \bar{\varphi}_{1} \end{array}\right) \quad \text { and } \quad X^{\prime}= \left(\begin{array}{cc} \Upsilon_{u} \varphi_{1}^{\prime} & \Upsilon_{d} \varphi_{2}^{\prime} \\ -\Upsilon_{u} \bar{\varphi}_{2}^{\prime} & \Upsilon_{d} \bar{\varphi}_{1}^{\prime} \end{array}\right) \\ Y= \left(\begin{array}{cc} \Upsilon_{\nu}^{*} \varphi_{1} & \Upsilon_{\nu}^{*} \varphi_{2} \\ -\Upsilon_{e}^{*} \bar{\varphi}_{2} & \Upsilon_{e}^{*} \bar{\varphi}_{1} \end{array}\right) \quad \text { and } \quad Y^{\prime}=\left(\begin{array}{cc} \Upsilon_{\nu} \varphi_{1}^{\prime} & \Upsilon_{e} \varphi_{2}^{\prime} \\ -\Upsilon_{\nu} \bar{\varphi}_{2}^{\prime} & \Upsilon_{e} \bar{\varphi}_{1}^{\prime} \end{array} \right) \end{gathered}$
cardona-qft-methods_ EQ0326 (13)	161	■ 0.613 • N|+1	`\begin{gathered} \eta_{a c} \omega^{c}{ }_{b}=-\eta_{b c} \omega^{c}{ }_{a} \\ \epsilon_{b_{1} a_{2} \cdots a_{D}} \omega_{a_{1}}^{b_{1}}+\epsilon_{a_{1} b_{2} \cdots a_{D}} \omega_{a_{2}}^{b_{2}}+\cdots+\epsilon_{a_{1} a_{2} \cdots b_{D}} \omega_{a_{D}}^{b_{D}}=0 . \end{gathered}`	$\begin{gathered} \eta_{a c} \omega^{c}{ }_{b}=-\eta_{b c} \omega^{c}{ }_{a} \\ \epsilon_{b_{1} a_{2} \cdots a_{D}} \omega_{a_{1}}^{b_{1}}+\epsilon_{a_{1} b_{2} \cdots a_{D}} \omega_{a_{2}}^{b_{2}}+\cdots+\epsilon_{a_{1} a_{2} \cdots b_{D}} \omega_{a_{D}}^{b_{D}}=0 . \end{gathered}$	$\begin{gathered} \eta_{a c} \omega^{c}{ }_{b}=-\eta_{b c} \omega^{c}{ }_{a}, \\ \epsilon_{b_{1} a_{2} \cdots a_{D}} \omega^{b_{1}}{}_{a_{1}}+\epsilon_{a_{1} b_{2} \cdots a_{D}} \omega^{b_{2}}{}_{a_{2}}+\cdots+\epsilon_{a_{1} a_{2} \cdots b_{D}} \omega^{b_{D}}{}_{a_{D}}=0 . \end{gathered}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_ EQ1000</td><td>374</td><td>■0.621 ● N|+0</td><td><code>\begin{aligned} \Delta\left(X^{\dagger} E\right) & = \Delta\left(X^{\dagger}\right) \Delta(E)=\left(X^{\dagger} \otimes 1+1 \otimes X^{\dagger}\right) \Delta(E) \\ & =\left(\left(X^{\dagger}(\cdot) \otimes \mathrm{Id}+ \mathrm{Id} \otimes X^{\dagger}(\cdot)\right) \circ \Delta\right)(E) . \end{aligned}</code></td><td>$$\begin{aligned} \Delta\left(X^{\dagger} E\right) & = \Delta\left(X^{\dagger}\right) \Delta(E)=\left(X^{\dagger} \otimes 1+1 \otimes X^{\dagger}\right) \Delta(E) \\ & =\left(\left(X^{\dagger}(\cdot) \otimes \mathrm{Id}+ \mathrm{Id} \otimes X^{\dagger}(\cdot)\right) \circ \Delta\right)(E) . \end{aligned}$$</td><td>$$\begin{aligned} \Delta(X^{\dagger} E) &= \Delta(X^{\dagger}) \Delta(E)=(X^{\dagger} \otimes 1+1 \otimes X^{\dagger}) \Delta(E) \\ &= \big((X^{\dagger}(\cdot) \otimes \mathrm{Id}+ \mathrm{Id} \otimes X^{\dagger}(\cdot)) \circ \Delta\big)(E). \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0041 (11)</td><td>032</td><td>■0.623 ● K|+0</td><td><code>c\left(\omega_{1}\right) c\left(\omega_{2}\right)+c\left(\omega_{2}\right) c\left(\omega_{1}\right)=2 g\left(\omega_{1}, \omega_{2}\right) \mathrm{id}_{E} .</code></td><td>$c\left(\omega_{1}\right) c\left(\omega_{2}\right)+c\left(\omega_{2}\right) c\left(\omega_{1}\right)=2 g\left(\omega_{1}, \omega_{2}\right) \mathrm{id}_{E} .$</td><td>$c(\omega_1)\, c(\omega_2) + c(\omega_2) c(\omega_1) = 2g(\omega_1, \omega_2)\, \mathrm{id}_E.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0916</td><td>351</td><td>■0.638 ● N|+0</td><td><code>C^{*}(X) \stackrel{L}{\otimes_{C^{*}}(X)^{e}} C^{*}(X) \# G \stackrel{\simeq}{\rightarrow} C^{*}\left(P_{G} X \right)</code></td><td>$C^{*}(X) \stackrel{L}{\otimes_{C^{*}}(X)^{e}} C^{*}(X) \# G \stackrel{\simeq}{\rightarrow} C^{*}\left(P_{G} X \right)$</td><td>$C^*(X) \overset{L}{\otimes}_{C^*(X)^e} C^*(X)\#G \stackrel{\simeq}{\rightarrow} C^*(P_G X)$</td></tr>
<tr><td>cardona-qft-methods_ EQ0217 (4)</td><td>099</td><td>■0.641 ● K|+0</td><td><code>c(\xi+v(x)): \pi^{*} E_{x, \xi}^{+} \rightarrow \pi^{*} E_{x, \xi}^{-} .</code></td><td>$c(\xi+v(x)): \pi^{*} E_{x, \xi}^{+} \rightarrow \pi^{*} E_{x, \xi}^{-} .$</td><td>$c\big(\xi + v(x)\big) : \pi^* E^+_{x,\xi} \;\to\; \pi^* E^-_{x,\xi}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0214 (2)</td><td>099</td><td>■0.644 ● N|+0</td><td><code>v(x):=\left.\frac{d}{d t}\right|_{t=0} \exp (t \mathbf{v}(x)) \cdot x .</code></td><td>$v(x):=\left.\frac{d}{d t}\right|_{t=0} \exp (t \mathbf{v}(x)) \cdot x .$</td><td>$v(x) \;:=\; \frac{d}{dt}\Big|_{t=0} \exp\left(t\mathbf{v}(x)\right)\cdot x.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0339 (7)</td><td>166</td><td>■0.644 ● N|+0</td><td><code>\Omega_{n} \int_{M^{2 n}} \mathbf{E}_{2 n} \in \mathbb{Z}, \quad \tilde{\Omega}_{k} \int_{M^{4 k}} \mathbf{P}_{4 k} \in \mathbb{Z},</code></td><td>$\Omega_{n} \int_{M^{2 n}} \mathbf{E}_{2 n} \in \mathbb{Z}, \quad \tilde{\Omega}_{k} \int_{M^{4 k}} \mathbf{P}_{4 k} \in \mathbb{Z},$</td><td>$\Omega_n \int_{M^{2n}} \mathbf{E}_{2n} \in \mathbb{Z}, \qquad \tilde{\Omega}_k \int_{M^{4k}} \mathbf{P}_{4k} \in \mathbb{Z}\,,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0846 (2)</td><td>329</td><td>■0.651 ● N|+1</td><td><code>\begin{aligned} & \frac{\delta \mathcal{L}}{\delta X}=d X+\left(\Pi^{\#} \circ X\right) \eta=0 \\ & \frac{\delta \mathcal{L}}{\delta \eta}=d \eta+\frac{1}{2}\left\langle \left(\partial \Pi^{\#} \circ X\right) \eta, \eta\right \rangle=0 \end{aligned}</code></td><td>$$\begin{aligned} & \frac{\delta \mathcal{L}}{\delta X}=d X+\left(\Pi^{\#} \circ X\right) \eta=0 \\ & \frac{\delta \mathcal{L}}{\delta \eta}=d \eta+\frac{1}{2}\left\langle \left(\partial \Pi^{\#} \circ X\right) \eta, \eta\right \rangle=0 \end{aligned}$$</td><td>$$\begin{aligned} &\frac{\delta\mathcal{L}}{\delta X} = dX + (\Pi^{\#}\circ X)\eta = 0 \\ &\frac{\delta\mathcal{L}}{\delta \eta} = d\eta + \frac{1}{2}\langle(\partial\Pi^{\#}\circ X)\eta, \eta\rangle = 0. \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0104</td><td>062</td><td>■0.660 ● N|+2</td><td><code>\begin{aligned} \gamma^{\alpha_{1} \alpha_{2}} & =\frac{1}{2}\left(\gamma^{\alpha_{1}} \gamma^{\alpha_{2}}-\gamma^{\alpha_{2}} \gamma^{\alpha_{1}}\right) \\ \Omega_{\alpha_{1} \alpha_{2}} & =\left[\delta_{\alpha_{1}}+\tilde{A}_{\alpha_{1}}, \delta_{\alpha_{2}}+\tilde{A}_{\alpha_{2}}\right]=L\left(F_{\alpha_{1} \alpha_{2}}\right)-R\left(F_{\alpha_{1} \alpha_{2}}\right) \\ F_{\alpha_{1} \alpha_{2}} & = \delta_{\alpha_{1}}\left(A_{\alpha_{2}}\right)-\delta_{\alpha_{2}}\left(A_{\alpha_{1}}\right)+\left[A_{\alpha_{1}}, A_{\alpha_{2}}\right] \end{aligned}</code></td><td>$$\begin{aligned} \gamma^{\alpha_{1} \alpha_{2}} & =\frac{1}{2}\left(\gamma^{\alpha_{1}} \gamma^{\alpha_{2}}-\gamma^{\alpha_{2}} \gamma^{\alpha_{1}}\right) \\ \Omega_{\alpha_{1} \alpha_{2}} & =\left[\delta_{\alpha_{1}}+\tilde{A}_{\alpha_{1}}, \delta_{\alpha_{2}}+\tilde{A}_{\alpha_{2}}\right]=L\left(F_{\alpha_{1} \alpha_{2}}\right)-R\left(F_{\alpha_{1} \alpha_{2}}\right) \\ F_{\alpha_{1} \alpha_{2}} & = \delta_{\alpha_{1}}\left(A_{\alpha_{2}}\right)-\delta_{\alpha_{2}}\left(A_{\alpha_{1}}\right)+\left[A_{\alpha_{1}}, A_{\alpha_{2}}\right] \end{aligned}$$</td><td>$$\begin{aligned} \gamma^{\alpha_1\alpha_2} &= \tfrac{1}{2}(\gamma^{\alpha_1}\gamma^{\alpha_2} - \gamma^{\alpha_2}\gamma^{\alpha_1}), \\ \Omega_{\alpha_1\alpha_2} &= [\delta_{\alpha_1} + \tilde{A}_{\alpha_1}, \delta_{\alpha_2} + \tilde{A}_{\alpha_2}] = L(F_{\alpha_1\alpha_2}) - R(F_{\alpha_1\alpha_2}) \\ F_{\alpha_1\alpha_2} &= \delta_{\alpha_1}(A_{\alpha_2}) - \delta_{\alpha_2}(A_{\alpha_1}) + [A_{\alpha_1}, A_{\alpha_2}]. \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0659 (9)</td><td>262</td><td>■0.662 ● K|+0</td><td><code>S=T \exp \left(-i \int d^{4} x H_{I}(x)\right) .</code></td><td>$S=T \exp \left(-i \int d^{4} x H_{I}(x)\right) .$</td><td>$S = T\exp\left(-i\int d^4x H_I(x)\right).$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0896	346	0.664 N|+1	`\begin{aligned} L M^{e v^{*}(-T M)} \wedge L M^{e v^{*}(-T M)}= & (L M \times L M)^{e v *(-2 T M)} \\ & \downarrow \\ & M a p(8, M)^{e v^{*}(-T M)} \rightarrow L M^{e v^{*}(-T M)} \end{aligned}`	$\begin{aligned} L M^{e v^{*}(-T M)} \wedge L M^{e v^{*}(-T M)}= & (L M \times L M)^{e v *(-2 T M)} \\ & \downarrow \\ & M a p(8, M)^{e v^{*}(-T M)} \rightarrow L M^{e v^{*}(-T M)} \end{aligned}$	$\begin{aligned} LM^{ev^*(-TM)} \wedge LM^{ev^*(-TM)} = (LM \times LM)^{ev*(-2TM)} & \\ \downarrow & \\ Map(8,M)^{ev^*(-TM)} & \rightarrow LM^{ev^*(-TM)} \end{aligned}$
cardona-qft-methods_EQ0467	217	0.689 W|+1	`\begin{gathered} D_{A}^{2}=\left(D^{1,0}\right)^{2}+1_{4} \otimes\left(D^{0,1}\right)^{2}-\gamma_{5}\left[D^{1,0}, 1_{4} \otimes D^{0,1}\right] \\ {\left[D^{1,0}, 1_{4} \otimes D^{0,1}\right]=\sqrt{-1} \gamma^{\mu}\left[\left(\nabla_{\mu}^{s}+\mathbb{A}_{\mu}\right), 1_{4} \otimes D^{0,1}\right]} \end{gathered}`	$\begin{gathered} D_{A}^{2}=\left(D^{1,0}\right)^{2}+1_{4} \otimes\left(D^{0,1}\right)^{2}-\gamma_{5}\left[D^{1,0}, 1_{4} \otimes D^{0,1}\right] \\ {\left[D^{1,0}, 1_{4} \otimes D^{0,1}\right]=\sqrt{-1} \gamma^{\mu}\left[\left(\nabla_{\mu}^{s}+\mathbb{A}_{\mu}\right), 1_{4} \otimes D^{0,1}\right]} \end{gathered}$	$D_A^2 = (D^{1,0})^2 + 1_4 \otimes (D^{0,1})^2 - \gamma_5\,[D^{1,0}, 1_4 \otimes D^{0,1}]$ $[D^{1,0}, 1_4 \otimes D^{0,1}] = \sqrt{-1}\,\gamma^\mu\,[(\nabla^s_\mu + \mathbb{A}_\mu), 1_4 \otimes D^{0,1}]$
cardona-qft-methods_EQ0808 (8)	307	0.696 N|+0	`[X+\zeta, Y+\eta]_{C}=[X, Y]+\mathcal{L}_{X} \eta-\mathcal{L}_{Y} \zeta-\frac{1}{2} \mathrm{~d}\left(\iota_{X} \eta-\iota_{Y} \zeta\right) .`	$[X+\zeta, Y+\eta]_{C}=[X, Y]+\mathcal{L}_{X} \eta-\mathcal{L}_{Y} \zeta-\frac{1}{2} \mathrm{~d}\left(\iota_{X} \eta-\iota_{Y} \zeta\right) .$	$[X+\zeta, Y+\eta]_C = [X,Y] + \mathcal{L}_X\eta - \mathcal{L}_Y\zeta - \frac{1}{2}\mathrm{d}(\iota_X\eta - \iota_Y\zeta)\ .$
cardona-qft-methods_EQ0330 (17)	162	0.706 W|+0	`R^{a b}=\frac{1}{2} e_{\alpha}^{a} e_{\beta}^{b} R_{\mu \nu}^{\alpha \beta} d x^{\mu} \wedge d x^{\nu} .`	$R^{a b}=\frac{1}{2} e_{\alpha}^{a} e_{\beta}^{b} R_{\mu \nu}^{\alpha \beta} d x^{\mu} \wedge d x^{\nu} .$	$R^{ab} = \frac{1}{2} e^a{}_\alpha e^b{}_\beta R^{\alpha\beta}{}_{\mu\nu} dx^\mu \wedge dx^\nu.$
cardona-qft-methods_EQ0070 (21)	042	0.716 W|-1	`a_{2}\left(\not D^{2}\right)=-\frac{1}{12(4 \pi)^{d / 2}} \int_{M} s(x) d v o l_{g}(x) .`	$a_{2}\left(\not D^{2}\right)=-\frac{1}{12(4 \pi)^{d / 2}} \int_{M} s(x) d v o l_{g}(x) .$	$a_2(\not D^2) = -\frac{1}{12(4\pi)^{d/2}} \int_M s(x)\,dvol_g(x).$
cardona-qft-methods_EQ0233	109	0.718 K|+0	`\chi(Y \cup Z)=\chi(Y)+\chi(Z)-\chi(Y \cap Z) .`	$\chi(Y \cup Z)=\chi(Y)+\chi(Z)-\chi(Y \cap Z) .$	$\chi(Y \cup Z) = \chi(Y) + \chi(Z) - \chi(Y \cap Z) \quad .$
cardona-qft-methods_EQ0017 (5)	022	0.722 W|-1	`c_{P}(x)=\frac{1}{(2 \pi)^{d}} \int_{\mathbb{S}^{d-1}} \operatorname{tr}\left(\sigma_{-d}^{P}(x, \xi)\right) d \xi .`	$c_{P}(x)=\frac{1}{(2 \pi)^{d}} \int_{\mathbb{S}^{d-1}} \operatorname{tr}\left(\sigma_{-d}^{P}(x, \xi)\right) d \xi .$	$c_P(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{S}^{d-1}} tr\big(\sigma^P_{-d}(x,\xi)\big)\,d\xi.$
cardona-qft-methods_EQ0043	032	0.725 N|+0	`\sigma_{1}^{d}(x, \xi)=\lim _{t \rightarrow \infty} \frac{1}{t}\left(e^{-i t h(x)} d e^{i t h(x)}\right)(x)=\lim _{t \rightarrow \infty} \frac{1}{t} i t d_{x} h=i d_{x} h=i \xi,`	$\sigma_{1}^{d}(x, \xi)=\lim _{t \rightarrow \infty} \frac{1}{t}\left(e^{-i t h(x)} d e^{i t h(x)}\right)(x)=\lim _{t \rightarrow \infty} \frac{1}{t} i t d_{x} h=i d_{x} h=i \xi,$	$\sigma_1^d(x,\xi) = \lim_{t\to\infty} \frac{1}{t}\big(e^{-ith(x)} d e^{ith(x)}\big)(x) = \lim_{t\to\infty} \frac{1}{t}\, it\, d_x h = i\, d_x h = i\,\xi,$
cardona-qft-methods_EQ0227	102	0.728 N|+0	`e^{i t} \cdot(z, \nu):=\left(e^{i t} z, \nu\right), \quad z \in M, t, \nu \in \mathbb{R} \text {. }`	$e^{i t} \cdot(z, \nu):=\left(e^{i t} z, \nu\right), \quad z \in M, t, \nu \in \mathbb{R} \text {. }$	$e^{it} \cdot (z,\nu) \ :=\ (e^{it}z, \nu), \qquad z \in M,\ t, \nu \in \mathbb{R}.$
cardona-qft-methods_EQ0855	331	0.737 W|+1	`G \times_{(s, t)} G \xrightarrow{\mu} G \xrightarrow{i} G \underset{t}{\stackrel{s}{\rightleftarrows}} M`	$G \times_{(s, t)} G \xrightarrow{\mu} G \xrightarrow{i} G \underset{t}{\stackrel{s}{\rightleftarrows}} M$	$G \times_{(s,t)} G \xrightarrow{\ \mu\ } G \xrightarrow{\ i\ } G \underset{t}{\overset{s}{\overset{\varepsilon}{\rightleftarrows}}} M$
cardona-qft-methods_EQ0794 (35)	301	0.738 N|+0	`\begin{aligned} \text { type IIB R - R : } & \mathbf{8}_{s} \otimes \mathbf{8}_{s}=\mathbf{1} \oplus \mathbf{2 8} \oplus \mathbf{3 5}_{+}=C_{0} \oplus C_{2} \oplus C_{4+} \\ \text { type IIA R - R : } & \mathbf{8}_{s} \otimes \mathbf{8}_{c}=\mathbf{8}_{v} \oplus \mathbf{5 6}_{t}=C_{1} \oplus C_{3} . \end{aligned}`	$\begin{aligned} \text { type IIB R - R : } & \mathbf{8}_{s} \otimes \mathbf{8}_{s}=\mathbf{1} \oplus \mathbf{2 8} \oplus \mathbf{3 5}_{+}=C_{0} \oplus C_{2} \oplus C_{4+} \\ \text { type IIA R - R : } & \mathbf{8}_{s} \otimes \mathbf{8}_{c}=\mathbf{8}_{v} \oplus \mathbf{5 6}_{t}=C_{1} \oplus C_{3} . \end{aligned}$	$\begin{aligned} \text{type IIB R} - \text{R}: &\ \mathbf{8}_s \otimes \mathbf{8}_s = \mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_+ = C_0 \oplus C_2 \oplus C_{4+} \\ \text{type IIA R} - \text{R}: &\ \mathbf{8}_s \otimes \mathbf{8}_c = \mathbf{8}_v \oplus \mathbf{56}_t = C_1 \oplus C_3\ . \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_ EQ0069</td><td>042</td><td>0.751 N|+2</td><td><code>\operatorname{Tr}_{\operatorname{Dix}}\left(D^{-d} \right)=\operatorname{Tr}_{\omega}\left(D^{-d}\right)= \frac{a_{0}\left(D^{2}\right)}{\Gamma(d / 2+1)}= \frac{1}{d} W \operatorname{Res}\left(D^{-d}\right) .</code></td><td>$\operatorname{Tr}_{\operatorname{Dix}}\left(D^{-d}\right)=\operatorname{Tr}_{\omega}\left(D^{-d}\right)=\frac{a_{0}\left(D^{2}\right)}{\Gamma(d / 2+1)}=\frac{1}{d} W \operatorname{Res}\left(D^{-d}\right) .$</td><td>$\mathrm{Tr}_{Dix}(D^{-d})=\mathrm{Tr}_{\omega}(D^{-d})=\frac{a_0(D^2)}{\Gamma(d/2+1)}=\frac{1}{d}\,WRes(D^{-d}).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0687</td><td>269</td><td>0.753 S|+0</td><td><code>\begin{array}{ll} S^{+}|\uparrow\rangle=0, & S^{-}| \uparrow\rangle=|\downarrow\rangle, \\ S^{+}|\downarrow \rangle=|\uparrow\rangle, & S^{-}|\downarrow\rangle=0 . \end{array}</code></td><td>$\begin{array}{ll} S^{+}|\uparrow\rangle=0, & S^{-}|\uparrow\rangle=|\downarrow\rangle, \\ S^{+}|\downarrow\rangle=|\uparrow\rangle, & S^{-}|\downarrow\rangle=0 . \end{array}$</td><td>$\begin{array}{ll} S^{+}|\uparrow\rangle=0, & S^{-}|\uparrow\rangle=|\downarrow\rangle, \\ S^{+}|\downarrow\rangle=|\uparrow\rangle, & S^{-}|\downarrow\rangle=0\,. \end{array}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0282</td><td>137</td><td>0.755 K|+0</td><td><code>\underline{t} \in Y_{G} \Longleftrightarrow \Psi_{G}(\underline{t}) \neq 0 \Longleftrightarrow \Psi_{G / e} \left(\underline{t}^{\prime}\right) \neq-t_{e} \Psi_{G \backslash e}\left(\underline{t}^{\prime}\right),</code></td><td>$\underline{t} \in Y_{G} \Longleftrightarrow \Psi_{G}(\underline{t}) \neq 0 \Longleftrightarrow \Psi_{G / e}\left(\underline{t}^{\prime}\right) \neq-t_{e} \Psi_{G \backslash e}\left(\underline{t}^{\prime}\right),$</td><td>$\underline{t}\in Y_G \iff \Psi_G(\underline{t})\neq 0 \iff \Psi_{G/e}(\underline{t}')\neq -t_e\Psi_{G\smallsetminus e}(\underline{t}')\quad ,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0652 (2)</td><td>261</td><td>0.755 N|+0</td><td><code>\begin{gathered} \exp \left(-\beta V_{S T A T} \left(x_{1}, x_{2}\right)\right)=\left\langle x_{1}, x_{2}\right| e^{-\beta H}\left|x_{1}, x_{2}\right\rangle, \\ H=\frac{1}{2 m}\left(p_{1}^{2}+p_{2}^{2}\right) . \end{gathered}</code></td><td>$\begin{gathered} \exp \left(-\beta V_{S T A T}\left(x_{1}, x_{2}\right)\right)=\left\langle x_{1}, x_{2}\right| e^{-\beta H}\left|x_{1}, x_{2}\right\rangle, \\ H=\frac{1}{2 m}\left(p_{1}^{2}+p_{2}^{2}\right) . \end{gathered}$</td><td>$\begin{gathered} \exp(-\beta V_{STAT}(x_1,x_2))=\langle x_1,x_2|e^{-\beta H}|x_1,x_2\rangle, \\ H=\frac{1}{2m}(p_1^2+p_2^2). \end{gathered}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0236 (2)</td><td>113</td><td>0.763 N|+0</td><td><code>[X]=a_{0}+a_{1} \mathbb{L}+\cdots+a_{r} \mathbb{L}^{r},</code></td><td>$[X]=a_{0}+a_{1} \mathbb{L}+\cdots+a_{r} \mathbb{L}^{r},$</td><td>$[X]=a_0+a_1\mathbb{L}+\cdots+a_r\mathbb{L}^r\quad ,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0883</td><td>343</td><td>0.769 N|+0</td><td><code>\begin{gathered} H^{n-p}(M) \times H^{p}(M) \rightarrow \mathbf{k} \\ (\alpha, \beta) \rightarrow\langle\alpha \cup \beta,[M]\rangle \end{gathered}</code></td><td>$\begin{gathered} H^{n-p}(M) \times H^{p}(M) \rightarrow \mathbf{k} \\ (\alpha, \beta) \rightarrow\langle\alpha \cup \beta,[M]\rangle \end{gathered}$</td><td>$\begin{gathered} H^{n-p}(M)\times H^p(M)\to\mathbf{k} \\ (\alpha,\beta)\to\langle\alpha\cup\beta,[M]\rangle \end{gathered}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0181 (7)</td><td>092</td><td>0.772 K|+0</td><td><code>\mathrm{t}-\operatorname{Ind}_{G}: K_{G}\left(T_{G}^{*} M \right) \longrightarrow \widehat{R}(G)</code></td><td>$\mathrm{t}-\operatorname{Ind}_{G}: K_{G}\left(T_{G}^{*} M\right) \longrightarrow \widehat{R}(G)$</td><td>$\text{t-Ind}_G:\ K_G(T_G^*M)\ \longrightarrow\ \widehat{R}(G)$</td></tr>
<tr><td>cardona-qft-methods_ EQ0408</td><td>186</td><td>0.772 N|+0</td><td><code>\delta B\left(\left.\phi^{r}\right|_{\partial M}\right)+ \int_{\partial M} \Theta(\phi, \delta \phi)=0 .</code></td><td>$\delta B\left(\left.\phi^{r}\right|_{\partial M}\right)+\int_{\partial M} \Theta(\phi, \delta \phi)=0 .$</td><td>$\delta B(\phi^r|_{\partial M})+\int_{\partial M}\Theta(\phi,\delta\phi)=0.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0022</td><td>025</td><td>0.776 N|-2</td><td><code>W \operatorname{Res}\left(P_{1} P_{2}\right)= \underset{s=0}{\operatorname{Res}} \operatorname{Tr} \left(P_{2}|\mathcal{D}|^{-s} P_{1}\right)= \underset{s=0}{\operatorname{Res}} \operatorname{Tr} \left(P_{2} P_{1}|\mathcal{D}|^{-s}\right)=W \operatorname{Res}\left(P_{2} P_{1}\right)</code></td><td>$W \operatorname{Res}\left(P_{1} P_{2}\right)=\underset{s=0}{\operatorname{Res}} \operatorname{Tr}\left(P_{2}|\mathcal{D}|^{-s} P_{1}\right)=\underset{s=0}{\operatorname{Res}} \operatorname{Tr}\left(P_{2} P_{1}|\mathcal{D}|^{-s}\right)=W \operatorname{Res}\left(P_{2} P_{1}\right)$</td><td>$WRes(P_1P_2)=\underset{s=0}{\mathrm{Res}}\,\mathrm{Tr}\big(P_2|\mathcal{D}|^{-s}P_1\big)=\underset{s=0}{\mathrm{Res}}\,\mathrm{Tr}\big(P_2P_1|\mathcal{D}|^{-s}\big)=WRes(P_2P_1)$</td></tr>
<tr><td>cardona-qft-methods_ EQ0928</td><td>353</td><td>0.782 N|+1</td><td><code>H H^{*}\left(C^{*}(M) \# G, C^{*}(M) \# G\right) \cong H_{*}^{\text {pro }}\left(\mathbb{L}\left(M \times_{G} E G\right)^{-T\left(M \times_{G} E G\right)}\right)</code></td><td>$H H^{*}\left(C^{*}(M) \# G, C^{*}(M) \# G\right) \cong H_{*}^{\text {pro }}\left(\mathbb{L}\left(M \times_{G} E G\right)^{-T\left(M \times_{G} E G\right)}\right)$</td><td>$HH^*(C^*(M)\#G,C^*(M)\#G)\cong H_*^{\mathrm{pro}}\Big(\mathbb{L}(M\times_G EG)^{-T(M\times_G EG)}\Big)\,.$</td></tr>
</table>

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0435	208	0.790 W|+1	`\begin{array}{r} \mathbf{2}_{L} \otimes \mathbf{1}^{0} \oplus \mathbf{2}_{R} \otimes \mathbf{1}^{0} \oplus \mathbf{2}_{L} \otimes \mathbf{3}^{0} \oplus \mathbf{2}_{R} \otimes \mathbf{3}^{0} \\ \oplus \mathbf{1} \otimes \mathbf{2}_{L}^{0} \oplus \mathbf{1} \otimes \mathbf{2}_{R}^{0} \oplus \mathbf{3} \otimes \mathbf{2}_{L}^{0} \oplus \mathbf{3} \otimes \mathbf{2}_{R}^{0} \end{array}`	$\begin{array}{r} \mathbf{2}_{L} \otimes \mathbf{1}^{0} \oplus \mathbf{2}_{R} \otimes \mathbf{1}^{0} \oplus \mathbf{2}_{L} \otimes \mathbf{3}^{0} \oplus \mathbf{2}_{R} \otimes \mathbf{3}^{0} \\ \oplus \mathbf{1} \otimes \mathbf{2}_{L}^{0} \oplus \mathbf{1} \otimes \mathbf{2}_{R}^{0} \oplus \mathbf{3} \otimes \mathbf{2}_{L}^{0} \oplus \mathbf{3} \otimes \mathbf{2}_{R}^{0} \end{array}$	$\mathbf{2}_L\otimes\mathbf{1}^0\oplus\mathbf{2}_R\otimes\mathbf{1}^0\oplus\mathbf{2}_L\otimes\mathbf{3}^0\oplus\mathbf{2}_R\otimes\mathbf{3}^0$ $\oplus\mathbf{1}\otimes\mathbf{2}^0_L\oplus\mathbf{1}\otimes\mathbf{2}^0_R\oplus\mathbf{3}\otimes\mathbf{2}^0_L\oplus\mathbf{3}\otimes\mathbf{2}^0_R$
cardona-qft-methods_ EQ0838 (17)	317	0.790 N|+0	`F^{(10)}=F+\operatorname{vol}_{4} \wedge \lambda(* F),`	$F^{(10)}=F+\operatorname{vol}_{4} \wedge \lambda(* F),$	$F^{(10)}=F+\mathrm{vol}_4\wedge\lambda(*F)\ ,$
cardona-qft-methods_ EQ0959 (11)	361	0.793 K|+0	`\mathcal{C}: X_{\Gamma} \backslash \Sigma_{n} \rightarrow X_{\Gamma \vee} \backslash \Sigma_{n},`	$\mathcal{C}: X_{\Gamma} \backslash \Sigma_{n} \rightarrow X_{\Gamma \vee} \backslash \Sigma_{n},$	$\mathcal{C}: X_\Gamma\setminus\Sigma_n\to X_{\Gamma^\vee}\setminus\Sigma_n\ ,$
cardona-qft-methods_ EQ0359 (30)	171	0.794 C|+3	`\begin{aligned} & d C_{3}^{L o r}=R^{a b} R_{a b}= \mathbf{P}_{4} \\ & d C_{3}^{T o r}=T^{a} T_{a}-e^{a} e^{b} R_{a b}=\mathbf{N}_{4} \end{aligned}`	$\begin{aligned} & d C_{3}^{L o r}=R^{a b} R_{a b}= \mathbf{P}_{4} \\ & d C_{3}^{T o r}=T^{a} T_{a}-e^{a} e^{b} R_{a b}=\mathbf{N}_{4} \end{aligned}$	$dC_3^{Lor}=R^{ab}R_{ab}=\mathbf{P}_4,$ $dC_3^{Tor}=T^aT_a-e^ae^bR_{ab}=\mathbf{N}_4.$
cardona-qft-methods_ EQ0818 (18)	309	0.794 S|+2	`\begin{aligned} 6 & \rightarrow 3+\overline{3} \\ 15 & \rightarrow 8+3+\overline{3}+1 \\ 10+10^{*} & \rightarrow 6+\overline{6}+3+\overline{3}+1+1 . \end{aligned}`	$\begin{aligned} 6 & \rightarrow 3+\overline{3} \\ 15 & \rightarrow 8+3+\overline{3}+1 \\ 10+10^{*} & \rightarrow 6+\overline{6}+3+\overline{3}+1+1 . \end{aligned}$	$\mathbf{6}\ \to\mathbf{3}+\bar{\mathbf{3}}\ ,$ $\mathbf{15}\ \to\mathbf{8}+\mathbf{3}+\bar{\mathbf{3}}+\mathbf{1}\ ,$ $\mathbf{10}+\mathbf{10}^*\ \to\mathbf{6}+\bar{\mathbf{6}}+\mathbf{3}+\bar{\mathbf{3}}+\mathbf{1}+\mathbf{1}\ .$
cardona-qft-methods_ EQ0906	349	0.798 W|+0	`\begin{aligned} G \times P_{G} X & \rightarrow P_{G} X \\ ((f, k), g) & \mapsto\left(f g, g^{-1} k g\right) . \end{aligned}`	$\begin{aligned} G \times P_{G} X & \rightarrow P_{G} X \\ ((f, k), g) & \mapsto\left(f g, g^{-1} k g\right) . \end{aligned}$	$G\times P_GX\to P_GX$ $((f,k),g)\mapsto(fg,g^{-1}kg).$
cardona-qft-methods_ EQ0362 (2)	174	0.805 W|+0	`\begin{aligned} \delta \omega_{b}^{a} & =d \lambda_{b}^{a}+\omega_{c}^{a} \lambda_{b}^{c}- \omega_{b}^{c} \lambda_{c}^{a} \\ & =D_{\omega} \lambda_{b}^{a}, \end{aligned}`	$\begin{aligned} \delta \omega_{b}^{a} & =d \lambda_{b}^{a}+\omega_{c}^{a} \lambda_{b}^{c}- \omega_{b}^{c} \lambda_{c}^{a} \\ & =D_{\omega} \lambda_{b}^{a}, \end{aligned}$	$\delta\omega^a{}_b=d\lambda^a{}_b+\omega^a{}_c\lambda^c{}_b-\omega^c{}_b\lambda^a{}_c$ $=D_\omega\lambda^a{}_b,$
cardona-qft-methods_ EQ0146	082	0.808 W|+0	`\sigma_{L}(\Delta)\left(\xi_{1}, \xi_{2},, \ldots, \xi_{n}\right)=\xi_{1}^{2}+\xi_{2}^{2}+\cdots+ \xi_{n}^{2},`	$\sigma_{L}(\Delta)\left(\xi_{1}, \xi_{2},, \ldots, \xi_{n}\right)=\xi_{1}^{2}+\xi_{2}^{2}+\cdots+ \xi_{n}^{2},$	$\sigma_L(\Delta)(\xi_1,\xi_2,,\ldots,\xi_n)\ =\ \xi_1^2+\xi_2^2+\cdots+\xi_n^2,$
cardona-qft-methods_ EQ0177 (4)	090	0.808 S|+1	`T_{G}^{*} M:=\left\{\xi \in T^{*} M \mid \xi \text { is perpendicular to the orbits of } G\right\} .`	$T_{G}^{*} M:=\left\{\xi \in T^{*} M \mid \xi \text { is perpendicular to the orbits of } G\right\} .$	$T^*_GM\ :=\ \big\{\xi\in T^*M\big\vert\,\xi\ \textit{is perpendicular to the orbits of}\ G\big\}.$
cardona-qft-methods_ EQ0059	039	0.810 N|+0	`\begin{aligned} k_{t}(x, y) & =\left\langle x, e^{t \Delta} y\right\rangle=\sum_{n, m=1}^{\infty}\left\langle x, \psi_{m}\right\rangle\left\langle\psi_{m}, e^{t \Delta} \psi_{n}\right\rangle\left\langle \psi_{n}, y\right\rangle \\ & =\sum_{n, m=1}^{\infty} \overline{\psi_{n}(x)} \psi_{n}(y) e^{t \lambda_{n}} . \end{aligned}`	$\begin{aligned} k_{t}(x, y) & =\left\langle x, e^{t \Delta} y\right\rangle=\sum_{n, m=1}^{\infty}\left\langle x, \psi_{m}\right\rangle\left\langle\psi_{m}, e^{t \Delta} \psi_{n}\right\rangle\left\langle \psi_{n}, y\right\rangle \\ & =\sum_{n, m=1}^{\infty} \overline{\psi_{n}(x)} \psi_{n}(y) e^{t \lambda_{n}} . \end{aligned}$	$k_t(x,y)=\langle x,e^{t\,\Delta}y\rangle=\sum_{n,m=1}^{\infty}\langle x,\psi_m\rangle\,\langle\psi_m,e^{t\,\Delta}\psi_n\rangle\,\langle\psi_n,y\rangle$ $=\sum_{n,m=1}^{\infty}\overline{\psi_n(x)}\,\psi_n(y)\,e^{t\,\lambda_n}.$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_ EQ0254</td><td>119</td><td>0.811 K|+0</td><td><code>P_{G}(\underline{t}, p)=\sum_{C} s_{C} \prod_{e \in C} t_{e},</code></td><td>$P_{G}(\underline{t}, p)=\sum_{C} s_{C} \prod_{e \in C} t_{e},$</td><td>$P_G(\underline{t},p)=\sum_C s_C \prod_{e\in C} t_e \quad ,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0758</td><td>286</td><td>0.813 S|+0</td><td><code>\begin{aligned} \text { string energy } & \longleftrightarrow \text { scaling dimension } \\ E(\lambda) & =\Delta(\lambda) . \end{aligned}</code></td><td>$\begin{aligned} \text { string energy } & \longleftrightarrow \text { scaling dimension } \\ E(\lambda) & =\Delta(\lambda) . \end{aligned}$</td><td>$\begin{aligned} \text{string energy} &\longleftrightarrow \text{scaling dimension} \\ E(\lambda) &= \Delta(\lambda)\,. \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0185 (1)</td><td>093</td><td>0.816 K|+0</td><td><code>c(v)^{2}=-|v|^{2} \text { Id. }</code></td><td>$c(v)^{2}=-|v|^{2} \text { Id. }$</td><td>$c(v)^2 \;=\; -|v|^2\, \mathrm{Id}\,.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0615 (23)</td><td>255</td><td>0.817 N|+0</td><td><code>\begin{gathered} S_{N}=\left\langle\tau_{i, i+1}\right| i \in|1,2, \ldots, N-1|, \tau_{i, i+1}^{2}=\mathbb{I}, \\ \left.\tau_{i, i+1} \tau_{i+1, i+2} \tau_{i, i+1}=\tau_{i+1, i+2} \tau_{i, i+1} \tau_{i+1, i+2}\right\rangle \end{gathered}</code></td><td>$\begin{gathered} S_{N}=\left\langle\tau_{i, i+1}\right| i \in|1,2, \ldots, N-1|, \tau_{i, i+1}^{2}=\mathbb{I}, \\ \left.\tau_{i, i+1} \tau_{i+1, i+2} \tau_{i, i+1}=\tau_{i+1, i+2} \tau_{i, i+1} \tau_{i+1, i+2}\right\rangle \end{gathered}$</td><td>$\begin{gathered} S_N = \langle \tau_{i,i+1} | i \in |1,2,\ldots,N-1|, \tau^2_{i,i+1} = \mathbb{I}, \\ \tau_{i,i+1}\tau_{i+1,i+2}\tau_{i,i+1} = \tau_{i+1,i+2}\tau_{i,i+1}\tau_{i+1,i+2}\rangle \end{gathered}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0320 (6)</td><td>160</td><td>0.820 W|+0</td><td><code>\begin{aligned} \phi_{\|}^{a}(x) & \equiv \phi^{a}(x+d x)+d x^{\mu} \omega_{b \mu}^{a}(x) \phi^{b}(x) \\ & = \phi^{a}(x)+d x^{\mu}\left[\partial_{\mu} \phi^{a}(x)+ \omega_{b \mu}^{a}(x) \phi^{b}(x)\right] . \end{aligned}</code></td><td>$\begin{aligned} \phi_{\|}^{a}(x) & \equiv \phi^{a}(x+d x)+d x^{\mu} \omega_{b \mu}^{a}(x) \phi^{b}(x) \\ & = \phi^{a}(x)+d x^{\mu}\left[\partial_{\mu} \phi^{a}(x)+ \omega_{b \mu}^{a}(x) \phi^{b}(x)\right] . \end{aligned}$</td><td>$\begin{aligned} \phi^a_{\|}(x) &\equiv \phi^a(x+dx) + dx^\mu \omega^a_{b\mu}(x)\phi^b(x) \\ &= \phi^a(x) + dx^\mu[\partial_\mu\phi^a(x) + \omega^a_{b\mu}(x)\phi^b(x)]. \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0650 (57)</td><td>260</td><td>0.831 N|+0</td><td><code>\varphi_{\theta}^{*}=\exp \frac{i}{2} \partial \wedge P \varphi_{0},</code></td><td>$\varphi_{\theta}^{*}=\exp \frac{i}{2} \partial \wedge P \varphi_{0},$</td><td>$\varphi^*_\theta = \exp\frac{i}{2}\partial\wedge P\varphi_0,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0029</td><td>028</td><td>0.834 N|+0</td><td><code>\operatorname{Tr}_{D i x}(P)=\frac{1}{d} W R e s(P)= \frac{1}{d} \int_{M} \int_{S^{*} M} \sigma_{-d}^{P}(x, \xi) d \xi|d x|</code></td><td>$\operatorname{Tr}_{D i x}(P)=\frac{1}{d} W R e s(P)= \frac{1}{d} \int_{M} \int_{S^{*} M} \sigma_{-d}^{P}(x, \xi) d \xi|d x|$</td><td>$\mathrm{Tr}_{Dix}(P) = \frac{1}{d}\; WRes(P) = \frac{1}{d}\int_M\int_{S^*M} \sigma^P_{-d}(x,\xi)\, d\xi |dx|$</td></tr>
<tr><td>cardona-qft-methods_ EQ0519 (5)</td><td>237</td><td>0.835 N|+1</td><td><code>f * g(x)=f e^{i / 2 \overleftarrow{\delta_{\mu}} \theta^{\mu \nu} \overrightarrow{\partial_{\nu}}} g</code></td><td>$f * g(x)=f e^{i / 2 \overleftarrow{\delta_{\mu}} \theta^{\mu \nu} \overrightarrow{\partial_{\nu}}} g$</td><td>$f * g(x) = f e^{i/2\overleftarrow{\partial_\mu}\theta^{\mu\nu}\overrightarrow{\partial_\nu}} g.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0613 (21)</td><td>254</td><td>0.835 N|+1</td><td><code>\begin{gathered} \int \prod_{i} d \mu\left(p_{i}\right) c_{\Delta p_{1}}^{\dagger} c_{\Delta p_{2}}^{\dagger} \ldots c_{\Delta p_{N}}^{\dagger}|0\rangle e_{\Delta p_{1}} \otimes e_{\Delta p_{1}} \otimes \ldots \otimes e_{\Delta p_{N}}= \\ \int \prod_{i} d \mu\left(p_{i}\right) c_{p_{1}}^{\dagger} c_{p_{2}}^{\dagger} \ldots c_{p_{N}}^{\dagger}|0\rangle e_{p_{1}} \otimes e_{p_{1}} \otimes \ldots \otimes e_{p_{N}} \end{gathered}</code></td><td>$\begin{gathered} \int \prod_{i} d \mu\left(p_{i}\right) c_{\Delta p_{1}}^{\dagger} c_{\Delta p_{2}}^{\dagger} \ldots c_{\Delta p_{N}}^{\dagger}|0\rangle e_{\Delta p_{1}} \otimes e_{\Delta p_{1}} \otimes \ldots \otimes e_{\Delta p_{N}}= \\ \int \prod_{i} d \mu\left(p_{i}\right) c_{p_{1}}^{\dagger} c_{p_{2}}^{\dagger} \ldots c_{p_{N}}^{\dagger}|0\rangle e_{p_{1}} \otimes e_{p_{1}} \otimes \ldots \otimes e_{p_{N}} \end{gathered}$</td><td>$\begin{gathered} \int\prod_i d\mu(p_i) c^\dagger_{\Delta p_1} c^\dagger_{\Delta p_2}\ldots c^\dagger_{\Delta p_N}|0\rangle e_{\Delta p_1}\otimes e_{\Delta p_1}\otimes\ldots\otimes e_{\Delta p_N} = \\ \int\prod_i d\mu(p_i) c^\dagger_{p_1} c^\dagger_{p_2}\ldots c^\dagger_{p_N}|0\rangle e_{p_1}\otimes e_{p_1}\otimes\ldots\otimes e_{p_N}, \end{gathered}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0512 (5)</td><td>230</td><td>0.840 N|+0</td><td><code>\partial_{t} x_{j}=\beta_{j}^{\mathrm{SM}}+\beta_{j}^{\mathrm{grav}},</code></td><td>$\partial_{t} x_{j}=\beta_{j}^{\mathrm{SM}}+\beta_{j}^{\mathrm{grav}},$</td><td>$\partial_t x_j = \beta_j^{\mathrm{SM}} + \beta_j^{\mathrm{grav}},$</td></tr>
<tr><td>cardona-qft-methods_ EQ0046</td><td>035</td><td>0.840 W|+0</td><td><code>\mathcal{C} \ell_{x} M=\left\{\begin{array}{l} M_{2^{m}}(\mathbb{C}) \text { when } d=2 m \text { is even }, \\ M_{2^{m}}(\mathbb{C}) \oplus M_{2^{m}}(\mathbb{C}) \text { when } d=2 m+1 . \end{array}\right.</code></td><td>$\mathcal{C} \ell_{x} M=\left\{\begin{array}{l} M_{2^{m}}(\mathbb{C}) \text { when } d=2 m \text { is even }, \\ M_{2^{m}}(\mathbb{C}) \oplus M_{2^{m}}(\mathbb{C}) \text { when } d=2 m+1 . \end{array}\right.$</td><td>$\mathcal{C}\ell_x M = \begin{cases} M_{2^m}(\mathbb{C}) \text{ when } d=2m \text{ is even,} \\ M_{2^m}(\mathbb{C})\oplus M_{2^m}(\mathbb{C}) \text{ when } d=2m+1. \end{cases}$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0749	284	■ 0.842 ● N|+1	`\text { Super }\left[A d S_{5} \times S^{5}\right]:= \frac{\widetilde{\operatorname{PSU}}(2,2 \mid 4)}{ \operatorname{SO}(1,4) \times \operatorname{SO}(5)}`	$\text { Super }\left[A d S_{5} \times S^{5}\right]:= \frac{\widetilde{\operatorname{PSU}}(2,2 \mid 4)}{ \operatorname{SO}(1,4) \times \operatorname{SO}(5)}$	$\mathrm{Super}\Big[AdS_5\times S^5\Big] := \frac{\widetilde{\mathrm{PSU}}(2,2\mid 4)}{\mathrm{SO}(1,4)\times\mathrm{SO}(5)}\,,$
cardona-qft-methods_EQ0335 (21)	164	■ 0.848 ● N|+0	`D\left(R^{a}{ }_{b} \phi^{b}\right)=R^{a}{ }_{b} \wedge D \phi^{b} .`	$D\left(R^{a}{ }_{b} \phi^{b}\right)=R^{a}{ }_{b} \wedge D \phi^{b} .$	$D(R^a_{\ b}\phi^b) = R^a_{\ b}\wedge D\phi^b\,.$
cardona-qft-methods_EQ0898	347	■ 0.850 ● W|+0	`\begin{gathered} A \stackrel{L}{\otimes_{A^{e}}} M=B(A) \stackrel{L}{\otimes_{A^{e}}} M \\ \mathcal{R} \mathcal{H o m}{ }_{A^{e}}(A, M)=\operatorname{Hom}_{A^{e}}(B(A), M) \end{gathered}`	$\begin{gathered} A \stackrel{L}{\otimes_{A^{e}}} M=B(A) \stackrel{L}{\otimes_{A^{e}}} M \\ \mathcal{R} \mathcal{H o m}{ }_{A^{e}}(A, M)=\operatorname{Hom}_{A^{e}}(B(A), M) \end{gathered}$	$\begin{gathered} A \overset{L}{\otimes}_{A^e} M = B(A)\overset{L}{\otimes}_{A^e} M \\ \mathcal{RH}om_{A^e}(A,M) = Hom_{A^e}(B(A),M) \end{gathered}$
cardona-qft-methods_EQ0100	058	■ 0.850 ● W|-2	`\operatorname{Tr}\left(e^{-t \not D^{2}}\right)= \frac{1}{t^{2}}\left(\frac{2}{3}+\frac{2}{3} t+ \sum_{k=0}^{n} a_{k} t^{k+2}+\mathcal{O}\left(t^{n+3}\right)\right),`	$\operatorname{Tr}\left(e^{-t \not D^{2}}\right)= \frac{1}{t^{2}}\left(\frac{2}{3}+\frac{2}{3} t+ \sum_{k=0}^{n} a_{k} t^{k+2}+\mathcal{O}\left(t^{n+3}\right)\right),$	$\mathrm{Tr}(e^{-t\not D^2}) = \frac{1}{t^2}\big(\frac{2}{3}+\frac{2}{3}\,t+\sum_{k=0}^{n} a_k\,t^{k+2}+\mathcal{O}(t^{n+3})\big),$
cardona-qft-methods_EQ0957 (9)	361	■ 0.850 ● N|+0	`\mathcal{C}\left(X_{\Gamma} \cap\left( \mathbb{P}_{k}^{n-1} \backslash \Sigma_{n}\right) \right)=X_{\Gamma^{\vee}} \cap\left(\mathbb{P}_{k}^{n-1} \backslash \Sigma_{n}\right)`	$\mathcal{C}\left(X_{\Gamma} \cap\left(\mathbb{P}_{k}^{n-1} \backslash \Sigma_{n}\right) \right)=X_{\Gamma^{\vee}} \cap\left(\mathbb{P}_{k}^{n-1} \backslash \Sigma_{n}\right)$	$\mathcal{C}\left(X_\Gamma\cap(\mathbb{P}^{n-1}_k\setminus\Sigma_n)\right) = X_{\Gamma^\vee}\cap(\mathbb{P}^{n-1}_k\setminus\Sigma_n)$
cardona-qft-methods_EQ0284	137	■ 0.856 ● N|+1	`\Psi_{G_{2 e}}=t_{e} t_{e^{\prime}} \Psi_{G \backslash e}+\left(t_{e}+t_{e^{\prime}}\right) \Psi_{G / e}`	$\Psi_{G_{2 e}}=t_{e} t_{e^{\prime}} \Psi_{G \backslash e}+\left(t_{e}+t_{e^{\prime}}\right) \Psi_{G / e}$	$\Psi_{G_{2e}} = t_e t_{e'}\Psi_{G\smallsetminus e} + (t_e+t_{e'})\Psi_{G/e}\quad .$
cardona-qft-methods_EQ0337 (23)	164	■ 0.856 ● N|+0	`e^{a} \equiv e_{\mu}^{a}(x) d x^{\mu} \quad \text { and } \quad \omega_{b}^{a} \equiv \omega_{b \mu}^{a}(x) d x^{ \mu} .`	$e^{a} \equiv e_{\mu}^{a}(x) d x^{\mu} \quad \text { and } \quad \omega_{b}^{a} \equiv \omega_{b \mu}^{a}(x) d x^{ \mu} .$	$e^a \equiv e^a_\mu(x)dx^\mu \quad\text{and}\quad \omega^a_{\ b}\equiv\omega^a_{b\mu}(x)dx^\mu.$
cardona-qft-methods_EQ0037	029	■ 0.857 ● N|+0	`\tau_{\lambda}\left(T_{1}+T_{2}\right)-\tau_{\lambda} \left(T_{1}\right)-\tau_{\lambda}\left(T_{2}\right) \underset{\lambda \rightarrow \infty}{\simeq} \mathcal{O} \left(\frac{\log (\log \lambda)}{\log \lambda}\right), \text { when } 0 \leq T_{1}, T_{2} \in \mathcal{L}^{1, \infty} .`	$\tau_{\lambda}\left(T_{1}+T_{2}\right)-\tau_{\lambda} \left(T_{1}\right)-\tau_{\lambda}\left(T_{2}\right) \underset{\lambda \rightarrow \infty}{\simeq} \mathcal{O} \left(\frac{\log (\log \lambda)}{\log \lambda}\right), \text { when } 0 \leq T_{1}, T_{2} \in \mathcal{L}^{1, \infty} .$	$\tau_\lambda(T_1+T_2)-\tau_\lambda(T_1)-\tau_\lambda(T_2) \underset{\lambda\to\infty}{\simeq} \mathcal{O}\big(\frac{\log(\log\lambda)}{\log\lambda}\big),\ \text{when}\ 0\le T_1,T_2\in\mathcal{L}^{1,\infty}.$
cardona-qft-methods_EQ0068	042	■ 0.858 ● W|+1	`\begin{aligned} \operatorname{WRes}\left(D^{-p}\right) & =\frac{2}{\Gamma(p / 2)} a_{d-p}\left(D^{2}\right) \\ & =\frac{2}{(4 \pi)^{d / 2} \Gamma(p / 2)} \int_{M} \operatorname{tr}\left(k_{(d-p) / 2}(x, x)\right) d \operatorname{vol}_{g}(x) \end{aligned}`	$\begin{aligned} \operatorname{WRes}\left(D^{-p}\right) & =\frac{2}{\Gamma(p / 2)} a_{d-p}\left(D^{2}\right) \\ & =\frac{2}{(4 \pi)^{d / 2} \Gamma(p / 2)} \int_{M} \operatorname{tr}\left(k_{(d-p) / 2}(x, x)\right) d \operatorname{vol}_{g}(x) \end{aligned}$	$\begin{aligned} WRes(D^{-p}) &= \frac{2}{\Gamma(p/2)}a_{d-p}(D^2) \\ &= \frac{2}{(4\pi)^{d/2}\Gamma(p/2)}\int_M tr\big(k_{(d-p)/2}(x,x)\big)\,dvol_g(x). \end{aligned}$
cardona-qft-methods_EQ0265	125	■ 0.863 ● N|+0	`\langle G\rangle=\text { 'propagator' · }\left\langle G_{1}\right\rangle \cdot\left\langle G_{2}\right\rangle,`	$\langle G\rangle=\text { 'propagator' · }\left\langle G_{1}\right\rangle \cdot\left\langle G_{2}\right\rangle,$	$\langle G\rangle = \text{‘propagator’}\cdot\langle G_1\rangle\cdot\langle G_2\rangle\quad ,$
cardona-qft-methods_EQ0261	123	■ 0.872 ● K|+0	`t_{1} t_{2}+t_{1} t_{3}+t_{1} t_{5}+t_{2} t_{4}+t_{2} t_{5}+t_{3} t_{4}+t_{3} t_{5}+t_{4} t_{5},`	$t_{1} t_{2}+t_{1} t_{3}+t_{1} t_{5}+t_{2} t_{4}+t_{2} t_{5}+t_{3} t_{4}+t_{3} t_{5}+t_{4} t_{5},$	$t_1t_2+t_1t_3+t_1t_5+t_2t_4+t_2t_5+t_3t_4+t_3t_5+t_4t_5\quad ,$
cardona-qft-methods_EQ0967 (7)	362	■ 0.873 ● S|+0	`\left[X_{\Gamma^{\vee}}\right]=\left[\Sigma_{n}\right]= \frac{(1+\mathbb{T})^{n}-1-\mathbb{T}^{n}}{\mathbb{T}}= \sum_{i=1}^{n-1}\binom{n}{i} \mathbb{T}^{n-1-i} .`	$\left[X_{\Gamma^{\vee}}\right]=\left[\Sigma_{n}\right]= \frac{(1+\mathbb{T})^{n}-1-\mathbb{T}^{n}}{\mathbb{T}}= \sum_{i=1}^{n-1}\binom{n}{i} \mathbb{T}^{n-1-i} .$	$[X_{\Gamma^\vee}] = [\Sigma_n] = \frac{(1+\mathbb{T})^n-1-\mathbb{T}^n}{\mathbb{T}} = \sum_{i=1}^{n-1}\binom{n}{i}\mathbb{T}^{n-1-i}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0338 (4)	165	■0.874 ● W|+2	\begin{aligned} \mathbf{P}_{2 k} & =: R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{a_{1}} \\ v_{k} & =: e_{a_{1}} R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{b} e^{b}, \text { odd } k \\ \tau_{k} & =: T_{a_{1}} R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{b} T^{b}, \text { even } k \\ \zeta_{k} & =: e_{a_{1}} R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{b} T^{b} \\ \mathbf{E}_{D} & =: \epsilon_{a_{1} a_{2} \cdots a_{D}} R^{a_{1} a_{2}} R^{a_{3} a_{4}} \cdots R^{a_{D-1} a_{D}}, \text { even } D \\ L_{p} & =: \epsilon_{a_{1} a_{2} \cdots a_{D}} R^{a_{1} a_{2}} R^{a_{3} a_{4}} \cdots R^{a_{2 p-1} a_{2 p}} e^{a_{2 p+1}} \cdots e^{a_{D}} . \end{aligned}	$\begin{aligned} \mathbf{P}_{2 k} & =: R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{a_{1}} \\ v_{k} & =: e_{a_{1}} R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{b} e^{b}, \text { odd } k \\ \tau_{k} & =: T_{a_{1}} R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{b} T^{b}, \text { even } k \\ \zeta_{k} & =: e_{a_{1}} R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{b} T^{b} \\ \mathbf{E}_{D} & =: \epsilon_{a_{1} a_{2} \cdots a_{D}} R^{a_{1} a_{2}} R^{a_{3} a_{4}} \cdots R^{a_{D-1} a_{D}}, \text { even } D \\ L_{p} & =: \epsilon_{a_{1} a_{2} \cdots a_{D}} R^{a_{1} a_{2}} R^{a_{3} a_{4}} \cdots R^{a_{2 p-1} a_{2 p}} e^{a_{2 p+1}} \cdots e^{a_{D}} . \end{aligned}$	$\begin{aligned} \mathbf{P}_{2 k} & =: R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{a_{1}} \\ v_{k} & =: e_{a_{1}} R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{b} e^{b}, \text { odd } k \\ \tau_{k} & =: T_{a_{1}} R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{b} T^{b}, \text { even } k \\ \zeta_{k} & =: e_{a_{1}} R^{a_{1}}{ }_{a_{2}} R^{a_{2}}{ }_{a_{3}} \cdots R^{a_{k}}{ }_{b} T^{b} \\ \mathbf{E}_{D} & =: \epsilon_{a_{1} a_{2} \cdots a_{D}} R^{a_{1} a_{2}} R^{a_{3} a_{4}} \cdots R^{a_{D-1} a_{D}}, \text { even } D \\ L_{p} & =: \epsilon_{a_{1} a_{2} \cdots a_{D}} R^{a_{1} a_{2}} R^{a_{3} a_{4}} \cdots R^{a_{2 p-1} a_{2 p}} e^{a_{2 p+1}} \cdots e^{a_{D}} . \end{aligned}$
cardona-qft-methods_EQ0915	351	■0.874 ● N|+0	H H_{*}\left(\left(C^{*}(X) \# G, C^{*}(X) \# G\right) \cong \operatorname{Tor}_{\mathbb{Z} G}^{*}\left(\mathbb{Z}, C^{*}(X) \stackrel{L}{\otimes_{C^{*}}(X)^{e}} C^{*}(X) \# G\right)\right.	$H H_{*}\left(\left(C^{*}(X) \# G, C^{*}(X) \# G\right) \cong \operatorname{Tor}_{\mathbb{Z} G}^{*}\left(\mathbb{Z}, C^{*}(X) \stackrel{L}{\otimes_{C^{*}}(X)^{e}} C^{*}(X) \# G\right)\right.$	$HH_*((C^*(X)\#G, C^*(X)\#G) \cong \operatorname{Tor}^*_{\mathbb{Z}G}(\mathbb{Z}, C^*(X) \overset{L}{\otimes}_{C^*(X)^e} C^*(X)\#G)$
cardona-qft-methods_EQ0870	337	■0.875 ● N|+1	\begin{aligned} \mathcal{G} & =G \\ L & =\left\{ \left(g_{1}, g_{2}, g_{3}\right) \mid\left(g_{1}, g_{2} \right) \in G \times_{(s, t)} G, g_{3}=\mu\left(g_{1}, g_{2}\right)\right\} . \\ I & =g \mapsto g^{-1}, g \in G . \end{aligned}	$\begin{aligned} \mathcal{G} & =G \\ L & =\left\{ \left(g_{1}, g_{2}, g_{3}\right) \mid\left(g_{1}, g_{2} \right) \in G \times_{(s, t)} G, g_{3}=\mu\left(g_{1}, g_{2}\right)\right\} . \\ I & =g \mapsto g^{-1}, g \in G . \end{aligned}$	$\begin{aligned} &\mathcal{G} = G. \\ &L = \{(g_1,g_2,g_3) \mid (g_1,g_2) \in G \times_{(s,t)} G,\ g_3 = \mu(g_1,g_2)\}. \\ &I = g \mapsto g^{-1},\ g \in G. \end{aligned}$
cardona-qft-methods_EQ0670	264	■0.875 ● N|+1	\begin{aligned} & \left\langle a_{l m} a_{l^{\prime} m^{ \prime}}^{*}\right\rangle_{\theta} \\ & =* \pi^{2} \int d k \sum_{l^{\prime \prime}=0, l^{\prime \prime}: \text { even }}^{\infty} i^{l+l^{\prime}}(-1)^{l+m}\left(2 l^{ \prime \prime}+1\right) k^{2} \Delta_{l}(k) \Delta_{l}^{ \prime}(k) P_{\Phi}(k) i_{l^{\prime \prime}}(\theta k H) \\ & \quad \times \sqrt{(2 l+1)\left(2 l^{\prime}+1 \right)}\left[\begin{array}{ccc} l & l^{\prime} & l^{ \prime \prime} \\ 0 & 0 & 0 \end{array}\right]\left[\begin{array}{ccc} l & l^{\prime} & l^{\prime \prime} \\ -m & m^{\prime} & 0 \end{array}\right] \end{aligned}	$\begin{aligned} & \left\langle a_{l m} a_{l^{\prime} m^{ \prime}}^{*}\right\rangle_{\theta} \\ & =* \pi^{2} \int d k \sum_{l^{\prime \prime}=0, l^{\prime \prime}: \text { even }}^{\infty} i^{l+l^{\prime}}(-1)^{l+m}\left(2 l^{ \prime \prime}+1\right) k^{2} \Delta_{l}(k) \Delta_{l}^{ \prime}(k) P_{\Phi}(k) i_{l^{\prime \prime}}(\theta k H) \\ & \quad \times \sqrt{(2 l+1)\left(2 l^{\prime}+1 \right)}\left[\begin{array}{ccc} l & l^{\prime} & l^{ \prime \prime} \\ 0 & 0 & 0 \end{array}\right]\left[\begin{array}{ccc} l & l^{\prime} & l^{\prime \prime} \\ -m & m^{\prime} & 0 \end{array}\right] \end{aligned}$	$\begin{aligned} & \langle a_{lm} a^*_{l'm'} \rangle_\theta \\ & = *\pi^2 \int dk \sum_{l''=0, l'':even}^{\infty} i^{l+l'} (-1)^{l+m} (2l''+1) k^2 \Delta_l(k) \Delta'_l(k) P_\Phi(k) i_{l''}(\theta k H) \\ & \quad \times \sqrt{(2l+1)(2l'+1)} \begin{bmatrix} l & l' & l'' \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} l & l' & l'' \\ -m & m' & 0 \end{bmatrix}, \end{aligned}$
cardona-qft-methods_EQ0357 (26)	171	■0.877 ● N|+2	\begin{aligned} & \left(\mathbf{N}_{4}\right)^{2}= \left(T^{a} T_{a}-e^{a} e^{b} R_{a b}\right)^{2} \\ & \mathbf{N}_{4} \mathbf{P}_{4}=\left(T^{a} T_{a}-e^{a} e^{b} R_{a b}\right)\left(R_{d}^{c} R_{c}^{d}\right) \end{aligned}	$\begin{aligned} & \left(\mathbf{N}_{4}\right)^{2}= \left(T^{a} T_{a}-e^{a} e^{b} R_{a b}\right)^{2} \\ & \mathbf{N}_{4} \mathbf{P}_{4}=\left(T^{a} T_{a}-e^{a} e^{b} R_{a b}\right)\left(R_{d}^{c} R_{c}^{d}\right) \end{aligned}$	$\begin{aligned} & (\mathbf{N}_4)^2 = (T^a T_a - e^a e^b R_{ab})^2, \\ & \mathbf{N}_4 \mathbf{P}_4 = (T^a T_a - e^a e^b R_{ab})(R^c{}_d R^d{}_c), \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0267	127	0.880 K|+0	`\Psi_{G_{1}}\left(t_{1}, \ldots, t_{n_{1}}\right) \Psi_{G_{2}}\left(u_{1}, \ldots, u_{n_{2}}\right) \neq 0,`	$\Psi_{G_{1}}\left(t_{1}, \ldots, t_{n_{1}}\right) \Psi_{G_{2}}\left(u_{1}, \ldots, u_{n_{2}}\right) \neq 0,$	$\Psi_{G_1}(t_1,\ldots,t_{n_1})\Psi_{G_2}(u_1,\ldots,u_{n_2}) \neq 0 \quad ,$
cardona-qft-methods_ EQ0314 (18)	156	0.882 W|+1	`\begin{aligned} & {\left[\mathcal{H}_{\perp}(x), \mathcal{H}_{\perp}(y)\right]=g^{i j}(x) \delta(x, y),{ }_{i} \mathcal{H}_{j}(y)-g^{i j}(y) \delta(y, x),{ }_{i} \mathcal{H}_{j}(x)} \\ & {\left[\mathcal{H}_{i}(x), \mathcal{H}_{j}(y)\right]=\delta(x, y),{ }_{i} \mathcal{H}_{j}(y)-\delta(x, y),{ }_{j} \mathcal{H}_{i}(y)} \\ & {\left[\mathcal{H}_{\perp}(x), \mathcal{H}_{i}(y)\right]=\delta(x, y),{ }_{i} \mathcal{H}_{\perp}(y)} \end{aligned}`	$\begin{aligned} & {\left[\mathcal{H}_{\perp}(x), \mathcal{H}_{\perp}(y)\right]=g^{i j}(x) \delta(x, y),{ }_{i} \mathcal{H}_{j}(y)-g^{i j}(y) \delta(y, x),{ }_{i} \mathcal{H}_{j}(x)} \\ & {\left[\mathcal{H}_{i}(x), \mathcal{H}_{j}(y)\right]=\delta(x, y),{ }_{i} \mathcal{H}_{j}(y)-\delta(x, y),{ }_{j} \mathcal{H}_{i}(y)} \\ & {\left[\mathcal{H}_{\perp}(x), \mathcal{H}_{i}(y)\right]=\delta(x, y),{ }_{i} \mathcal{H}_{\perp}(y)} \end{aligned}$	$\begin{aligned} [\mathcal{H}_\perp(x),\mathcal{H}_\perp(y)] &= g^{ij}(x)\delta(x,y)_{,i}\mathcal{H}_j(y) - g^{ij}(y)\delta(y,x)_{,i}\mathcal{H}_j(x) \\ [\mathcal{H}_i(x),\mathcal{H}_j(y)] &= \delta(x,y)_{,i}\mathcal{H}_j(y) - \delta(x,y)_{,j}\mathcal{H}_i(y) \quad , \\ [\mathcal{H}_\perp(x),\mathcal{H}_i(y)] &= \delta(x,y)_{,i}\mathcal{H}_\perp(y) \end{aligned}$
cardona-qft-methods_ EQ0175 (3)	090	0.890 N|+0	`\operatorname{Ind}_{G}(\tilde{\mathcal{D}}):=\bigoplus_{V \in \operatorname{Irr}(G)}(\operatorname{dim} V)( \operatorname{dimKer}(\mathcal{D})-\operatorname{dimKer}( \mathcal{D})) V .`	$\operatorname{Ind}_{G}(\tilde{\mathcal{D}}):=\bigoplus_{V \in \operatorname{Irr}(G)}(\operatorname{dim} V)(\operatorname{dimKer}(\mathcal{D})-\operatorname{dimKer}(\mathcal{D})) V .$	$\operatorname{Ind}_G(\tilde{\mathcal{D}}) \;:=\; \bigoplus_{V\in\operatorname{Irr}(G)} (\dim V)\big(\dim\operatorname{Ker}(\mathcal{D}) - \dim\operatorname{Ker}(\mathcal{D})\big)\, V.$
cardona-qft-methods_ EQ0820 (20)	310	0.890 N|+0	`\begin{aligned} & \Phi_{+}=\eta_{+}^{1} \otimes \eta_{+}^{2 \dagger}=-\frac{\mathrm{i}}{8} \omega \wedge e^{-\mathrm{i} v \wedge v^{\prime}}, \\ & \Phi_{-}= \eta_{+}^{1} \otimes \eta_{-}^{2 \dagger}=-\frac{1}{8} e^{-\mathrm{i} j} \wedge\left(v+\mathrm{i} v^{\prime} \right) . \end{aligned}`	$\begin{aligned} & \Phi_{+}=\eta_{+}^{1} \otimes \eta_{+}^{2 \dagger}=-\frac{\mathrm{i}}{8} \omega \wedge e^{-\mathrm{i} v \wedge v^{\prime}}, \\ & \Phi_{-}= \eta_{+}^{1} \otimes \eta_{-}^{2 \dagger}=-\frac{1}{8} e^{-\mathrm{i} j} \wedge\left(v+\mathrm{i} v^{\prime} \right) . \end{aligned}$	$\begin{aligned} \Phi_+ &= \eta_+^1 \otimes \eta_+^{2\dagger} = -\frac{\mathrm{i}}{8}\omega \wedge e^{-\mathrm{i}v\wedge v'} \,, \\ \Phi_- &= \eta_+^1 \otimes \eta_-^{2\dagger} = -\frac{1}{8} e^{-\mathrm{i}j} \wedge (v + \mathrm{i}v') \,. \end{aligned}$
cardona-qft-methods_ EQ0449	211	0.896 N|+0	`Y_{(\downarrow 1)}^{\prime}=V_{1} Y_{(\downarrow 1)} V_{3}^{*}, \quad Y_{(\uparrow 1)}^{\prime}=V_{2} Y_{(\uparrow 1)} V_{3}^{*}, \quad Y_{R}^{\prime}=V_{2} Y_{R} \bar{V}_{2}^{*},`	$Y_{(\downarrow 1)}^{\prime}=V_{1} Y_{(\downarrow 1)} V_{3}^{*}, \quad Y_{(\uparrow 1)}^{\prime}=V_{2} Y_{(\uparrow 1)} V_{3}^{*}, \quad Y_{R}^{\prime}=V_{2} Y_{R} \bar{V}_{2}^{*},$	$Y'_{(\downarrow 1)} = V_1\, Y_{(\downarrow 1)} V_3^*, \quad Y'_{(\uparrow 1)} = V_2\, Y_{(\uparrow 1)}\, V_3^*, \quad Y'_R = V_2\, Y_R\, \bar{V}_2^*,$
cardona-qft-methods_ EQ0427	205	0.896 K|+0	`J^{2}=\varepsilon, \quad J D=\varepsilon^{\prime} D J, \quad \text { and } \quad J \gamma=\varepsilon^{\prime \prime} \gamma J,`	$J^{2}=\varepsilon, \quad J D=\varepsilon^{\prime} D J, \quad \text { and } \quad J \gamma=\varepsilon^{\prime \prime} \gamma J,$	$J^2 = \varepsilon, \quad JD = \varepsilon' DJ, \quad \text{and} \quad J\gamma = \varepsilon''\gamma J,$
cardona-qft-methods_ EQ0728	277	0.905 W|+0	`M_{a}(u)=\left(\begin{array}{ll} \hat{A}(u) & \hat{B}(u) \\ \hat{C}(u) & \hat{D}(u) \end{array}\right),`	$M_{a}(u)=\left(\begin{array}{ll} \hat{A}(u) & \hat{B}(u) \\ \hat{C}(u) & \hat{D}(u) \end{array}\right),$	$M_a(u) = \begin{pmatrix} \hat{A}(u) & \hat{B}(u) \\ \hat{C}(u) & \hat{D}(u) \end{pmatrix},$
cardona-qft-methods_ EQ0900	347	0.907 N|+0	`C_{*}\left(L M^{-T M}\right) \stackrel{\simeq}{ \rightarrow} \mathcal{R} \mathcal{H} o m_{C^{*}(M)^{e}} \left(C^{*}(M), C^{*}(M)\right)`	$C_{*}\left(L M^{-T M}\right) \stackrel{\simeq}{ \rightarrow} \mathcal{R} \mathcal{H} o m_{C^{*}(M)^{e}} \left(C^{*}(M), C^{*}(M)\right)$	$C_*(LM^{-TM}) \stackrel{\simeq}{\rightarrow} \mathcal{R}\mathcal{H}om_{C^*(M)^e}(C^*(M), C^*(M))$
cardona-qft-methods_ EQ0979 (19)	364	0.907 W|+0	`\left[X_{\Gamma^{\vee}}\right]=\frac{( \mathbb{T}+1)^{n}-1}{\mathbb{T}}-\frac{ \mathbb{T}^{n}-(-1)^{n}}{\mathbb{T}}-n \mathbb{T}^{n-2} .`	$\left[X_{\Gamma^{\vee}}\right]=\frac{(\mathbb{T}+1)^{n}-1}{\mathbb{T}}-\frac{ \mathbb{T}^{n}-(-1)^{n}}{\mathbb{T}}-n \mathbb{T}^{n-2} .$	$[X_{\Gamma^\vee}] = \frac{(\mathbb{T}+1)^n - 1}{\mathbb{T}} - \frac{\mathbb{T}^n - (-1)^n}{\mathbb{T}} - n\mathbb{T}^{n-2}\,.$
cardona-qft-methods_ EQ0549 (18)	244	0.911 W|+0	`\begin{aligned} m_{r}[(S \otimes \mathrm{Id} \otimes \mathrm{Id})(\mathrm{Id} \otimes \Delta) \Delta(\alpha)] & =e \otimes \alpha \\ m_{r}^{\prime}[(\mathrm{Id} \otimes S \otimes \mathrm{Id})(\mathrm{Id} \otimes \Delta) \Delta(\alpha)] & =\alpha \otimes e, \quad \alpha \in H . \end{aligned}`	$\begin{aligned} m_{r}[(S \otimes \mathrm{Id} \otimes \mathrm{Id})(\mathrm{Id} \otimes \Delta) \Delta(\alpha)] & =e \otimes \alpha \\ m_{r}^{\prime}[(\mathrm{Id} \otimes S \otimes \mathrm{Id})(\mathrm{Id} \otimes \Delta) \Delta(\alpha)] & =\alpha \otimes e, \quad \alpha \in H . \end{aligned}$	$\begin{aligned} m_r[(S\otimes \mathrm{Id}\otimes \mathrm{Id})(\mathrm{Id}\otimes\Delta)\Delta(\alpha)] &= e\otimes\alpha \\ m'_r[(\mathrm{Id}\otimes S\otimes \mathrm{Id})(\mathrm{Id}\otimes\Delta)\Delta(\alpha)] &= \alpha\otimes e, \quad \alpha\in H. \end{aligned}$

Conf. MathPix confidence: ■ $\geq$0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0492	222	■ 0.913 ● C|+3	`\begin{array}{ll} \frac{1}{2 \kappa_{0}^{2}}= \frac{96 f_{2} \Lambda^{2}-f_{0} \mathfrak{c}}{24 \pi^{2}} \gamma_{0}=\frac{1}{\pi^{2}}\left(48 f_{4} \Lambda^{4}-f_{2} \Lambda^{2} \mathfrak{c}+ \frac{f_{0}}{4} \mathfrak{d}\right) \\ \alpha_{0}=- \frac{3 f_{0}}{10 \pi^{2}} \quad \tau_{0}=\frac{11 f_{0}}{60 \pi^{2}} \\ \mu_{0}^{2}=2 \frac{f_{2} \Lambda^{2}}{f_{0}}-\frac{\mathfrak{e}}{\mathfrak{a}} \quad \xi_{0}=\frac{1}{12} \\ \lambda_{0}=\frac{\pi^{2} \mathfrak{b}}{2 f_{0} \mathfrak{a}^{2}} \end{array}`	$\begin{array}{ll} \frac{1}{2 \kappa_{0}^{2}}= \frac{96 f_{2} \Lambda^{2}-f_{0} \mathfrak{c}}{24 \pi^{2}} \gamma_{0}=\frac{1}{\pi^{2}}\left(48 f_{4} \Lambda^{4}-f_{2} \Lambda^{2} \mathfrak{c}+ \frac{f_{0}}{4} \mathfrak{d}\right) \\ \alpha_{0}=- \frac{3 f_{0}}{10 \pi^{2}} \quad \tau_{0}=\frac{11 f_{0}}{60 \pi^{2}} \\ \mu_{0}^{2}=2 \frac{f_{2} \Lambda^{2}}{f_{0}}-\frac{\mathfrak{e}}{\mathfrak{a}} \quad \xi_{0}=\frac{1}{12} \\ \lambda_{0}=\frac{\pi^{2} \mathfrak{b}}{2 f_{0} \mathfrak{a}^{2}} \end{array}$	$\begin{array}{ll} \frac{1}{2 \kappa_{0}^{2}}= \frac{96 f_{2} \Lambda^{2}-f_{0} \mathfrak{c}}{24 \pi^{2}} & \gamma_{0}=\frac{1}{\pi^{2}}(48 f_{4} \Lambda^{4}-f_{2} \Lambda^{2} \mathfrak{c}+ \frac{f_{0}}{4} \mathfrak{d}) \\ \alpha_{0}=- \frac{3 f_{0}}{10 \pi^{2}} & \tau_{0}=\frac{11 f_{0}}{60 \pi^{2}} \\ \mu_{0}^{2}=2 \frac{f_{2} \Lambda^{2}}{f_{0}}-\frac{\mathfrak{e}}{\mathfrak{a}} & \xi_{0}=\frac{1}{12} \\ \lambda_{0}=\frac{\pi^{2} \mathfrak{b}}{2 f_{0} \mathfrak{a}^{2}} & \end{array}$
cardona-qft-methods_ EQ0105	063	■ 0.913 ● N|+0	`\begin{aligned} & f\left|D_{A} \right|^{-n}=f|D|^{-n}=2^{m+1} \pi^{n / 2} \Gamma \left(\frac{n}{2}\right)^{-1}, \\ & f\left|D_{A} \right|^{-n+k}=0 \text { for } k \text { odd, } \\ & f\left|D_{A}\right|^{-n+2}=0 . \end{aligned}`	$\begin{aligned} & f\left\|D_{A} \right\|^{-n}=f\|D\|^{-n}=2^{m+1} \pi^{n / 2} \Gamma \left(\frac{n}{2}\right)^{-1}, \\ & f\left\|D_{A} \right\|^{-n+k}=0 \text { for } k \text { odd, } \\ & f\left\|D_{A}\right\|^{-n+2}=0 . \end{aligned}$	$\begin{aligned} & \fint \|D_{A}\|^{-n} = \fint \|D\|^{-n}=2^{m+1} \pi^{n / 2} \Gamma (\tfrac{n}{2})^{-1}, \\ & \fint \|D_{A}\|^{-n+k}=0 \text { for } k \text { odd, } \\ & \fint \|D_{A}\|^{-n+2}=0 . \end{aligned}$
cardona-qft-methods_ EQ0096	056	■ 0.914 ● N|+1	`\operatorname{Tr}\left(g\left(t \mathcal{D}^{2} \right)\right) \underset{t \downarrow 0}{\sim} \sum_{\alpha \in \mathrm{Sp}^{-}}\left[\frac{1}{2} \operatorname{Res}_{s=-2 \alpha} \zeta_{\mathcal{D}}(s) \int_{0}^{\infty} g(y) y^{-\alpha-1} d y\right] t^{ \alpha}`	$\operatorname{Tr}\left(g\left(t \mathcal{D}^{2} \right)\right) \underset{t \downarrow 0}{\sim} \sum_{\alpha \in \mathrm{Sp}^{-}}\left[\frac{1}{2} \operatorname{Res}_{s=-2 \alpha} \zeta_{\mathcal{D}}(s) \int_{0}^{\infty} g(y) y^{-\alpha-1} d y\right] t^{ \alpha}$	$\operatorname{Tr}\big(g(t \mathcal{D}^{2})\big) \underset{t \downarrow 0}{\sim} \sum_{\alpha \in \mathrm{Sp}^{-}} \big[\tfrac{1}{2} \underset{s=-2\alpha}{\operatorname{Res}} \zeta_{\mathcal{D}}(s) \int_{0}^{\infty} g(y)\, y^{-\alpha-1} dy\big]\, t^{\alpha}.$
cardona-qft-methods_ EQ0386 (26)	179	■ 0.918 ● K|+0	`\mathbf{E}_{4}=d L_{3}^{A d S} .`	$\mathbf{E}_{4}=d L_{3}^{A d S} .$	$\mathbf{E}_{4}=d L_{3}^{AdS}.$
cardona-qft-methods_ EQ0397 (32)	181	■ 0.919 ● N|+0	`L_{3}^{E x o t i c}=L_{3}^{L o r} \pm \frac{2}{l^{2}} L_{3}^{T o r} .`	$L_{3}^{E x o t i c}=L_{3}^{L o r} \pm \frac{2}{l^{2}} L_{3}^{T o r} .$	$L_{3}^{Exotic}=L_{3}^{Lor} \pm \frac{2}{l^{2}} L_{3}^{Tor}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_EQ0779 (20)</td><td>297</td><td>■ 0.922
● C|+4</td><td><code>\begin{aligned} \beta_{M N}^{G} & =\alpha^{\prime} \left(R_{M N}+2 \nabla_{M} \nabla_{N} \Phi-\frac{1}{4} H_{M P Q} H_{N}^{P Q}\right)+O\left(\alpha^{\prime 2} \right), \\ \beta_{M N}^{B} & =\alpha^{\prime}\left(- \frac{1}{2} \nabla^{P} H_{P M N}+\nabla^{P} \Phi H_{P M N}\right)+O\left(\alpha^{\prime 2}\right), \\ \beta^{ \Phi} & =\alpha^{\prime}\left(\frac{D-26}{6 \alpha^{ \prime}}-\frac{1}{2} \nabla^{2} \Phi+\nabla_{M} \Phi \nabla^{M} \Phi-\frac{1}{24} H_{M N P} H^{M N P}\right)+O \left(\alpha^{\prime 2}\right), \end{aligned}</code></td><td>$$\begin{aligned} \beta_{M N}^{G} & =\alpha^{\prime} \left(R_{M N}+2 \nabla_{M} \nabla_{N} \Phi-\frac{1}{4} H_{M P Q} H_{N}^{P Q}\right)+O\left(\alpha^{\prime 2} \right), \\ \beta_{M N}^{B} & =\alpha^{\prime}\left(- \frac{1}{2} \nabla^{P} H_{P M N}+\nabla^{P} \Phi H_{P M N}\right)+O\left(\alpha^{\prime 2}\right), \\ \beta^{ \Phi} & =\alpha^{\prime}\left(\frac{D-26}{6 \alpha^{ \prime}}-\frac{1}{2} \nabla^{2} \Phi+\nabla_{M} \Phi \nabla^{M} \Phi-\frac{1}{24} H_{M N P} H^{M N P}\right)+O \left(\alpha^{\prime 2}\right), \end{aligned}$$</td><td>$$\begin{aligned} \beta_{MN}^{G} &= \alpha'\left(R_{MN}+2\nabla_M\nabla_N\Phi-\frac{1}{4}H_{MPQ}H_N{}^{PQ}\right)+O(\alpha'^2), \\ \beta_{MN}^{B} &= \alpha'\left(-\frac{1}{2}\nabla^P H_{PMN}+\nabla^P\Phi H_{PMN}\right)+O(\alpha'^2), \\ \beta^{\Phi} &= \alpha'\left(\frac{D-26}{6\alpha'}-\frac{1}{2}\nabla^2\Phi+\nabla_M\Phi\nabla^M\Phi-\frac{1}{24}H_{MNP}H^{MNP}\right)+O(\alpha'^2), \end{aligned} \qquad (20)$$</td></tr>
<tr><td>cardona-qft-methods_EQ0238</td><td>114</td><td>■ 0.922
● K|+0</td><td><code>c(E)=1+c_{1}(E)+c_{2}(E)+\cdots+c_{\mathrm{rk}} E(E) .</code></td><td>$c(E)=1+c_{1}(E)+c_{2}(E)+\cdots+c_{\mathrm{rk}} E(E).$</td><td>$c(E)=1+c_1(E)+c_2(E)+\cdots+c_{\mathrm{rk}\,E}(E) \quad .$</td></tr>
<tr><td>cardona-qft-methods_EQ0264</td><td>125</td><td>■ 0.926
● N|+0</td><td><code>\sum_{\text {graphs } G}\langle G\rangle=\exp \left(\sum_{\text {connected graphs } G}\langle G\rangle\right) .</code></td><td>$$\sum_{\text {graphs } G}\langle G\rangle=\exp \left(\sum_{\text {connected graphs } G}\langle G\rangle\right) .$$</td><td>$$\sum_{\text{graphs } G}\langle G\rangle=\exp\left(\sum_{\textit{connected}\text{ graphs } G}\langle G\rangle\right) \quad .$$</td></tr>
<tr><td>cardona-qft-methods_EQ0768 (9)</td><td>294</td><td>■ 0.927
● K|+0</td><td><code>\left[\alpha_{n}^{P}, \alpha_{m}^{Q}\right]=\left[\tilde{\alpha}_{n}^{P}, \tilde{\alpha}_{m}^{Q}\right]=n \delta_{n+m} \eta^{P Q}, \quad\left[x^{P}, p^{Q}\right]=i \eta^{P Q} .</code></td><td>$\left[\alpha_{n}^{P}, \alpha_{m}^{Q}\right]=\left[\tilde{\alpha}_{n}^{P}, \tilde{\alpha}_{m}^{Q}\right]=n \delta_{n+m} \eta^{P Q}, \quad\left[x^{P}, p^{Q}\right]=i \eta^{P Q} .$</td><td>$[\alpha_n^P,\alpha_m^Q]=[\tilde{\alpha}_n^P,\tilde{\alpha}_m^Q]=n\delta_{n+m}\eta^{PQ} \;, \quad [x^P,p^Q]=i\eta^{PQ} \;.$</td></tr>
<tr><td>cardona-qft-methods_EQ0098 (35)</td><td>056</td><td>■ 0.928
● N|+1</td><td><code>\begin{aligned} & a_{k}\left(\mathcal{D}^{2}\right)= \frac{1}{2} \Gamma\left(\frac{d-k}{2}\right) f| \mathcal{D}|^{-d+k} \quad \text { for } k=0, \cdots, d-1, \\ & a_{d}\left(\mathcal{D}^{2}\right)=\operatorname{dim} \mathcal{D}+\zeta_{\mathcal{D}^{2}}(0) . \end{aligned}</code></td><td>$$\begin{aligned} & a_{k}\left(\mathcal{D}^{2}\right)= \frac{1}{2} \Gamma\left(\frac{d-k}{2}\right) f| \mathcal{D}|^{-d+k} \quad \text { for } k=0, \cdots, d-1, \\ & a_{d}\left(\mathcal{D}^{2}\right)=\operatorname{dim} \mathcal{D}+\zeta_{\mathcal{D}^{2}}(0) . \end{aligned}$$</td><td>$$\begin{aligned} & a_k(\mathcal{D}^2)=\tfrac{1}{2}\,\Gamma(\tfrac{d-k}{2}) ⨍ |\mathcal{D}|^{-d+k} \qquad \textit{for } k=0,\cdots,d-1, \\ & a_d(\mathcal{D}^2)=\dim\,\mathcal{D}+\zeta_{\mathcal{D}^2}(0). \end{aligned}$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0734 (1)	279	■0.929 ● C|+3	`\begin{aligned} S=\frac{N}{\lambda} \int \frac{ \mathrm{~d}^{4} x}{(2 \pi)^{4}} \operatorname{Tr} & \left(\frac{1}{4} F^{\mu \nu} F_{\mu \nu}+\frac{1}{2} \left(D^{\mu}\right)^{*} \varphi_{m} D_{\mu} \varphi_{m}+ \bar{\psi}_{\dot{\alpha}}^{a}\left(\sigma_{\mu}\right)^{ \dot{\alpha} \beta} D^{\mu} \psi_{a \beta}\right. \ \ & -\frac{1}{4}\left[\varphi^{m}, \varphi^{n}\right] \left[\varphi_{m}, \varphi_{n}\right] \\ & \left.- \frac{i}{2} \psi_{a \alpha}\left(\Sigma_{m}\right)^{a b} \varepsilon^{\alpha \beta}\left[\varphi^{m} \psi_{a \beta}\right]-\frac{i}{2} \bar{\psi}_{\dot{\alpha}}^{a} \left(\Sigma_{m}\right)^{a b} \varepsilon^{\dot{\alpha} \dot{\beta}}\left[\varphi^{m} \bar{\psi}_{a \dot{\beta}} \right]\right) \\ & + \text { ghosts }+ \text { gauge fixing }+ \text { topological term } \end{aligned}`	$\begin{aligned} S=\frac{N}{\lambda} \int \frac{ \mathrm{~d}^{4} x}{(2 \pi)^{4}} \operatorname{Tr} & \left(\frac{1}{4} F^{\mu \nu} F_{\mu \nu}+\frac{1}{2} \left(D^{\mu}\right)^{*} \varphi_{m} D_{\mu} \varphi_{m}+ \bar{\psi}_{\dot{\alpha}}^{a}\left(\sigma_{\mu}\right)^{ \dot{\alpha} \beta} D^{\mu} \psi_{a \beta}\right. \\ & -\frac{1}{4}\left[\varphi^{m}, \varphi^{n}\right] \left[\varphi_{m}, \varphi_{n}\right] \\ & \left.- \frac{i}{2} \psi_{a \alpha}\left(\Sigma_{m}\right)^{a b} \varepsilon^{\alpha \beta}\left[\varphi^{m} \psi_{a \beta}\right]-\frac{i}{2} \bar{\psi}_{\dot{\alpha}}^{a} \left(\Sigma_{m}\right)^{a b} \varepsilon^{\dot{\alpha} \dot{\beta}}\left[\varphi^{m} \bar{\psi}_{a \dot{\beta}} \right]\right) \\ & + \text { ghosts }+ \text { gauge fixing }+ \text { topological term } \end{aligned}$	$\begin{aligned} S=\frac{N}{\lambda}\int\frac{\mathrm{d}^4x}{(2\pi)^4}\mathrm{Tr}\Big(&\frac{1}{4}F^{\mu\nu}F_{\mu\nu}+\frac{1}{2}(D^\mu)^*\varphi_m D_\mu\varphi_m+\bar\psi^a_{\dot\alpha}(\sigma_\mu)^{\dot\alpha\beta}D^\mu\psi_{a\beta} \\ &-\frac{1}{4}[\varphi^m,\varphi^n][\varphi_m,\varphi_n] \\ &-\frac{i}{2}\psi_{a\alpha}(\Sigma_m)^{ab}\varepsilon^{\alpha\beta}[\varphi^m\psi_{a\beta}]-\frac{i}{2}\bar\psi^a_{\dot\alpha}(\Sigma_m)^{ab}\varepsilon^{\dot\alpha\dot\beta}[\varphi^m\bar\psi_{a\dot\beta}]\Big) \\ &+\text{ghosts}+\text{gauge fixing}+\text{topological term}\,. \qquad (1)\end{aligned}$
cardona-qft-methods_ EQ0972 (12)	363	■0.929 ● K|+0	`\left[X_{\Gamma} \backslash \Sigma_{n}\right]=\left[X_{ \Gamma^{\vee}} \backslash \Sigma_{n}\right] .`	$\left[X_{\Gamma} \backslash \Sigma_{n}\right]=\left[X_{ \Gamma^{\vee}} \backslash \Sigma_{n}\right] .$	$[X_\Gamma \setminus \Sigma_n] = [X_{\Gamma^\vee} \setminus \Sigma_n]\,.$
cardona-qft-methods_ EQ0398	181	■0.929 ● S|+1	`\begin{aligned} d L_{3}^{\text {Exotic }} & =R_{b}^{a} R_{a}^{b} \pm \frac{2}{l^{2}}\left(T^{a} T_{a}-e^{a} e^{b} R_{a b}\right) \\ & =F_{B}^{A} F_{A}^{B} \end{aligned}`	$\begin{aligned} d L_{3}^{\text {Exotic }} & =R_{b}^{a} R_{a}^{b} \pm \frac{2}{l^{2}}\left(T^{a} T_{a}-e^{a} e^{b} R_{a b}\right) \\ & =F_{B}^{A} F_{A}^{B} \end{aligned}$	$\begin{aligned} dL_3^{Exotic} &= R^a_b R^b_a \pm \frac{2}{l^2}\left(T^a T_a - e^a e^b R_{ab}\right) \\ &= F^A_B F^B_A. \end{aligned}$
cardona-qft-methods_ EQ0044	033	■0.933 ● N|+0	`i[D, f]=i\left[d+d^{*}, f\right]=i\left(i \sigma_{1}^{d}(d f)-i \sigma_{1}^{d^{*}}(d f)\right)=-i( \epsilon+\iota)(d f) .`	$i[D, f]=i\left[d+d^{*}, f\right]=i\left(i \sigma_{1}^{d}(d f)-i \sigma_{1}^{d^{*}}(d f)\right)=-i(\epsilon+\iota)(d f) .$	$i[D,f]=i[d+d^*,\,f]=i\big(i\sigma_1^d(df)-i\sigma_1^{d^*}(df)\big)=-i(\epsilon+\iota)(df).$
cardona-qft-methods_ EQ0112	073	■0.933 ● K|+0	`\operatorname{Ind}(A)=-\operatorname{Ind}\left(A^{*} \right)=\operatorname{dimKer}(A)-\operatorname{dimKer} \left(A^{*}\right) .`	$\operatorname{Ind}(A)=-\operatorname{Ind}\left(A^{*} \right)=\operatorname{dimKer}(A)-\operatorname{dimKer} \left(A^{*}\right) .$	$\operatorname{Ind}(A) \;=\; -\operatorname{Ind}(A^*) \;=\; \dim\operatorname{Ker}(A) \;-\; \dim\operatorname{Ker}(A^*).$
cardona-qft-methods_ EQ0845 (3)	329	■0.933 ● N|+2	`\delta S=\int_{\Sigma} \frac{\delta \mathcal{L}}{\delta X} \delta X+\frac{\delta \mathcal{L}}{\delta \eta} \delta \eta+\text { boundary terms }`	$\delta S=\int_{\Sigma} \frac{\delta \mathcal{L}}{\delta X} \delta X+\frac{\delta \mathcal{L}}{\delta \eta} \delta \eta+\text { boundary terms }$	$\delta S=\int_\Sigma \frac{\delta\mathcal{L}}{\delta X}\delta X+\frac{\delta\mathcal{L}}{\delta\eta}\delta\eta+\text{boundary terms.}$
cardona-qft-methods_ EQ0040	032	■0.936 ● K|+0	`\begin{gathered} \epsilon\left(\omega_{1}\right) \omega_{2}=\omega_{1} \wedge \omega_{2}, \\ \iota(\omega) \left(\omega_{1} \wedge \cdots \wedge \omega_{m}\right)= \sum_{j=1}^{m}(-1)^{j-1} g\left(\omega, \omega_{j}\right) \omega_{1} \wedge \cdots \wedge \widehat{\omega_{j}} \wedge \cdots \wedge \omega_{m} \end{gathered}`	$\begin{gathered} \epsilon\left(\omega_{1}\right) \omega_{2}=\omega_{1} \wedge \omega_{2}, \\ \iota(\omega) \left(\omega_{1} \wedge \cdots \wedge \omega_{m}\right)= \sum_{j=1}^{m}(-1)^{j-1} g\left(\omega, \omega_{j}\right) \omega_{1} \wedge \cdots \wedge \widehat{\omega_{j}} \wedge \cdots \wedge \omega_{m} \end{gathered}$	$\begin{gathered} \epsilon(\omega_1)\,\omega_2=\omega_1\wedge\omega_2, \\ \iota(\omega)\,(\omega_1\wedge\cdots\wedge\omega_m)=\sum_{j=1}^m(-1)^{j-1}g(\omega,\omega_j)\,\omega_1\wedge\cdots\wedge\widehat{\omega_j}\wedge\cdots\wedge\omega_m \end{gathered}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0745	283	0.941 N|+1	`\langle\mathcal{O}(x), \overline{\mathcal{O}}(y)\rangle= \frac{1}{(x-y)^{2 \Delta}} \sim \frac{1}{(x-y)^{2 \Delta_{0}}}\left(1-g_{\mathrm{YM}}^{2} \gamma_{1} \log \left[(x-y)^{2} \Lambda^{2}\right]\right),`	$\langle\mathcal{O}(x), \overline{\mathcal{O}}(y)\rangle= \frac{1}{(x-y)^{2 \Delta}} \sim \frac{1}{(x-y)^{2 \Delta_{0}}}\left(1-g_{\mathrm{YM}}^{2} \gamma_{1} \log \left[(x-y)^{2} \Lambda^{2}\right]\right),$	$\langle\mathcal{O}(x),\overline{\mathcal{O}}(y)\rangle=\frac{1}{(x-y)^{2\Delta}}\sim\frac{1}{(x-y)^{2\Delta_0}}(1-g_{\mathrm{YM}}^2\gamma_1\log[(x-y)^2\Lambda^2])\,,$
cardona-qft-methods_ EQ0661 (11)	262	0.943 W|+2	`\begin{aligned} S^{(2)} & =\frac{-i^{2}}{2!} \int d_{x}^{4} d^{4} y T\left(H_{I}(x), H_{I}(y)\right) \\ & =\frac{-i^{2}}{2!} \int d_{x}^{4} d^{4} y\left(\theta\left(x_{0}-y_{0}\right)\left[H_{I}(x), H_{I}(y)\right]+H_{I}(y) H_{I}(x)\right) \end{aligned}`	$\begin{aligned} S^{(2)} & =\frac{-i^{2}}{2!} \int d_{x}^{4} d^{4} y T\left(H_{I}(x), H_{I}(y)\right) \\ & =\frac{-i^{2}}{2!} \int d_{x}^{4} d^{4} y\left(\theta\left(x_{0}-y_{0}\right)\left[H_{I}(x), H_{I}(y)\right]+H_{I}(y) H_{I}(x)\right) \end{aligned}$	$\begin{aligned} S^{(2)} & =\frac{-i^2}{2!}\int d_x^4d^4yT(H_I(x),H_I(y)) \\ & =\frac{-i^2}{2!}\int d_x^4d^4y\,(\theta(x_0-y_0)[H_I(x),H_I(y)]+H_I(y)H_I(x))\,, \end{aligned}$
cardona-qft-methods_ EQ0308 (12)	154	0.944 N|+0	`d\left(\delta_{\text {gauge }} \mathcal{C}_{2 k-1}=0\right) .`	$d\left(\delta_{\text {gauge }} \mathcal{C}_{2 k-1}=0\right) .$	$d(\delta_{gauge}\mathcal{C}_{2k-1}=0).$
cardona-qft-methods_ EQ0333 (19)	163	0.947 W|+0	`R_{b}^{a}=\bar{R}_{b}^{a}+\bar{D} \kappa_{b}^{a}+\kappa_{c}^{a} \kappa_{b}^{c},`	$R_{b}^{a}=\bar{R}_{b}^{a}+\bar{D} \kappa_{b}^{a}+\kappa_{c}^{a} \kappa_{b}^{c},$	$R^a_b=\bar{R}^a{}_b+\bar{D}\kappa^a{}_b+\kappa^a{}_c\kappa^c{}_b,$
cardona-qft-methods_ EQ0432	208	0.947 N|+0	`\mathcal{B}=\oplus^{4-\text { times }} M_{2}(\mathbb{C}) \oplus M_{6}(\mathbb{C}) .`	$\mathcal{B}=\oplus^{4-\text { times }} M_{2}(\mathbb{C}) \oplus M_{6}(\mathbb{C}) .$	$\mathcal{B}=\oplus^{4-times}M_2(\mathbb{C})\oplus M_6(\mathbb{C}).$
cardona-qft-methods_ EQ0278	135	0.947 S|+0	`(\mathbb{T}-1) \mathbb{T}(\mathbb{T}+1)^{2}+\mathbb{T}(\mathbb{T}+1)^{2}+(\mathbb{T}+1) \mathbb{T}(\mathbb{T}+1)=\mathbb{T}(\mathbb{T}+1)^{3} .`	$(\mathbb{T}-1) \mathbb{T}(\mathbb{T}+1)^{2}+\mathbb{T}(\mathbb{T}+1)^{2}+(\mathbb{T}+1) \mathbb{T}(\mathbb{T}+1)=\mathbb{T}(\mathbb{T}+1)^{3} .$	$(\mathbb{T}-1)\mathbb{T}(\mathbb{T}+1)^2+\mathbb{T}(\mathbb{T}+1)^2+(\mathbb{T}+1)\mathbb{T}(\mathbb{T}+1)=\mathbb{T}(\mathbb{T}+1)^3\quad.$
cardona-qft-methods_ EQ0132	078	0.948 N|+0	`f=\left(f_{1}, \ldots, f_{N}\right): \mathbb{R}^{n} \longrightarrow \mathbb{R}^{N}, \quad f_{j} \in C^{\infty}\left(\mathbb{R}^{n}\right) .`	$f=\left(f_{1}, \ldots, f_{N}\right): \mathbb{R}^{n} \longrightarrow \mathbb{R}^{N}, \quad f_{j} \in C^{\infty}\left(\mathbb{R}^{n}\right) .$	$f\ =\ (f_1,\ldots,f_N):\mathbb{R}^n\ \longrightarrow\ \mathbb{R}^N,\qquad f_j\in C^\infty(\mathbb{R}^n).$
cardona-qft-methods_ EQ0237 (3)	113	0.948 N|+0	`N_{q}(X)=a_{0}+a_{1} q+\cdots+a_{r} q^{r},`	$N_{q}(X)=a_{0}+a_{1} q+\cdots+a_{r} q^{r},$	$N_q(X)=a_0+a_1q+\cdots+a_rq^r\quad,$
cardona-qft-methods_ EQ0316 (2)	159	0.949 N|+0	`\begin{aligned} d s^{2} & =\eta_{a b} d z^{a} d z^{b} \\ & =\eta_{a b} e_{\mu}^{a}(x) d x^{\mu} e_{\nu}^{b}(x) d x^{\nu} \\ & =g_{\mu \nu}(x) d x^{\mu} d x^{\nu}, \end{aligned}`	$\begin{aligned} d s^{2} & =\eta_{a b} d z^{a} d z^{b} \\ & =\eta_{a b} e_{\mu}^{a}(x) d x^{\mu} e_{\nu}^{b}(x) d x^{\nu} \\ & =g_{\mu \nu}(x) d x^{\mu} d x^{\nu}, \end{aligned}$	$\begin{aligned} ds^2 & =\eta_{ab}dz^adz^b \\ & =\eta_{ab}\,e^a_\mu(x)dx^\mu\,e^b_\nu(x)dx^\nu \\ & =g_{\mu\nu}(x)dx^\mu dx^\nu, \end{aligned}$
cardona-qft-methods_ EQ0363	174	0.949 N|+1	`\begin{aligned} \delta e^{a} & =-\lambda^{a}{ }_{c} e^{c} \\ \delta R^{a b} & =-\left(\lambda^{a}{ }_{c} R^{c b}+\lambda^{b}{ }_{c} R^{a c}\right) \\ \delta \epsilon_{a b c} & =-\left(\lambda^{d}{ }_{a} \epsilon_{d b c}+\lambda^{d}{ }_{b} \epsilon_{a d c}+\lambda^{d}{ }_{c} \epsilon_{a b d}\right) \equiv 0 . \end{aligned}`	$\begin{aligned} \delta e^{a} & =-\lambda^{a}{ }_{c} e^{c} \\ \delta R^{a b} & =-\left(\lambda^{a}{ }_{c} R^{c b}+\lambda^{b}{ }_{c} R^{a c}\right) \\ \delta \epsilon_{a b c} & =-\left(\lambda^{d}{ }_{a} \epsilon_{d b c}+\lambda^{d}{ }_{b} \epsilon_{a d c}+\lambda^{d}{ }_{c} \epsilon_{a b d}\right) \equiv 0 . \end{aligned}$	$\begin{aligned} \delta e^a & =-\lambda^a{}_ce^c \\ \delta R^{ab} & =-(\lambda^a{}_cR^{cb}+\lambda^b{}_cR^{ac}), \\ \delta\epsilon_{abc} & =-(\lambda^d{}_a\epsilon_{dbc}+\lambda^d{}_b\epsilon_{adc}+\lambda^d{}_c\epsilon_{abd})\equiv 0. \end{aligned}$
cardona-qft-methods_ EQ0837 (16)	317	0.950 K|+0	`F_{\mathrm{IIA} \pm}=F_{0} \pm F_{2}+F_{4} \pm F_{6}, \quad F_{\mathrm{IIB} \pm}=F_{1} \pm F_{3}+F_{5} .`	$F_{\mathrm{IIA} \pm}=F_{0} \pm F_{2}+F_{4} \pm F_{6}, \quad F_{\mathrm{IIB} \pm}=F_{1} \pm F_{3}+F_{5} .$	$F_{\mathrm{IIA}\,\pm}=F_0\pm F_2+F_4\pm F_6\ ,\qquad F_{\mathrm{IIB}\,\pm}=F_1\pm F_3+F_5\ .$
cardona-qft-methods_ EQ0186	093	0.950 K|+0	`c(v) c(u)+c(u) c(v)=-2\langle v, u\rangle \mathrm{Id}, \quad \text { for all } u, v \in V .`	$c(v) c(u)+c(u) c(v)=-2\langle v, u\rangle \mathrm{Id}, \quad \text { for all } u, v \in V .$	$c(v)\,c(u)+c(u)\,c(v)\ =\ -2\langle v,u\rangle\ \mathrm{Id},\qquad \text{for all } u,v\in V.$
cardona-qft-methods_ EQ0931	353	0.950 N|+0	`E G_{n} / C(g) \times E G_{n} / C(h) \stackrel{\Delta}{\longleftarrow} E G_{n} /(C(g) \cap C(h)) \longrightarrow E G_{n} / C(g h)`	$E G_{n} / C(g) \times E G_{n} / C(h) \stackrel{\Delta}{\longleftarrow} E G_{n} /(C(g) \cap C(h)) \longrightarrow E G_{n} / C(g h)$	$EG_n/C(g)\times EG_n/C(h)\xleftarrow{\ \Delta\ }EG_n/(C(g)\cap C(h))\longrightarrow EG_n/C(gh)$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0360 (31)	171	■ 0.951 ● N|+2	`C_{7}^{L o r}=\operatorname{Tr}\left[\omega(d \omega)^{3}+\frac{8}{5} \omega^{3}(d \omega)^{2}+ \frac{4}{5} \omega^{2}(d \omega) \omega(d \omega)+2 \omega^{5}(d \omega)+\frac{4}{7} \omega^{7}\right]`	$C_{7}^{L o r}=\operatorname{Tr}\left[\omega(d \omega)^{3}+\frac{8}{5} \omega^{3}(d \omega)^{2}+\frac{4}{5} \omega^{2}(d \omega) \omega(d \omega)+2 \omega^{5}(d \omega)+\frac{4}{7} \omega^{7}\right]$	$C_7^{Lor}=Tr[\omega(d\omega)^3+\frac{8}{5}\omega^3(d\omega)^2+\frac{4}{5}\omega^2(d\omega)\omega(d\omega)+2\omega^5(d\omega)+\frac{4}{7}\omega^7],$
cardona-qft-methods_ EQ0315 (1)	159	■ 0.953 ● K|+0	`d z^{a}=e_{\mu}^{a}(x) d x^{\mu},`	$d z^{a}=e_{\mu}^{a}(x) d x^{\mu},$	$dz^a=e^a_\mu(x)dx^\mu,$
cardona-qft-methods_ EQ0765 (6)	293	■ 0.953 ● W|+0	`\begin{aligned} X^{M}(\tau, \sigma) & =X_{R}^{M}\left(\sigma^{-}\right)+X_{L}^{M}\left(\sigma^{+}\right) \\ X_{R}^{M}\left(\sigma^{-}\right) & =\frac{1}{2} x^{M}+ \alpha^{\prime} p^{M} \sigma^{-}+\mathrm{i}\left(\frac{ \alpha^{\prime}}{2}\right)^{1 / 2} \sum_{n \neq 0} \frac{1}{n} \alpha_{n}^{M} e^{-2 \mathrm{i} n \sigma^{-}} \\ X_{L}^{M}\left(\sigma^{+}\right) & =\frac{1}{2} x^{M}+ \alpha^{\prime} p^{M} \sigma^{+}+\mathrm{i}\left(\frac{ \alpha^{\prime}}{2}\right)^{1 / 2} \sum_{n \neq 0} \frac{1}{n} \tilde{\alpha}_{n}^{M} e^{-2 \mathrm{i} n \sigma^{+}}, \end{aligned}`	$\begin{aligned} X^{M}(\tau, \sigma) & =X_{R}^{M}\left(\sigma^{-}\right)+X_{L}^{M}\left(\sigma^{+}\right) \\ X_{R}^{M}\left(\sigma^{-}\right) & =\frac{1}{2} x^{M}+\alpha^{\prime} p^{M} \sigma^{-}+\mathrm{i}\left(\frac{\alpha^{\prime}}{2}\right)^{1 / 2} \sum_{n \neq 0} \frac{1}{n} \alpha_{n}^{M} e^{-2 \mathrm{i} n \sigma^{-}} \\ X_{L}^{M}\left(\sigma^{+}\right) & =\frac{1}{2} x^{M}+\alpha^{\prime} p^{M} \sigma^{+}+\mathrm{i}\left(\frac{\alpha^{\prime}}{2}\right)^{1 / 2} \sum_{n \neq 0} \frac{1}{n} \tilde{\alpha}_{n}^{M} e^{-2 \mathrm{i} n \sigma^{+}}, \end{aligned}$	$\begin{aligned} X^M(\tau,\sigma)&=X^M_R(\sigma^-)+X^M_L(\sigma^+) \\ X^M_R(\sigma^-)&=\frac{1}{2}x^M+\alpha' p^M\sigma^-+\mathrm{i}\left(\frac{\alpha'}{2}\right)^{1/2}\sum_{n\neq 0}\frac{1}{n}\alpha^M_n e^{-2\mathrm{i}n\sigma^-} \\ X^M_L(\sigma^+)&=\frac{1}{2}x^M+\alpha' p^M\sigma^++\mathrm{i}\left(\frac{\alpha'}{2}\right)^{1/2}\sum_{n\neq 0}\frac{1}{n}\tilde{\alpha}^M_n e^{-2\mathrm{i}n\sigma^+}\ , \end{aligned}$
cardona-qft-methods_ EQ0736 (6)	280	■ 0.954 ● N|+1	`\begin{aligned} {\left[M_{\mu \nu}, K_{\rho}\right] } & =i\left(\eta_{\rho \nu} K_{\mu}-\eta_{\mu \rho} K_{ \nu}\right), \\ {\left[P_{\mu}, K_{\nu}\right] } & =2 i \left(\eta_{\mu \nu} \mathcal{D}-M_{\mu \nu}\right) . \end{aligned}`	$\begin{aligned} {\left[M_{\mu \nu}, K_{\rho}\right] } & =i\left(\eta_{\rho \nu} K_{\mu}-\eta_{\mu \rho} K_{\nu}\right), \\ {\left[P_{\mu}, K_{\nu}\right] } & =2 i\left(\eta_{\mu \nu} \mathcal{D}-M_{\mu \nu}\right) . \end{aligned}$	$\begin{aligned}[M_{\mu\nu},K_\rho]&=i(\eta_{\rho\nu}K_\mu-\eta_{\mu\rho}K_\nu)\,, \\ [P_\mu,K_\nu]&=2i(\eta_{\mu\nu}\mathcal{D}-M_{\mu\nu})\,. \end{aligned}$
cardona-qft-methods_ EQ0323 (9)	161	■ 0.954 ● N|+0	`\varphi_{\\|}^{\mu}(x)=\varphi^{\mu}(x)+d x^{\lambda} \left[\partial_{\lambda} \varphi^{\mu}(x)+\Gamma_{\lambda \rho}^{\mu}(x) \varphi^{\rho}(x)\right] .`	$\varphi_{\Vert}^{\mu}(x)=\varphi^{\mu}(x)+d x^{\lambda}\left[\partial_{\lambda} \varphi^{\mu}(x)+\Gamma_{\lambda \rho}^{\mu}(x) \varphi^{\rho}(x)\right] .$	$\varphi^\mu_{\Vert}(x)=\varphi^\mu(x)+dx^\lambda[\partial_\lambda\varphi^\mu(x)+\Gamma^\mu_{\lambda\rho}(x)\varphi^\rho(x)].$
cardona-qft-methods_ EQ0147	083	■ 0.957 ● N|+0	`\mathcal{D}:=-i \frac{d}{d t}+\sin t: C^{\infty}(M) \rightarrow C^{\infty}(M) .`	$\mathcal{D}:=-i \frac{d}{d t}+\sin t: C^{\infty}(M) \rightarrow C^{\infty}(M) .$	$\mathcal{D}\ :=\ -i\,\frac{d}{dt}+\sin t:\ C^\infty(M)\ \rightarrow\ C^\infty(M).$
cardona-qft-methods_ EQ0065	040	■ 0.957 ● N|+1	`\begin{aligned} \omega_{\nu} & =\frac{1}{2} g_{\nu \mu} \left(\mathbb{A}^{\mu}+g^{\sigma \varepsilon} \Gamma_{ \sigma \varepsilon}^{\mu} \mathrm{id}_{V}\right) \\ E & = \mathbb{B}-g^{\nu \mu}\left(\partial_{\nu} \omega_{\mu}+ \omega_{\nu} \omega_{\mu}-\omega_{\sigma} \Gamma_{\nu \mu}^{\sigma}\right) . \end{aligned}`	$\begin{aligned} \omega_{\nu} & =\frac{1}{2} g_{\nu \mu}\left(\mathbb{A}^{\mu}+g^{\sigma \varepsilon} \Gamma_{\sigma \varepsilon}^{\mu} \mathrm{id}_{V}\right) \\ E & =\mathbb{B}-g^{\nu \mu}\left(\partial_{\nu} \omega_{\mu}+\omega_{\nu} \omega_{\mu}-\omega_{\sigma} \Gamma_{\nu \mu}^{\sigma}\right) . \end{aligned}$	$\begin{aligned}\omega_\nu&=\tfrac{1}{2}g_{\nu\mu}(\mathbb{A}^\mu+g^{\sigma\varepsilon}\Gamma^\mu_{\sigma\varepsilon}\ \mathrm{id}_V)\,, \\ E&=\mathbb{B}-g^{\nu\mu}(\partial_\nu\omega_\mu+\omega_\nu\omega_\mu-\omega_\sigma\Gamma^\sigma_{\nu\mu})\,. \end{aligned}$
cardona-qft-methods_ EQ0618 (26)	256	■ 0.957 ● N|+0	`\left(f_{1} f_{2}\right)(p)=f_{1}(p) f_{2}(p) .`	$\left(f_{1} f_{2}\right)(p)=f_{1}(p) f_{2}(p) .$	$(f_1f_2)(p)=f_1(p)f_2(p).$
cardona-qft-methods_ EQ0751	284	■ 0.957 ● N|+0	`\left.\frac{30 \mid 32}{10|0 \oplus 10| 0}=10 \right \rvert\, 32,`	$\left.\frac{30 \mid 32}{10\vert 0 \oplus 10\vert 0}=10 \right\rvert\, 32,$	$\frac{30\vert 32}{10\vert 0\oplus 10\vert 0}=10\vert 32\,,$
cardona-qft-methods_ EQ0816 (16)	309	■ 0.958 ● N|+0	`J_{m n}=\mp 2 \mathrm{i} \eta_{ \pm}^{\dagger} \gamma_{m n} \eta_{ \pm} \quad \Omega_{m n p}=-2 \mathrm{i} \eta_{-}^{\dagger} \gamma_{m n p} \eta_{+} .`	$J_{m n}=\mp 2 \mathrm{i} \eta_{\pm}^{\dagger} \gamma_{m n} \eta_{\pm} \quad \Omega_{m n p}=-2 \mathrm{i} \eta_{-}^{\dagger} \gamma_{m n p} \eta_{+} .$	$J_{mn}=\mp 2\mathrm{i}\,\eta^\dagger_\pm\gamma_{mn}\eta_\pm \qquad \Omega_{mnp}=-2\mathrm{i}\,\eta^\dagger_-\gamma_{mnp}\eta_+\ .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_EQ0488</td><td>221</td><td>■0.958 ● W|-1</td><td>\begin{aligned} a_{0}(x, P) & =(4 \pi)^{-m / 2} \operatorname{Tr}(\mathrm{id}) \\ a_{2}(x, P) & =(4 \pi)^{-m / 2} \operatorname{Tr}\left(-\frac{R}{6} \mathrm{id}+E\right) \\ a_{4}(x, P) & =(4 \pi)^{-m / 2} \frac{1}{360} \operatorname{Tr}\left(-12 R_{; \mu}{ }^{\mu}+5 R^{2}-2 R_{\mu \nu} R^{\mu \nu}\right. \\ & +2 R_{\mu \nu \rho \sigma} R^{\mu \nu \rho \sigma}-60 R E+180 E^{2}+60 E_{; \mu}{ }^{\mu} \\ & \left.+30 \Omega_{\mu \nu} \Omega^{\mu \nu}\right) . \end{aligned}</td><td>$\begin{aligned} a_{0}(x, P) & =(4 \pi)^{-m / 2} \operatorname{Tr}(\mathrm{id}) \\ a_{2}(x, P) & =(4 \pi)^{-m / 2} \operatorname{Tr}\left(-\frac{R}{6} \mathrm{id}+E\right) \\ a_{4}(x, P) & =(4 \pi)^{-m / 2} \frac{1}{360} \operatorname{Tr}\left(-12 R_{; \mu}{ }^{\mu}+5 R^{2}-2 R_{\mu \nu} R^{\mu \nu}\right. \\ & +2 R_{\mu \nu \rho \sigma} R^{\mu \nu \rho \sigma}-60 R E+180 E^{2}+60 E_{; \mu}{ }^{\mu} \\ & \left.+30 \Omega_{\mu \nu} \Omega^{\mu \nu}\right) . \end{aligned}$</td><td>$\begin{aligned} a_0(x,P) &= (4\pi)^{-m/2}\operatorname{Tr}(\mathrm{id}) \\ a_2(x,P) &= (4\pi)^{-m/2}\operatorname{Tr}\left(-\tfrac{R}{6}\,\mathrm{id}+E\right) \\ a_4(x,P) &= (4\pi)^{-m/2}\tfrac{1}{360}\operatorname{Tr}(-12R_{;\mu}{}^{\mu}+5R^2-2R_{\mu\nu}R^{\mu\nu} \\ &\quad +2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}-60\,R\,E+180\,E^2+60\,E_{;\mu}{}^{\mu} \\ &\quad +30\,\Omega_{\mu\nu}\,\Omega^{\mu\nu}). \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_EQ0500</td><td>226</td><td>■0.959 ● N|+0</td><td>\frac{2 f_{2} \Lambda_{u n i f}^{2}}{f_{0}} \leq \mathfrak{c}\left(\Lambda_{u n i f}\right) \leq \frac{6 f_{2} \Lambda_{u n i f}^{2}}{f_{0}}</td><td>$\frac{2 f_{2} \Lambda_{u n i f}^{2}}{f_{0}} \leq \mathfrak{c}\left(\Lambda_{u n i f}\right) \leq \frac{6 f_{2} \Lambda_{u n i f}^{2}}{f_{0}}$</td><td>$\frac{2f_2\Lambda_{unif}^2}{f_0} \leq \mathfrak{c}(\Lambda_{unif}) \leq \frac{6f_2\Lambda_{unif}^2}{f_0}$</td></tr>
<tr><td>cardona-qft-methods_EQ0234 (1)</td><td>112</td><td>■0.961 ● K|+0</td><td>[\widetilde{X}]-[E]=[X]-[Z]</td><td>$[\widetilde{X}]-[E]=[X]-[Z]$</td><td>$[\widetilde{X}]-[E]=[X]-[Z]$</td></tr>
<tr><td>cardona-qft-methods_EQ0262</td><td>123</td><td>■0.961 ● C|-3</td><td>\hat{Y}_{G} \text { mixed-Tate } \Longrightarrow \left[\hat{Y}_{G}\right] \in \mathbb{Z}[\mathbb{L}] \Longrightarrow N_{q}(G) \text { is a polynomial in } q \text {. }</td><td>$\hat{Y}_{G} \text { mixed-Tate } \Longrightarrow \left[\hat{Y}_{G}\right] \in \mathbb{Z}[\mathbb{L}] \Longrightarrow N_{q}(G) \text { is a polynomial in } q \text {. }$</td><td>$\hat{Y}_G \text{ mixed-Tate } \Longrightarrow [\hat{Y}_G] \in \mathbb{Z}[\mathbb{L}] \Longrightarrow N_q(G) \text{ is a polynomial in } q.$</td></tr>
<tr><td>cardona-qft-methods_EQ0332</td><td>163</td><td>■0.963 ● N|+0</td><td>d e^{a}+\bar{\omega}_{b}^{a} \wedge e^{b} \equiv 0, \quad \text { and } \quad T^{a}=\kappa_{b}^{a} \wedge e^{b} .</td><td>$d e^{a}+\bar{\omega}_{b}^{a} \wedge e^{b} \equiv 0, \quad \text { and } \quad T^{a}=\kappa_{b}^{a} \wedge e^{b} .$</td><td>$de^a+\bar{\omega}_b^a\wedge e^b \equiv 0, \quad \text{and} \quad T^a=\kappa_b^a\wedge e^b.$</td></tr>
<tr><td>cardona-qft-methods_EQ0142 (6)</td><td>081</td><td>■0.963 ● N|+0</td><td>\sigma_{L}(\mathcal{D})\left(x_{0}, \xi\right)=\sum_{|\alpha|=k} a_{\alpha}\left(x_{0}\right) \xi_{1}^{\alpha_{1}} \cdots \xi_{n}^{\alpha_{n}} .</td><td>$\sigma_{L}(\mathcal{D})\left(x_{0}, \xi\right)=\sum_{|\alpha|=k} a_{\alpha}\left(x_{0}\right) \xi_{1}^{\alpha_{1}} \cdots \xi_{n}^{\alpha_{n}} .$</td><td>$\sigma_L(\mathcal{D})(x_0,\xi) = \sum_{|\alpha|=k} a_\alpha(x_0)\,\xi_1^{\alpha_1}\cdots\xi_n^{\alpha_n}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0454</td><td>214</td><td>■0.964 ● N|+0</td><td>\left(C^{\infty}(M, \mathcal{E}), L^{2}(M, \mathcal{E} \otimes S), \mathcal{D}_{\mathcal{E}}\right),</td><td>$\left(C^{\infty}(M, \mathcal{E}), L^{2}(M, \mathcal{E} \otimes S), \mathcal{D}_{\mathcal{E}}\right),$</td><td>$(C^\infty(M,\mathcal{E}),L^2(M,\mathcal{E}\otimes S),\mathcal{D}_{\mathcal{E}}),$</td></tr>
<tr><td>cardona-qft-methods_EQ0086</td><td>050</td><td>■0.965 ● W|+1</td><td>\begin{gathered} \operatorname{Tad}_{\mathcal{D}+A}(d-k)=-(d-k) f A \mathcal{D}|\mathcal{D}|^{-(d-k)-2}, \quad \forall k \neq d \\ \operatorname{Tad}_{\mathcal{D}+A}(0)=-f A \mathcal{D}^{-1} . \end{gathered}</td><td>$\begin{gathered} \operatorname{Tad}_{\mathcal{D}+A}(d-k)=-(d-k) f A \mathcal{D}|\mathcal{D}|^{-(d-k)-2}, \quad \forall k \neq d \\ \operatorname{Tad}_{\mathcal{D}+A}(0)=-f A \mathcal{D}^{-1} . \end{gathered}$</td><td>$\begin{gathered} \operatorname{Tad}_{\mathcal{D}+A}(d-k) = -(d-k){\textstyle\int\!\!\!\!-}\, A\mathcal{D}|\mathcal{D}|^{-(d-k)-2}, \quad \forall k \neq d, \\ \operatorname{Tad}_{\mathcal{D}+A}(0) = -{\textstyle\int\!\!\!\!-}\, A\mathcal{D}^{-1}. \end{gathered}$</td></tr>
<tr><td>cardona-qft-methods_EQ0106 (1)</td><td>072</td><td>■0.965 ● N|+0</td><td>\operatorname{Ind}(A):=\operatorname{dimKer}(A)-\operatorname{dim} \operatorname{Coker}(A) \in \mathbb{Z},</td><td>$\operatorname{Ind}(A):=\operatorname{dimKer}(A)-\operatorname{dim} \operatorname{Coker}(A) \in \mathbb{Z},$</td><td>$\operatorname{Ind}(A) \ := \ \dim\operatorname{Ker}(A) \ - \ \dim\operatorname{Coker}(A) \ \in \ \mathbb{Z},$</td></tr>
<tr><td>cardona-qft-methods_EQ0272</td><td>131</td><td>■0.966 ● K|+0</td><td>C_{G_{1} \amalg G_{2}}(T)=\chi\left(Y_{G_{1}}\right) \chi\left(Y_{G_{2}}\right) T^{2}+\text { h.o.t : }</td><td>$C_{G_{1} \amalg G_{2}}(T)=\chi\left(Y_{G_{1}}\right) \chi\left(Y_{G_{2}}\right) T^{2}+\text { h.o.t : }$</td><td>$C_{G_1\amalg G_2}(T) = \chi(Y_{G_1})\chi(Y_{G_2})\,T^2 + \text{h.o.t} \quad :$</td></tr>
<tr><td>cardona-qft-methods_EQ0055</td><td>038</td><td>■0.970 ● C|+7</td><td>\sigma_{d}^{\not D_{g^{\prime}}}(x, \xi)=e^{-h(x) / 2} U_{g^{\prime}, g}^{-1}(x) \sigma_{d}^{\not D_{g}}(x, \xi) U_{g^{\prime}, g}(x) e^{-h(x) / 2}, \quad \xi \in T_{x}^{*} M .</td><td>$\sigma_{d}^{\not D_{g^{\prime}}}(x, \xi)=e^{-h(x) / 2} U_{g^{\prime}, g}^{-1}(x) \sigma_{d}^{\not D_{g}}(x, \xi) U_{g^{\prime}, g}(x) e^{-h(x) / 2}, \quad \xi \in T_{x}^{*} M .$</td><td>$\sigma_d^{\not D_{g'}}(x,\xi) = e^{-h(x)/2}\,U_{g',g}^{-1}(x)\,\sigma_d^{\not D_g}(x,\xi)\,U_{g',g}(x)\,e^{-h(x)/2}, \quad \xi \in T_x^*M.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0321 (7)	161	■0.971 ● N|+0	\begin{aligned} d x^{\mu}\left[\partial_{\mu} \phi^{a}(x)+\omega_{b \mu}^{a}(x) \phi^{b}(x)\right] & =d x^{\mu} D_{\mu} \phi^{a}(x) \\ & =D \phi^{a}(x) . \end{aligned}	$\begin{aligned} d x^{\mu}\left[\partial_{\mu} \phi^{a}(x)+\omega_{b \mu}^{a}(x) \phi^{b}(x)\right] & =d x^{\mu} D_{\mu} \phi^{a}(x) \\ & =D \phi^{a}(x) . \end{aligned}$	$\begin{aligned} dx^{\mu}[\partial_{\mu}\phi^{a}(x)+\omega^{a}_{b\mu}(x)\phi^{b}(x)] &= dx^{\mu}D_{\mu}\phi^{a}(x) \\ &= D\phi^{a}(x). \end{aligned}$
cardona-qft-methods_ EQ0969 (9)	363	■0.971 ● K|+0	X_{\Gamma}=\left\{t \in \mathbb{P}_{k}^{n-1} \mid t_{1}+t_{2}+\cdots+t_{n}=0\right\}=: \mathcal{L} .	$X_{\Gamma}=\left\{t \in \mathbb{P}_{k}^{n-1} \mid t_{1}+t_{2}+\cdots+t_{n}=0\right\}=: \mathcal{L} .$	$X_{\Gamma}=\{t\in\mathbb{P}_{k}^{n-1}\|t_{1}+t_{2}+\cdots+t_{n}=0\}=:\mathcal{L}\,.$
cardona-qft-methods_ EQ0452	213	■0.972 ● W|+1	\begin{gathered} \mathcal{A}=\mathcal{A}_{1} \otimes \mathcal{A}_{2} \quad \mathcal{H}=\mathcal{H}_{1} \otimes \mathcal{H}_{2} \\ D=D_{1} \otimes 1+\gamma_{1} \otimes D_{2} \\ \gamma=\gamma_{1} \otimes \gamma_{2} \quad J=J_{1} \otimes J_{2} \end{gathered}	$\begin{gathered} \mathcal{A}=\mathcal{A}_{1} \otimes \mathcal{A}_{2} \quad \mathcal{H}=\mathcal{H}_{1} \otimes \mathcal{H}_{2} \\ D=D_{1} \otimes 1+\gamma_{1} \otimes D_{2} \\ \gamma=\gamma_{1} \otimes \gamma_{2} \quad J=J_{1} \otimes J_{2} \end{gathered}$	$\mathcal{A}=\mathcal{A}_{1}\otimes\mathcal{A}_{2} \quad \mathcal{H}=\mathcal{H}_{1}\otimes\mathcal{H}_{2}$ $D=D_{1}\otimes 1+\gamma_{1}\otimes D_{2}$ $\gamma=\gamma_{1}\otimes\gamma_{2} \quad J=J_{1}\otimes J_{2}.$
cardona-qft-methods_ EQ0971 (11)	363	■0.974 ● K|+0	\Psi_{\Gamma^{\vee}}(t)=t_{2} t_{3} \cdots t_{n}+t_{1} t_{3} \cdots t_{n}+\cdots+t_{1} t_{2} \cdots \hat{t_{i}} \cdots t_{n}+\cdots+t_{1} t_{2} \cdots t_{n-1},	$\Psi_{\Gamma^{\vee}}(t)=t_{2} t_{3} \cdots t_{n}+t_{1} t_{3} \cdots t_{n}+\cdots+t_{1} t_{2} \cdots \hat{t_{i}} \cdots t_{n}+\cdots+t_{1} t_{2} \cdots t_{n-1},$	$\Psi_{\Gamma^{\vee}}(t)=t_{2}t_{3}\cdots t_{n}+t_{1}t_{3}\cdots t_{n}+\cdots+t_{1}t_{2}\cdots\hat{t_{i}}\cdots t_{n}+\cdots+t_{1}t_{2}\cdots t_{n-1}\,,$
cardona-qft-methods_ EQ0591 (25)	251	■0.974 ● N|+1	\begin{aligned} \phi \otimes_{S_{\theta}} \chi & \equiv \left(\frac{1+\tau_{0}}{2}\right)(\phi \otimes \chi), \\ \phi \otimes_{A_{\theta}} \chi & \equiv\left(\frac{1-\tau_{0}}{2}\right)(\phi \otimes \chi), \end{aligned}	$\begin{aligned} \phi \otimes_{S_{\theta}} \chi & \equiv \left(\frac{1+\tau_{0}}{2}\right)(\phi \otimes \chi), \\ \phi \otimes_{A_{\theta}} \chi & \equiv\left(\frac{1-\tau_{0}}{2}\right)(\phi \otimes \chi), \end{aligned}$	$\phi\otimes_{S_{\theta}}\chi\equiv\left(\frac{1+\tau_{0}}{2}\right)(\phi\otimes\chi),$ $\phi\otimes_{A_{\theta}}\chi\equiv\left(\frac{1-\tau_{0}}{2}\right)(\phi\otimes\chi),$
cardona-qft-methods_ EQ0960 (12)	361	■0.974 ● K|+0	\pi_{2}: \pi_{1}^{-1}\left(X_{\Gamma} \backslash \Sigma_{n}\right) \rightarrow X_{\Gamma^{\vee}} \backslash \Sigma_{n} .	$\pi_{2}: \pi_{1}^{-1}\left(X_{\Gamma} \backslash \Sigma_{n}\right) \rightarrow X_{\Gamma^{\vee}} \backslash \Sigma_{n} .$	$\pi_{2}:\pi_{1}^{-1}(X_{\Gamma}\setminus\Sigma_{n})\rightarrow X_{\Gamma^{\vee}}\setminus\Sigma_{n}\,.$
cardona-qft-methods_ EQ0302 (6)	151	■0.975 ● K|+0	\delta I=\int_{M} \delta \mathcal{C} \wedge \star j=\int_{M}(d \Omega) \wedge \star j=\int_{M} d(\Omega \wedge \star j)-\int_{M} \Omega \wedge d \star j,	$\delta I=\int_{M} \delta \mathcal{C} \wedge \star j=\int_{M}(d \Omega) \wedge \star j=\int_{M} d(\Omega \wedge \star j)-\int_{M} \Omega \wedge d \star j,$	$\delta I=\int_{M}\delta\mathcal{C}\wedge\star j=\int_{M}(d\Omega)\wedge\star j=\int_{M}d(\Omega\wedge\star j)-\int_{M}\Omega\wedge d\star j,$
cardona-qft-methods_ EQ0358 (28)	171	■0.976 ● N|+1	\begin{aligned} C_{3}^{\text {Lor }} & =\omega_{b}^{a} d \omega_{a}^{b}+\frac{2}{3} \omega_{b}^{a} \omega_{c}^{b} \omega_{a}^{c} \\ C_{3}^{\text {Tor }} & =e^{a} T_{a} . \end{aligned}	$\begin{aligned} C_{3}^{\text {Lor }} & =\omega_{b}^{a} d \omega_{a}^{b}+\frac{2}{3} \omega_{b}^{a} \omega_{c}^{b} \omega_{a}^{c} \\ C_{3}^{\text {Tor }} & =e^{a} T_{a} . \end{aligned}$	$C_{3}^{Lor}=\omega^{a}_{\ b}d\omega^{b}_{\ a}+\frac{2}{3}\omega^{a}_{\ b}\omega^{b}_{\ c}\omega^{c}_{\ a}$ $C_{3}^{Tor}=e^{a}T_{a}.$
cardona-qft-methods_ EQ0027 (8)	027	■0.976 ● N|+1	\operatorname{WRes}\left((1+\Delta)^{-d / 2}\right)= \operatorname{Vol}\left(\mathbb{S}^{\mathrm{d}-1}\right)= \frac{2 \pi^{\mathrm{d} / 2}}{\Gamma(\mathrm{~d} / 2)} .	$\operatorname{WRes}\left((1+\Delta)^{-d / 2}\right)= \operatorname{Vol}\left(\mathbb{S}^{\mathrm{d}-1}\right)= \frac{2 \pi^{\mathrm{d} / 2}}{\Gamma(\mathrm{~d} / 2)} .$	$\operatorname{WRes}\big((1+\Delta)^{-d/2}\big)=\operatorname{Vol}\big(\mathbb{S}^{\mathrm{d}-1}\big)=\frac{2\pi^{\mathrm{d}/2}}{\Gamma(\mathrm{d}/2)}\,.$
cardona-qft-methods_ EQ0336 (22)	164	■0.978 ● N|+0	D T^{a}=R_{b}^{a} \wedge e^{b} .	$D T^{a}=R_{b}^{a} \wedge e^{b} .$	$DT^{a}=R^{a}_{\ b}\wedge e^{b}.$
cardona-qft-methods_ EQ0925	352	■0.978 ● K|+0	\left(P_{G} M \times_{G} E G_{n}\right)^{-e_{0}^{*} T M_{n}} .	$\left(P_{G} M \times_{G} E G_{n}\right)^{-e_{0}^{*} T M_{n}} .$	$(P_{G}M\times_{G}EG_{n})^{-e_{0}^{*}TM_{n}}\,.$
cardona-qft-methods_ EQ0303 (7)	152	■0.979 ● N|+1	\begin{aligned} \mathbf{A} & =\mathbf{A}_{\mu} d x^{\mu} \\ & =A_{\mu}^{a} \mathbf{K}_{a} d x^{\mu} \end{aligned}	$\begin{aligned} \mathbf{A} & =\mathbf{A}_{\mu} d x^{\mu} \\ & =A_{\mu}^{a} \mathbf{K}_{a} d x^{\mu} \end{aligned}$	$\begin{aligned}\mathbf{A} &= \mathbf{A}_{\mu}dx^{\mu} \\ &= A^{a}_{\mu}\mathbf{K}_{a}dx^{\mu},\end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_EQ0489</td><td>221</td><td>■ 0.980
● N|+0</td><td><code>\int \frac{1}{2}\left|D_{\mu} \mathbf{H}\right|^{2} \sqrt{g} d^{4} x ;</code></td><td>$\int \frac{1}{2}\left|D_{\mu} \mathbf{H}\right|^{2} \sqrt{g} d^{4} x ;$</td><td>$\int \frac{1}{2}|D_{\mu}\mathbf{H}|^{2} \sqrt{g}\, d^{4}x;$</td></tr>
<tr><td>cardona-qft-methods_EQ0268</td><td>127</td><td>■ 0.981
● K|+0</td><td><code>\Psi_{G_{1}}\left(t_{1}, \ldots, t_{n_{1}}\right) \neq 0 \quad \text { and } \quad \Psi_{G_{2}}\left(u_{1}, \ldots, u_{n_{2}}\right) \neq 0,</code></td><td>$\Psi_{G_{1}}\left(t_{1}, \ldots, t_{n_{1}}\right) \neq 0 \quad \text { and } \quad \Psi_{G_{2}}\left(u_{1}, \ldots, u_{n_{2}}\right) \neq 0,$</td><td>$\Psi_{G_1}(t_1,\ldots,t_{n_1}) \neq 0 \quad \text{and} \quad \Psi_{G_2}(u_1,\ldots,u_{n_2}) \neq 0 \quad ,$</td></tr>
<tr><td>cardona-qft-methods_EQ0095</td><td>056</td><td>■ 0.983
● W|+2</td><td><code>g\left(t \mathcal{D}^{2}\right)=\int_{0}^{\infty} e^{-s t \mathcal{D}^{2}} \phi(s) d s, \operatorname{Tr}\left(g \left(t \mathcal{D}^{2}\right)\right) \underset{t \downarrow 0}{\sim} \sum_{\alpha \in \mathrm{Sp}^{+}} a_{\alpha} t^{\alpha} \int_{0}^{\infty} s^{\alpha} g(s) d s</code></td><td>$g\left(t \mathcal{D}^{2}\right)=\int_{0}^{\infty} e^{-s t \mathcal{D}^{2}} \phi(s) d s, \operatorname{Tr}\left(g \left(t \mathcal{D}^{2}\right)\right) \underset{t \downarrow 0}{\sim} \sum_{\alpha \in \mathrm{Sp}^{+}} a_{\alpha} t^{\alpha} \int_{0}^{\infty} s^{\alpha} g(s) d s$</td><td>$g(t\mathcal{D}^2) = \int_0^\infty e^{-st\mathcal{D}^2}\, \phi(s)\, ds,\ \mathrm{Tr}\big(g(t\mathcal{D}^2)\big) \underset{t\downarrow 0}{\sim} \sum_{\alpha\in\mathrm{Sp}^+} a_\alpha\, t^\alpha \int_0^\infty s^\alpha\, g(s)\, ds.$</td></tr>
<tr><td>cardona-qft-methods_EQ0379 (20)</td><td>177</td><td>■ 0.983
● K|+0</td><td><code>L_{4}=\epsilon_{a b c d} R^{a b} e^{c} e^{d} .</code></td><td>$L_{4}=\epsilon_{a b c d} R^{a b} e^{c} e^{d} .$</td><td>$L_4 = \epsilon_{abcd} R^{ab} e^c e^d.$</td></tr>
<tr><td>cardona-qft-methods_EQ0490</td><td>221</td><td>■ 0.983
● N|+1</td><td><code>\frac{1}{4} G_{\mu \nu}^{i} \bar{G}^{\mu \nu i}+ \frac{1}{4} F_{\mu \nu}^{\alpha} \bar{F}^{\mu \nu \alpha}+\frac{1}{4} B_{\mu \nu} \bar{B}^{\mu \nu} .</code></td><td>$\frac{1}{4} G_{\mu \nu}^{i} \bar{G}^{\mu \nu i}+ \frac{1}{4} F_{\mu \nu}^{\alpha} \bar{F}^{\mu \nu \alpha}+\frac{1}{4} B_{\mu \nu} \bar{B}^{\mu \nu} .$</td><td>$\frac{1}{4} G^i_{\mu\nu} \overline{G}^{\mu\nu i} + \frac{1}{4} F^\alpha_{\mu\nu} \overline{F}^{\mu\nu\alpha} + \frac{1}{4} B_{\mu\nu} \overline{B}^{\mu\nu}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0698</td><td>271</td><td>■ 0.983
● N|+0</td><td><code>\hat{H}=\sum_{l=1}^{L} \mathcal{H}_{l, l+1} .</code></td><td>$\hat{H}=\sum_{l=1}^{L} \mathcal{H}_{l, l+1} .$</td><td>$\hat{H} = \sum_{l=1}^{L} \mathcal{H}_{l,l+1}\,.$</td></tr>
<tr><td>cardona-qft-methods_EQ0368 (7)</td><td>175</td><td>■ 0.984
● N|+0</td><td><code>\begin{aligned} \delta \omega^{a b} & =d \lambda^{a b}+ \omega_{c}^{a} \lambda^{c b}+\omega_{c}^{b} \lambda^{a c} \pm\left[e^{a} \lambda^{b}-\lambda^{a} e^{b}\right] l^{-2} \\ \delta e^{a} & =d \lambda^{a}+\omega_{b}^{a} \lambda^{b}-\lambda_{b}^{a} e^{b} . \end{aligned}</code></td><td>$\begin{aligned} \delta \omega^{a b} & =d \lambda^{a b}+ \omega_{c}^{a} \lambda^{c b}+\omega_{c}^{b} \lambda^{a c} \pm\left[e^{a} \lambda^{b}-\lambda^{a} e^{b}\right] l^{-2} \\ \delta e^{a} & =d \lambda^{a}+\omega_{b}^{a} \lambda^{b}-\lambda_{b}^{a} e^{b} . \end{aligned}$</td><td>$\begin{aligned} \delta\omega^{ab} &= d\lambda^{ab} + \omega^a{}_c\lambda^{cb} + \omega^b{}_c\lambda^{ac} \pm [e^a\lambda^b - \lambda^a e^b] l^{-2} \\ \delta e^a &= d\lambda^a + \omega^a{}_b\lambda^b - \lambda^a{}_b e^b. \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_EQ0730</td><td>277</td><td>■ 0.984
● N|+0</td><td><code>\left(\frac{u_{k}+\frac{i}{2}}{u_{k}-\frac{i}{2}} \right)^{L}=\prod_{\substack{j=1 \\ j \neq k}}^{M} \frac{u_{k}-u_{j}+i}{u_{k}-u_{j}-i} .</code></td><td>$\left(\frac{u_{k}+\frac{i}{2}}{u_{k}-\frac{i}{2}} \right)^{L}=\prod_{\substack{j=1 \\ j \neq k}}^{M} \frac{u_{k}-u_{j}+i}{u_{k}-u_{j}-i} .$</td><td>$\left(\frac{u_k + \frac{i}{2}}{u_k - \frac{i}{2}}\right)^L = \prod_{\substack{j=1\\ j\neq k}}^{M} \frac{u_k - u_j + i}{u_k - u_j - i}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0781 (22)</td><td>298</td><td>■ 0.984
● N|+0</td><td><code>X^{M}(\tau, \sigma)=x^{M}+2 \alpha^{\prime} p^{M} \tau+ \mathrm{i}\left(2 \alpha^{\prime}\right)^{1 / 2} \sum_{n \neq 0} \frac{1}{n} \alpha_{n}^{M} e^{-\mathrm{i} n \tau} \cos (n \sigma) .</code></td><td>$X^{M}(\tau, \sigma)=x^{M}+2 \alpha^{\prime} p^{M} \tau+ \mathrm{i}\left(2 \alpha^{\prime}\right)^{1 / 2} \sum_{n \neq 0} \frac{1}{n} \alpha_{n}^{M} e^{-\mathrm{i} n \tau} \cos (n \sigma) .$</td><td>$X^M(\tau,\sigma) = x^M + 2\alpha' p^M \tau + \mathrm{i}(2\alpha')^{1/2} \sum_{n\neq 0} \frac{1}{n}\alpha_n^M e^{-\mathrm{i}n\tau} \cos(n\sigma)\ .$</td></tr>
<tr><td>cardona-qft-methods_EQ0434</td><td>208</td><td>■ 0.985
● K|+0</td><td><code>\mathcal{E}=\mathbf{2}_{L} \otimes \mathbf{1}^{0} \oplus \mathbf{2}_{R} \otimes \mathbf{1}^{0} \oplus \mathbf{2}_{L} \otimes \mathbf{3}^{0} \oplus \mathbf{2}_{R} \otimes \mathbf{3}^{0},</code></td><td>$\mathcal{E}=\mathbf{2}_{L} \otimes \mathbf{1}^{0} \oplus \mathbf{2}_{R} \otimes \mathbf{1}^{0} \oplus \mathbf{2}_{L} \otimes \mathbf{3}^{0} \oplus \mathbf{2}_{R} \otimes \mathbf{3}^{0},$</td><td>$\mathcal{E} = \mathbf{2}_L \otimes \mathbf{1}^0 \oplus \mathbf{2}_R \otimes \mathbf{1}^0 \oplus \mathbf{2}_L \otimes \mathbf{3}^0 \oplus \mathbf{2}_R \otimes \mathbf{3}^0,$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>cardona-qft-methods_EQ0485</td><td>221</td><td>■0.985
● C|-4</td><td><code>\begin{aligned} & \mathfrak{a}=\operatorname{Tr} \left(Y_{\nu}^{\dagger} Y_{\nu}+Y_{e}^{\dagger} Y_{e}+3 \left(Y_{u}^{\dagger} Y_{u}+Y_{d}^{\dagger} Y_{d} \right)\right) \\ & \mathfrak{b}=\operatorname{Tr}\left(\left(Y_{\nu}^{\dagger} Y_{\nu}\right)^{2}+\left(Y_{e}^{\dagger} Y_{e}\right)^{2}+3\left(Y_{u}^{\dagger} Y_{u} \right)^{2}+3\left(Y_{d}^{\dagger} Y_{d}\right)^{2} \right) \\ & \mathfrak{c}=\operatorname{Tr}\left(M M^{\dagger}\right) \\ & \mathfrak{d}=\operatorname{Tr}\left(\left(M M^{\dagger}\right)^{2}\right) \\ & \mathfrak{e}= \operatorname{Tr}\left(M M^{\dagger} Y_{\nu}^{\dagger} Y_{\nu}\right) . \end{aligned}</code></td><td>$$\begin{aligned} & \mathfrak{a}=\operatorname{Tr}\left(Y_{\nu}^{\dagger} Y_{\nu}+Y_{e}^{\dagger} Y_{e}+3\left(Y_{u}^{\dagger} Y_{u}+Y_{d}^{\dagger} Y_{d}\right)\right) \\ & \mathfrak{b}=\operatorname{Tr}\left(\left(Y_{\nu}^{\dagger} Y_{\nu}\right)^{2}+\left(Y_{e}^{\dagger} Y_{e}\right)^{2}+3\left(Y_{u}^{\dagger} Y_{u}\right)^{2}+3\left(Y_{d}^{\dagger} Y_{d}\right)^{2}\right) \\ & \mathfrak{c}=\operatorname{Tr}\left(M M^{\dagger}\right) \\ & \mathfrak{d}=\operatorname{Tr}\left(\left(M M^{\dagger}\right)^{2}\right) \\ & \mathfrak{e}=\operatorname{Tr}\left(M M^{\dagger} Y_{\nu}^{\dagger} Y_{\nu}\right) . \end{aligned}$$</td><td>$$\begin{aligned} \mathfrak{a} &= \operatorname{Tr}(Y_{\nu}^{\dagger}Y_{\nu}+Y_{e}^{\dagger}Y_{e}+3(Y_{u}^{\dagger}Y_{u}+Y_{d}^{\dagger}Y_{d})) \\ \mathfrak{b} &= \operatorname{Tr}((Y_{\nu}^{\dagger}Y_{\nu})^{2}+(Y_{e}^{\dagger}Y_{e})^{2}+3(Y_{u}^{\dagger}Y_{u})^{2}+3(Y_{d}^{\dagger}Y_{d})^{2}) \\ \mathfrak{c} &= \operatorname{Tr}(MM^{\dagger}) \\ \mathfrak{d} &= \operatorname{Tr}((MM^{\dagger})^{2}) \\ \mathfrak{e} &= \operatorname{Tr}(MM^{\dagger}Y_{\nu}^{\dagger}Y_{\nu}). \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_EQ0085 (30)</td><td>049</td><td>■0.985
● C|+4</td><td><code>\begin{aligned} & f\left|D_{A}\right|^{-d}=f|D|^{-d}, \\ & f\left|D_{A}\right|^{-(d-1)}=f|D|^{-(d-1)}-\left(\frac{d-1}{2}\right) f X|D|^{-d-1}, \\ & f\left|D_{A} \right|^{-(d-2)}=f|D|^{-(d-2)}+\frac{d-2}{2}\left(-f X|D|^{-d}+\frac{d}{4} f X^{2}|D|^{-2-d}\right) . \end{aligned}</code></td><td>$$\begin{aligned} & f\left|D_{A}\right|^{-d}=f|D|^{-d}, \\ & f\left|D_{A}\right|^{-(d-1)}=f|D|^{-(d-1)}-\left(\frac{d-1}{2}\right) f X|D|^{-d-1}, \\ & f\left|D_{A}\right|^{-(d-2)}=f|D|^{-(d-2)}+\frac{d-2}{2}\left(-f X|D|^{-d}+\frac{d}{4} f X^{2}|D|^{-2-d}\right) . \end{aligned}$$</td><td>$$\begin{aligned} & \fint |D_A|^{-d}=\fint |D|^{-d}, \qquad (3 \\ & \fint |D_A|^{-(d-1)}=\fint |D|^{-(d-1)}-(\tfrac{d-1}{2})\fint X|D|^{-d-1}, \\ & \fint |D_A|^{-(d-2)}=\fint |D|^{-(d-2)}+\tfrac{d-2}{2}(-\fint X|D|^{-d}+\tfrac{d}{4}\fint X^{2}|D|^{-2-d}). \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_EQ0250 (1)</td><td>118</td><td>■0.986
● N|-1</td><td><code>S_{\mathrm{eff}}(\phi)=\sum_{G}\langle G\rangle</code></td><td>$S_{\mathrm{eff}}(\phi)=\sum_{G}\langle G\rangle$</td><td>$S_{\mathrm{eff}}(\phi)=\sum_{G}\langle G\rangle$</td></tr>
<tr><td>cardona-qft-methods_EQ0102</td><td>061</td><td>■0.986
● N|+0</td><td><code>U_{k} U_{q}=e^{-\frac{i}{2} k \cdot \Theta q} U_{k+q}, \text { and } U_{k} U_{q}=e^{-i k \cdot \Theta q} U_{q} U_{k}</code></td><td>$U_{k} U_{q}=e^{-\frac{i}{2} k \cdot \Theta q} U_{k+q}, \text { and } U_{k} U_{q}=e^{-i k \cdot \Theta q} U_{q} U_{k}$</td><td>$U_{k}U_{q}=e^{-\frac{i}{2}k.\Theta q}U_{k+q}, \text{ and } U_{k}U_{q}=e^{-ik.\Theta q}U_{q}U_{k}$</td></tr>
<tr><td>cardona-qft-methods_EQ0445</td><td>211</td><td>■0.986
● N|+0</td><td><code>S_{3}=\left(\begin{array}{cccc} 0 & 0 & Y_{(\uparrow 3)}^{*} & 0 \\ 0 & 0 & 0 & Y_{(\downarrow 3)}^{*} \\ Y_{(\uparrow 3)} & 0 & 0 & 0 \\ 0 & Y_{(\downarrow 3)} & 0 & 0 \end{array}\right)</code></td><td>$S_{3}=\left(\begin{array}{cccc} 0 & 0 & Y_{(\uparrow 3)}^{*} & 0 \\ 0 & 0 & 0 & Y_{(\downarrow 3)}^{*} \\ Y_{(\uparrow 3)} & 0 & 0 & 0 \\ 0 & Y_{(\downarrow 3)} & 0 & 0 \end{array}\right)$</td><td>$S_{3}=\left(\begin{array}{cccc} 0 & 0 & Y_{(\uparrow 3)}^{*} & 0 \\ 0 & 0 & 0 & Y_{(\downarrow 3)}^{*} \\ Y_{(\uparrow 3)} & 0 & 0 & 0 \\ 0 & Y_{(\downarrow 3)} & 0 & 0 \end{array}\right)$</td></tr>
<tr><td>cardona-qft-methods_EQ0441</td><td>210</td><td>■0.987
● W|+1</td><td><code>\begin{array}{ccccc} & \uparrow \otimes \mathbf{1}^{0} \downarrow \otimes \mathbf{1}^{0} \uparrow \otimes \mathbf{3}^{0} \downarrow \otimes \mathbf{3}^{0} \\ \mathbf{2}_{L} & -1 & -1 & \frac{1}{3} & \frac{1}{3} \\ \mathbf{2}_{R} & 0 & -2 & \frac{4}{3} & -\frac{2}{3} \end{array}</code></td><td>$\begin{array}{ccccc} & \uparrow \otimes \mathbf{1}^{0} \downarrow \otimes \mathbf{1}^{0} \uparrow \otimes \mathbf{3}^{0} \downarrow \otimes \mathbf{3}^{0} \\ \mathbf{2}_{L} & -1 & -1 & \frac{1}{3} & \frac{1}{3} \\ \mathbf{2}_{R} & 0 & -2 & \frac{4}{3} & -\frac{2}{3} \end{array}$</td><td>$\begin{array}{ccccc} & \uparrow \otimes \mathbf{1}^{0} & \downarrow \otimes \mathbf{1}^{0} & \uparrow \otimes \mathbf{3}^{0} & \downarrow \otimes \mathbf{3}^{0} \\ \mathbf{2}_{L} & -1 & -1 & \frac{1}{3} & \frac{1}{3} \\ \mathbf{2}_{R} & 0 & -2 & \frac{4}{3} & -\frac{2}{3} \end{array}$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_ EQ0078 (26)</td><td>046</td><td>■0.987 ● N|+0</td><td><code>\mathcal{D}_{\widetilde{A}}=\mathcal{D}+\widetilde{A}, \quad \widetilde{A}=A+\epsilon J A J^{-1}, \quad A=A^{*} .</code></td><td>$\mathcal{D}_{\widetilde{A}}=\mathcal{D}+\widetilde{A}, \quad \widetilde{A}=A+\epsilon J A J^{-1}, \quad A=A^{*} .$</td><td>$\mathcal{D}_{\widetilde{A}}=\mathcal{D}+\widetilde{A}, \quad \widetilde{A}=A+\epsilon J A J^{-1}, \quad A=A^{*}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0164</td><td>088</td><td>■0.988 ● N|+0</td><td><code>\mathrm{t}-\operatorname{Ind}_{G}: K_{G}(M) \rightarrow K_{G}(\{p t\}) \simeq R(G) .</code></td><td>$\mathrm{t}-\operatorname{Ind}_{G}: K_{G}(M) \rightarrow K_{G}(\{p t\}) \simeq R(G) .$</td><td>$\text{t-Ind}_{G}: K_{G}(M) \rightarrow K_{G}(\{pt\}) \simeq R(G).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0956 (8)</td><td>361</td><td>■0.988 ● K|+0</td><td><code>\Psi_{\Gamma}(t)=\left(\prod_{i=1}^{n} t_{i}\right) \Psi_{\Gamma^{\vee}}\left(\frac{1}{t}\right),</code></td><td>$\Psi_{\Gamma}(t)=\left(\prod_{i=1}^{n} t_{i}\right) \Psi_{\Gamma^{\vee}}\left(\frac{1}{t}\right),$</td><td>$\Psi_{\Gamma}(t)=\left(\prod_{i=1}^{n} t_{i}\right) \Psi_{\Gamma^{\vee}}\left(\frac{1}{t}\right),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0655 (5)</td><td>262</td><td>■0.988 ● C|-6</td><td><code>\begin{aligned} \langle 0| \rho_{\theta}(x) \eta_{ \theta}(y)|0\rangle & =e^{-\frac{1}{2} \frac{\partial}{ \partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{ \partial y^{\nu}}}\langle 0| \rho_{0}(x) \eta_{0}(y)|0 \rangle \\ & =e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} F(x-y)=F(x-y) \\ & =\langle 0| \rho_{0}(x) \eta_{0}(y)|0\rangle \end{aligned}</code></td><td>$\begin{aligned} \langle 0| \rho_{\theta}(x) \eta_{\theta}(y)|0\rangle & =e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}}\langle 0| \rho_{0}(x) \eta_{0}(y)|0 \rangle \\ & =e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} F(x-y)=F(x-y) \\ & =\langle 0| \rho_{0}(x) \eta_{0}(y)|0\rangle \end{aligned}$</td><td>$\begin{aligned} \langle 0| \rho_{\theta}(x) \eta_{\theta}(y)|0\rangle & =e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}}\langle 0| \rho_{0}(x) \eta_{0}(y)|0 \rangle \\ & =e^{-\frac{1}{2} \frac{\partial}{\partial x^{\mu}} \theta^{\mu \nu} \frac{\partial}{\partial y^{\nu}}} F(x-y)=F(x-y) \\ & =\langle 0| \rho_{0}(x) \eta_{0}(y)|0\rangle \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0375 (16)</td><td>176</td><td>■0.989 ● N|+0</td><td><code>\Upsilon_{0}^{A B}=\left[\begin{array}{cc} : \eta^{a b} & 0 \\ 0 & 0 \end{array}\right],</code></td><td>$\Upsilon_{0}^{A B}=\left[\begin{array}{cc} : \eta^{a b} & 0 \\ 0 & 0 \end{array}\right],$</td><td>$\Upsilon_{0}^{AB}=\left[\begin{array}{cc} : \eta^{ab} & 0 \\ 0 & 0 \end{array}\right],$</td></tr>
<tr><td>cardona-qft-methods_ EQ0884</td><td>343</td><td>■0.989 ● N|+0</td><td><code>P . D: H_{p}(M) \cong\left(H^{p}(M)\right)^{*} \cong H^{n-p}(M) .</code></td><td>$P . D: H_{p}(M) \cong\left(H^{p}(M)\right)^{*} \cong H^{n-p}(M) .$</td><td>$P.D: H_{p}(M) \cong (H^{p}(M))^{*} \cong H^{n-p}(M).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0937</td><td>354</td><td>■0.989 ● N|+0</td><td><code>H_{*}^{p r o}\left(L B G^{-T B G}\right) \cong H^{*}(L B G) .</code></td><td>$H_{*}^{p r o}\left(L B G^{-T B G}\right) \cong H^{*}(L B G) .$</td><td>$H_{*}^{pro}(LBG^{-TBG}) \cong H^{*}(LBG).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0897</td><td>347</td><td>■0.989 ● W|+0</td><td><code>\begin{aligned} & H H_{*}(A, M) \cong \operatorname{Tor}_{A \otimes A^{o p}}^{*}(A, M) \\ & H H^{*}(A, M) \cong \operatorname{Ext}_{A \otimes A^{o p}}^{*}(A, M) \end{aligned}</code></td><td>$\begin{aligned} & H H_{*}(A, M) \cong \operatorname{Tor}_{A \otimes A^{o p}}^{*}(A, M) \\ & H H^{*}(A, M) \cong \operatorname{Ext}_{A \otimes A^{o p}}^{*}(A, M) \end{aligned}$</td><td>$\begin{aligned} & HH_{*}(A, M) \cong \operatorname{Tor}_{A \otimes A^{op}}^{*}(A, M) \\ & HH^{*}(A, M) \cong \operatorname{Ext}_{A \otimes A^{op}}^{*}(A, M) \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0513 (6)</td><td>230</td><td>■0.990 ● N|+0</td><td><code>\beta_{j}^{\text {grav }}=\frac{a_{j}}{8 \pi} \frac{ \Lambda^{2}}{M_{P}^{2}(\Lambda)} x_{j},</code></td><td>$\beta_{j}^{\text {grav }}=\frac{a_{j}}{8 \pi} \frac{\Lambda^{2}}{M_{P}^{2}(\Lambda)} x_{j},$</td><td>$\beta_{j}^{\text{grav}}=\frac{a_{j}}{8 \pi} \frac{\Lambda^{2}}{M_{P}^{2}(\Lambda)} x_{j},$</td></tr>
<tr><td>cardona-qft-methods_ EQ0002</td><td>018</td><td>■0.990 ● W|+1</td><td><code>\left|\partial_{x}^{\alpha} \partial_{y}^{\gamma} \partial_{\xi}^{\beta} a(x, y, \xi)\right| \leq C_{K \alpha \beta \gamma}(1+|\xi|)^{\Re(m)-|\beta|}, \quad \forall x, y \in K, \xi \in \mathbb{R}^{d}</code></td><td>$\left|\partial_{x}^{\alpha} \partial_{y}^{\gamma} \partial_{\xi}^{\beta} a(x, y, \xi)\right| \leq C_{K \alpha \beta \gamma}(1+|\xi|)^{\Re(m)-|\beta|}, \quad \forall x, y \in K, \xi \in \mathbb{R}^{d}$</td><td>$|\partial_{x}^{\alpha} \partial_{y}^{\gamma} \partial_{\xi}^{\beta} a(x, y, \xi)| \leq C_{K \alpha \beta \gamma}(1+|\xi|)^{\Re(m)-|\beta|}, \quad \forall x, y \in K, \xi \in \mathbb{R}^{d}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0199 (9)</td><td>095</td><td>■0.990 ● N|+0</td><td><code>\mathcal{D}=\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) \frac{1}{i} \frac{\partial}{ \partial x_{1}}+\left(\begin{array}{cc} 0 & -i \\ i & 0 \end{array}\right) \frac{1}{i} \frac{\partial}{ \partial x_{2}}+\left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right) \frac{1}{i} \frac{\partial}{\partial x_{3}} .</code></td><td>$\mathcal{D}=\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) \frac{1}{i} \frac{\partial}{\partial x_{1}}+\left(\begin{array}{cc} 0 & -i \\ i & 0 \end{array}\right) \frac{1}{i} \frac{\partial}{\partial x_{2}}+\left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right) \frac{1}{i} \frac{\partial}{\partial x_{3}} .$</td><td>$\mathcal{D}=\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{i} \frac{\partial}{\partial x_{1}}+\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \frac{1}{i} \frac{\partial}{\partial x_{2}}+\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{i} \frac{\partial}{\partial x_{3}}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0301 (5)</td><td>151</td><td>■0.990 ● K|+0</td><td><code>I=\int \mathcal{C} \wedge \star j,</code></td><td>$I=\int \mathcal{C} \wedge \star j,$</td><td>$I=\int \mathcal{C} \wedge \star j,$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0409 (39)	187	■ 0.991 ● S|+1	`B_{2 n}=-\kappa n \int_{0}^{1} d t \int_{0}^{t} d s \epsilon \theta e\left(\widetilde{R}+t^{2} \theta^{2}+s^{2} e^{2}\right)^{n-1}`	$B_{2 n}=-\kappa n \int_{0}^{1} d t \int_{0}^{t} d s \epsilon \theta e\left(\widetilde{R}+t^{2} \theta^{2}+s^{2} e^{2}\right)^{n-1}$	$B_{2n} = -\kappa n \int\limits_0^1 dt \int\limits_0^t ds\ \epsilon\theta e \left(\widetilde{R}+t^2\theta^2+s^2e^2\right)^{n-1},$
cardona-qft-methods_ EQ0787 (28)	299	■ 0.991 ● S|+0	`\begin{array}{rc} \mathrm{R}: & \Psi^{M}(\tau, 0)=\Psi^{M}(\tau, 2 \pi) \\ \mathrm{NS}: & \Psi^{M}(\tau, 0)=-\Psi^{M}(\tau, 2 \pi) \end{array}`	$\begin{array}{rc} \mathrm{R}: & \Psi^{M}(\tau, 0)=\Psi^{M}(\tau, 2 \pi) \\ \mathrm{NS}: & \Psi^{M}(\tau, 0)=-\Psi^{M}(\tau, 2 \pi) \end{array}$	$\begin{array}{ll} \mathrm{R}: & \Psi^{M}(\tau, 0)=\Psi^{M}(\tau, 2 \pi) \\ \mathrm{NS}: & \Psi^{M}(\tau, 0)=-\Psi^{M}(\tau, 2 \pi) \end{array}$
cardona-qft-methods_ EQ0995	374	■ 0.991 ● N|+1	`\Delta X^{\alpha}=\sum_{\beta \leq \alpha}\binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta}`	$\Delta X^{\alpha}=\sum_{\beta \leq \alpha}\binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta}$	$\Delta X^{\alpha}=\sum_{\beta \leq \alpha}\binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta},$
cardona-qft-methods_ EQ1003	375	■ 0.991 ● C|+3	`\begin{aligned} f_{a b}(x) & =\sum_{\alpha} \frac{x^{\alpha}}{\alpha!} a b\left(X^{\alpha}\right)=\sum_{\alpha} \frac{x^{\alpha}}{\alpha!}(a \otimes b)\left(\Delta X^{\alpha}\right) \\ & =\sum_{\alpha} \frac{x^{\alpha}}{\alpha!}(a \otimes b)\left(\sum_{\beta \leq \alpha}\binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta}\right) \\ & =\sum_{\beta} \sum_{\gamma} \frac{x^{\beta} x^{\gamma}}{\beta!\gamma!} a\left(X^{\beta}\right) b\left(X^{\gamma}\right)=f_{a}(x) f_{b}(x) . \end{aligned}`	$\begin{aligned} f_{a b}(x) & =\sum_{\alpha} \frac{x^{\alpha}}{\alpha!} a b\left(X^{\alpha}\right)=\sum_{\alpha} \frac{x^{\alpha}}{\alpha!}(a \otimes b)\left(\Delta X^{\alpha}\right) \\ & =\sum_{\alpha} \frac{x^{\alpha}}{\alpha!}(a \otimes b)\left(\sum_{\beta \leq \alpha}\binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta}\right) \\ & =\sum_{\beta} \sum_{\gamma} \frac{x^{\beta} x^{\gamma}}{\beta!\gamma!} a\left(X^{\beta}\right) b\left(X^{\gamma}\right)=f_{a}(x) f_{b}(x) . \end{aligned}$	$\begin{aligned} f_{ab}(x) & =\sum_{\alpha} \frac{x^{\alpha}}{\alpha!} ab(X^{\alpha})=\sum_{\alpha} \frac{x^{\alpha}}{\alpha!}(a \otimes b)(\Delta X^{\alpha}) \\ & =\sum_{\alpha} \frac{x^{\alpha}}{\alpha!}(a \otimes b)\left(\sum_{\beta \leq \alpha}\binom{\alpha}{\beta} X^{\beta} \otimes X^{\alpha-\beta}\right) \\ & =\sum_{\beta} \sum_{\gamma} \frac{x^{\beta} x^{\gamma}}{\beta!\gamma!} a(X^{\beta}) b(X^{\gamma})=f_{a}(x) f_{b}(x). \end{aligned}$
cardona-qft-methods_ EQ0118 (6)	076	■ 0.991 ● N|+1	`\left\{\bigoplus_{V \in \operatorname{Irr}(G)} m_{V} V \mid m_{V} \in \mathbb{Z}, \text { only finitely many numbers } m_{V} \neq 0\right\} .`	$\left\{\bigoplus_{V \in \operatorname{Irr}(G)} m_{V} V \mid m_{V} \in \mathbb{Z}, \text { only finitely many numbers } m_{V} \neq 0\right\} .$	$\left\{ \bigoplus_{V\in \mathrm{Irr}\,(G)} m_V V \middle\| m_V \in \mathbb{Z},\ \text{only finitely many numbers } m_V \neq 0 \right\}.$
cardona-qft-methods_ EQ0835 (14)	317	■ 0.991 ● N|+1	`\begin{aligned} & e^{-2 A+\phi}(\mathrm{d}-H \wedge)\left(e^{2 A-\phi} \Phi_{+}\right)=\mathrm{d} A \wedge \bar{\Phi}_{+}-\frac{\mathrm{i}}{16} e^{\phi+A} * F_{\mathrm{IIB}-} \\ & e^{-2 A+\phi}(\mathrm{d}-H \wedge)\left(e^{2 A-\phi} \Phi_{-}\right)=0 \end{aligned}`	$\begin{aligned} & e^{-2 A+\phi}(\mathrm{d}-H \wedge)\left(e^{2 A-\phi} \Phi_{+}\right)=\mathrm{d} A \wedge \bar{\Phi}_{+}-\frac{\mathrm{i}}{16} e^{\phi+A} * F_{\mathrm{IIB}-} \\ & e^{-2 A+\phi}(\mathrm{d}-H \wedge)\left(e^{2 A-\phi} \Phi_{-}\right)=0 \end{aligned}$	$\begin{aligned} & e^{-2A+\phi}(\mathrm{d}-H\wedge)(e^{2A-\phi}\Phi_{+})=\mathrm{d}A\wedge\bar{\Phi}_{+}-\frac{\mathrm{i}}{16}e^{\phi+A}*F_{\mathrm{IIB}-} \\ & e^{-2A+\phi}(\mathrm{d}-H\wedge)(e^{2A-\phi}\Phi_{-})=0\,, \end{aligned}$
cardona-qft-methods_ EQ0088	050	■ 0.992 ● C|+6	`\operatorname{Tr}\left(P e^{-t \not D^{2}}\right) \underset{t \downarrow 0^{+}}{\sim} \sum_{k=0}^{\infty} a_{k} t^{(k-\operatorname{ord}(P)-d) / 2}+\sum_{k=0}^{\infty}\left(-a_{k}^{\prime} \log t+b_{k}\right) t^{k}`	$\operatorname{Tr}\left(P e^{-t \not D^{2}}\right) \underset{t \downarrow 0^{+}}{\sim} \sum_{k=0}^{\infty} a_{k} t^{(k-\operatorname{ord}(P)-d) / 2}+\sum_{k=0}^{\infty}\left(-a_{k}^{\prime} \log t+b_{k}\right) t^{k}$	$\mathrm{Tr}(P\,e^{-t\,\not D^2}) \underset{t\downarrow 0+}{\sim} \sum_{k=0}^{\infty} a_k\, t^{(k-ord(P)-d)/2}+\sum_{k=0}^{\infty}(-a_k'\,\log t+b_k)\,t^k,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0502	226	0.992 N|+0	`\begin{gathered} \zeta=\exp (2 \pi i / 3) \\ U_{P M N S}\left(\Lambda_{u n i f}\right)=\frac{1}{3}\left( \begin{array}{ccc} 1 & \zeta & \zeta^{2} \\ \zeta & 1 & \zeta \\ \zeta^{2} & \zeta & 1 \end{array}\right) \end{gathered}`	$\begin{gathered} \zeta=\exp (2 \pi i / 3) \\ U_{P M N S}\left(\Lambda_{u n i f}\right)=\frac{1}{3}\left(\begin{array}{ccc} 1 & \zeta & \zeta^{2} \\ \zeta & 1 & \zeta \\ \zeta^{2} & \zeta & 1 \end{array}\right) \end{gathered}$	$\zeta=\exp(2\pi i/3)$ $U_{PMNS}(\Lambda_{unif})=\frac{1}{3}\begin{pmatrix} 1 & \zeta & \zeta^2 \\ \zeta & 1 & \zeta \\ \zeta^2 & \zeta & 1 \end{pmatrix}$
cardona-qft-methods_EQ0141 (5)	081	0.992 N|+0	`\sigma_{L}(\mathcal{D})\left(x_{0}, \xi\right) e:=\left. \lim _{t \rightarrow \infty} t^{-k} \mathcal{D}\left(e^{i t f(x)} s(x)\right)\right|_{x=x_{0}} .`	$\sigma_{L}(\mathcal{D})\left(x_{0}, \xi\right) e:=\left. \lim _{t \rightarrow \infty} t^{-k} \mathcal{D}\left(e^{i t f(x)} s(x)\right)\right\rvert_{x=x_{0}} .$	$\sigma_L(\mathcal{D})(x_0,\xi)\,e \ := \ \lim_{t\to\infty} t^{-k}\,\mathcal{D}\big(e^{itf(x)}s(x)\big)\big\rvert_{x=x_0}.$
cardona-qft-methods_EQ0461	216	0.992 K|+0	`V^{\prime}=-V-\frac{1}{3}\left(\begin{array}{ccc} \Lambda & 0 & 0 \\ 0 & \Lambda & 0 \\ 0 & 0 & \Lambda \end{array} \right)=-V-\frac{1}{3} \Lambda 1_{3} .`	$V^{\prime}=-V-\frac{1}{3}\left(\begin{array}{ccc} \Lambda & 0 & 0 \\ 0 & \Lambda & 0 \\ 0 & 0 & \Lambda \end{array} \right)=-V-\frac{1}{3} \Lambda 1_{3} .$	$V'=-V-\frac{1}{3}\begin{pmatrix} \Lambda & 0 & 0 \\ 0 & \Lambda & 0 \\ 0 & 0 & \Lambda \end{pmatrix}=-V-\frac{1}{3}\Lambda 1_3.$
cardona-qft-methods_EQ0372 (12)	176	0.993 N|+0	`\Upsilon^{A B}=\left[\begin{array}{cc} \eta^{a b} & 0 \\ 0 & \mp 1 \end{array}\right],`	$\Upsilon^{A B}=\left[\begin{array}{cc} \eta^{a b} & 0 \\ 0 & \mp 1 \end{array}\right],$	$\Upsilon^{AB}=\begin{bmatrix} \eta^{ab} & 0 \\ 0 & \mp 1 \end{bmatrix},$
cardona-qft-methods_EQ0374 (14)	176	0.993 N|+1	`\begin{aligned} \delta \omega^{a b} & =d \lambda^{a b}+\omega_{c}^{a} \lambda^{c b}+\omega_{c}^{b} \lambda^{a c}=D_{\omega} \lambda^{a b} \\ \delta e^{a} & =d \lambda^{a}+\omega_{b}^{a} \lambda^{b}-\lambda_{b}^{a} e^{b}=D_{\omega} \lambda^{a} \end{aligned}`	$\begin{aligned} \delta \omega^{a b} & =d \lambda^{a b}+\omega_{c}^{a} \lambda^{c b}+\omega_{c}^{b} \lambda^{a c}=D_{\omega} \lambda^{a b} \\ \delta e^{a} & =d \lambda^{a}+\omega_{b}^{a} \lambda^{b}-\lambda_{b}^{a} e^{b}=D_{\omega} \lambda^{a} \end{aligned}$	$\begin{aligned} \delta\omega^{ab} &= d\lambda^{ab}+\omega^a{}_c\lambda^{cb}+\omega^b{}_c\lambda^{ac}=D_\omega\lambda^{ab} \\ \delta e^a &= d\lambda^a+\omega^a{}_b\lambda^b-\lambda^a{}_b e^b=D_\omega\lambda^a. \end{aligned}$
cardona-qft-methods_EQ0605 (14)	253	0.993 K|+0	`e_{p}(x)=\exp -i p \cdot x, \quad\left|p_{0}\right|= \left(\mathbf{p}^{2}+m^{2}\right)^{\frac{1}{2}}, \quad d \mu(p)=\frac{d^{d} p}{2\left|p_{0}\right|},`	$e_{p}(x)=\exp -i p \cdot x, \quad\left\lvert p_{0}\right\rvert= \left(\mathbf{p}^{2}+m^{2}\right)^{\frac{1}{2}}, \quad d \mu(p)=\frac{d^{d} p}{2\left\lvert p_{0}\right\rvert},$	$e_p(x)=\exp -ip\cdot x, \quad \lvert p_0\rvert=(\mathbf{p}^2+m^2)^{\frac{1}{2}}, \quad d\mu(p)=\frac{d^dp}{2\lvert p_0\rvert},$
cardona-qft-methods_EQ0740 (8)	282	0.993 N|+0	`\begin{aligned} Q \cdot \varphi \sim \psi & , & \bar{Q} \cdot \varphi \sim \bar{\psi} \\ Q \cdot \psi \sim F+g_{Y M}[\varphi, \varphi] & , & \bar{Q} \cdot \psi \sim \mathcal{D} \varphi \end{aligned}`	$\begin{aligned} Q \cdot \varphi \sim \psi & , & \bar{Q} \cdot \varphi \sim \bar{\psi} \\ Q \cdot \psi \sim F+g_{Y M}[\varphi, \varphi] & , & \bar{Q} \cdot \psi \sim \mathcal{D} \varphi \end{aligned}$	$\begin{array}{ccc} Q\cdot\varphi\sim\psi & , & \bar{Q}\cdot\varphi\sim\bar{\psi} \\ Q\cdot\psi\sim F+g_{YM}[\varphi,\varphi] & , & \bar{Q}\cdot\psi\sim\mathcal{D}\varphi \end{array}$
cardona-qft-methods_EQ0075	045	0.993 K|+0	`\Psi(\mathcal{A})=\left\{T \mid \forall N \in \mathbb{N} \exists P \in \mathcal{D}(\mathcal{A}), R \in O P^{-N}, p \in \mathbb{N} \text { s.t. } T=P|\mathcal{D}|^{-p}+R \right\} .`	$\Psi(\mathcal{A})=\left\{T \mid \forall N \in \mathbb{N} \exists P \in \mathcal{D}(\mathcal{A}), R \in O P^{-N}, p \in \mathbb{N} \text { s.t. } T=P\lvert\mathcal{D}\rvert^{-p}+R \right\} .$	$\Psi(\mathcal{A})=\{\,T \mid \forall N\in\mathbb{N}\exists P\in\mathcal{D}(\mathcal{A}),\ R\in OP^{-N},\ p\in\mathbb{N}\text{ s.t. } T=P\lvert\mathcal{D}\rvert^{-p}+R\,\}.$
cardona-qft-methods_EQ0927	352	0.993 N|+1	`H_{*}^{p r o}\left(\mathbb{L}\left(M \times_{G} E G \right)^{-T\left(M \times_{G} E G\right)}\right):=\lim _{ \leftarrow n} H_{*}\left(\left(P_{G} M \times_{G} E G_{n} \right)^{-e_{0}^{*} T\left(M \times_{G} E G_{n}\right)} \right)`	$H_{*}^{p r o}\left(\mathbb{L}\left(M \times_{G} E G \right)^{-T\left(M \times_{G} E G\right)}\right):=\lim _{ \leftarrow n} H_{*}\left(\left(P_{G} M \times_{G} E G_{n} \right)^{-e_{0}^{*} T\left(M \times_{G} E G_{n}\right)} \right)$	$H_*^{pro}(\mathbb{L}(M\times_G EG)^{-T(M\times_G EG)}):=\lim_{\leftarrow n} H_*((P_GM\times_G EG_n)^{-e_0^*T(M\times_G EG_n)})$
cardona-qft-methods_EQ0899	347	0.993 N|+0	`\begin{gathered} H H_{*}(A, M)=H^{*}\left(A \stackrel{L}{ \otimes_{A^{e}}} M\right) \\ H H^{*}(A, M)=H_{*}\left( \mathcal{R} \mathcal{H} o m_{A^{e}}(A, M)\right) \end{gathered}`	$\begin{gathered} H H_{*}(A, M)=H^{*}\left(A \stackrel{L}{ \otimes_{A^{e}}} M\right) \\ H H^{*}(A, M)=H_{*}\left(\mathcal{R} \mathcal{H} o m_{A^{e}}(A, M)\right) \end{gathered}$	$HH_*(A,M)=H^*(A\overset{L}{\otimes}_{A^e}M)$ $HH^*(A,M)=H_*(\mathcal{RH}om_{A^e}(A,M))$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0138	080	0.994 N|+0	\phi_{i}: U_{i} \xrightarrow{\approx} V_{i} \subset \mathbb{R}^{n} \quad(i=1, \ldots, m)	$\phi_{i}: U_{i} \xrightarrow{\approx} V_{i} \subset \mathbb{R}^{n} \quad(i=1, \ldots, m)$	$\phi_i : U_i \xrightarrow{\approx} V_i \subset \mathbb{R}^n \qquad (i=1,\ldots,m)$
cardona-qft-methods_EQ0292	142	0.994 K|+0	\left[B \ell_{B} X\right]_{S B}-[E]_{S B}=[X]_{S B}-[B]_{S B},	$\left[B \ell_{B} X\right]_{S B}-[E]_{S B}=[X]_{S B}-[B]_{S B},$	$[B\ell_B X]_{SB} - [E]_{SB} = [X]_{SB} - [B]_{SB} \quad ,$
cardona-qft-methods_EQ0917	351	0.994 N|+0	\operatorname{Tor}_{C^{*}(X)^{e}}\left(C^{*}(X), C^{*}(X) \# G\right) \cong H^{*}\left(P_{G} X\right) .	$\operatorname{Tor}_{C^{*}(X)^{e}}\left(C^{*}(X), C^{*}(X) \# G\right) \cong H^{*}\left(P_{G} X\right) .$	$\operatorname{Tor}_{C^*(X)^e}(C^*(X), C^*(X)\#G) \cong H^*(P_G X).$
cardona-qft-methods_EQ0209	097	0.994 N|+0	E^{+}:=\bigoplus_{j \text { even }} \Lambda^{j} \left(T^{*} M \otimes \mathcal{C}\right), \quad E^{-}:= \bigoplus_{j \text { odd }} \Lambda^{j}\left(T^{*} M \otimes \mathcal{C}\right) .	$E^{+}:=\bigoplus_{j \text { even }} \Lambda^{j} \left(T^{*} M \otimes \mathcal{C}\right), \quad E^{-}:= \bigoplus_{j \text { odd }} \Lambda^{j}\left(T^{*} M \otimes \mathcal{C}\right) .$	$E^+ \;:=\; \bigoplus_{j\,\text{even}} \Lambda^j(T^*M \otimes \mathcal{C}), \qquad E^- \;:=\; \bigoplus_{j\,\text{odd}} \Lambda^j(T^*M \otimes \mathcal{C}).$
cardona-qft-methods_EQ0263	125	0.994 N|+1	\langle G\rangle=\prod_{i}\left\langle G_{i}\right\rangle / \text { extra symmetries }	$\langle G\rangle=\prod_{i}\left\langle G_{i}\right\rangle / \text { extra symmetries }$	$\langle G\rangle = \prod_i \langle G_i\rangle/\text{extra symmetries} \quad .$
cardona-qft-methods_EQ0422 (7)	193	0.994 N|+0	\int_{M} \Theta \operatorname{Tr}[F \wedge F]=\int_{ \partial \tilde{M}} j \wedge \operatorname{Tr}[A \wedge d A],	$\int_{M} \Theta \operatorname{Tr}[F \wedge F]=\int_{ \partial \tilde{M}} j \wedge \operatorname{Tr}[A \wedge d A],$	$\int_M \Theta Tr[F \wedge F] = \int_{\partial\tilde{M}} j \wedge Tr[A \wedge dA],$
cardona-qft-methods_EQ0888	345	0.994 N|+0	\circ: H_{p}(L M) \otimes H_{q}(L M) \rightarrow H_{p+q-n}(L M) .	$\circ: H_{p}(L M) \otimes H_{q}(L M) \rightarrow H_{p+q-n}(L M) .$	$\circ : H_p(LM) \otimes H_q(LM) \to H_{p+q-n}(LM).$
cardona-qft-methods_EQ0364 (3)	174	0.994 N|+1	\begin{aligned} \delta e^{a} & =d \lambda^{a}+ \omega_{b}^{a} \lambda^{b} \\ & =D_{\omega} \lambda^{a} \end{aligned}	$\begin{aligned} \delta e^{a} & =d \lambda^{a}+ \omega_{b}^{a} \lambda^{b} \\ & =D_{\omega} \lambda^{a} \end{aligned}$	$\begin{aligned} \delta e^a &= d\lambda^a + \omega^a{}_b \lambda^b, \\ &= D_\omega \lambda^a \end{aligned}$
cardona-qft-methods_EQ0412 (42)	188	0.995 K|+0	\mathcal{T}_{2 n+1}(A, \bar{A})=\mathcal{C}_{2 n+1}(A)- \mathcal{C}_{2 n+1}(\bar{A})+d \mathcal{B}_{2 n}(A, \bar{A}) .	$\mathcal{T}_{2 n+1}(A, \bar{A})=\mathcal{C}_{2 n+1}(A)- \mathcal{C}_{2 n+1}(\bar{A})+d \mathcal{B}_{2 n}(A, \bar{A}) .$	$\mathcal{T}_{2n+1}(A,\bar{A}) = \mathcal{C}_{2n+1}(A) - \mathcal{C}_{2n+1}(\bar{A}) + d\mathcal{B}_{2n}(A,\bar{A}).$
cardona-qft-methods_EQ0942 (2)	358	0.995 K|+0	[\emptyset]=0 .	$[\emptyset]=0 .$	$[\emptyset] = 0\,.$
cardona-qft-methods_EQ0349 (17)	168	0.995 N|+0	\begin{aligned} & \mathcal{E}_{a}^{(p)}:=\epsilon_{a b_{2} \cdots b_{D}} R^{b_{2} b_{3}} \cdots R^{b_{2 p} b_{2 p+1}} e^{b_{2 p+2}} \cdots e^{b_{D}}, \\ & \mathcal{E}_{a b}^{(p)}:=\epsilon_{a b a_{3} \cdots a_{D}} R^{a_{3} a_{4}} \cdots R^{a_{2 p-1} a_{2 p}} T^{a_{2 p+1}} e^{a_{2 p+2}} \cdots e^{a_{D}} . \end{aligned}	$\begin{aligned} & \mathcal{E}_{a}^{(p)}:=\epsilon_{a b_{2} \cdots b_{D}} R^{b_{2} b_{3}} \cdots R^{b_{2 p} b_{2 p+1}} e^{b_{2 p+2}} \cdots e^{b_{D}}, \\ & \mathcal{E}_{a b}^{(p)}:=\epsilon_{a b a_{3} \cdots a_{D}} R^{a_{3} a_{4}} \cdots R^{a_{2 p-1} a_{2 p}} T^{a_{2 p+1}} e^{a_{2 p+2}} \cdots e^{a_{D}} . \end{aligned}$	$\begin{aligned} \mathcal{E}^{(p)}_a &:= \epsilon_{ab_2\cdots b_D} R^{b_2b_3} \cdots R^{b_{2p}b_{2p+1}} e^{b_{2p+2}} \cdots e^{b_D}, \\ \mathcal{E}^{(p)}_{ab} &:= \epsilon_{aba_3\cdots a_D} R^{a_3a_4} \cdots R^{a_{2p-1}a_{2p}} T^{a_{2p+1}} e^{a_{2p+2}} \cdots e^{a_D}. \end{aligned}$
cardona-qft-methods_EQ0796 (37)	302	0.995 K|+0	N S-N S: \text { type IIA and IIB } \mathbf{8}_{ \mathrm{v}} \otimes \mathbf{8}_{\mathrm{v}}=\mathbf{1} \oplus \mathbf{2 8} \oplus \mathbf{3 5}=\Phi \oplus \mathrm{B}_{2} \oplus \mathrm{G}_{\mathrm{MN}}	$N S-N S: \text { type IIA and IIB } \mathbf{8}_{ \mathrm{v}} \otimes \mathbf{8}_{\mathrm{v}}=\mathbf{1} \oplus \mathbf{2 8} \oplus \mathbf{3 5}=\Phi \oplus \mathrm{B}_{2} \oplus \mathrm{G}_{\mathrm{MN}}$	$NS-NS:\ \text{type IIA and IIB}\ \mathbf{8}_{\mathrm{v}} \otimes \mathbf{8}_{\mathrm{v}} = \mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35} = \Phi \oplus \mathrm{B}_2 \oplus \mathrm{G}_{\mathrm{MN}}$
cardona-qft-methods_EQ0471	218	0.995 N|-1	\operatorname{Tr}(f(D / \Lambda)) \sim \sum_{k} f_{k} \Lambda^{k} f|D|^{-k}+f(0) \zeta_{D}(0)+o(1),	$\operatorname{Tr}(f(D / \Lambda)) \sim \sum_{k} f_{k} \Lambda^{k} f\|D\|^{-k}+f(0) \zeta_{D}(0)+o(1),$	$\operatorname{Tr}(f(D/\Lambda)) \sim \sum_k f_k \Lambda^k \fint \|D\|^{-k} + f(0)\,\zeta_D(0) + o(1),$
cardona-qft-methods_EQ0574 (9)	249	0.995 N|+0	\widehat{x}_{\mu} \star \widehat{x}_{\nu}-\widehat{x}_{ \nu} \star \widehat{x}_{\mu}=\left[\widehat{x}_{\mu}, \widehat{x}_{\nu}\right]_{\star}=i \theta_{\mu \nu} \quad \mu, \nu=0,1, \ldots d .	$\widehat{x}_{\mu} \star \widehat{x}_{\nu}-\widehat{x}_{ \nu} \star \widehat{x}_{\mu}=\left[\widehat{x}_{\mu}, \widehat{x}_{\nu}\right]_{\star}=i \theta_{\mu \nu} \quad \mu, \nu=0,1, \ldots d .$	$\widehat{x}_\mu \star \widehat{x}_\nu - \widehat{x}_\nu \star \widehat{x}_\mu = [\widehat{x}_\mu, \widehat{x}_\nu]_\star = i\theta_{\mu\nu} \quad \mu,\nu = 0,1,\ldots d.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0111	073	■ 0.996 ● K|+0	`\operatorname{Ind}(B A)=\operatorname{Ind}(A)+ \operatorname{Ind}(B) .`	$\operatorname{Ind}(BA)=\operatorname{Ind}(A)+\operatorname{Ind}(B).$	$\operatorname{Ind}(BA) = \operatorname{Ind}(A) + \operatorname{Ind}(B).$
cardona-qft-methods_ EQ0429	206	■ 0.996 ● N|+0	`\mathcal{A}=\oplus_{i=1}^{N} M_{n_{i}}\left( \mathbb{K}_{i}\right)`	$\mathcal{A}=\oplus_{i=1}^{N} M_{n_{i}}\left(\mathbb{K}_{i}\right)$	$\mathcal{A}=\oplus_{i=1}^{N} M_{n_{i}}(\mathbb{K}_{i})$
cardona-qft-methods_ EQ0494	223	■ 0.996 ● C|+3	`\begin{gathered} 16 \pi^{2} \beta_{g_{i}}=b_{i} g_{i}^{3} \quad \text { with }\left(b_{S U(3)}, b_{S U(2)}, b_{U(1)}\right)=\left(-7,-\frac{19}{6}, \frac{41}{10} \right) \\ 16 \pi^{2} \beta_{Y_{u}}=Y_{u}\left(\frac{3}{2} Y_{u}^{\dagger} Y_{u}-\frac{3}{2} Y_{d}^{ \dagger} Y_{d}+\mathfrak{a}-\frac{17}{20} g_{1}^{2}- \frac{9}{4} g_{2}^{2}-8 g_{3}^{2}\right) \\ 16 \pi^{2} \beta_{Y_{d}}=Y_{d}\left(\frac{3}{2} Y_{d}^{\dagger} Y_{d}-\frac{3}{2} Y_{u}^{\dagger} Y_{u}+\mathfrak{a}- \frac{1}{4} g_{1}^{2}-\frac{9}{4} g_{2}^{2}-8 g_{3}^{2} \right) \\ 16 \pi^{2} \beta_{Y_{\nu}}=Y_{\nu}\left(\frac{3}{2} Y_{\nu}^{\dagger} Y_{\nu}-\frac{3}{2} Y_{e}^{\dagger} Y_{e}+\mathfrak{a}-\frac{9}{20} g_{1}^{2}-\frac{9}{4} g_{2}^{2}\right) \\ 16 \pi^{2} \beta_{Y_{e}}=Y_{e}\left(\frac{3}{2} Y_{e}^{\dagger} Y_{e}-\frac{3}{2} Y_{\nu}^{\dagger} Y_{\nu}+\mathfrak{a}- \frac{9}{4} g_{1}^{2}-\frac{9}{4} g_{2}^{2}\right) \ \ 16 \pi^{2} \beta_{M}=Y_{\nu} Y_{\nu}^{\dagger} M+M \left(Y_{\nu} Y_{\nu}^{\dagger}\right)^{T} \\ 16 \pi^{2} \beta_{\lambda}=6 \lambda^{2}-3 \lambda\left(3 g_{2}^{2}+ \frac{3}{5} g_{1}^{2}\right)+3 g_{2}^{4}+\frac{3}{2} \left(\frac{3}{5} g_{1}^{2}+g_{2}^{2}\right)^{2}+4 \lambda \mathfrak{a}-8 \mathfrak{b} \end{gathered}`	$\begin{gathered} 16 \pi^{2} \beta_{g_{i}}=b_{i} g_{i}^{3} \quad \text { with }\left(b_{S U(3)}, b_{S U(2)}, b_{U(1)}\right)=\left(-7,-\frac{19}{6}, \frac{41}{10} \right) \\ 16 \pi^{2} \beta_{Y_{u}}=Y_{u}\left(\frac{3}{2} Y_{u}^{\dagger} Y_{u}-\frac{3}{2} Y_{d}^{ \dagger} Y_{d}+\mathfrak{a}-\frac{17}{20} g_{1}^{2}- \frac{9}{4} g_{2}^{2}-8 g_{3}^{2}\right) \\ 16 \pi^{2} \beta_{Y_{d}}=Y_{d}\left(\frac{3}{2} Y_{d}^{\dagger} Y_{d}-\frac{3}{2} Y_{u}^{\dagger} Y_{u}+\mathfrak{a}- \frac{1}{4} g_{1}^{2}-\frac{9}{4} g_{2}^{2}-8 g_{3}^{2} \right) \\ 16 \pi^{2} \beta_{Y_{\nu}}=Y_{\nu}\left(\frac{3}{2} Y_{\nu}^{\dagger} Y_{\nu}-\frac{3}{2} Y_{e}^{\dagger} Y_{e}+\mathfrak{a}-\frac{9}{20} g_{1}^{2}-\frac{9}{4} g_{2}^{2}\right) \\ 16 \pi^{2} \beta_{Y_{e}}=Y_{e}\left(\frac{3}{2} Y_{e}^{\dagger} Y_{e}-\frac{3}{2} Y_{\nu}^{\dagger} Y_{\nu}+\mathfrak{a}- \frac{9}{4} g_{1}^{2}-\frac{9}{4} g_{2}^{2}\right) \\ 16 \pi^{2} \beta_{M}=Y_{\nu} Y_{\nu}^{\dagger} M+M \left(Y_{\nu} Y_{\nu}^{\dagger}\right)^{T} \\ 16 \pi^{2} \beta_{\lambda}=6 \lambda^{2}-3 \lambda\left(3 g_{2}^{2}+ \frac{3}{5} g_{1}^{2}\right)+3 g_{2}^{4}+\frac{3}{2} \left(\frac{3}{5} g_{1}^{2}+g_{2}^{2}\right)^{2}+4 \lambda \mathfrak{a}-8 \mathfrak{b} \end{gathered}$	$\begin{gathered} 16\pi^2\ \beta_{g_i} = b_i\, g_i^3 \quad \text{with } (b_{SU(3)}, b_{SU(2)}, b_{U(1)}) = (-7, -\frac{19}{6}, \frac{41}{10}) \\ 16\pi^2\ \beta_{Y_u} = Y_u(\frac{3}{2}Y_u^\dagger Y_u - \frac{3}{2}Y_d^\dagger Y_d + \mathfrak{a} - \frac{17}{20}g_1^2 - \frac{9}{4}g_2^2 - 8g_3^2) \\ 16\pi^2\ \beta_{Y_d} = Y_d(\frac{3}{2}Y_d^\dagger Y_d - \frac{3}{2}Y_u^\dagger Y_u + \mathfrak{a} - \frac{1}{4}g_1^2 - \frac{9}{4}g_2^2 - 8g_3^2) \\ 16\pi^2\ \beta_{Y_\nu} = Y_\nu(\frac{3}{2}Y_\nu^\dagger Y_\nu - \frac{3}{2}Y_e^\dagger Y_e + \mathfrak{a} - \frac{9}{20}g_1^2 - \frac{9}{4}g_2^2) \\ 16\pi^2\ \beta_{Y_e} = Y_e(\frac{3}{2}Y_e^\dagger Y_e - \frac{3}{2}Y_\nu^\dagger Y_\nu + \mathfrak{a} - \frac{9}{4}g_1^2 - \frac{9}{4}g_2^2) \\ 16\pi^2\ \beta_M = Y_\nu Y_\nu^\dagger M + M(Y_\nu Y_\nu^\dagger)^T \\ 16\pi^2\ \beta_\lambda = 6\lambda^2 - 3\lambda(3g_2^2 + \frac{3}{5}g_1^2) + 3g_2^4 + \frac{3}{2}(\frac{3}{5}g_1^2 + g_2^2)^2 + 4\lambda\mathfrak{a} - 8\mathfrak{b}. \end{gathered}$
cardona-qft-methods_ EQ0241	115	■ 0.996 ● N|+0	`\kappa_{*}: A_{*} X \rightarrow A_{*}\{\mathrm{pt}\}= \mathbb{Z},`	$\kappa_{*}: A_{*} X \rightarrow A_{*}\{\mathrm{pt}\}=\mathbb{Z},$	$\kappa_{*}: A_{*} X \rightarrow A_{*}\{\mathrm{pt}\}=\mathbb{Z} \quad ,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0279	136	0.996 W|+2	`\begin{aligned} \sum_{m \geq 0} \mathbb{U}\left(G_{m e}\right) \frac{s^{m}}{m!}= & \frac{e^{\mathbb{T} s}-e^{-s}}{\mathbb{T}+1} \mathbb{U}(G)+\frac{e^{\mathbb{T} s}+\mathbb{T} e^{-s}}{\mathbb{T}+1} \mathbb{U}(G \backslash e) \\ & +\left(s e^{\mathbb{T} s}-\frac{e^{\mathbb{T} s}-e^{-s}}{\mathbb{T}+1}\right) \mathbb{U}(G / e) \end{aligned}`	$\begin{aligned} \sum_{m \geq 0} \mathbb{U}\left(G_{m e}\right) \frac{s^{m}}{m!}= & \frac{e^{\mathbb{T} s}-e^{-s}}{\mathbb{T}+1} \mathbb{U}(G)+\frac{e^{\mathbb{T} s}+\mathbb{T} e^{-s}}{\mathbb{T}+1} \mathbb{U}(G \backslash e) \\ & +\left(s e^{\mathbb{T} s}-\frac{e^{\mathbb{T} s}-e^{-s}}{\mathbb{T}+1}\right) \mathbb{U}(G / e) \end{aligned}$	$\begin{aligned} \sum_{m \geq 0} \mathbb{U}(G_{me}) \frac{s^{m}}{m!} &= \frac{e^{\mathbb{T} s}-e^{-s}}{\mathbb{T}+1} \mathbb{U}(G)+\frac{e^{\mathbb{T} s}+\mathbb{T} e^{-s}}{\mathbb{T}+1} \mathbb{U}(G \smallsetminus e) \\ & +\left(s\, e^{\mathbb{T} s}-\frac{e^{\mathbb{T} s}-e^{-s}}{\mathbb{T}+1}\right) \mathbb{U}(G / e). \end{aligned}$
cardona-qft-methods_EQ0588 (21)	251	0.996 K|+0	`\tau_{0} \Delta_{\theta}(\Lambda) \neq \Delta_{\theta}(\Lambda) \tau_{0},`	$\tau_{0} \Delta_{\theta}(\Lambda) \neq \Delta_{\theta}(\Lambda) \tau_{0},$	$\tau_0\Delta_\theta(\Lambda) \neq \Delta_\theta(\Lambda)\tau_0,$
cardona-qft-methods_EQ0709	273	0.996 N|+0	`e^{i p_{k} L}=\prod_{j \neq k}^{M} S\left(p_{k}, p_{j}\right) \quad \text { for } \quad k=1, \ldots, M,`	$e^{i p_{k} L}=\prod_{j \neq k}^{M} S\left(p_{k}, p_{j}\right) \quad \text { for } \quad k=1, \ldots, M,$	$e^{ip_kL} = \prod_{j\neq k}^{M} S(p_k,p_j) \quad \text{for} \quad k=1,\ldots,M\,,$
cardona-qft-methods_EQ0803 (3)	306	0.996 K|+0	`0 \longrightarrow T^{*} M \longrightarrow E \xrightarrow{\pi} T M \longrightarrow 0 .`	$0 \longrightarrow T^{*} M \longrightarrow E \xrightarrow{\pi} T M \longrightarrow 0 .$	$0 \longrightarrow T^*M \longrightarrow E \xrightarrow{\pi} TM \longrightarrow 0\,.$
cardona-qft-methods_EQ0202 (10)	096	0.997 K|+0	`\mathcal{D}=d+d^{*} .`	$\mathcal{D}=d+d^{*} .$	$\mathcal{D} \;=\; d \;+\; d^*.$
cardona-qft-methods_EQ0290	139	0.997 S|+1	`\begin{aligned} & \sum_{m \geq 0} C_{G_{(m+1) e}}(t) \frac{s^{m}}{m!}=\left(e^{t s}-e^{(t-1) s}\right) C_{G_{2 e}}(t) \\ & \quad-\left((t-1) e^{t s}-t e^{(t-1) s}\right) C_{G}(t)+t\left((s-1) e^{t s}+e^{(t-1) s}\right) C_{G / e}(t) \end{aligned}`	$\begin{aligned} & \sum_{m \geq 0} C_{G_{(m+1) e}}(t) \frac{s^{m}}{m!}=\left(e^{t s}-e^{(t-1) s}\right) C_{G_{2 e}}(t) \\ & \quad-\left((t-1) e^{t s}-t e^{(t-1) s}\right) C_{G}(t)+t\left((s-1) e^{t s}+e^{(t-1) s}\right) C_{G / e}(t) \end{aligned}$	$\begin{aligned} & \sum_{m \geq 0} C_{G_{(m+1)e}}(t) \frac{s^{m}}{m!}=\left(e^{ts}-e^{(t-1)s}\right) C_{G_{2e}}(t) \\ & \quad-\left((t-1)\, e^{ts}-t\, e^{(t-1)s}\right) C_{G}(t)+t\left((s-1)\, e^{ts}+e^{(t-1)s}\right) C_{G/e}(t)\,. \end{aligned}$
cardona-qft-methods_EQ0439	210	0.997 W|+0	`\begin{gathered} \mathcal{A}_{F}=\left\{\left(\lambda, q_{L}, \lambda, m\right) \mid \lambda \in \mathbb{C}, q_{L} \in \mathbb{H}, m \in M_{3}(\mathbb{C})\right\} \\ \sim \mathbb{C} \oplus \mathbb{H} \oplus M_{3}(\mathbb{C}), \end{gathered}`	$\begin{gathered} \mathcal{A}_{F}=\left\{\left(\lambda, q_{L}, \lambda, m\right) \mid \lambda \in \mathbb{C}, q_{L} \in \mathbb{H}, m \in M_{3}(\mathbb{C})\right\} \\ \sim \mathbb{C} \oplus \mathbb{H} \oplus M_{3}(\mathbb{C}), \end{gathered}$	$\begin{gathered} \mathcal{A}_F = \{(\lambda, q_L, \lambda, m) \mid \lambda \in \mathbb{C},\ q_L \in \mathbb{H},\ m \in M_3(\mathbb{C})\} \\ \sim \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}), \end{gathered}$
cardona-qft-methods_EQ0958 (10)	361	0.997 K|+0	`X_{\Gamma^{\vee}}=\pi_{2}\left(\pi_{1}^{-1}\left(X_{\Gamma}\right)\right),`	$X_{\Gamma^{\vee}}=\pi_{2}\left(\pi_{1}^{-1}\left(X_{\Gamma}\right)\right),$	$X_{\Gamma^\vee} = \pi_2(\pi_1^{-1}(X_\Gamma))\,,$
cardona-qft-methods_EQ0804 (4)	306	0.997 N|+0	`x_{(\alpha)}+\xi_{(\alpha)}=a_{(\alpha \beta)} \cdot x_{(\beta)}+\left[a_{(\alpha \beta)}^{-T} \xi_{(\beta)}-\iota_{a_{(\alpha \beta)} x_{(\beta)}} b_{(\alpha \beta)}\right],`	$x_{(\alpha)}+\xi_{(\alpha)}=a_{(\alpha \beta)} \cdot x_{(\beta)}+\left[a_{(\alpha \beta)}^{-T} \xi_{(\beta)}-\iota_{a_{(\alpha \beta)} x_{(\beta)}} b_{(\alpha \beta)}\right],$	$x_{(\alpha)}+\xi_{(\alpha)}=a_{(\alpha \beta)} \cdot x_{(\beta)}+\left[a_{(\alpha \beta)}^{-T} \xi_{(\beta)}-\iota_{a_{(\alpha \beta)} x_{(\beta)}} b_{(\alpha \beta)}\right]\,,$
cardona-qft-methods_EQ0157	086	0.997 N|+0	`j: T^{*} M \hookrightarrow \mathbb{R}^{N} \oplus \mathbb{R}^{N} \simeq \mathcal{C}^{N} .`	$j: T^{*} M \hookrightarrow \mathbb{R}^{N} \oplus \mathbb{R}^{N} \simeq \mathcal{C}^{N} .$	$j:\, T^*M \;\hookrightarrow \mathbb{R}^N \oplus \mathbb{R}^N \;\simeq\; \mathcal{C}^N.$
cardona-qft-methods_EQ0286	138	0.997 N|+0	`C_{G_{2 e}}(T)=(2 T-1) C_{G}(T)-T(T-1) C_{G \backslash e}(T)+C_{G / e}(T) .`	$C_{G_{2 e}}(T)=(2 T-1) C_{G}(T)-T(T-1) C_{G \backslash e}(T)+C_{G / e}(T) .$	$C_{G_{2e}}(T) = (2T-1)C_G(T) - T(T-1)C_{G\smallsetminus e}(T) + C_{G/e}(T) \quad .$
cardona-qft-methods_EQ0543	243	0.997 N|+1	`\Delta \circ m(a \otimes b)=\left(m_{13} \otimes m_{24}\right)(\Delta \otimes \Delta)(a \otimes b) .`	$\Delta \circ m(a \otimes b)=\left(m_{13} \otimes m_{24}\right)(\Delta \otimes \Delta)(a \otimes b) .$	$\Delta \circ m(a \otimes b) = (m_{13} \otimes m_{24})(\Delta \otimes \Delta)(a \otimes b).$
cardona-qft-methods_EQ0811 (11)	308	0.997 N|+0	`\Phi_{ \pm}=\eta_{+}^{1} \otimes \eta_{ \pm}^{2 \dagger} .`	$\Phi_{ \pm}=\eta_{+}^{1} \otimes \eta_{ \pm}^{2 \dagger} .$	$\Phi_\pm = \eta_+^1 \otimes \eta_\pm^{2\dagger}\;.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>cardona-qft-methods_ EQ0987</td><td>370</td><td>■0.997 ● W|+1</td><td><code>\mathbb{D} \pi(X) \xi=X \xi=\lim _{t \rightarrow 0} \frac{1}{t}(\exp (t X) \xi-\xi),</code></td><td>$\mathbb{D} \pi(X) \xi=X \xi=\lim _{t \rightarrow 0} \frac{1}{t}(\exp (t X) \xi-\xi),$</td><td>$\mathbb{D} \pi(X) \xi=X \xi=\lim _{t \rightarrow 0} \frac{1}{t}(\exp (t X) \xi-\xi),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0081 (28)</td><td>048</td><td>■0.998 ● N|+0</td><td><code>\mathcal{D}_{A}=\mathcal{D}+\widetilde{A} \text { where } \widetilde{A}=A+\varepsilon J A J^{-1}, \quad D_{A}= \mathcal{D}_{A}+P_{A}</code></td><td>$\mathcal{D}_{A}=\mathcal{D}+\widetilde{A} \text { where } \widetilde{A}=A+\varepsilon J A J^{-1}, \quad D_{A}= \mathcal{D}_{A}+P_{A}$</td><td>$\mathcal{D}_{A}=\mathcal{D}+\widetilde{A} \text { where } \widetilde{A}=A+\varepsilon J A J^{-1}, \qquad D_{A}= \mathcal{D}_{A}+P_{A}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0169</td><td>089</td><td>■0.998 ● K|+0</td><td><code>g_{1} \cdot\left(x, g_{2}\right):=\left(x, g_{1} g_{2} \right), \quad x \in N, g_{1}, g_{2} \in G .</code></td><td>$g_{1} \cdot\left(x, g_{2}\right):=\left(x, g_{1} g_{2} \right), \quad x \in N, g_{1}, g_{2} \in G .$</td><td>$g_{1} \cdot (x, g_{2}) \;:=\; (x, g_{1} g_{2}), \qquad x \in N,\; g_{1}, g_{2} \in G.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0542</td><td>243</td><td>■0.998 ● N|+0</td><td><code>m\left(\int d \mu(g) \alpha(g) g \otimes \int d \mu(h) \beta(h) h\right)=\int d \mu(g) d \mu(h) \alpha(g) \beta(h) g h .</code></td><td>$m\left(\int d \mu(g) \alpha(g) g \otimes \int d \mu(h) \beta(h) h\right)=\int d \mu(g) d \mu(h) \alpha(g) \beta(h) g h .$</td><td>$m\left(\int d \mu(g) \alpha(g) g \otimes \int d \mu(h) \beta(h) h\right)=\int d \mu(g) d \mu(h) \alpha(g) \beta(h) g h .$</td></tr>
<tr><td>cardona-qft-methods_ EQ0617 (25)</td><td>256</td><td>■0.998 ● N|+0</td><td><code>f_{1} f_{2} \in C^{\infty}(M),</code></td><td>$f_{1} f_{2} \in C^{\infty}(M),$</td><td>$f_{1} f_{2} \in C^{\infty}(M),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0638 (45)</td><td>258</td><td>■0.998 ● N|+0</td><td><code>\varphi_{\theta}=\varphi_{0} e^{-\frac{i}{2} \overleftarrow{\partial} \wedge P} .</code></td><td>$\varphi_{\theta}=\varphi_{0} e^{-\frac{i}{2} \overleftarrow{\partial} \wedge P} .$</td><td>$\varphi_{\theta}=\varphi_{0} e^{-\frac{i}{2} \overleftarrow{\partial} \wedge P} .$</td></tr>
<tr><td>cardona-qft-methods_ EQ0664 (15)</td><td>263</td><td>■0.998 ● N|+0</td><td><code>\theta\left(x_{0}-y_{0}\right)=\theta\left((\Lambda x)_{0}-(\Lambda y)_{0}\right),</code></td><td>$\theta\left(x_{0}-y_{0}\right)=\theta\left((\Lambda x)_{0}-(\Lambda y)_{0}\right),$</td><td>$\theta(x_{0}-y_{0})=\theta((\Lambda x)_{0}-(\Lambda y)_{0}),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0983 (3)</td><td>367</td><td>■0.998 ● N|+0</td><td><code>Q=\left\{t_{i} \mid \text { such that } t_{i} \text { appears in } q(t)\right\} .</code></td><td>$Q=\left\{t_{i} \mid \text { such that } t_{i} \text { appears in } q(t)\right\} .$</td><td>$Q=\{t_{i} \mid \textit{ such that } t_{i} \textit{ appears in } q(t)\} .$</td></tr>
<tr><td>cardona-qft-methods_ EQ0052</td><td>037</td><td>■0.998 ● N|+1</td><td><code>\begin{aligned} \left\langle U_{g^{\prime}, g} \psi^{ \prime}, U_{g^{\prime}, g} \psi^{\prime}\right\rangle_{ \mathcal{H}_{g}} & =\int_{M}\left|U_{g, g^{\prime}} \psi^{\prime}\right|_{g}^{2} d v o l_{g}=\int_{M} \left|I_{g^{\prime}, g} \psi^{\prime}\right|_{g}^{2} f_{g^{\prime}, g} d v o l_{g} \\ & =\int_{M}\left|\psi^{ \prime}\right|_{g^{\prime}}^{2} d v o l_{g^{\prime}}= \left\langle\psi_{g^{\prime}}, \psi_{g^{\prime}}\right\rangle_{\mathcal{H}_{g^{\prime}}} \end{aligned}</code></td><td>$\begin{aligned} \left\langle U_{g^{\prime}, g} \psi^{ \prime}, U_{g^{\prime}, g} \psi^{\prime}\right\rangle_{ \mathcal{H}_{g}} & =\int_{M}\left|U_{g, g^{\prime}} \psi^{\prime}\right|_{g}^{2} d v o l_{g}=\int_{M} \left|I_{g^{\prime}, g} \psi^{\prime}\right|_{g}^{2} f_{g^{\prime}, g} d v o l_{g} \\ & =\int_{M}\left|\psi^{ \prime}\right|_{g^{\prime}}^{2} d v o l_{g^{\prime}}= \left\langle\psi_{g^{\prime}}, \psi_{g^{\prime}}\right\rangle_{\mathcal{H}_{g^{\prime}}} \end{aligned}$</td><td>$\begin{aligned} \langle U_{g^{\prime}, g} \psi^{\prime}, U_{g^{\prime}, g} \psi^{\prime}\rangle_{\mathcal{H}_{g}} & =\int_{M}|U_{g, g^{\prime}} \psi^{\prime}|_{g}^{2}\, dvol_{g}=\int_{M} |I_{g^{\prime}, g} \psi^{\prime}|_{g}^{2} f_{g^{\prime}, g}\, dvol_{g} \\ & =\int_{M}|\psi^{\prime}|_{g^{\prime}}^{2}\, dvol_{g^{\prime}}= \langle\psi_{g^{\prime}}, \psi_{g^{\prime}}\rangle_{\mathcal{H}_{g^{\prime}}}. \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0006</td><td>019</td><td>■0.998 ● N|+0</td><td><code>\begin{aligned} \sigma(x, \xi) \in S^{m}\left(U \times \mathbb{R}^{d}\right) & \longleftrightarrow k_{ \sigma}(x, y) \in C_{c}^{\infty}\left(U \times U \times \mathbb{R}^{d}\right) \\ & \longleftrightarrow A=O p(\sigma) \in \Psi D O^{m} \end{aligned}</code></td><td>$\begin{aligned} \sigma(x, \xi) \in S^{m}\left(U \times \mathbb{R}^{d}\right) & \longleftrightarrow k_{ \sigma}(x, y) \in C_{c}^{\infty}\left(U \times U \times \mathbb{R}^{d}\right) \\ & \longleftrightarrow A=O p(\sigma) \in \Psi D O^{m} \end{aligned}$</td><td>$\begin{aligned} \sigma(x, \xi) \in S^{m}(U \times \mathbb{R}^{d}) & \longleftrightarrow k_{\sigma}(x, y) \in C_{c}^{\infty}(U \times U \times \mathbb{R}^{d}) \\ & \longleftrightarrow A=Op(\sigma) \in \Psi DO^{m} \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0010</td><td>019</td><td>■0.998 ● N|+0</td><td><code>\sigma^{\phi_{*} P}(x, \xi)=\sigma_{m}^{P}\left(\phi^{-1}(x),(d \phi)^{t} \xi\right)+\sum_{|\alpha|>0} \frac{(-i)^{\alpha}}{\alpha!} \phi_{\alpha}(x, \xi) \partial_{\xi}^{\alpha} \sigma^{P}\left(\phi^{-1}(x),(d \phi)^{t} \xi\right)</code></td><td>$\sigma^{\phi_{*} P}(x, \xi)=\sigma_{m}^{P}\left(\phi^{-1}(x),(d \phi)^{t} \xi\right)+\sum_{|\alpha|>0} \frac{(-i)^{\alpha}}{\alpha!} \phi_{\alpha}(x, \xi) \partial_{\xi}^{\alpha} \sigma^{P}\left(\phi^{-1}(x),(d \phi)^{t} \xi\right)$</td><td>$\sigma^{\phi_{*} P}(x, \xi)=\sigma_{m}^{P}(\phi^{-1}(x),(d \phi)^{t} \xi)+\sum_{|\alpha|>0} \frac{(-i)^{\alpha}}{\alpha!}\, \phi_{\alpha}(x, \xi)\, \partial_{\xi}^{\alpha} \sigma^{P}(\phi^{-1}(x),(d \phi)^{t} \xi)$</td></tr>
<tr><td>cardona-qft-methods_ EQ0074</td><td>045</td><td>■0.998 ● K|+0</td><td><code>\nabla\left(O P^{\alpha}\right) \subset O P^{\alpha+1} .</code></td><td>$\nabla\left(O P^{\alpha}\right) \subset O P^{\alpha+1} .$</td><td>$\nabla(OP^{\alpha}) \subset OP^{\alpha+1}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0083 (29)</td><td>048</td><td>■0.998 ● N|+0</td><td><code>\left|D_{A}\right|^{-s} \sim|D|^{-s}+\sum_{p=1}^{N} K_{p}(Y, s)|D|^{-s} \quad \bmod O P^{-N-1-\Re(s)}</code></td><td>$\left|D_{A}\right|^{-s} \sim|D|^{-s}+\sum_{p=1}^{N} K_{p}(Y, s)|D|^{-s} \quad \bmod O P^{-N-1-\Re(s)}$</td><td>$|D_{A}|^{-s} \sim|D|^{-s}+\sum_{p=1}^{N} K_{p}(Y, s)|D|^{-s} \quad \bmod OP^{-N-1-\Re(s)}$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>cardona-qft-methods_ EQ0413 (43)</td><td>188</td><td>■0.998 ● K|+0</td><td><code>\begin{aligned} & A \rightarrow A^{\prime}=\Lambda^{-1} A \Lambda+\Lambda^{-1} d \Lambda \\ & \bar{A} \rightarrow \bar{A}^{\prime}=\bar{\Lambda}^{-1} \bar{A} \bar{\Lambda}+\bar{\Lambda}^{-1} d \bar{\Lambda} \end{aligned}</code></td><td>$\begin{aligned} & A \rightarrow A^{\prime}=\Lambda^{-1} A \Lambda+\Lambda^{-1} d \Lambda \\ & \bar{A} \rightarrow \bar{A}^{\prime}=\bar{\Lambda}^{-1} \bar{A} \bar{\Lambda}+\bar{\Lambda}^{-1} d \bar{\Lambda} \end{aligned}$</td><td>$\begin{aligned} & A \rightarrow A'=\Lambda^{-1} A \Lambda+\Lambda^{-1} d \Lambda \\ & \bar{A} \rightarrow \bar{A}'=\bar{\Lambda}^{-1} \bar{A} \bar{\Lambda}+\bar{\Lambda}^{-1} d \bar{\Lambda} \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0437</td><td>209</td><td>■0.998 ● K|+0</td><td><code>J_{F}^{2}=1, \quad J_{F} \gamma_{F}=-\gamma_{F} J_{F} .</code></td><td>$J_{F}^{2}=1, \quad J_{F} \gamma_{F}=-\gamma_{F} J_{F} .$</td><td>$J_F^2 = 1\,, \quad J_F\, \gamma_F = -\,\gamma_F\, J_F .$</td></tr>
<tr><td>cardona-qft-methods_ EQ0438</td><td>209</td><td>■0.998 ● K|+0</td><td><code>q(\lambda)=\left(\begin{array}{ll} \lambda & 0 \\ 0 & \bar{\lambda} \end{array}\right) \quad q(\lambda)|\uparrow\rangle=\lambda|\uparrow\rangle, \quad q(\lambda)|\downarrow\rangle=\bar{\lambda}|\downarrow\rangle .</code></td><td>$q(\lambda)=\left(\begin{array}{ll} \lambda & 0 \\ 0 & \bar{\lambda} \end{array}\right) \quad q(\lambda)|\uparrow\rangle=\lambda|\uparrow\rangle, \quad q(\lambda)|\downarrow\rangle=\bar{\lambda}|\downarrow\rangle .$</td><td>$q(\lambda) = \begin{pmatrix} \lambda\ 0 \\ 0\ \bar{\lambda} \end{pmatrix} \qquad q(\lambda)|\uparrow\rangle = \lambda|\uparrow\rangle, \qquad q(\lambda)|\downarrow\rangle = \bar{\lambda}|\downarrow\rangle.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0530 (10)</td><td>241</td><td>■0.998 ● K|+0</td><td><code>U^{(2)}(g)=U^{(1)}(g) \otimes U^{(1)}(g),</code></td><td>$U^{(2)}(g)=U^{(1)}(g) \otimes U^{(1)}(g),$</td><td>$U^{(2)}(g) = U^{(1)}(g) \otimes U^{(1)}(g),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0580</td><td>250</td><td>■0.998 ● N|+0</td><td><code>\Lambda: x \rightarrow \Lambda(x) \in \mathbb{R}^{N} .</code></td><td>$\Lambda: x \rightarrow \Lambda(x) \in \mathbb{R}^{N} .$</td><td>$\Lambda : x \to \Lambda(x) \in \mathbb{R}^N.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0874</td><td>338</td><td>■0.998 ● N|+0</td><td><code>\begin{aligned} \mathcal{G} & =T^{*}(P M) . \\ L & = \left\{\left(X_{1}, \eta_{1}\right),\left(X_{2}, \eta_{2} \right),\left(X_{3}, \eta_{3}\right) \in C_{\Pi}^{3} \mid\left(X_{1} * X_{2}, \eta_{1} * \eta_{2}\right) \sim \left(X_{3} * \eta_{3}\right)\right\} . \\ I & \left.=(X, \eta) \mapsto\left(\phi^{*} X, \phi^{*} \eta\right)\right \} . \end{aligned}</code></td><td>$\begin{aligned} \mathcal{G} & =T^{*}(P M) . \\ L & = \left\{\left(X_{1}, \eta_{1}\right),\left(X_{2}, \eta_{2} \right),\left(X_{3}, \eta_{3}\right) \in C_{\Pi}^{3} \mid\left(X_{1} * X_{2}, \eta_{1} * \eta_{2}\right) \sim \left(X_{3} * \eta_{3}\right)\right\} . \\ I & \left.=(X, \eta) \mapsto\left(\phi^{*} X, \phi^{*} \eta\right)\right \} . \end{aligned}$</td><td>$\begin{aligned} & \mathcal{G} = T^*(PM). \\ & L = \{(X_1,\eta_1),(X_2,\eta_2),(X_3,\eta_3) \in C_\Pi^3 | (X_1 * X_2, \eta_1 * \eta_2) \sim (X_3 * \eta_3)\}. \\ & I = (X,\eta) \mapsto (\phi^* X, \phi^* \eta)\}. \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0965 (5)</td><td>362</td><td>■0.998 ● K|+0</td><td><code>X_{\Gamma} \backslash \Sigma_{n}=\emptyset=X_{\Gamma^{\vee}} \backslash \Sigma_{n} .</code></td><td>$X_{\Gamma} \backslash \Sigma_{n}=\emptyset=X_{\Gamma^{\vee}} \backslash \Sigma_{n} .$</td><td>$X_\Gamma \setminus \Sigma_n = \emptyset = X_{\Gamma^\vee} \setminus \Sigma_n\,.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0377 (18)</td><td>177</td><td>■0.998 ● N|+0</td><td><code>S O(D-1,1) \hookrightarrow\left\{\begin{array}{cc} S O(D, 1), & \Upsilon^{A B}=\operatorname{diag}\left(\eta^{\mathrm{ab}},+1\right) \\ S O(D-1,2), & \Upsilon^{A B}= \operatorname{diag}\left(\eta^{\mathrm{ab}},-1\right) \end{array}\right.</code></td><td>$S O(D-1,1) \hookrightarrow\left\{\begin{array}{cc} S O(D, 1), & \Upsilon^{A B}=\operatorname{diag}\left(\eta^{\mathrm{ab}},+1\right) \\ S O(D-1,2), & \Upsilon^{A B}= \operatorname{diag}\left(\eta^{\mathrm{ab}},-1\right) \end{array}\right.$</td><td>$SO(D-1,1) \hookrightarrow \begin{cases} SO(D,1), & \Upsilon^{AB} = \mathrm{diag}(\eta^{\mathrm{ab}},+1) \\ SO(D-1,2), & \Upsilon^{AB} = \mathrm{diag}(\eta^{\mathrm{ab}},-1) \end{cases}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0116</td><td>075</td><td>■0.999 ● N|+1</td><td><code>\operatorname{Ker}(A)=\bigoplus_{V \in \operatorname{Irr}(G)} m_{V}^{+} V ; \quad \operatorname{Coker}(A)=\bigoplus_{V \in \operatorname{Irr}(G)} m_{V}^{-} V .</code></td><td>$\operatorname{Ker}(A)=\bigoplus_{V \in \operatorname{Irr}(G)} m_{V}^{+} V ; \quad \operatorname{Coker}(A)=\bigoplus_{V \in \operatorname{Irr}(G)} m_{V}^{-} V .$</td><td>$\mathrm{Ker}(A) \;=\; \bigoplus_{V \in \mathrm{Irr}\,(G)} m_V^+\, V; \qquad \mathrm{Coker}(A) \;=\; \bigoplus_{V \in \mathrm{Irr}\,(G)} m_V^-\, V.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0411 (41)</td><td>188</td><td>■0.999 ● N|+0</td><td><code>d \mathcal{T}_{2 n-1}(A, \bar{A})=\mathcal{P}_{2 n}(A)-\mathcal{P}_{2 n}(\bar{A}),</code></td><td>$d \mathcal{T}_{2 n-1}(A, \bar{A})=\mathcal{P}_{2 n}(A)-\mathcal{P}_{2 n}(\bar{A}),$</td><td>$d\mathcal{T}_{2n-1}(A,\bar{A}) = \mathcal{P}_{2n}(A) - \mathcal{P}_{2n}(\bar{A}),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0415 (46)</td><td>189</td><td>■0.999 ● N|+0</td><td><code>I[A, \bar{A}]=\int_{M} \mathcal{C}(A)+\int_{\bar{M}} \mathcal{C}(\bar{A})+\int_{\partial M} \mathcal{B}(A, \bar{A}),</code></td><td>$I[A, \bar{A}]=\int_{M} \mathcal{C}(A)+\int_{\bar{M}} \mathcal{C}(\bar{A})+\int_{\partial M} \mathcal{B}(A, \bar{A}),$</td><td>$I[A,\bar{A}] = \int_M \mathcal{C}(A) + \int_{\bar{M}} \mathcal{C}(\bar{A}) + \int_{\partial M} \mathcal{B}(A,\bar{A}),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0448</td><td>211</td><td>■0.999 ● N|+0</td><td><code>\mathcal{C}_{3}=(K \times K) \backslash(G \times G) / K,</code></td><td>$\mathcal{C}_{3}=(K \times K) \backslash(G \times G) / K,$</td><td>$\mathcal{C}_3 = (K \times K)\backslash(G \times G)/K,$</td></tr>
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Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0612 (20)	254	■ 0.999 ● K|+0	`\left(\Lambda e_{p}\right)(x)=e_{p}\left(\Lambda^{-1} x \right)=e_{\Lambda p(x)},`	$\left(\Lambda e_{p}\right)(x)=e_{p}\left(\Lambda^{-1} x \right)=e_{\Lambda p(x)},$	$(\Lambda e_p)(x) = e_p(\Lambda^{-1}x) = e_{\Lambda p(x)},$
cardona-qft-methods_ EQ0624 (32)	257	■ 0.999 ● N|+1	`\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta} ; \quad F_{\theta}=\exp ^{\frac{i}{2} \hat{P}_{\mu} \otimes \theta_{\mu \nu} \hat{P}_{\nu}},`	$\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta} ; \quad F_{\theta}=\exp ^{\frac{i}{2} \hat{P}_{\mu} \otimes \theta_{\mu \nu} \hat{P}_{\nu}},$	$\Delta_\theta(g) = F_\theta^{-1}(g \otimes g)F_\theta; \quad F_\theta = \exp^{\frac{i}{2}\hat{P}_\mu\otimes\theta_{\mu\nu}\hat{P}_\nu},$
cardona-qft-methods_ EQ0744	283	■ 0.999 ● N|+0	`[K, \mathcal{O}]=0 \quad, \quad[S, \mathcal{O}]=0 \quad, \quad[\bar{S}, \mathcal{O}]=0 .`	$[K, \mathcal{O}]=0 \quad, \quad[S, \mathcal{O}]=0 \quad, \quad[\bar{S}, \mathcal{O}]=0 .$	$[K, \mathcal{O}] = 0 \qquad , \qquad [S, \mathcal{O}] = 0 \qquad , \qquad [\bar{S}, \mathcal{O}] = 0.$
cardona-qft-methods_ EQ0767 (8)	293	■ 0.999 ● K|+0	`\eta_{M N} \partial_{\sigma^{+}} X_{L}^{M} \partial_{\sigma^{+}} X_{L}^{N}=\eta_{M N} \partial_{\sigma^{-}} X_{R}^{M} \partial_{\sigma^{-}} X_{R}^{N}=0 .`	$\eta_{M N} \partial_{\sigma^{+}} X_{L}^{M} \partial_{\sigma^{+}} X_{L}^{N}=\eta_{M N} \partial_{\sigma^{-}} X_{R}^{M} \partial_{\sigma^{-}} X_{R}^{N}=0 .$	$\eta_{MN}\partial_{\sigma^+}X_L^M\partial_{\sigma^+}X_L^N = \eta_{MN}\partial_{\sigma^-}X_R^M\partial_{\sigma^-}X_R^N = 0 \; .$
cardona-qft-methods_ EQ0943 (3)	358	■ 0.999 ● K|+0	`[X] \cdot[Y]=[X \times Y] .`	$[X] \cdot[Y]=[X \times Y] .$	$[X] \cdot [Y] = [X \times Y].$
cardona-qft-methods_ EQ0989	371	■ 0.999 ● K|+0	`A B=\text { the closure of }\left.A B\right|_{\mathcal{D}}, \quad \mathcal{D}=B^{-1} \operatorname{dom} A .`	$A B=\text { the closure of }\left.A B\right\|_{\mathcal{D}}, \quad \mathcal{D}=B^{-1} \operatorname{dom} A .$	$AB = \text{the closure of } AB\|_{\mathcal{D}}, \quad \mathcal{D} = B^{-1}\,\mathrm{dom}\,A.$
cardona-qft-methods_ EQ0289	138	■ 0.999 ● N|+1	`\begin{aligned} C_{G_{(m+3) e}}(T)= & (3 T-1) C_{G_{(m+2) e}}(T)-\left(3 T^{2}-2 T\right) C_{G_{(m+1) e}}(T) \\ & +\left(T^{3}-T^{2}\right) C_{G_{m e}}(T) \end{aligned}`	$\begin{aligned} C_{G_{(m+3) e}}(T)= & (3 T-1) C_{G_{(m+2) e}}(T)-\left(3 T^{2}-2 T\right) C_{G_{(m+1) e}}(T) \\ & +\left(T^{3}-T^{2}\right) C_{G_{m e}}(T) \end{aligned}$	$\begin{aligned} C_{G_{(m+3)e}}(T) = {} & (3T-1)\,C_{G_{(m+2)e}}(T) - (3T^2-2T)\,C_{G_{(m+1)e}}(T) \\ & +(T^3-T^2)\,C_{G_{me}}(T) \end{aligned}$
cardona-qft-methods_ EQ0389 (27)	179	■ 0.999 ● K|+0	`\mathbf{E}_{2 n}=\epsilon_{A_{1} \cdots A_{2 n}} F^{A_{1} A_{2}} \cdots F^{A_{2 n-1} A_{2 n}} .`	$\mathbf{E}_{2 n}=\epsilon_{A_{1} \cdots A_{2 n}} F^{A_{1} A_{2}} \cdots F^{A_{2 n-1} A_{2 n}} .$	$\mathbf{E}_{2n} = \epsilon_{A_1\cdots A_{2n}}F^{A_1A_2}\ldots F^{A_{2n-1}A_{2n}}.$
cardona-qft-methods_ EQ0392 (30)	180	■ 0.999 ● N|+1	`I_{2 n-1}=\frac{\kappa}{l} \int_{M} \int_{0}^{1} d t \epsilon_{a_{1} \cdots a_{2 n-1}} R_{t}^{a_{1} a_{2}} \cdots R_{t}^{a_{2 n-3} a_{2 n-2}} e^{a_{2 n-1}},`	$I_{2 n-1}=\frac{\kappa}{l} \int_{M} \int_{0}^{1} d t \epsilon_{a_{1} \cdots a_{2 n-1}} R_{t}^{a_{1} a_{2}} \cdots R_{t}^{a_{2 n-3} a_{2 n-2}} e^{a_{2 n-1}},$	$I_{2n-1} = \frac{\kappa}{l}\int\limits_M\int\limits_0^1 dt\; \epsilon_{a_1\cdots a_{2n-1}}R_t^{a_1a_2}\cdots R_t^{a_{2n-3}a_{2n-2}}e^{a_{2n-1}} \; ,$
cardona-qft-methods_ EQ0405 (37)	184	■ 0.999 ● N|-1	`\begin{aligned} L_{2 n}^{B I} & =\epsilon_{a_{1} b_{1} a_{2} b_{2} \cdots a_{n} b_{n}}\left[R^{a_{1} b_{1}}+\frac{1}{l^{2}} e^{a_{1}} e^{b_{1}}\right] \cdots\left[R^{a_{n} b_{n}}+\frac{1}{l^{2}} e^{a_{n}} e^{b_{n}}\right] \\ & =\operatorname{Pfaff}\left[R^{a b}+\frac{1}{l^{2}} e^{a} e^{b}\right] . \end{aligned}`	$\begin{aligned} L_{2 n}^{B I} & =\epsilon_{a_{1} b_{1} a_{2} b_{2} \cdots a_{n} b_{n}}\left[R^{a_{1} b_{1}}+\frac{1}{l^{2}} e^{a_{1}} e^{b_{1}}\right] \cdots\left[R^{a_{n} b_{n}}+\frac{1}{l^{2}} e^{a_{n}} e^{b_{n}}\right] \\ & =\operatorname{Pfaff}\left[R^{a b}+\frac{1}{l^{2}} e^{a} e^{b}\right] . \end{aligned}$	$\begin{aligned} L_{2n}^{BI} & = \epsilon_{a_1b_1a_2b_2\cdots a_nb_n}\left[R^{a_1b_1}+\frac{1}{l^2}e^{a_1}e^{b_1}\right]\cdots\left[R^{a_nb_n}+\frac{1}{l^2}e^{a_n}e^{b_n}\right] \\ & = \mathrm{Pfaff}\left[R^{ab}+\frac{1}{l^2}e^ae^b\right]. \end{aligned}$
cardona-qft-methods_ EQ0436	209	■ 0.999 ● N|+0	`J_{F}^{2}=1, \quad \xi b=J_{F} b^{*} J_{F} \xi \quad \xi \in \mathcal{M}_{F}, b \in \mathcal{A}_{L R} .`	$J_{F}^{2}=1, \quad \xi b=J_{F} b^{*} J_{F} \xi \quad \xi \in \mathcal{M}_{F}, b \in \mathcal{A}_{L R} .$	$J_F^2 = 1\,, \quad \xi\, b = J_F b^* J_F\, \xi \qquad \xi \in \mathcal{M}_F\,,\; b \in \mathcal{A}_{LR}.$
cardona-qft-methods_ EQ0498	226	■ 0.999 ● W|+0	`\lambda\left(\Lambda_{\text {unif }}\right)=\frac{\pi^{2}}{2 f_{0}} \frac{\mathfrak{b}\left(\Lambda_{\text {unif }}\right)}{\mathfrak{a}\left(\Lambda_{\text {unif }}\right)^{2}}`	$\lambda\left(\Lambda_{\text {unif }}\right)=\frac{\pi^{2}}{2 f_{0}} \frac{\mathfrak{b}\left(\Lambda_{\text {unif }}\right)}{\mathfrak{a}\left(\Lambda_{\text {unif }}\right)^{2}}$	$\lambda(\Lambda_{unif}) = \frac{\pi^2}{2f_0}\frac{\mathfrak{b}(\Lambda_{unif})}{\mathfrak{a}(\Lambda_{unif})^2}$
cardona-qft-methods_ EQ0520 (6)	237	■ 0.999 ● N|+0	`\hat{x}_{\mu} \star \hat{x}_{\nu}-\hat{x}_{\nu} \star \hat{x}_{\mu}=\left[\hat{x}_{\mu}, \hat{x}_{\nu}\right]_{\star}=i \theta_{\mu \nu}, \quad \mu, \nu=0,1, \ldots, d .`	$\hat{x}_{\mu} \star \hat{x}_{\nu}-\hat{x}_{\nu} \star \hat{x}_{\mu}=\left[\hat{x}_{\mu}, \hat{x}_{\nu}\right]_{\star}=i \theta_{\mu \nu}, \quad \mu, \nu=0,1, \ldots, d .$	$\hat{x}_\mu \star \hat{x}_\nu - \hat{x}_\nu \star \hat{x}_\mu = [\hat{x}_\mu, \hat{x}_\nu]_\star = i\theta_{\mu\nu}, \quad \mu,\nu = 0,1,\ldots,d.$
cardona-qft-methods_ EQ0592 (1)	252	■ 0.999 ● N|+0	`(a, \Lambda) \in \mathcal{P}:(a, \Lambda) x=\Lambda x+a .`	$(a, \Lambda) \in \mathcal{P}:(a, \Lambda) x=\Lambda x+a .$	$(a,\Lambda) \in \mathcal{P} : (a,\Lambda)x = \Lambda x + a.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0598 (7)	252	0.999 K|+0	(\operatorname{Ad} U \otimes V) \Delta_{0}((a, \Lambda)) \varphi=\varphi	$(\operatorname{Ad} U \otimes V) \Delta_{0}((a, \Lambda)) \varphi=\varphi$	$(\mathrm{Ad} U \otimes V)\Delta_0((a,\Lambda))\varphi=\varphi$
cardona-qft-methods_EQ0623 (31)	257	0.999 N|+1	\mathcal{P} \ni(a, \Delta): \alpha \rightarrow(a, \Delta) \alpha, \quad((a, \Delta) \alpha)(x)=\alpha \left(\left((a, \Delta)^{-1} x\right) .\right.	$\mathcal{P} \ni(a, \Delta): \alpha \rightarrow(a, \Delta) \alpha, \quad((a, \Delta) \alpha)(x)=\alpha \left(\left((a, \Delta)^{-1} x\right) .\right.$	$\mathcal{P} \ni (a,\Delta):\alpha\to(a,\Delta)\alpha, \quad ((a,\Delta)\alpha)(x)=\alpha(((a,\Delta)^{-1}x).$
cardona-qft-methods_EQ0678 (3)	269	0.999 N|+0	\left[S^{i}, S^{j}\right]=i \epsilon^{i j k} S^{k} .	$\left[S^{i}, S^{j}\right]=i \epsilon^{i j k} S^{k} .$	$[S^i,S^j]=i\epsilon^{ijk}S^k\,.$
cardona-qft-methods_EQ0697	271	0.999 N|+0	\mathcal{H}_{l, l+1}=2\left(\mathbb{I}_{l, l+1}-\mathbb{P}_{l, l+1}\right)=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\\\ 0 & +2 & -2 & 0 \\\\ 0 & -2 & +2 & 0 \\\\ 0 & 0 & 0 & 0 \end{array}\right),	$\mathcal{H}_{l, l+1}=2\left(\mathbb{I}_{l, l+1}-\mathbb{P}_{l, l+1}\right)=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & +2 & -2 & 0 \\ 0 & -2 & +2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right),$	$\mathcal{H}_{l,l+1}=2\left(\mathbb{I}_{l,l+1}-\mathbb{P}_{l,l+1}\right)=\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & +2 & -2 & 0 \\ 0 & -2 & +2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\,,$
cardona-qft-methods_EQ0699	271	0.999 K|+0	\left[\vec{S}_{\mathrm{tot}}, \hat{H}\right]=0,	$\left[\vec{S}_{\mathrm{tot}}, \hat{H}\right]=0,$	$[\vec{S}_{\mathrm{tot}},\hat{H}]=0\,,$
cardona-qft-methods_EQ0823 (1)	312	0.999 N|+0	\mathrm{d} s^{2}=e^{2 A(y)} \eta_{\mu \nu} \mathrm{d} x^{\mu} \mathrm{d} x^{\nu}+g_{m n} \mathrm{~d} y^{m} \mathrm{~d} y^{n}, \quad \mu=0,1,2,3 \quad m=1, \ldots, 6	$\mathrm{d} s^{2}=e^{2 A(y)} \eta_{\mu \nu} \mathrm{d} x^{\mu} \mathrm{d} x^{\nu}+g_{m n} \mathrm{~d} y^{m} \mathrm{~d} y^{n}, \quad \mu=0,1,2,3 \quad m=1, \ldots, 6$	$\mathrm{d}s^2=e^{2A(y)}\eta_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu+g_{mn}\mathrm{d}y^m\mathrm{d}y^n\,, \quad \mu=0,1,2,3 \quad m=1,...,6$
cardona-qft-methods_EQ0842 (1)	329	0.999 N|+0	\Pi^{s r}(x)\left(\partial_{r}\right) \Pi^{l k}(x)+\Pi^{k r}(x)\left(\partial_{r}\right) \Pi^{s l}(x)+\Pi^{l r}(x)\left(\partial_{r}\right) \Pi^{k s}(x)=0,	$\Pi^{s r}(x)\left(\partial_{r}\right) \Pi^{l k}(x)+\Pi^{k r}(x)\left(\partial_{r}\right) \Pi^{s l}(x)+\Pi^{l r}(x)\left(\partial_{r}\right) \Pi^{k s}(x)=0,$	$\Pi^{sr}(x)(\partial_r)\Pi^{lk}(x)+\Pi^{kr}(x)(\partial_r)\Pi^{sl}(x)+\Pi^{lr}(x)(\partial_r)\Pi^{ks}(x)=0,$
cardona-qft-methods_EQ0887	344	0.999 N|+0	H_{p}(M) \otimes H_{q}(M) \longrightarrow H_{p+q}(M \times M) \xrightarrow{\tau_{\Delta *}} H_{p+q}\left(M^{T M}\right) \xrightarrow{T h o m} H_{p+q-n}(M)	$H_{p}(M) \otimes H_{q}(M) \longrightarrow H_{p+q}(M \times M) \xrightarrow{\tau_{\Delta *}} H_{p+q}\left(M^{T M}\right) \xrightarrow{T h o m} H_{p+q-n}(M)$	$H_p(M)\otimes H_q(M)\longrightarrow H_{p+q}(M\times M)\xrightarrow{\tau_{\Delta *}} H_{p+q}(M^{TM})\xrightarrow{Thom} H_{p+q-n}(M)$
cardona-qft-methods_EQ0988	371	0.999 K|+0	A+B=\text { the closure of }\left.A\right|_{\mathcal{D}}+\left.B\right|_{\mathcal{D}}, \quad \mathcal{D}=\operatorname{dom} A \cap \operatorname{dom} B .	$A+B=\text { the closure of }\left.A\right\vert_{\mathcal{D}}+\left.B\right\vert_{\mathcal{D}}, \quad \mathcal{D}=\operatorname{dom} A \cap \operatorname{dom} B .$	$A+B=\text{the closure of } A\vert_{\mathcal{D}}+B\vert_{\mathcal{D}}, \quad \mathcal{D}=\operatorname{dom} A\cap\operatorname{dom} B.$
cardona-qft-methods_EQ0759	287	0.999 N|+1	\begin{aligned} & \uparrow=Z=\text { nothing }, \\\\ & \downarrow=M_{A \dot{A}}=8+8 \text { magnons transforming under } \mathfrak{s u}(2 \mid 2) \otimes \overline{\mathfrak{s u}}(2 \mid 2), \end{aligned}	$\begin{aligned} & \uparrow=Z=\text { nothing }, \\ & \downarrow=M_{A \dot{A}}=8+8 \text { magnons transforming under } \mathfrak{s u}(2 \mid 2) \otimes \overline{\mathfrak{s u}}(2 \mid 2), \end{aligned}$	$\begin{aligned} \uparrow &= Z = \text{nothing}\,, \\ \downarrow &= M_{A\dot{A}} = 8+8 \text{ magnons transforming under } \mathfrak{su}(2\vert 2)\otimes\overline{\mathfrak{su}}(2\vert 2)\,, \end{aligned}$
cardona-qft-methods_EQ0039	031	1.000 W|-1	\lim _{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^{N} \mu_{n}\left((1+\Delta)^{-d / 2}\right)=\frac{\alpha}{\Gamma(d / 2+1)}=\frac{\pi^{d / 2}}{\Gamma(d / 2+1)} .	$\lim _{N \rightarrow \infty} \frac{1}{\log N} \sum_{n=0}^{N} \mu_{n}\left((1+\Delta)^{-d / 2}\right)=\frac{\alpha}{\Gamma(d / 2+1)}=\frac{\pi^{d / 2}}{\Gamma(d / 2+1)} .$	$\lim_{N\to\infty}\tfrac{1}{\log N}\sum_{n=0}^{N}\mu_n\big((1+\Delta)^{-d/2}\big)=\tfrac{\alpha}{\Gamma(d/2+1)}=\tfrac{\pi^{d/2}}{\Gamma(d/2+1)}.$
cardona-qft-methods_EQ0047 (13)	035	1.000 K|+0	\not D=-i c \circ \nabla^{S} .	$\not D=-i c \circ \nabla^{S} .$	$\not D=-i\,c\circ\nabla^S.$
cardona-qft-methods_EQ0087	050	1.000 N|+0	f A \mathcal{D}^{-1}=0, \forall A \in \Omega_{\mathcal{D}}^{1}(\mathcal{A}) \Longleftrightarrow f a b=f a \alpha(b), \forall a, b \in \mathcal{A},	$f A \mathcal{D}^{-1}=0, \forall A \in \Omega_{\mathcal{D}}^{1}(\mathcal{A}) \Longleftrightarrow f a b=f a \alpha(b), \forall a, b \in \mathcal{A},$	$\fint A\mathcal{D}^{-1}=0,\ \forall A\in\Omega^1_{\mathcal{D}}(\mathcal{A}) \Longleftrightarrow \fint ab=\fint a\alpha(b),\ \forall a,b\in\mathcal{A},$
cardona-qft-methods_EQ0125 (1)	077	1.000 N|+0	D^{\alpha}:=D_{1}^{\alpha_{1}} D_{2}^{\alpha_{1}} \cdots D_{n}^{\alpha_{n}}: C^{\infty}\left(\mathbb{R}^{n}\right) \longrightarrow C^{\infty}\left(\mathbb{R}^{n}\right) .	$D^{\alpha}:=D_{1}^{\alpha_{1}} D_{2}^{\alpha_{1}} \cdots D_{n}^{\alpha_{n}}: C^{\infty}\left(\mathbb{R}^{n}\right) \longrightarrow C^{\infty}\left(\mathbb{R}^{n}\right) .$	$D^\alpha \;:=\; D_1^{\alpha_1}D_2^{\alpha_1}\cdots D_n^{\alpha_n}\,:\,C^\infty(\mathbb{R}^n) \;\longrightarrow\; C^\infty(\mathbb{R}^n).$
cardona-qft-methods_EQ0135	079	1.000 N|+0	\mathcal{D}: H_{K}^{m}\left(\mathbb{R}^{n}, \mathbb{R}^{N}\right) \longrightarrow L^{2}\left(\mathbb{R}^{n}, \mathbb{R}^{N}\right), \quad \text { for } m>k .	$\mathcal{D}: H_{K}^{m}\left(\mathbb{R}^{n}, \mathbb{R}^{N}\right) \longrightarrow L^{2}\left(\mathbb{R}^{n}, \mathbb{R}^{N}\right), \quad \text { for } m>k .$	$\mathcal{D}:\ H_K^m(\mathbb{R}^n,\mathbb{R}^N) \;\longrightarrow\; L^2(\mathbb{R}^n,\mathbb{R}^N), \qquad \text{for } \; m>k.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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cardona-qft-methods_ EQ0162	087	1.000 N|+0	`\mathcal{D}: C^{\infty}(M, E) \longrightarrow C^{\infty}(M, E),`	$\mathcal{D}: C^{\infty}(M, E) \longrightarrow C^{\infty}(M, E),$	$\mathcal{D}: C^{\infty}(M, E) \longrightarrow C^{\infty}(M, E),$
cardona-qft-methods_ EQ0187 (2)	093	1.000 K|+0	`\sigma_{1}=\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right), \quad \sigma_{2}=\left( \begin{array}{cc} 0 & -i \\ i & 0 \end{array}\right), \quad \sigma_{3}=\left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right) .`	$\sigma_{1}=\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right), \quad \sigma_{2}=\left(\begin{array}{cc} 0 & -i \\ i & 0 \end{array}\right), \quad \sigma_{3}=\left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right) .$	$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$
cardona-qft-methods_ EQ0283	137	1.000 N|+1	`\mathbb{L} \cdot\left[\left(Y_{G \backslash e} \cup Y_{G / e}\right) \backslash Y_{G \backslash e}\right],`	$\mathbb{L} \cdot\left[\left(Y_{G \backslash e} \cup Y_{G / e}\right) \backslash Y_{G \backslash e}\right],$	$\mathbb{L} \cdot [(Y_{G\smallsetminus e} \cup Y_{G/e}) \smallsetminus Y_{G\smallsetminus e}] \quad ,$
cardona-qft-methods_ EQ0287	138	1.000 K|+0	`T(1+T)^{2} \quad, \quad(1+T)^{2} \quad, \quad T(1+T)`	$T(1+T)^{2} \quad, \quad(1+T)^{2} \quad, \quad T(1+T)$	$T(1+T)^{2} \quad , \quad (1+T)^{2} \quad , \quad T(1+T)$
cardona-qft-methods_ EQ0296	143	1.000 W|+0	`\mathbb{U}(G)=\left\{\begin{array}{lll} 0 & \bmod ( \mathbb{L}) & \text { if } G \text { has edges that are not looping edges } \\ (-1)^{n} & \bmod (\mathbb{L}) & \text { if } G \text { has } n \text { looping edges and no other edge. } \end{array}\right.`	$\mathbb{U}(G)=\left\{\begin{array}{lll} 0 & \bmod (\mathbb{L}) & \text { if } G \text { has edges that are not looping edges } \\ (-1)^{n} & \bmod (\mathbb{L}) & \text { if } G \text { has } n \text { looping edges and no other edge. } \end{array}\right.$	$\mathbb{U}(G) = \begin{cases} 0 & \bmod (\mathbb{L}) \quad \text{if } G \text{ has edges that are not looping edges} \\ (-1)^n & \bmod (\mathbb{L}) \quad \text{if } G \text{ has } n \text{ looping edges and no other edge.} \end{cases}$
cardona-qft-methods_ EQ0329 (16)	162	1.000 N|+0	`\begin{aligned} R_{b}^{a} & =d \omega_{b}^{a}+ \omega_{c}^{a} \wedge \omega_{b}^{c} \\ & =\frac{1}{2} R_{b \mu \nu}^{a} d x^{\mu} \wedge d x^{\nu} \end{aligned}`	$\begin{aligned} R_{b}^{a} & =d \omega_{b}^{a}+ \omega_{c}^{a} \wedge \omega_{b}^{c} \\ & =\frac{1}{2} R_{b \mu \nu}^{a} d x^{\mu} \wedge d x^{\nu} \end{aligned}$	$\begin{aligned} R^a_{\ b} &= d\omega^a_{\ b} + \omega^a_{\ c} \wedge \omega^c_{\ b} \\ &= \frac{1}{2} R^a_{\ b\mu\nu} dx^\mu \wedge dx^\nu. \end{aligned}$
cardona-qft-methods_ EQ0334 (20)	163	1.000 W|+0	`D R_{b}^{a}=d R_{b}^{a}+\omega_{c}^{a} \wedge R_{b}^{c}- \omega_{b}^{c} \wedge R_{c}^{a} \equiv 0 .`	$D R_{b}^{a}=d R_{b}^{a}+\omega_{c}^{a} \wedge R_{b}^{c}- \omega_{b}^{c} \wedge R_{c}^{a} \equiv 0 .$	$DR^a_{\ b} = dR^a_{\ b} + \omega^a_{\ c} \wedge R^c_{\ b} - \omega^c_{\ b} \wedge R^a_{\ c} \equiv 0 \,.$
cardona-qft-methods_ EQ0343 (11)	167	1.000 S|+2	`\begin{aligned} I_{2} & =\kappa \int_{M} \epsilon_{a b} \left[\alpha_{1} R^{a b}+\alpha_{0} e^{a} e^{b}\right] \\ & =\kappa \int_{M} \sqrt{|g|}\left(\alpha_{1} R+2 \alpha_{0}\right) d^{2} x \\ & =\kappa \alpha_{1} \cdot \mathbf{E}_{2}+2 \kappa \alpha_{0} \cdot V_{2} \end{aligned}`	$\begin{aligned} I_{2} & =\kappa \int_{M} \epsilon_{a b} \left[\alpha_{1} R^{a b}+\alpha_{0} e^{a} e^{b}\right] \\ & =\kappa \int_{M} \sqrt{\|g\|}\left(\alpha_{1} R+2 \alpha_{0}\right) d^{2} x \\ & =\kappa \alpha_{1} \cdot \mathbf{E}_{2}+2 \kappa \alpha_{0} \cdot V_{2} \end{aligned}$	$\begin{aligned} I_2 &= \kappa \int_M \epsilon_{ab}[\alpha_1 R^{ab} + \alpha_0 e^a e^b] \\ &= \kappa \int_M \sqrt{\|g\|}\,(\alpha_1 R + 2\alpha_0)\, d^2x \\ &= \kappa\alpha_1 \cdot \mathbf{E}_2 + 2\kappa\alpha_0 \cdot V_2. \end{aligned}$
cardona-qft-methods_ EQ0351 (20)	168	1.000 N|+0	`\omega^{a}{ }_{b \mu}=-E_{b}^{\nu}\left(\partial_{\mu} e_{\nu}^{a}-\Gamma_{\mu \nu}^{\lambda} e_{\lambda}^{a} \right),`	$\omega^{a}{ }_{b \mu}=-E_{b}^{\nu}\left(\partial_{\mu} e_{\nu}^{a}-\Gamma_{\mu \nu}^{\lambda} e_{\lambda}^{a} \right),$	$\omega^a{}_{b\mu} = -E^\nu_b(\partial_\mu e^a_\nu - \Gamma^\lambda_{\mu\nu} e^a_\lambda),$
cardona-qft-methods_ EQ0354 (22)	170	1.000 K|+0	`v_{1}=e^{a} e^{b} R_{a b}, \quad \tau_{0}=T^{a} T_{a}, \quad \mathbf{P}_{4}=R^{a b} R_{a b} .`	$v_{1}=e^{a} e^{b} R_{a b}, \quad \tau_{0}=T^{a} T_{a}, \quad \mathbf{P}_{4}=R^{a b} R_{a b} .$	$v_1 = e^a e^b R_{ab}, \quad \tau_0 = T^a T_a, \quad \mathbf{P}_4 = R^{ab} R_{ab}.$
cardona-qft-methods_ EQ0355 (23)	170	1.000 K|+0	`\mathbf{N}_{4}=T^{a} T_{a}-e^{a} e^{b} R_{a b},`	$\mathbf{N}_{4}=T^{a} T_{a}-e^{a} e^{b} R_{a b},$	$\mathbf{N}_4 = T^a T_a - e^a e^b R_{ab},$
cardona-qft-methods_ EQ0378 (19)	177	1.000 K|+0	`S O(D-1,1) \hookrightarrow I S O(D-1,1) .`	$S O(D-1,1) \hookrightarrow I S O(D-1,1) .$	$SO(D-1,1) \hookrightarrow ISO(D-1,1).$
cardona-qft-methods_ EQ0380 (21)	178	1.000 N|+1	`\begin{aligned} \delta L_{4} & =2 \epsilon_{a b c d} R^{a b} e^{c} \delta e^{d} \\ & =d\left(2 \epsilon_{a b c d} R^{a b} e^{c} \lambda^{d}\right)-2 \epsilon_{a b c d} R^{a b} T^{c} \lambda^{d} \end{aligned}`	$\begin{aligned} \delta L_{4} & =2 \epsilon_{a b c d} R^{a b} e^{c} \delta e^{d} \\ & =d\left(2 \epsilon_{a b c d} R^{a b} e^{c} \lambda^{d}\right)-2 \epsilon_{a b c d} R^{a b} T^{c} \lambda^{d} \end{aligned}$	$\begin{aligned} \delta L_4 &= 2\epsilon_{abcd} R^{ab} e^c \delta e^d \\ &= d(2\epsilon_{abcd} R^{ab} e^c \lambda^d) - 2\epsilon_{abcd} R^{ab} T^c \lambda^d. \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0388	179	1.000 K|+0	`d\left(\delta L_{3}^{A d S}\right)=0,`	$d\left(\delta L_{3}^{A d S}\right)=0,$	$d\left(\delta L_{3}^{AdS}\right)=0,$
cardona-qft-methods_ EQ0414 (45)	188	1.000 N|+0	`\bar{\Lambda}(x)-\Lambda(x)=0, \quad \text { for } \quad x \in \partial M .`	$\bar{\Lambda}(x)-\Lambda(x)=0, \quad \text { for } \quad x \in \partial M .$	$\bar{\Lambda}(x)-\Lambda(x)=0, \quad \text{for} \quad x \in \partial M.$
cardona-qft-methods_ EQ0443	211	1.000 N|+1	`D(Y)=\left(\begin{array}{cc} S & T^{*} \\ T & \bar{S} \end{array}\right) \quad \text { with } \quad S=S_{1} \oplus\left(S_{3} \otimes 1_{3}\right)`	$D(Y)=\left(\begin{array}{cc} S & T^{*} \\ T & \bar{S} \end{array}\right) \quad \text { with } \quad S=S_{1} \oplus\left(S_{3} \otimes 1_{3}\right)$	$D(Y)=\begin{pmatrix} S & T^{*} \\ T & \bar{S} \end{pmatrix} \quad \text{with} \quad S=S_{1} \oplus\left(S_{3} \otimes 1_{3}\right),$
cardona-qft-methods_ EQ0444	211	1.000 N|+0	`S_{1}=\left(\begin{array}{cccc} 0 & 0 & Y_{(\uparrow 1)}^{*} & 0 \\ 0 & 0 & 0 & Y_{(\downarrow 1)}^{*} \\ Y_{(\uparrow 1)} & 0 & 0 & 0 \\ 0 & Y_{(\downarrow 1)} & 0 & 0 \end{array}\right)`	$S_{1}=\left(\begin{array}{cccc} 0 & 0 & Y_{(\uparrow 1)}^{*} & 0 \\ 0 & 0 & 0 & Y_{(\downarrow 1)}^{*} \\ Y_{(\uparrow 1)} & 0 & 0 & 0 \\ 0 & Y_{(\downarrow 1)} & 0 & 0 \end{array}\right)$	$S_{1}=\begin{pmatrix} 0 & 0 & Y_{(\uparrow 1)}^{*} & 0 \\ 0 & 0 & 0 & Y_{(\downarrow 1)}^{*} \\ Y_{(\uparrow 1)} & 0 & 0 & 0 \\ 0 & Y_{(\downarrow 1)} & 0 & 0 \end{pmatrix}$
cardona-qft-methods_ EQ0514 (7)	230	1.000 N|+0	`M_{P}^{2}(\Lambda)=M_{P}^{2}+2 \rho_{0} \Lambda^{2},`	$M_{P}^{2}(\Lambda)=M_{P}^{2}+2 \rho_{0} \Lambda^{2},$	$M_{P}^{2}(\Lambda)=M_{P}^{2}+2 \rho_{0} \Lambda^{2},$
cardona-qft-methods_ EQ0547 (17)	244	1.000 K|+0	`(\mathrm{Id} \otimes \epsilon) \Delta=\mathrm{Id}=( \epsilon \otimes \mathrm{Id}) \Delta .`	$(\mathrm{Id} \otimes \epsilon) \Delta=\mathrm{Id}=(\epsilon \otimes \mathrm{Id}) \Delta .$	$(\mathrm{Id} \otimes \epsilon) \Delta=\mathrm{Id}=(\epsilon \otimes \mathrm{Id}) \Delta .$
cardona-qft-methods_ EQ0568 (2)	248	1.000 N|+1	`\widetilde{R}(\omega)=\int_{-\infty}^{\infty} d t e^{i \omega t} R(t)=\int_{0}^{\infty} e^{i \omega t} R(t)`	$\widetilde{R}(\omega)=\int_{-\infty}^{\infty} d t e^{i \omega t} R(t)=\int_{0}^{\infty} e^{i \omega t} R(t)$	$\widetilde{R}(\omega)=\int_{-\infty}^{\infty} d t e^{i \omega t} R(t)=\int_{0}^{\infty} e^{i \omega t} R(t)$
cardona-qft-methods_ EQ0573 (8)	249	1.000 W|-2	`f * g(x)=f e^{\frac{i}{2} \vec{\partial}_{\mu} \theta^{ \mu N} \vec{\partial}_{\nu}} g`	$f * g(x)=f e^{\frac{i}{2} \vec{\partial}_{\mu} \theta^{ \mu N} \vec{\partial}_{\nu}} g$	$f * g(x)=f e^{\frac{i}{2} \overleftarrow{\partial}_{\mu} \theta^{\mu N} \overrightarrow{\partial}_{\nu}} g$
cardona-qft-methods_ EQ0621 (29)	256	1.000 N|+0	`m_{\theta}(\alpha \otimes \beta)=m_{0} \mathcal{F}_{ \theta}(\alpha \otimes \beta) \quad \alpha, \beta \in \mathcal{A}_{\theta}\left(M^{d+1}\right), \quad \mathcal{F}_{\theta}=e^{\frac{i}{2} \theta_{\mu \nu} \partial_{\mu} \otimes \partial \nu},`	$m_{\theta}(\alpha \otimes \beta)=m_{0} \mathcal{F}_{ \theta}(\alpha \otimes \beta) \quad \alpha, \beta \in \mathcal{A}_{\theta}\left(M^{d+1}\right), \quad \mathcal{F}_{\theta}=e^{\frac{i}{2} \theta_{\mu \nu} \partial_{\mu} \otimes \partial \nu},$	$m_{\theta}(\alpha \otimes \beta)=m_{0} \mathcal{F}_{\theta}(\alpha \otimes \beta) \quad \alpha, \beta \in \mathcal{A}_{\theta}(M^{d+1}), \quad \mathcal{F}_{\theta}=e^{\frac{i}{2} \theta_{\mu \nu} \partial_{\mu} \otimes \partial \nu},$
cardona-qft-methods_ EQ0634	258	1.000 S|+0	`\begin{array}{r} \int \prod_{i} d \mu\left(p_{i}\right) U_{\theta}(\Lambda) a_{p_{1}}^{\dagger} a_{p_{2}}^{ \dagger}|0\rangle \exp ^{\frac{i}{2}\left(\Lambda p_{1}\right)} \wedge\left(\Lambda p_{2}\right) \exp ^{\frac{i}{2}\left(p_{1}\right) \wedge\left(p_{2}\right)} e_{\Lambda p_{1}} \otimes e_{\Lambda p_{2}}= \\ \int \prod_{i} d \mu\left(p_{i}\right) U_{\theta}(\Lambda) a_{p_{1}}^{\dagger} a_{p_{2}}^{\dagger}|0\rangle e_{ \Lambda p_{1}} \otimes e_{\Lambda p_{2}} \end{array}`	$\begin{array}{r} \int \prod_{i} d \mu\left(p_{i}\right) U_{\theta}(\Lambda) a_{p_{1}}^{\dagger} a_{p_{2}}^{ \dagger}\|0\rangle \exp ^{\frac{i}{2}\left(\Lambda p_{1}\right)} \wedge\left(\Lambda p_{2}\right) \exp ^{\frac{i}{2}\left(p_{1}\right) \wedge\left(p_{2}\right)} e_{\Lambda p_{1}} \otimes e_{\Lambda p_{2}}= \\ \int \prod_{i} d \mu\left(p_{i}\right) U_{\theta}(\Lambda) a_{p_{1}}^{\dagger} a_{p_{2}}^{\dagger}\|0\rangle e_{ \Lambda p_{1}} \otimes e_{\Lambda p_{2}} \end{array}$	$\begin{array}{r} \int \prod_{i} d \mu(p_{i}) U_{\theta}(\Lambda) a_{p_{1}}^{\dagger} a_{p_{2}}^{\dagger}\|0\rangle \exp^{\frac{i}{2}(\Lambda p_{1})} \wedge(\Lambda p_{2}) \exp^{\frac{i}{2}(p_{1}) \wedge(p_{2})} e_{\Lambda p_{1}} \otimes e_{\Lambda p_{2}}= \\ \int \prod_{i} d \mu(p_{i}) U_{\theta}(\Lambda) a_{p_{1}}^{\dagger} a_{p_{2}}^{\dagger}\|0\rangle e_{\Lambda p_{1}} \otimes e_{\Lambda p_{2}} \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0643 (51)	259	■1.000 • N|+0	`\begin{aligned} e_{p_{1}} \otimes_{S_{\theta}} e_{p_{2}} & =\frac{1}{2}\left[e_{p_{1}} \otimes e_{p_{2}}+ \mathcal{F}_{\theta}^{-2} e_{p_{2}} \otimes e_{p_{1}} \right] \\ & =\frac{1}{2}\left[e_{p_{1}} \otimes e_{p_{2}}+e^{i p_{2} \wedge p_{1}} e_{p_{2}} \otimes e_{p_{1}}\right] \\ & =e^{i p_{2} \wedge p_{1}} e_{p_{2}} \otimes e_{p_{1}} . \end{aligned}`	$\begin{aligned} e_{p_{1}} \otimes_{S_{\theta}} e_{p_{2}} & =\frac{1}{2}\left[e_{p_{1}} \otimes e_{p_{2}}+ \mathcal{F}_{\theta}^{-2} e_{p_{2}} \otimes e_{p_{1}} \right] \\ & =\frac{1}{2}\left[e_{p_{1}} \otimes e_{p_{2}}+e^{i p_{2} \wedge p_{1}} e_{p_{2}} \otimes e_{p_{1}}\right] \\ & =e^{i p_{2} \wedge p_{1}} e_{p_{2}} \otimes e_{p_{1}} . \end{aligned}$	$\begin{aligned} e_{p_1} \otimes_{S_\theta} e_{p_2} & =\frac{1}{2}[e_{p_1} \otimes e_{p_2}+\mathcal{F}_\theta^{-2} e_{p_2} \otimes e_{p_1}] \\ & =\frac{1}{2}[e_{p_1} \otimes e_{p_2}+e^{ip_2 \wedge p_1} e_{p_2} \otimes e_{p_1}] \\ & =e^{ip_2 \wedge p_1} e_{p_2} \otimes e_{p_1} . \end{aligned}$
cardona-qft-methods_ EQ0646 (53)	260	■1.000 • K|+0	`a_{p_{1}}^{\dagger} a_{p_{2}}^{\dagger}=e^{i p_{1} \wedge p_{2}} a_{p_{2}}^{\dagger} a_{p_{1}}^{\dagger} .`	$a_{p_{1}}^{\dagger} a_{p_{2}}^{\dagger}=e^{i p_{1} \wedge p_{2}} a_{p_{2}}^{\dagger} a_{p_{1}}^{\dagger} .$	$a_{p_1}^{\dagger} a_{p_2}^{\dagger}=e^{ip_1 \wedge p_2} a_{p_2}^{\dagger} a_{p_1}^{\dagger} .$
cardona-qft-methods_ EQ0660 (10)	262	■1.000 • N|+0	`\left[H_{I}(x), H_{I}(y)\right]=0,`	$\left[H_{I}(x), H_{I}(y)\right]=0,$	$[H_I(x), H_I(y)]=0,$
cardona-qft-methods_ EQ0684	269	■1.000 • N|+0	`S^{+}=\frac{1}{2} \sigma^{+}=\left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right), \quad S^{-}=\frac{1}{2} \sigma^{-}=\left(\begin{array}{ll} 0 & 0 \\ 1 & 0 \end{array}\right),`	$S^{+}=\frac{1}{2} \sigma^{+}=\left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right), \quad S^{-}=\frac{1}{2} \sigma^{-}=\left(\begin{array}{ll} 0 & 0 \\ 1 & 0 \end{array}\right),$	$S^{+}=\frac{1}{2} \sigma^{+}=\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad S^{-}=\frac{1}{2} \sigma^{-}=\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$
cardona-qft-methods_ EQ0695 (10)	271	■1.000 • N|+0	`\begin{aligned} \hat{H} & =\sum_{l=1}^{L}\left(1-\vec{\sigma}_{l} \cdot \vec{\sigma}_{l+1}\right) \\ & = \sum_{l=1}^{L}\left(1-\sigma_{l}^{3} \sigma_{l+1}^{3}-\frac{1}{2} \sigma_{l}^{+} \sigma_{l+1}^{-}-\frac{1}{2} \sigma_{l}^{-} \sigma_{l+1}^{+}\right) \\ & =2 \sum_{l=1}^{L}\left(\mathbb{I}_{l, l+1}-\mathbb{P}_{l, l+1}\right) . \end{aligned}`	$\begin{aligned} \hat{H} & =\sum_{l=1}^{L}\left(1-\vec{\sigma}_{l} \cdot \vec{\sigma}_{l+1}\right) \\ & = \sum_{l=1}^{L}\left(1-\sigma_{l}^{3} \sigma_{l+1}^{3}-\frac{1}{2} \sigma_{l}^{+} \sigma_{l+1}^{-}-\frac{1}{2} \sigma_{l}^{-} \sigma_{l+1}^{+}\right) \\ & =2 \sum_{l=1}^{L}\left(\mathbb{I}_{l, l+1}-\mathbb{P}_{l, l+1}\right) . \end{aligned}$	$\begin{aligned} \hat{H} & =\sum_{l=1}^{L}(1-\vec{\sigma}_{l} \cdot \vec{\sigma}_{l+1}) \\ & = \sum_{l=1}^{L}\left(1-\sigma_{l}^{3} \sigma_{l+1}^{3}-\frac{1}{2} \sigma_{l}^{+} \sigma_{l+1}^{-}-\frac{1}{2} \sigma_{l}^{-} \sigma_{l+1}^{+}\right) \\ & =2 \sum_{l=1}^{L}(\mathbb{I}_{l, l+1}-\mathbb{P}_{l, l+1}) . \end{aligned}$
cardona-qft-methods_ EQ0718	275	■1.000 • N|+0	`R_{a, b}\left(u-u^{\prime}\right) M_{a}(u) M_{b}\left(u^{\prime}\right)=M_{b}\left(u^{\prime}\right) M_{a}(u) R_{a, b}\left(u-u^{\prime}\right) .`	$R_{a, b}\left(u-u^{\prime}\right) M_{a}(u) M_{b}\left(u^{\prime}\right)=M_{b}\left(u^{\prime}\right) M_{a}(u) R_{a, b}\left(u-u^{\prime}\right) .$	$R_{a,b}(u-u') M_a(u) M_b(u')=M_b(u') M_a(u) R_{a,b}(u-u').$
cardona-qft-methods_ EQ0766 (7)	293	■1.000 • W|+1	`T^{\alpha \beta} \equiv-\frac{2 \pi}{\sqrt{\gamma}} \frac{\delta S}{\delta \gamma_{\alpha \beta}}=-\frac{1}{ \alpha^{\prime}}\left(\partial^{\alpha} X^{M} \partial^{ \beta} X_{M}-\frac{1}{2} \gamma^{\alpha \beta} \gamma_{ \lambda \rho} \partial^{\lambda} X^{M} \partial^{\rho} X_{M}\right)=0 .`	$T^{\alpha \beta} \equiv-\frac{2 \pi}{\sqrt{\gamma}} \frac{\delta S}{\delta \gamma_{\alpha \beta}}=-\frac{1}{ \alpha^{\prime}}\left(\partial^{\alpha} X^{M} \partial^{ \beta} X_{M}-\frac{1}{2} \gamma^{\alpha \beta} \gamma_{ \lambda \rho} \partial^{\lambda} X^{M} \partial^{\rho} X_{M}\right)=0 .$	$T^{\alpha\beta} \equiv -\frac{2\pi}{\sqrt{\gamma}} \frac{\delta S}{\delta \gamma_{\alpha\beta}} = -\frac{1}{\alpha'}\left(\partial^\alpha X^M \partial^\beta X_M - \frac{1}{2}\gamma^{\alpha\beta}\gamma_{\lambda\rho}\partial^\lambda X^M \partial^\rho X_M\right) = 0\ .$
cardona-qft-methods_ EQ0775 (16)	295	■1.000 • C|+6	`\begin{aligned} Z & =\int \mathcal{D} X \mathcal{D} \gamma e^{-S} \\ & =\int \mathcal{D} X \mathcal{D} \gamma e^{-S_{0}}\left(1+\frac{1}{4 \pi \alpha^{\prime}} \int \mathrm{d} \sigma \mathrm{~d} \tau \sqrt{\gamma} \gamma^{ \alpha \beta} h_{M N} \partial_{\alpha} X^{M} \partial_{ \beta} X^{N}+\cdots\right) \end{aligned}`	$\begin{aligned} Z & =\int \mathcal{D} X \mathcal{D} \gamma e^{-S} \\ & =\int \mathcal{D} X \mathcal{D} \gamma e^{-S_{0}}\left(1+\frac{1}{4 \pi \alpha^{\prime}} \int \mathrm{d} \sigma \mathrm{~d} \tau \sqrt{\gamma} \gamma^{ \alpha \beta} h_{M N} \partial_{\alpha} X^{M} \partial_{ \beta} X^{N}+\cdots\right) \end{aligned}$	$\begin{aligned} Z & =\int \mathcal{D}X\mathcal{D}\gamma e^{-S} \qquad (16) \\ & =\int \mathcal{D}X\mathcal{D}\gamma e^{-S_0}\left(1+\frac{1}{4\pi\alpha'}\int \mathrm{d}\sigma\mathrm{d}\tau\sqrt{\gamma}\gamma^{\alpha\beta}h_{MN}\partial_\alpha X^M \partial_\beta X^N+\cdots\right), \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_ EQ0780 (21)</td><td>297</td><td>■ 1.000 ● C|+3</td><td>\begin{aligned} \mathrm{S}= & \frac{1}{2 \kappa_{0}^{2}} \int \mathrm{~d}^{D} X(-G)^{1 / 2} e^{-2 \Phi}\left[R+4 \nabla_{M} \Phi \nabla^{M} \Phi-\frac{1}{12} H_{M N P} H^{M N P}\right. \\ & \left.-\frac{2(D-26)}{3 \alpha^{\prime}}+O\left(\alpha^{\prime}\right)\right] \end{aligned}</td><td>$$\begin{aligned} \mathrm{S}= & \frac{1}{2 \kappa_{0}^{2}} \int \mathrm{~d}^{D} X(-G)^{1 / 2} e^{-2 \Phi}\left[R+4 \nabla_{M} \Phi \nabla^{M} \Phi-\frac{1}{12} H_{M N P} H^{M N P}\right. \\ & \left.-\frac{2(D-26)}{3 \alpha^{\prime}}+O\left(\alpha^{\prime}\right)\right] \end{aligned}$$</td><td>$$\mathrm{S}=\frac{1}{2\kappa_0^2}\int \mathrm{d}^D X(-G)^{1/2}e^{-2\Phi}\Big[R+4\nabla_M\Phi\nabla^M\Phi-\frac{1}{12}H_{MNP}H^{MNP} -\frac{2(D-26)}{3\alpha'}+O(\alpha')\Big]\,. \qquad ($$</td></tr>
<tr><td>cardona-qft-methods_ EQ0819 (19)</td><td>310</td><td>■ 1.000 ● N|+0</td><td>\begin{aligned} & \Phi_{-}=-\frac{\mathrm{i}}{8} \Omega \leftrightarrow \mathcal{J}_{1} \\ & \Phi_{+}=\frac{1}{8} e^{-\mathrm{i} J} \leftrightarrow \mathcal{J}_{2} \end{aligned}</td><td>$$\begin{aligned} & \Phi_{-}=-\frac{\mathrm{i}}{8} \Omega \leftrightarrow \mathcal{J}_{1} \\ & \Phi_{+}=\frac{1}{8} e^{-\mathrm{i} J} \leftrightarrow \mathcal{J}_{2} \end{aligned}$$</td><td>$$\Phi_- = -\frac{\mathrm{i}}{8}\,\Omega \ \leftrightarrow \mathcal{J}_1$$ $$\Phi_+ = \frac{1}{8}\,e^{-\mathrm{i}J} \ \leftrightarrow \mathcal{J}_2$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0851</td><td>330</td><td>■ 1.000 ● K|+0</td><td>[d f, d g]:=d\{f, g\}, \forall f, g \in \mathcal{C}^{\infty}(M),</td><td>$[d f, d g]:=d\{f, g\}, \forall f, g \in \mathcal{C}^{\infty}(M),$</td><td>$[df,dg]:=d\{f,g\},\forall f,g\in\mathcal{C}^{\infty}(M),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0863</td><td>334</td><td>■ 1.000 ● K|+0</td><td>L_{3} \circ\left(L_{1} \times L_{1}\right)=L_{1} .</td><td>$L_{3} \circ\left(L_{1} \times L_{1}\right)=L_{1} .$</td><td>$L_3\circ(L_1\times L_1)=L_1.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0947 (7)</td><td>359</td><td>■ 1.000 ● K|+0</td><td>[X]-[X \cap Y]=[X \backslash(X \cap Y)]=[(X \cup Y) \backslash Y]=[X \cup Y]-[Y],</td><td>$[X]-[X \cap Y]=[X \backslash(X \cap Y)]=[(X \cup Y) \backslash Y]=[X \cup Y]-[Y],$</td><td>$[X]-[X\cap Y]=[X\setminus(X\cap Y)]=[(X\cup Y)\setminus Y]=[X\cup Y]-[Y]\,,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0992</td><td>372</td><td>■ 1.000 ● N|+0</td><td>\mathcal{S}=\text { the multiplicative set generated by } \left\{: 1+E^{\dagger} E \mid E \in \mathcal{E}:\right\}</td><td>$\mathcal{S}=\text { the multiplicative set generated by } \left\{: 1+E^{\dagger} E \mid E \in \mathcal{E}:\right\}$</td><td>$\mathcal{S}=\text{the multiplicative set generated by }\left\{:1+E^{\dagger}E \,\middle\vert\, E\in\mathcal{E}:\right\}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0001</td><td>018</td><td>■ 1.000 ● N|+1</td><td>\left|\partial_{x}^{\alpha} \partial_{\xi}^{\beta}\left(\sigma-\sum_{j<N} \sigma_{m-j}\right)(x, \xi)\right| \leq C_{N \alpha \beta}(x)|\xi|^{\Re(m)-N-|\beta|}</td><td>$$\left\vert\partial_{x}^{\alpha} \partial_{\xi}^{\beta}\left(\sigma-\sum_{j < N} \sigma_{m-j}\right)(x, \xi)\right\vert \leq C_{N \alpha \beta}(x)\vert\xi\vert^{\Re(m)-N-\vert\beta\vert}$$</td><td>$$\vert\partial_x^\alpha\partial_\xi^\beta(\sigma-\sum_{j < N}\sigma_{m-j})(x,\xi)\vert\le C_{N\alpha\beta}(x)\,\vert\xi\vert^{\Re(m)-N-\vert\beta\vert}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0003 (1)</td><td>018</td><td>■ 1.000 ● N|+0</td><td>\sigma(\cdot, D)(u)(x)=\sigma(x, D)(u)=\frac{1}{(2 \pi)^{d}} \int_{\mathbb{R}^{d}} \sigma(x, \xi) \widehat{u}(\xi) e^{i x \cdot \xi} d \xi</td><td>$$\sigma(\cdot, D)(u)(x)=\sigma(x, D)(u)=\frac{1}{(2 \pi)^{d}} \int_{\mathbb{R}^{d}} \sigma(x, \xi) \widehat{u}(\xi) e^{i x \cdot \xi} d \xi$$</td><td>$$\sigma(\cdot,D)(u)(x)=\sigma(x,D)(u)=\tfrac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\sigma(x,\xi)\,\widehat{u}(\xi)\,e^{ix\cdot\xi}\,d\xi$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0004</td><td>018</td><td>■ 1.000 ● N|+0</td><td>k^{\sigma(x, D)}(x, y)=\frac{1}{(2 \pi)^{d}} \int_{ \mathbb{R}^{d}} \sigma(x, \xi) e^{i(x-y) \cdot \xi} d \xi=\check{\sigma}_{\xi \rightarrow y}(x, x-y)</td><td>$$k^{\sigma(x, D)}(x, y)=\frac{1}{(2 \pi)^{d}} \int_{ \mathbb{R}^{d}} \sigma(x, \xi) e^{i(x-y) \cdot \xi} d \xi=\check{\sigma}_{\xi \rightarrow y}(x, x-y)$$</td><td>$$k^{\sigma(x,D)}(x,y)=\tfrac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\sigma(x,\xi)\,e^{i(x-y)\cdot\xi}\,d\xi=\check{\sigma}_{\xi\to y}(x,x-y)$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0005 (2)</td><td>018</td><td>■ 1.000 ● N|+2</td><td>k^{a}(x, y)=\frac{1}{(2 \pi)^{d}} \int_{\mathbb{R}^{d}} a(x, y, \xi) e^{i(x-y) \cdot \xi} d \xi</td><td>$$k^{a}(x, y)=\frac{1}{(2 \pi)^{d}} \int_{\mathbb{R}^{d}} a(x, y, \xi) e^{i(x-y) \cdot \xi} d \xi$$</td><td>$$k^a(x,y)=\tfrac{1}{(2\pi)^d}\int_{\mathbb{R}^d}a(x,y,\xi)\,e^{i(x-y)\cdot\xi}\,d\xi.$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0007</td><td>019</td><td>■ 1.000 ● N|+0</td><td>\sigma^{A} \simeq \sum_{\alpha} \frac{(-i)^{\alpha}}{ \alpha!} \partial_{\xi}^{\alpha} \partial_{y}^{\alpha} k_{\sigma}^{A}(x, y, \xi)_{\mid y=x}</td><td>$$\sigma^{A} \simeq \sum_{\alpha} \frac{(-i)^{\alpha}}{ \alpha!} \partial_{\xi}^{\alpha} \partial_{y}^{\alpha} k_{\sigma}^{A}(x, y, \xi)_{\mid y=x}$$</td><td>$$\sigma^A\simeq\sum_\alpha\tfrac{(-i)^\alpha}{\alpha!}\,\partial_\xi^\alpha\partial_y^\alpha\,k_\sigma^A(x,y,\xi)_{\vert y=x}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0008</td><td>019</td><td>■ 1.000 ● W|+1</td><td>k_{\sigma}^{A}(x, y)=\frac{1}{(2 \pi)^{d}} \int_{ \mathbb{R}^{d}} e^{i(x-y) \cdot \xi} k^{A}(x, y, \xi) d \xi</td><td>$$k_{\sigma}^{A}(x, y)=\frac{1}{(2 \pi)^{d}} \int_{ \mathbb{R}^{d}} e^{i(x-y) \cdot \xi} k^{A}(x, y, \xi) d \xi$$</td><td>$$k_\sigma^A(x,y)=\tfrac{1}{(2\pi)^d}\int_{\mathbb{R}^d}e^{i(x-y)\cdot\xi}\,k^A(x,y,\xi)\,d\xi\,,$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0009	019	1.000 N|+0	`\sigma^{P_{1} P_{2}}(x, \xi) \simeq \sum_{\alpha \in \mathbb{N}^{d}} \frac{(-i)^{\alpha}}{\alpha!} \partial_{\xi}^{\alpha} \sigma^{P_{1}}(x, \xi) \partial_{x}^{\alpha} \sigma^{P_{2}}(x, \xi) .`	$\sigma^{P_{1} P_{2}}(x, \xi) \simeq \sum_{\alpha \in \mathbb{N}^{d}} \frac{(-i)^{\alpha}}{\alpha!} \partial_{\xi}^{\alpha} \sigma^{P_{1}}(x, \xi) \partial_{x}^{\alpha} \sigma^{P_{2}}(x, \xi) .$	$\sigma^{P_1 P_2}(x,\xi) \simeq \sum_{\alpha\in\mathbb{N}^d} \frac{(-i)^\alpha}{\alpha!}\, \partial_\xi^\alpha \sigma^{P_1}(x,\xi)\, \partial_x^\alpha \sigma^{P_2}(x,\xi).$
cardona-qft-methods_ EQ0011	020	1.000 N|+0	`\sigma_{m}^{\phi_{*} P}(x, \xi)=\sigma_{m}^{P}\left(\phi^{-1}(x),(d \phi)^{t} \xi\right) .`	$\sigma_{m}^{\phi_{*} P}(x, \xi)=\sigma_{m}^{P}\left(\phi^{-1}(x),(d \phi)^{t} \xi\right) .$	$\sigma_m^{\phi_* P}(x,\xi) = \sigma_m^P\big(\phi^{-1}(x), (d\phi)^t\xi\big).$
cardona-qft-methods_ EQ0013 (3)	021	1.000 N|+0	`\sigma_{-m}^{P}(x, \xi)=\lim _{t \rightarrow \infty} t^{-m}\left(e^{-i t h} \cdot P \cdot e^{i t h}\right)(x) \text { for } x \in M, \xi \in T_{x}^{*} M`	$\sigma_{-m}^{P}(x, \xi)=\lim _{t \rightarrow \infty} t^{-m}\left(e^{-i t h} \cdot P \cdot e^{i t h}\right)(x) \text { for } x \in M, \xi \in T_{x}^{*} M$	$\sigma_{-m}^P(x,\xi) = \lim_{t\to\infty} t^{-m}\big(e^{-ith}\cdot P\cdot e^{ith}\big)(x) \quad \text{for } x\in M,\, \xi\in T_x^*M$
cardona-qft-methods_ EQ0014	022	1.000 N|+0	`k^{P}(x, y)=\sum_{-(m+d) \leq j \leq 0} a_{j}(x, x-y)-c_{P}(x) \log |x-y|+\mathcal{O}(1)`	$k^{P}(x, y)=\sum_{-(m+d) \leq j \leq 0} a_{j}(x, x-y)-c_{P}(x) \log \|x-y\|+\mathcal{O}(1)$	$k^P(x,y) = \sum_{-(m+d)\le j\le 0} a_j(x,x-y) - c_P(x)\log\|x-y\| + \mathcal{O}(1)$
cardona-qft-methods_ EQ0015 (4)	022	1.000 N|+1	`c_{P}(x)=\frac{1}{(2 \pi)^{d}} \int_{\mathbb{S}^{d-1}} \sigma_{-d}^{P}(x, \xi) d \xi`	$c_{P}(x)=\frac{1}{(2 \pi)^{d}} \int_{\mathbb{S}^{d-1}} \sigma_{-d}^{P}(x, \xi) d \xi$	$c_P(x) = \frac{1}{(2\pi)^d}\int_{\mathbb{S}^{d-1}} \sigma_{-d}^P(x,\xi)\, d\xi.$
cardona-qft-methods_ EQ0016	022	1.000 N|+0	`\operatorname{tr}\left(k^{P}(x, y)\right)= \sum_{j=-(m+d)}^{0} a_{j}(x, x-y)-c_{P}(x) \log |x-y|+ \mathcal{O}(1),`	$\operatorname{tr}\left(k^{P}(x, y)\right)= \sum_{j=-(m+d)}^{0} a_{j}(x, x-y)-c_{P}(x) \log \|x-y\|+ \mathcal{O}(1),$	$tr\big(k^P(x,y)\big) = \sum_{j=-(m+d)}^{0} a_j(x,x-y) - c_P(x)\log\|x-y\| + \mathcal{O}(1),$
cardona-qft-methods_ EQ0018 (6)	022	1.000 N|+0	`c_{\phi_{*} P}(x)=\phi_{*}\left(c_{p}(x)\right) .`	$c_{\phi_{*} P}(x)=\phi_{*}\left(c_{p}(x)\right) .$	$c_{\phi_* P}(x) = \phi_*\big(c_p(x)\big).$
cardona-qft-methods_ EQ0020 (8)	024	1.000 K|+0	`\text { WRes } P=-\int_{M} c_{P}(x)|d x|`	$\text { WRes } P=-\int_{M} c_{P}(x)\|d x\|$	$WRes\ P = -\int_M c_P(x)\,\|dx\|$
cardona-qft-methods_ EQ0021	025	1.000 N|+0	`\begin{aligned} \operatorname{Tr}(\phi P) & =\int_{U} \phi(x) k^{P}(x, x)|d x|=\frac{1}{(2 \pi)^{d}} \int_{U} \phi(x) \sigma(x, \xi) d \xi|d x| \\ & =\int_{U} \phi(x) L(\sigma(x, \cdot))|d x| \end{aligned}`	$\begin{aligned} \operatorname{Tr}(\phi P) & =\int_{U} \phi(x) k^{P}(x, x)\|d x\|=\frac{1}{(2 \pi)^{d}} \int_{U} \phi(x) \sigma(x, \xi) d \xi\|d x\| \\ & =\int_{U} \phi(x) L(\sigma(x, \cdot))\|d x\| \end{aligned}$	$\begin{aligned}\mathrm{Tr}(\phi\, P) &= \int_U \phi(x)\, k^P(x,x)\,\|dx\| = \frac{1}{(2\pi)^d}\int_U \phi(x)\,\sigma(x,\xi)\,d\xi\,\|dx\| \\ &= \int_U \phi(x)\, L(\sigma(x,\cdot))\,\|dx\|,\end{aligned}$
cardona-qft-methods_ EQ0023 (9)	026	1.000 N|-1	`\text { if } \phi \in \operatorname{Diff}(M), W \operatorname{Res}(P)=W \operatorname{Res}\left(\phi_{*} P\right)`	$\text { if } \phi \in \operatorname{Diff}(M), W \operatorname{Res}(P)=W \operatorname{Res}\left(\phi_{*} P\right)$	$\text{if } \ \phi\in \mathrm{Diff}(M),\ WRes(P) = WRes(\phi_* P)$
cardona-qft-methods_ EQ0024	026	1.000 N|+0	`\left[x^{j}, \sigma\right]=i \partial_{\xi_{j}} \sigma, \quad\left[\xi_{j}, \sigma\right]=-i \partial_{x^{j}} \sigma .`	$\left[x^{j}, \sigma\right]=i \partial_{\xi_{j}} \sigma, \quad\left[\xi_{j}, \sigma\right]=-i \partial_{x^{j}} \sigma .$	$[x^j,\sigma] = i\partial_{\xi_j}\sigma, \quad [\xi_j,\sigma] = -i\partial_{x^j}\sigma.$
cardona-qft-methods_ EQ0025	026	1.000 N|+0	`\sigma_{-d}^{P}=\sum_{k=1}^{d} \partial_{\xi_{k}}^{2} \tilde{h}=-i \sum_{k=1}^{d}\left[\partial_{\xi_{k}} \tilde{h}, x^{k}\right]`	$\sigma_{-d}^{P}=\sum_{k=1}^{d} \partial_{\xi_{k}}^{2} \tilde{h}=-i \sum_{k=1}^{d}\left[\partial_{\xi_{k}} \tilde{h}, x^{k}\right]$	$\sigma_{-d}^P = \sum_{k=1}^{d}\partial_{\xi_k}^2\tilde h = -i\sum_{k=1}^{d}[\partial_{\xi_k}\tilde h,\ x^k]$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_ EQ0030</td><td>028</td><td>1.000 W|+1</td><td>\begin{aligned} & \mu_{n}(T)=\mu_{n}\left(T^{*}\right)= \mu_{n}(|T|) . \\ & T \in \mathcal{L}^{1}(\mathcal{H}) \left(\text { meaning }\|T\|_{1}=\operatorname{Tr}(|T|)< \infty\right) \Longleftrightarrow \sum_{n \in \mathbb{N}} \mu_{n}(T)<\infty . \\ & \mu_{n}(A T B) \leq\|A\| \mu_{n}(T)\|B\| \text { when } A, B \in \mathcal{B}(\mathcal{H}) . \\ & \mu_{N}\left(U T U^{*}\right)= \mu_{N}(T) \text { when } U \text { is a unitary. } \end{aligned}</td><td>$\begin{aligned} & \mu_{n}(T)=\mu_{n}\left(T^{*}\right)=\mu_{n}(|T|) . \\ & T \in \mathcal{L}^{1}(\mathcal{H})\left(\text { meaning }\|T\|_{1}=\operatorname{Tr}(|T|)<\infty\right) \Longleftrightarrow \sum_{n \in \mathbb{N}} \mu_{n}(T)<\infty . \\ & \mu_{n}(A T B) \leq\|A\| \mu_{n}(T)\|B\| \text { when } A, B \in \mathcal{B}(\mathcal{H}) . \\ & \mu_{N}\left(U T U^{*}\right)=\mu_{N}(T) \text { when } U \text { is a unitary. } \end{aligned}$</td><td>$\mu_n(T)=\mu_n(T^*)=\mu_n(|T|).$
$T\in\mathcal{L}^1(\mathcal{H})$ (meaning $\|T\|_1=\operatorname{Tr}(|T|)<\infty) \Longleftrightarrow \sum_{n\in\mathbb{N}}\mu_n(T)<\infty.$
$\mu_n(A\,T\,B)\leq\|A\|\,\mu_n(T)\,\|B\|$ when $A,B\in\mathcal{B}(\mathcal{H})$.
$\mu_N(U\,T\,U^*)=\mu_N(T)$ when U is a unitary.</td></tr>
<tr><td>cardona-qft-methods_ EQ0032</td><td>029</td><td>1.000 K|+0</td><td>\sigma_{N}(T)=\inf \left\{\|x\|_{1}+N\|y\| \mid T=x+y \text { with } x \in \mathcal{L}^{1}(\mathcal{H}), y \in \mathcal{K}(\mathcal{H})\right\} .</td><td>$\sigma_{N}(T)=\inf \left\{\|x\|_{1}+N\|y\| \mid T=x+y \text { with } x \in \mathcal{L}^{1}(\mathcal{H}), y \in \mathcal{K}(\mathcal{H})\right\} .$</td><td>$\sigma_N(T)=\inf\{\,\|x\|_1+N\,\|y\|\mid T=x+y \text{ with } x\in\mathcal{L}^1(\mathcal{H}),\ y\in\mathcal{K}(\mathcal{H})\,\}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0033</td><td>029</td><td>1.000 N|+0</td><td>\sigma_{\lambda}(T)=\inf \left\{\|x\|_{1}+\lambda\|y \| \mid T=x+y \text { with } x \in \mathcal{L}^{1}(\mathcal{H}), y \in \mathcal{K}(\mathcal{H})\right\} .</td><td>$\sigma_{\lambda}(T)=\inf \left\{\|x\|_{1}+\lambda\|y\| \mid T=x+y \text { with } x \in \mathcal{L}^{1}(\mathcal{H}), y \in \mathcal{K}(\mathcal{H})\right\} .$</td><td>$\sigma_\lambda(T)=\inf\{\,\|x\|_1+\lambda\|y\|\,|\,T=x+y \text{ with } x\in\mathcal{L}^1(\mathcal{H}),\ y\in\mathcal{K}(\mathcal{H})\,\}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0034</td><td>029</td><td>1.000 N|+0</td><td>\mathcal{L}^{1, \infty}=\left\{T \in \mathcal{K}(\mathcal{H}) \left\lvert\,\|T\|_{1, \infty}=\sup _{ \lambda \geq e} \frac{\sigma_{\lambda}(T)}{\log \lambda}< \infty\right.\right\} .</td><td>$\mathcal{L}^{1, \infty}=\left\{T \in \mathcal{K}(\mathcal{H}) \left\lvert\,\|T\|_{1, \infty}=\sup _{\lambda \geq e} \frac{\sigma_{\lambda}(T)}{\log \lambda}<\infty\right.\right\} .$</td><td>$\mathcal{L}^{1,\infty}=\{\,T\in\mathcal{K}(\mathcal{H})\mid\|T\|_{1,\infty}=\sup_{\lambda\geq e}\frac{\sigma_\lambda(T)}{\log\,\lambda}<\infty\,\}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0036</td><td>029</td><td>1.000 K|+0</td><td>\mathcal{L}^{1,+}=\left\{T \in \mathcal{K}(\mathcal{H}) \left\lvert\,\|T\|_{1,+}=\sup _{N \geq 2} \frac{ \sigma_{N}(T)}{\log (N)}<\infty\right.\right\} .</td><td>$\mathcal{L}^{1,+}=\left\{T \in \mathcal{K}(\mathcal{H}) \left\lvert\,\|T\|_{1,+}=\sup _{N \geq 2} \frac{\sigma_{N}(T)}{\log (N)}<\infty\right.\right\} .$</td><td>$\mathcal{L}^{1,+}=\{\,T\in\mathcal{K}(\mathcal{H})\mid\|T\|_{1,+}=\sup_{N\geq 2}\frac{\sigma_N(T)}{\log(N)}<\infty\,\}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0038</td><td>029</td><td>1.000 N|+1</td><td>\left|\tau_{\lambda}\left(T_{1}+T_{2}\right)-\tau_{ \lambda}\left(T_{1}\right)-\tau_{\lambda}\left(T_{2} \right)\right| \leq\left(\frac{\log 2(2+\log \log \lambda)}{\log \lambda}\right)\left\|T_{1}+T_{2}\right \|_{1, \infty},</td><td>$\left|\tau_{\lambda}\left(T_{1}+T_{2}\right)-\tau_{\lambda}\left(T_{1}\right)-\tau_{\lambda}\left(T_{2}\right)\right| \leq\left(\frac{\log 2(2+\log \log \lambda)}{\log \lambda}\right)\left\|T_{1}+T_{2}\right\|_{1, \infty},$</td><td>$|\,\tau_\lambda(T_1+T_2)-\tau_\lambda(T_1)-\tau_\lambda(T_2)\,|\leq\left(\frac{\log\,2(2+\log\,\log\,\lambda)}{\log\,\lambda}\right)\|T_1+T_2\|_{1,\infty},$</td></tr>
<tr><td>cardona-qft-methods_ EQ0042 (12)</td><td>032</td><td>1.000 K|+0</td><td>\sigma_{1}^{d}(\omega)=i \epsilon(\omega), \quad \sigma_{1}^{d^{*}}(\omega)=-i \iota(\omega) .</td><td>$\sigma_{1}^{d}(\omega)=i \epsilon(\omega), \quad \sigma_{1}^{d^{*}}(\omega)=-i \iota(\omega) .$</td><td>$\sigma_1^d(\omega)=i\epsilon(\omega), \qquad \sigma_1^{d^*}(\omega)=-i\iota(\omega).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0045</td><td>033</td><td>1.000 N|+0</td><td>D_{\nabla}=-i c \circ \nabla, \quad \Gamma(M, E) \xrightarrow{\nabla} \Gamma\left(M, T^{*} M \otimes E \right) \xrightarrow{c \otimes 1} \Gamma(M, E),</td><td>$D_{\nabla}=-i c \circ \nabla, \quad \Gamma(M, E) \xrightarrow{\nabla} \Gamma\left(M, T^{*} M \otimes E\right) \xrightarrow{c \otimes 1} \Gamma(M, E),$</td><td>$D_\nabla=-i\,c\circ\nabla, \qquad \Gamma(M,E)\xrightarrow{\nabla}\Gamma(M,T^*M\otimes E)\xrightarrow{c\otimes 1}\Gamma(M,E),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0048</td><td>035</td><td>1.000 N|+0</td><td>c\left(d x^{j}\right)\left(x_{0}\right)=\gamma^{j}, \quad \sigma_{1}^{D}\left(x_{0}, \xi\right)=c(\xi)=\gamma^{j} \xi_{j}</td><td>$c\left(d x^{j}\right)\left(x_{0}\right)=\gamma^{j}, \quad \sigma_{1}^{D}\left(x_{0}, \xi\right)=c(\xi)=\gamma^{j} \xi_{j}$</td><td>$c(dx^j)(x_0)=\gamma^j, \qquad \sigma_1^D(x_0,\xi)=c(\xi)=\gamma^j\xi_j$</td></tr>
<tr><td>cardona-qft-methods_ EQ0050</td><td>036</td><td>1.000 N|-1</td><td>\Delta^{S}=-\operatorname{Tr}_{g}\left(\nabla^{S} \circ \nabla^{S}\right): \Gamma^{\infty}(M, S) \rightarrow \Gamma^{\infty}(M, S) .</td><td>$\Delta^{S}=-\operatorname{Tr}_{g}\left(\nabla^{S} \circ \nabla^{S}\right): \Gamma^{\infty}(M, S) \rightarrow \Gamma^{\infty}(M, S) .$</td><td>$\Delta^S=-\operatorname{Tr}_g(\nabla^S\circ\nabla^S):\Gamma^\infty(M,S)\to\Gamma^\infty(M,S).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0051</td><td>037</td><td>1.000 K|+0</td><td>U_{g^{\prime}, g}:=\sqrt{f_{g, g^{\prime}}} I_{g^{ \prime}, g}: \mathcal{H}_{g^{\prime}} \rightarrow \mathcal{H}_{g} .</td><td>$U_{g^{\prime}, g}:=\sqrt{f_{g, g^{\prime}}} I_{g^{\prime}, g}: \mathcal{H}_{g^{\prime}} \rightarrow \mathcal{H}_{g} .$</td><td>$U_{g',g}:=\sqrt{f_{g,g'}}\,I_{g',g}:\mathcal{H}_{g'}\to\mathcal{H}_g\,.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0053 (15)</td><td>037</td><td>1.000 N|+0</td><td>D_{g^{\prime}}: \mathcal{H}_{g} \rightarrow \mathcal{H}_{g}, \quad D_{g^{\prime}}:=U_{g, g^{ \prime}}^{-1} \not D_{g^{\prime}} U_{g, g^{\prime}} .</td><td>$D_{g^{\prime}}: \mathcal{H}_{g} \rightarrow \mathcal{H}_{g}, \quad D_{g^{\prime}}:=U_{g, g^{\prime}}^{-1} \not D_{g^{\prime}} U_{g, g^{\prime}} .$</td><td>$D_{g'}:\mathcal{H}_g\to\mathcal{H}_g, \qquad D_{g'}:=U_{g,g'}^{-1}\,\not D_{g'}\,U_{g,g'}.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>cardona-qft-methods_EQ0054</td><td>037</td><td>■ 1.000 ● W|+0</td><td>\begin{aligned} & \not D_{g^{\prime}} I_{g, g^{\prime}} \psi=e^{-h} I_{g, g^{\prime}}\left(\not D_{g} \psi-i \frac{d-1}{2} c_{g}(\operatorname{grad} h) \psi\right), \\ & \not D_{g^{\prime}}=e^{-\frac{d+1}{2} h} I_{g, g^{\prime}} \not D_{g} I_{g, g^{\prime}}^{-1} e^{ \frac{d-1}{2} h}, \\ & D_{g^{\prime}}=e^{-h / 2} \not D_{g} e^{-h / 2} . \end{aligned}</td><td>$\begin{aligned} & \not D_{g^{\prime}} I_{g, g^{\prime}} \psi=e^{-h} I_{g, g^{\prime}}\left(\not D_{g} \psi-i \frac{d-1}{2} c_{g}(\operatorname{grad} h) \psi\right), \\ & \not D_{g^{\prime}}=e^{-\frac{d+1}{2} h} I_{g, g^{\prime}} \not D_{g} I_{g, g^{\prime}}^{-1} e^{ \frac{d-1}{2} h}, \\ & D_{g^{\prime}}=e^{-h / 2} \not D_{g} e^{-h / 2} . \end{aligned}$</td><td>$\not D_{g'} I_{g,g'}\psi = e^{-h} I_{g,g'}\left(\not D_g \psi - i\frac{d-1}{2} c_g(grad\ h)\psi\right),$
$\not D_{g'} = e^{-\frac{d+1}{2}h} I_{g,g'} \not D_g I_{g,g'}^{-1} e^{\frac{d-1}{2}h},$
$D_{g'} = e^{-h/2} \not D_g e^{-h/2}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0056 (16)</td><td>038</td><td>■ 1.000 ● N|+0</td><td>k_{t}(x, y)=\frac{1}{(4 \pi t)^{d / 2}} e^{-|x-y|^{2} / 4 t} \text { for } x \neq y</td><td>$k_{t}(x, y)=\frac{1}{(4 \pi t)^{d / 2}} e^{-|x-y|^{2} / 4 t} \text { for } x \neq y$</td><td>$k_t(x,y) = \frac{1}{(4\pi t)^{d/2}} e^{-|x-y|^2/4t}$ for $x \neq y$</td></tr>
<tr><td>cardona-qft-methods_EQ0057</td><td>038</td><td>■ 1.000 ● N|+1</td><td>\partial_{t} k_{t}(x, y)=\Delta_{x} k_{t}(x, y), \quad \forall t>0, x, y \in \mathbb{R}^{d}</td><td>$\partial_{t} k_{t}(x, y)=\Delta_{x} k_{t}(x, y), \quad \forall t>0, x, y \in \mathbb{R}^{d}$</td><td>$\partial_t k_t(x,y) = \Delta_x k_t(x,y), \quad \forall t > 0,\, x, y \in \mathbb{R}^d,$</td></tr>
<tr><td>cardona-qft-methods_EQ0058</td><td>039</td><td>■ 1.000 ● N|+0</td><td>-\Delta \phi=\lambda \psi \text { in } U, \quad \psi=0 \text { on } \partial U .</td><td>$-\Delta \phi=\lambda \psi \text { in } U, \quad \psi=0 \text { on } \partial U .$</td><td>$-\Delta\phi = \lambda\psi$ in $U, \qquad \psi = 0$ on $\partial U.$</td></tr>
<tr><td>cardona-qft-methods_EQ0060</td><td>039</td><td>■ 1.000 ● N|+0</td><td>k_{t}(x, x) \underset{t \downarrow 0^{+}}{\sim} \sum_{k=0}^{\infty} a_{k}(x) t^{(-d+k) / m}</td><td>$k_{t}(x, x) \underset{t \downarrow 0^{+}}{\sim} \sum_{k=0}^{\infty} a_{k}(x) t^{(-d+k) / m}$</td><td>$k_t(x,x) \underset{t\downarrow 0^+}{\sim} \sum_{k=0}^{\infty} a_k(x)\, t^{(-d+k)/m}$</td></tr>
<tr><td>cardona-qft-methods_EQ0061 (17)</td><td>039</td><td>■ 1.000 ● N|+0</td><td>k(t, f, P) \underset{t \downarrow 0^{+}}{\sim} \sum_{k=0}^{\infty} a_{k}(f, P) t^{(-d+k) / m} .</td><td>$k(t, f, P) \underset{t \downarrow 0^{+}}{\sim} \sum_{k=0}^{\infty} a_{k}(f, P) t^{(-d+k) / m} .$</td><td>$k(t,f,P) \underset{t\downarrow 0^+}{\sim} \sum_{k=0}^{\infty} a_k(f,P)\, t^{(-d+k)/m}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0062 (18)</td><td>040</td><td>■ 1.000 ● N|+0</td><td>P=-\left(g^{\mu \nu} \partial_{\mu} \partial_{\nu}+\mathbb{A}^{\mu} \partial_{\mu}+\mathbb{B}\right)</td><td>$P=-\left(g^{\mu \nu} \partial_{\mu} \partial_{\nu}+\mathbb{A}^{\mu} \partial_{\mu}+\mathbb{B}\right)$</td><td>$P = -(g^{\mu\nu}\partial_\mu\partial_\nu + \mathbb{A}^\mu\partial_\mu + \mathbb{B})$</td></tr>
<tr><td>cardona-qft-methods_EQ0063</td><td>040</td><td>■ 1.000 ● N|-1</td><td>P=-\left(\operatorname{Tr}_{g} \nabla^{2}+E\right), \quad \nabla^{2}(X, Y):=\left[\nabla_{X}, \nabla_{Y}\right]-\nabla_{\nabla_{X}^{L C} Y},</td><td>$P=-\left(\operatorname{Tr}_{g} \nabla^{2}+E\right), \quad \nabla^{2}(X, Y):=\left[\nabla_{X}, \nabla_{Y}\right]-\nabla_{\nabla_{X}^{L C} Y},$</td><td>$P = -(\mathrm{Tr}_g \nabla^2 + E), \quad \nabla^2(X,Y) := [\nabla_X, \nabla_Y] - \nabla_{\nabla_X^{LC} Y},$</td></tr>
<tr><td>cardona-qft-methods_EQ0064</td><td>040</td><td>■ 1.000 ● C|+3</td><td>\operatorname{Tr}_{g} \nabla^{2}:=g^{\mu \nu}\left(\nabla_{\mu} \nabla_{\nu}-\Gamma_{\mu \nu}^{\rho} \nabla_{\rho}\right)</td><td>$\operatorname{Tr}_{g} \nabla^{2}:=g^{\mu \nu}\left(\nabla_{\mu} \nabla_{\nu}-\Gamma_{\mu \nu}^{\rho} \nabla_{\rho}\right)$</td><td>$\mathrm{Tr}_g \nabla^2 := g^{\mu\nu}(\nabla_\mu\nabla_\nu - \Gamma^\rho_{\mu\nu}\nabla_\rho)$</td></tr>
<tr><td>cardona-qft-methods_EQ0067 (20)</td><td>041</td><td>■ 1.000 ● N|-1</td><td>\operatorname{Tr}\left(e^{-t D^{2}}\right) \underset{t \downarrow 0^{+}}{\sim} \sum_{j \geq 0} t^{(j-d) / 2} a_{j}\left(D^{2}\right)</td><td>$\operatorname{Tr}\left(e^{-t D^{2}}\right) \underset{t \downarrow 0^{+}}{\sim} \sum_{j \geq 0} t^{(j-d) / 2} a_{j}\left(D^{2}\right)$</td><td>$\mathrm{Tr}\big(e^{-tD^2}\big) \underset{t\downarrow 0^+}{\sim} \sum_{j\geq 0} t^{(j-d)/2} a_j(D^2)$</td></tr>
<tr><td>cardona-qft-methods_EQ0071 (23)</td><td>043</td><td>■ 1.000 ● K|+0</td><td>\left[\pi(a), \pi(b)^{\circ}\right]=0, \quad\left[[\mathcal{D}, \pi(a)], \pi(b)^{\circ}\right]=0, \forall a, b \in \mathcal{A}</td><td>$\left[\pi(a), \pi(b)^{\circ}\right]=0, \quad\left[[\mathcal{D}, \pi(a)], \pi(b)^{\circ}\right]=0, \forall a, b \in \mathcal{A}$</td><td>$[\pi(a), \pi(b)^\circ] = 0, \qquad \big[[\mathcal{D}, \pi(a)], \pi(b)^\circ\big] = 0, \forall a, b \in \mathcal{A}$</td></tr>
<tr><td>cardona-qft-methods_EQ0072 (24)</td><td>044</td><td>■ 1.000 ● N|+0</td><td>d\left(\phi_{1}, \phi_{2}\right)=\sup \left\{\left| \phi_{1}(a)-\phi_{2}(a)\right| \mid\|[\mathcal{D}, \pi(a)]\| \leq 1, a \in \mathcal{A}\right\}</td><td>$d\left(\phi_{1}, \phi_{2}\right)=\sup \left\{\left| \phi_{1}(a)-\phi_{2}(a)\right| \mid\|[\mathcal{D}, \pi(a)]\| \leq 1, a \in \mathcal{A}\right\}$</td><td>$d(\phi_1, \phi_2) = \sup\{\, |\phi_1(a) - \phi_2(a)| \mid \|[\mathcal{D}, \pi(a)]\| \leq 1, a \in \mathcal{A}\,\}$</td></tr>
<tr><td>cardona-qft-methods_EQ0073</td><td>045</td><td>■ 1.000 ● N|+0</td><td>O P^{\alpha} O P^{\beta} \subset O P^{\alpha+\beta}, \quad O P^{\alpha} \subset O P^{\beta} \text { if } \alpha \leq \beta, \quad \delta\left(O P^{\alpha}\right) \subset O P^{\alpha}</td><td>$O P^{\alpha} O P^{\beta} \subset O P^{\alpha+\beta}, \quad O P^{\alpha} \subset O P^{\beta} \text { if } \alpha \leq \beta, \quad \delta\left(O P^{\alpha}\right) \subset O P^{\alpha}$</td><td>$OP^\alpha OP^\beta \subset OP^{\alpha+\beta}, \quad OP^\alpha \subset OP^\beta$ if $\alpha \leq \beta, \quad \delta(OP^\alpha) \subset OP^\alpha$</td></tr>
<tr><td>cardona-qft-methods_EQ0076 (25)</td><td>045</td><td>■ 1.000 ● N|-1</td><td>\zeta_{\mathcal{D}}^{P}: s \in \mathbb{C} \rightarrow \operatorname{Tr}\left(P|\mathcal{D}|^{-s}\right)</td><td>$\zeta_{\mathcal{D}}^{P}: s \in \mathbb{C} \rightarrow \operatorname{Tr}\left(P|\mathcal{D}|^{-s}\right)$</td><td>$\zeta^P_{\mathcal{D}} : s \in \mathbb{C} \to \mathrm{Tr}\big(P|\mathcal{D}|^{-s}\big)$</td></tr>
<tr><td>cardona-qft-methods_EQ0079</td><td>047</td><td>■ 1.000 ● N|+1</td><td>\sigma_{z}(T)=|D|^{z} T|D|^{-z}, z \in \mathbb{C}, \quad \epsilon(T)=\nabla(T) D^{-2}</td><td>$\sigma_{z}(T)=|D|^{z} T|D|^{-z}, z \in \mathbb{C}, \quad \epsilon(T)=\nabla(T) D^{-2}$</td><td>$\sigma_z(T) = |D|^z T |D|^{-z}, z \in \mathbb{C}, \qquad \epsilon(T) = \nabla(T) D^{-2},$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
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N|+0</td><td><code>\sigma_{z}(T) \sim \sum_{r=0}^{N} g(z, r) \varepsilon^{r}(T) \quad \bmod O P^{-N-1+q}</code></td><td>$\sigma_{z}(T) \sim \sum_{r=0}^{N} g(z, r) \varepsilon^{r}(T) \quad \bmod O P^{-N-1+q}$</td><td>$\sigma_z(T) \sim \sum_{r=0}^{N} g(z,r)\,\varepsilon^r(T) \quad \bmod OP^{-N-1+q}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0082</td><td>048</td><td>1.000
W|+0</td><td><code>Y \sim \sum_{p=1}^{N} \sum_{k_{1}, \cdots, k_{p}=0}^{N-p} \frac{(-1)^{|k|_{1}+p+1}}{|k|_{1}+p} \nabla^{k_{p}} \left(X \nabla^{k_{p-1}}\left(\cdots X \nabla^{k_{1}}(X) \cdots\right)\right) D^{-2\left(|k|_{1}+p\right)},</code></td><td>$Y \sim \sum_{p=1}^{N} \sum_{k_{1}, \cdots, k_{p}=0}^{N-p} \frac{(-1)^{|k|_{1}+p+1}}{|k|_{1}+p} \nabla^{k_{p}} \left(X \nabla^{k_{p-1}}\left(\cdots X \nabla^{k_{1}}(X) \cdots\right)\right) D^{-2\left(|k|_{1}+p\right)},$</td><td>$Y \sim \sum_{p=1}^{N}\sum_{k_1,\cdots,k_p=0}^{N-p} \frac{(-1)^{|k|_1+p+1}}{|k|_1+p}\nabla^{k_p}(X\nabla^{k_{p-1}}(\cdots X\nabla^{k_1}(X)\cdots))D^{-2(|k|_1+p)},$</td></tr>
<tr><td>cardona-qft-methods_ EQ0084</td><td>049</td><td>1.000
W|+0</td><td><code>\sum_{p=1}^{k} \sum_{r_{1}, \cdots, r_{p}=0}^{k-p} \operatorname{Res}_{s=d-k} h(s, r, p) \operatorname{Tr}\left(\varepsilon^{r_{1}}(Y) \cdots \varepsilon^{r_{p}}(Y)|D|^{-s}\right),</code></td><td>$\sum_{p=1}^{k} \sum_{r_{1}, \cdots, r_{p}=0}^{k-p} \operatorname{Res}_{s=d-k} h(s, r, p) \operatorname{Tr}\left(\varepsilon^{r_{1}}(Y) \cdots \varepsilon^{r_{p}}(Y)|D|^{-s}\right),$</td><td>$\sum_{p=1}^{k}\sum_{r_1,\cdots,r_p=0}^{k-p} \operatorname*{Res}_{s=d-k} h(s,r,p)\operatorname{Tr}(\varepsilon^{r_1}(Y)\cdots\varepsilon^{r_p}(Y)|D|^{-s}),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0089</td><td>051</td><td>1.000
N|+1</td><td><code>\zeta_{\mathcal{D}_{A}}(0)=c_{1} \int_{M}\left(5 R^{2}-8 R \mu \nu r^{\mu \nu}-7 R_{\mu \nu \rho \sigma} R^{\mu \nu \rho \sigma} d v o l+c_{2} \int_{M} \operatorname{tr}\left(F_{\mu \nu} F^{\mu \nu}\right) d v o l\right.</code></td><td>$\zeta_{\mathcal{D}_{A}}(0)=c_{1} \int_{M}\left(5 R^{2}-8 R \mu \nu r^{\mu \nu}-7 R_{\mu \nu \rho \sigma} R^{\mu \nu \rho \sigma} d v o l+c_{2} \int_{M} \operatorname{tr}\left(F_{\mu \nu} F^{\mu \nu}\right) d v o l\right.$</td><td>$\zeta_{\mathcal{D}_A}(0) = c_1\int_M (5R^2 - 8R\mu\nu r^{\mu\nu} - 7R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\, dvol + c_2\int_M \operatorname{tr}(F_{\mu\nu}F^{\mu\nu})\, dvol,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0090</td><td>052</td><td>1.000
N|+1</td><td><code>\mathcal{A}:=\mathcal{A}_{1} \otimes \mathcal{A}_{2}, \quad \mathcal{D}:=\mathcal{D}_{1} \otimes 1+ \chi_{1} \otimes \mathcal{D}_{2}, \quad \mathcal{H}:= \mathcal{H}_{1} \otimes \mathcal{H}_{2} .</code></td><td>$\mathcal{A}:=\mathcal{A}_{1} \otimes \mathcal{A}_{2}, \quad \mathcal{D}:=\mathcal{D}_{1} \otimes 1+ \chi_{1} \otimes \mathcal{D}_{2}, \quad \mathcal{H}:= \mathcal{H}_{1} \otimes \mathcal{H}_{2} .$</td><td>$\mathcal{A} := \mathcal{A}_1\otimes\mathcal{A}_2, \quad \mathcal{D} := \mathcal{D}_1\otimes 1 + \chi_1\otimes\mathcal{D}_2, \quad \mathcal{H} := \mathcal{H}_1\otimes\mathcal{H}_2.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0091</td><td>052</td><td>1.000
N|+0</td><td><code>\operatorname{Res}_{s=d_{1}+d_{2}}\left(\zeta_{ \mathcal{D}}(s)\right)=\frac{1}{2} \frac{\Gamma \left(d_{1} / 2\right) \Gamma\left(d_{2} / 2\right)}{ \Gamma(d / 2)} \operatorname{Res}_{s=d_{1}}\left(\zeta_{ \mathcal{D}_{1}}(s)\right) \operatorname{Res}_{s=d_{2}} \left(\zeta_{\mathcal{D}_{2}}(s)\right) .</code></td><td>$\operatorname{Res}_{s=d_{1}+d_{2}}\left(\zeta_{ \mathcal{D}}(s)\right)=\frac{1}{2} \frac{\Gamma \left(d_{1} / 2\right) \Gamma\left(d_{2} / 2\right)}{ \Gamma(d / 2)} \operatorname{Res}_{s=d_{1}}\left(\zeta_{ \mathcal{D}_{1}}(s)\right) \operatorname{Res}_{s=d_{2}} \left(\zeta_{\mathcal{D}_{2}}(s)\right) .$</td><td>$\operatorname{Res}_{s=d_1+d_2}(\zeta_{\mathcal{D}}(s)) = \frac{1}{2}\frac{\Gamma(d_1/2)\Gamma(d_2/2)}{\Gamma(d/2)}\operatorname{Res}_{s=d_1}(\zeta_{\mathcal{D}_1}(s))\operatorname{Res}_{s=d_2}(\zeta_{\mathcal{D}_2}(s)).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0092 (31)</td><td>055</td><td>1.000
N|+0</td><td><code>\operatorname{Tr}\left(e^{-t \mathcal{D}^{2}}\right) \underset{t \downarrow 0}{\sim} \sum_{\alpha \in S d} a_{ \alpha} t^{\alpha} \quad \text { with } a_{\alpha} \neq 0 .</code></td><td>$\operatorname{Tr}\left(e^{-t \mathcal{D}^{2}}\right) \underset{t \downarrow 0}{\sim} \sum_{\alpha \in S d} a_{ \alpha} t^{\alpha} \quad \text { with } a_{\alpha} \neq 0 .$</td><td>$\operatorname{Tr}(e^{-t\mathcal{D}^2}) \underset{t\downarrow 0}{\sim} \sum_{\alpha\in Sd} a_\alpha t^\alpha \quad with\ a_\alpha \neq 0.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0093 (32)</td><td>055</td><td>1.000
N|+0</td><td><code>a_{\alpha}=\frac{1}{2} \operatorname{Res}_{s=-2 \alpha} \left(\Gamma(s / 2) \zeta_{\mathcal{D}}(s)\right) .</code></td><td>$a_{\alpha}=\frac{1}{2} \operatorname{Res}_{s=-2 \alpha} \left(\Gamma(s / 2) \zeta_{\mathcal{D}}(s)\right) .$</td><td>$a_\alpha = \frac{1}{2}\operatorname{Res}_{s=-2\alpha}(\Gamma(s/2)\zeta_{\mathcal{D}}(s)).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0094 (33)</td><td>055</td><td>1.000
N|+0</td><td><code>\operatorname{Tr}(f(\mathcal{D} / \Lambda)) \underset{ \Lambda \rightarrow+\infty}{\sim} \sum_{\beta \in S d^{+}} f_{\beta} \Lambda^{\beta} f|\mathcal{D}|^{ \beta}+f(0) \zeta_{\mathcal{D}}(0)+\cdots</code></td><td>$\operatorname{Tr}(f(\mathcal{D} / \Lambda)) \underset{ \Lambda \rightarrow+\infty}{\sim} \sum_{\beta \in S d^{+}} f_{\beta} \Lambda^{\beta} f|\mathcal{D}|^{ \beta}+f(0) \zeta_{\mathcal{D}}(0)+\cdots$</td><td>$\operatorname{Tr}(f(\mathcal{D}/\Lambda)) \underset{\Lambda\to+\infty}{\sim} \sum_{\beta\in Sd^+} f_\beta\Lambda^\beta \fint |\mathcal{D}|^{\beta} + f(0)\,\zeta_{\mathcal{D}}(0) + \cdots$</td></tr>
<tr><td>cardona-qft-methods_ EQ0097 (34)</td><td>056</td><td>1.000
N|-1</td><td><code>\operatorname{Tr}\left(e^{-t \mathcal{D}^{2}}\right) \underset{t \downarrow 0}{\sim} \sum_{k \in\{0, \cdots, d\}} t^{(k-d) / 2} a_{k}\left(\mathcal{D}^{2}\right)+ \cdots,</code></td><td>$\operatorname{Tr}\left(e^{-t \mathcal{D}^{2}}\right) \underset{t \downarrow 0}{\sim} \sum_{k \in\{0, \cdots, d\}} t^{(k-d) / 2} a_{k}\left(\mathcal{D}^{2}\right)+ \cdots,$</td><td>$\operatorname{Tr}(e^{-t\mathcal{D}^2}) \underset{t\downarrow 0}{\sim} \sum_{k\in\{0,\cdots,d\}} t^{(k-d)/2} a_k(\mathcal{D}^2) + \cdots,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0101</td><td>058</td><td>1.000
N|+0</td><td><code>a_{k}:=\frac{(-1)^{k} 4}{3 k!}\left(\frac{B_{2 k+2}}{2 k+2}-\frac{B_{2 k+4}}{2 k+4}\right)</code></td><td>$a_{k}:=\frac{(-1)^{k} 4}{3 k!}\left(\frac{B_{2 k+2}}{2 k+2}-\frac{B_{2 k+4}}{2 k+4}\right)$</td><td>$a_k := \frac{(-1)^k 4}{3\,k!}\left(\frac{B_{2k+2}}{2k+2} - \frac{B_{2k+4}}{2k+4}\right)$</td></tr>
<tr><td>cardona-qft-methods_ EQ0103</td><td>061</td><td>1.000
K|+0</td><td><code>\mathcal{D}=-i \delta_{\mu} \otimes \gamma^{\mu},</code></td><td>$\mathcal{D}=-i \delta_{\mu} \otimes \gamma^{\mu},$</td><td>$\mathcal{D} = -i\,\delta_\mu\otimes\gamma^\mu,$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0107 (2)	072	■1.000 ● K|+0	`\operatorname{Ind}(A)=\operatorname{dim} \mathcal{H}_{1}-\operatorname{dim} \mathcal{H}_{2}`	$\operatorname{Ind}(A)=\operatorname{dim} \mathcal{H}_{1}-\operatorname{dim} \mathcal{H}_{2}$	$\operatorname{Ind}(A) = \dim \mathcal{H}_1 - \dim \mathcal{H}_2$
cardona-qft-methods_ EQ0108 (3)	073	■1.000 ● K|+0	`\mathcal{H}=l_{2}:=\left\{\left(x_{1}, x_{2},\left.\ldots \left|x_{j} \in \mathcal{C}, \sum_{j=1}^{\infty}\right| x_{j}\right|^{2}<\infty\right\},\right.`	$\mathcal{H}=l_{2}:=\left\{\left(x_{1}, x_{2},\left.\ldots \left\vert x_{j} \in \mathcal{C}, \sum_{j=1}^{\infty}\right\vert x_{j}\right\vert^{2}<\infty\right\},\right.$	$\mathcal{H} = l_2 := \{(x_1, x_2, \ldots \vert x_j \in \mathcal{C}, \sum_{j=1}^{\infty} \vert x_j\vert^2 < \infty\},$
cardona-qft-methods_ EQ0109 (4)	073	■1.000 ● K|+0	`T\left(x_{1}, x_{2}, x_{3}, \ldots\right):=\left(x_{3}, x_{4}, \ldots\right) .`	$T\left(x_{1}, x_{2}, x_{3}, \ldots\right):=\left(x_{3}, x_{4}, \ldots\right) .$	$T(x_1, x_2, x_3, ...) := (x_3, x_4, ...).$
cardona-qft-methods_ EQ0110	073	■1.000 ● N|+0	`\operatorname{Ker}(T)=\left\{\left(x_{1}, x_{2}, 0,0, \ldots\right) \in \mathcal{H} \mid x_{1}, x_{2} \in \mathbb{C}\right\} .`	$\operatorname{Ker}(T)=\left\{\left(x_{1}, x_{2}, 0,0, \ldots\right) \in \mathcal{H} \mid x_{1}, x_{2} \in \mathbb{C}\right\} .$	$\operatorname{Ker}(T) = \{(x_1, x_2, 0, 0, ...) \in \mathcal{H} \mid x_1, x_2 \in \mathbb{C}\}.$
cardona-qft-methods_ EQ0113 (5)	073	■1.000 ● N|+0	`(A+K) x=0`	$(A+K) x=0$	$(A+K)x = 0$
cardona-qft-methods_ EQ0114	074	■1.000 ● N|+0	`U=\bigoplus_{V \in \operatorname{Irr}(G)} m_{V} V .`	$U=\bigoplus_{V \in \operatorname{Irr}(G)} m_{V} V .$	$U = \bigoplus_{V \in \operatorname{Irr}(G)} m_V V.$
cardona-qft-methods_ EQ0115	074	■1.000 ● K|+0	`g A x=A g x, \quad \text { for all } \quad x \in \mathcal{H}_{1} .`	$g A x=A g x, \quad \text { for all } \quad x \in \mathcal{H}_{1} .$	$gAx = Agx, \quad \text{for all} \quad x \in \mathcal{H}_1.$
cardona-qft-methods_ EQ0117	075	■1.000 ● K|+0	`V-U \sim A-B \Longleftrightarrow V \oplus B \simeq A \oplus U,`	$V-U \sim A-B \Longleftrightarrow V \oplus B \simeq A \oplus U,$	$V - U \sim A - B \iff V \oplus B \simeq A \oplus U,$
cardona-qft-methods_ EQ0119 (7)	076	■1.000 ● N|+0	`\widehat{R}(G)=\left\{\bigoplus_{V \in \operatorname{Irr}(G)} m_{V} V \mid m_{V} \in \mathbb{Z} \right\} .`	$\widehat{R}(G)=\left\{\bigoplus_{V \in \operatorname{Irr}(G)} m_{V} V \mid m_{V} \in \mathbb{Z} \right\} .$	$\widehat{R}(G) = \{\bigoplus_{V \in \operatorname{Irr}(G)} m_V V \vert m_V \in \mathbb{Z}\}.$
cardona-qft-methods_ EQ0120 (8)	076	■1.000 ● N|+0	`\operatorname{Ind}_{G}(A):=\operatorname{Ker}(A)-\operatorname{Coker}(A)=\bigoplus_{V \in \operatorname{Irr}(G)}\left(m_{V}^{+}-m_{V}^{-}\right) V \in R(G) .`	$\operatorname{Ind}_{G}(A):=\operatorname{Ker}(A)-\operatorname{Coker}(A)=\bigoplus_{V \in \operatorname{Irr}(G)}\left(m_{V}^{+}-m_{V}^{-}\right) V \in R(G) .$	$\operatorname{Ind}_G(A) := \operatorname{Ker}(A) - \operatorname{Coker}(A) = \bigoplus_{V \in \operatorname{Irr}(G)} (m_V^+ - m_V^-) V \in R(G).$
cardona-qft-methods_ EQ0121	076	■1.000 ● K|+0	`G:=\mathbb{Z}_{2}=\{1,-1\} .`	$G:=\mathbb{Z}_{2}=\{1,-1\} .$	$G := \mathbb{Z}_2 = \{1, -1\}.$
cardona-qft-methods_ EQ0122	076	■1.000 ● N|+1	`q\left(x_{1}, x_{2}, \ldots\right)=\left(x_{2}, x_{1}, x_{4}, x_{3}, \ldots\right) .`	$q\left(x_{1}, x_{2}, \ldots\right)=\left(x_{2}, x_{1}, x_{4}, x_{3}, \ldots\right) .$	$q(x_1, x_2, ...) = (x_2, x_1, x_4, x_3, ...).$
cardona-qft-methods_ EQ0123	077	■1.000 ● N|+0	`D_{j} f=\frac{1}{i} \frac{\partial f}{\partial x_{j}}, \quad j=1, \ldots, n .`	$D_{j} f=\frac{1}{i} \frac{\partial f}{\partial x_{j}}, \quad j=1, \ldots, n .$	$D_j f = \frac{1}{i} \frac{\partial f}{\partial x_j}, \qquad j = 1, \ldots, n.$
cardona-qft-methods_ EQ0124	077	■1.000 ● K|+0	`|\alpha|:=\alpha_{1}+\cdots+\alpha_{n}`	$\vert\alpha\vert:=\alpha_{1}+\cdots+\alpha_{n}$	$\vert\alpha\vert := \alpha_1 + \cdots + \alpha_n$
cardona-qft-methods_ EQ0126 (1)	077	■1.000 ● N|+0	`\mathcal{D}=\sum_{|\alpha| \leq k} a_{\alpha}(x) D^{\alpha}: C^{\infty}\left(\mathbb{R}^{n}\right) \longrightarrow C^{\infty}\left(\mathbb{R}^{n}\right),`	$\mathcal{D}=\sum_{\vert\alpha\vert \leq k} a_{\alpha}(x) D^{\alpha}: C^{\infty}\left(\mathbb{R}^{n}\right) \longrightarrow C^{\infty}\left(\mathbb{R}^{n}\right),$	$\mathcal{D} = \sum_{\vert\alpha\vert \leq k} a_\alpha(x) D^\alpha : C^\infty(\mathbb{R}^n) \longrightarrow C^\infty(\mathbb{R}^n),$
cardona-qft-methods_ EQ0127	077	■1.000 ● N|+1	`\langle f, g\rangle:=\int_{\mathbb{R}^{n}} f(x) \bar{g}(x) d x`	$\langle f, g\rangle:=\int_{\mathbb{R}^{n}} f(x) \bar{g}(x) d x$	$\langle f, g\rangle := \int_{\mathbb{R}^n} f(x)\overline{g}(x)dx.$
cardona-qft-methods_ EQ0128 (2)	077	■1.000 ● N|+0	`\langle f, g\rangle_{k}:=\sum_{|\alpha|=k}\left\langle D^{\alpha} f, D^{\alpha} g\right\rangle .`	$\langle f, g\rangle_{k}:=\sum_{\vert\alpha\vert=k}\left\langle D^{\alpha} f, D^{\alpha} g\right\rangle .$	$\langle f, g\rangle_k := \sum_{\vert\alpha\vert=k} \langle D^\alpha f, D^\alpha g\rangle.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>cardona-qft-methods_EQ0129 (3)</td><td>077</td><td>■ 1.000 ● N|+0</td><td>\|f\|_{k}:=\sqrt{\langle f, f\rangle_{k}}</td><td>$\|f\|_{k}:=\sqrt{\langle f, f\rangle_{k}}$</td><td>$\|f\|_k := \sqrt{\langle f,f\rangle_k}$</td></tr>
<tr><td>cardona-qft-methods_EQ0130</td><td>078</td><td>■ 1.000 ● N|+0</td><td>H^{k}\left(\mathbb{R}^{n}\right)=\left\{f: \mathbb{R}^{n} \rightarrow \mathcal{C} \mid D^{\alpha} f \in L^{2} \left(\mathbb{R}^{n}\right) \text { for all } \alpha \in \mathbb{Z}_{+}^{n} \text { with }|\alpha| \leq k\right\} .</td><td>$H^{k}\left(\mathbb{R}^{n}\right)=\left\{f: \mathbb{R}^{n} \rightarrow \mathcal{C} \mid D^{\alpha} f \in L^{2} \left(\mathbb{R}^{n}\right) \text { for all } \alpha \in \mathbb{Z}_{+}^{n} \text { with }|\alpha| \leq k\right\} .$</td><td>$H^k(\mathbb{R}^n) = \left\{ f:\mathbb{R}^n\to\mathcal{C} \,\middle|\, D^\alpha f\in L^2(\mathbb{R}^n) \text{ for all } \alpha\in\mathbb{Z}^n_+ \text{ with } |\alpha|\le k\right\}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0131</td><td>078</td><td>■ 1.000 ● N|+0</td><td>H_{K}^{k}\left(\mathbb{R}^{n}\right)=\left\{f \in H^{k} \left(\mathbb{R}^{n}\right) \mid \operatorname{supp}(f) \subset K\right\} .</td><td>$H_{K}^{k}\left(\mathbb{R}^{n}\right)=\left\{f \in H^{k} \left(\mathbb{R}^{n}\right) \mid \operatorname{supp}(f) \subset K\right\} .$</td><td>$H^k_K(\mathbb{R}^n) = \left\{ f\in H^k(\mathbb{R}^n) \,\middle|\, \operatorname{supp}(f)\subset K\right\}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0133</td><td>079</td><td>■ 1.000 ● N|+0</td><td>a_{\alpha}(x)=\left\{a_{\alpha, i j}(x)\right\}_{ \substack{1 \leq i \leq N_{2} \\ 1 \leq j \leq N_{1}}} \in \operatorname{Mat}_{N_{1} \times N_{2}}, \quad a_{ \alpha, i j}(x) \in C^{\infty}\left(\mathbb{R}^{n}\right) .</td><td>$a_{\alpha}(x)=\left\{a_{\alpha, i j}(x)\right\}_{ \substack{1 \leq i \leq N_{2} \\ 1 \leq j \leq N_{1}}} \in \operatorname{Mat}_{N_{1} \times N_{2}}, \quad a_{ \alpha, i j}(x) \in C^{\infty}\left(\mathbb{R}^{n}\right) .$</td><td>$a_\alpha(x) = \left\{a_{\alpha,ij}(x)\right\}_{\substack{1\le i\le N_2\\1\le j\le N_1}} \in \mathrm{Mat}_{N_1\times N_2}, \qquad a_{\alpha,ij}(x)\in C^\infty(\mathbb{R}^n).$</td></tr>
<tr><td>cardona-qft-methods_EQ0134</td><td>079</td><td>■ 1.000 ● N|+0</td><td>\mathcal{D}: H^{k}\left(\mathbb{R}^{n}, \mathbb{R}^{N} \right) \longrightarrow L^{2}\left(\mathbb{R}^{n}, \mathbb{R}^{N}\right)</td><td>$\mathcal{D}: H^{k}\left(\mathbb{R}^{n}, \mathbb{R}^{N} \right) \longrightarrow L^{2}\left(\mathbb{R}^{n}, \mathbb{R}^{N}\right)$</td><td>$\mathcal{D}:\, H^k(\mathbb{R}^n,\mathbb{R}^N) \longrightarrow L^2(\mathbb{R}^n,\mathbb{R}^N)$</td></tr>
<tr><td>cardona-qft-methods_EQ0136</td><td>079</td><td>■ 1.000 ● N|+0</td><td>\pi(s(x))=x, \quad \text { for all } x \in M .</td><td>$\pi(s(x))=x, \quad \text { for all } x \in M .$</td><td>$\pi\big(s(x)\big) = x, \qquad \text{for all } x\in M.$</td></tr>
<tr><td>cardona-qft-methods_EQ0137 (4)</td><td>080</td><td>■ 1.000 ● N|+1</td><td>\mathcal{D}: C^{\infty}(M, E) \longrightarrow C^{ \infty}(M, F)</td><td>$\mathcal{D}: C^{\infty}(M, E) \longrightarrow C^{ \infty}(M, F)$</td><td>$\mathcal{D}:\, C^\infty(M,E) \longrightarrow C^\infty(M,F)$</td></tr>
<tr><td>cardona-qft-methods_EQ0139</td><td>080</td><td>■ 1.000 ● N|+0</td><td>\mathcal{D}_{i}: C^{\infty}\left(V_{i}, \mathbb{R}^{N_{1}}\right) \longrightarrow C^{\infty} \left(V_{i}, \mathbb{R}^{N_{2}}\right) .</td><td>$\mathcal{D}_{i}: C^{\infty}\left(V_{i}, \mathbb{R}^{N_{1}}\right) \longrightarrow C^{\infty} \left(V_{i}, \mathbb{R}^{N_{2}}\right) .$</td><td>$\mathcal{D}_i:\, C^\infty(V_i,\mathbb{R}^{N_1}) \longrightarrow C^\infty(V_i,\mathbb{R}^{N_2}).$</td></tr>
<tr><td>cardona-qft-methods_EQ0140</td><td>081</td><td>■ 1.000 ● N|+0</td><td>\mathcal{D}: H_{K}^{m}(M, E) \longrightarrow L^{2}(M, F) .</td><td>$\mathcal{D}: H_{K}^{m}(M, E) \longrightarrow L^{2}(M, F) .$</td><td>$\mathcal{D}:\, H^m_K(M,E) \longrightarrow L^2(M,F).$</td></tr>
<tr><td>cardona-qft-methods_EQ0143</td><td>081</td><td>■ 1.000 ● N|+0</td><td>\sigma(\mathcal{D})\left(x_{0}, \xi\right)=\sum_{|\alpha| \leq k} a_{\alpha}\left(x_{0}\right) \xi_{1}^{\alpha_{1}} \cdots \xi_{n}^{\alpha_{n}} .</td><td>$\sigma(\mathcal{D})\left(x_{0}, \xi\right)=\sum_{|\alpha| \leq k} a_{\alpha}\left(x_{0}\right) \xi_{1}^{\alpha_{1}} \cdots \xi_{n}^{\alpha_{n}} .$</td><td>$\sigma(\mathcal{D})(x_0,\xi) = \sum_{|\alpha|\le k} a_\alpha(x_0)\, \xi_1^{\alpha_1}\cdots\xi_n^{\alpha_n}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0144</td><td>082</td><td>■ 1.000 ● N|+0</td><td>\sigma_{L}(\mathcal{D})(x, \xi): \pi^{*} E_{(x, \xi)} \rightarrow \pi^{*} F_{(x, \xi)} .</td><td>$\sigma_{L}(\mathcal{D})(x, \xi): \pi^{*} E_{(x, \xi)} \rightarrow \pi^{*} F_{(x, \xi)} .$</td><td>$\sigma_L(\mathcal{D})(x,\xi):\, \pi^*E_{(x,\xi)} \;\to\; \pi^*F_{(x,\xi)}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0145</td><td>082</td><td>■ 1.000 ● N|+0</td><td>\Delta=-\frac{\partial^{2}}{\partial x_{1}^{2}}-\frac{\partial^{2}}{\partial x_{2}^{2}}-\cdots-\frac{ \partial^{2}}{\partial x_{n}^{2}}</td><td>$\Delta=-\frac{\partial^{2}}{\partial x_{1}^{2}}-\frac{\partial^{2}}{\partial x_{2}^{2}}-\cdots-\frac{ \partial^{2}}{\partial x_{n}^{2}}$</td><td>$\Delta = -\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2} - \cdots - \frac{\partial^2}{\partial x_n^2}$</td></tr>
<tr><td>cardona-qft-methods_EQ0150 (2)</td><td>085</td><td>■ 1.000 ● N|+0</td><td>\sigma_{1} \oplus \mathrm{Id}: E_{1} \oplus \mathcal{C}^{k_{1}} \rightarrow F_{1} \oplus \mathcal{C}^{k_{1}}, \quad \text { and } \quad \sigma_{2} \oplus \mathrm{Id}: E_{2} \oplus \mathcal{C}^{k_{2}} \rightarrow F_{2} \oplus \mathcal{C}^{k_{2}},</td><td>$\sigma_{1} \oplus \mathrm{Id}: E_{1} \oplus \mathcal{C}^{k_{1}} \rightarrow F_{1} \oplus \mathcal{C}^{k_{1}}, \quad \text { and } \quad \sigma_{2} \oplus \mathrm{Id}: E_{2} \oplus \mathcal{C}^{k_{2}} \rightarrow F_{2} \oplus \mathcal{C}^{k_{2}},$</td><td>$\sigma_1\oplus\mathrm{Id}:\, E_1\oplus\mathcal{C}^{k_1} \;\to\; F_1\oplus\mathcal{C}^{k_1}, \quad \text{and} \quad \sigma_2\oplus\mathrm{Id}:\, E_2\oplus\mathcal{C}^{k_2} \;\to\; F_2\oplus\mathcal{C}^{k_2},$</td></tr>
<tr><td>cardona-qft-methods_EQ0151</td><td>085</td><td>■ 1.000 ● K|+0</td><td>K(X)=\{\sigma: E \rightarrow F \mid \sigma(x) \text { is invertible outside of a compact set }\} / \sim</td><td>$K(X)=\{\sigma: E \rightarrow F \mid \sigma(x) \text { is invertible outside of a compact set }\} / \sim$</td><td>$K(X) = \{\sigma:E\to F \mid \sigma(x) \text{ is invertible outside of a compact set}\}/\sim$</td></tr>
<tr><td>cardona-qft-methods_EQ0152</td><td>085</td><td>■ 1.000 ● K|+0</td><td>\operatorname{dim} E-\operatorname{dim} F \in \mathbb{Z} .</td><td>$\operatorname{dim} E-\operatorname{dim} F \in \mathbb{Z} .$</td><td>$\dim E - \dim F \in \mathbb{Z}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0153</td><td>085</td><td>■ 1.000 ● N|+1</td><td>K(\{p t\}) \cong \mathbb{Z}</td><td>$K(\{p t\}) \cong \mathbb{Z}$</td><td>$K(\{pt\}) \cong \mathbb{Z}.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0155	086	1.000 N|+0	`j_{!}: K(Y) \rightarrow K(X),`	$j_{!}: K(Y) \rightarrow K(X),$	$j_{!}: K(Y) \rightarrow K(X),$
cardona-qft-methods_ EQ0156	086	1.000 N|+1	`i_{!}: K(p t) \longrightarrow K\left(\mathcal{C}^{N} \right)`	$i_{!}: K(p t) \longrightarrow K\left(\mathcal{C}^{N}\right)$	$i_{!}: K(pt) \longrightarrow K(\mathcal{C}^{N})$
cardona-qft-methods_ EQ0159	087	1.000 K|+0	`g \mapsto \psi(g): E \rightarrow E, \quad g \in G,`	$g \mapsto \psi(g): E \rightarrow E, \quad g \in G,$	$g \mapsto \psi(g): E \rightarrow E, \quad g \in G,$
cardona-qft-methods_ EQ0160	087	1.000 N|+0	`\pi \circ \psi(g)=\phi \circ \pi(g) .`	$\pi \circ \psi(g)=\phi \circ \pi(g) .$	$\pi \circ \psi(g)=\phi \circ \pi(g).$
cardona-qft-methods_ EQ0161 (5)	087	1.000 N|+0	`g \mapsto l_{g}: C^{\infty}(M, E) \longrightarrow C^{ \infty}(M, E), \quad l_{g}(f)(x):=g \cdot f\left(g^{-1} \cdot x\right) .`	$g \mapsto l_{g}: C^{\infty}(M, E) \longrightarrow C^{\infty}(M, E), \quad l_{g}(f)(x):=g \cdot f\left(g^{-1} \cdot x\right) .$	$g \mapsto l_{g}: C^{\infty}(M, E) \longrightarrow C^{\infty}(M, E), \quad l_{g}(f)(x):=g \cdot f(g^{-1} \cdot x).$
cardona-qft-methods_ EQ0163 (6)	088	1.000 K|+0	`K_{G}(\{p t\})=R(G),`	$K_{G}(\{p t\})=R(G),$	$K_{G}(\{pt\})=R(G),$
cardona-qft-methods_ EQ0165	088	1.000 K|+0	`\mathcal{D}: C^{\infty}(M, E) \rightarrow C^{\infty}(M, F)`	$\mathcal{D}: C^{\infty}(M, E) \rightarrow C^{\infty}(M, F)$	$\mathcal{D}: C^{\infty}(M, E) \rightarrow C^{\infty}(M, F)$
cardona-qft-methods_ EQ0166	088	1.000 N|+0	`\sigma_{L}(\mathcal{D}): \pi^{*} E \longrightarrow \pi^{*} F,`	$\sigma_{L}(\mathcal{D}): \pi^{*} E \longrightarrow \pi^{*} F,$	$\sigma_{L}(\mathcal{D}): \pi^{*} E \longrightarrow \pi^{*} F,$
cardona-qft-methods_ EQ0167	089	1.000 K|+0	`\operatorname{Ind}_{G}(\mathcal{D})=\mathrm{t}- \operatorname{Ind}_{G}(\sigma) .`	$\operatorname{Ind}_{G}(\mathcal{D})=\mathrm{t}-\operatorname{Ind}_{G}(\sigma) .$	$\operatorname{Ind}_{G}(\mathcal{D})=\text{t-Ind}_{G}(\sigma).$
cardona-qft-methods_ EQ0168	089	1.000 K|+0	`M:=N \times G`	$M:=N \times G$	$M:=N \times G$
cardona-qft-methods_ EQ0170	089	1.000 N|+0	`\tilde{\mathcal{D}}: C^{\infty}(M) \longrightarrow C^{ \infty}(M)`	$\tilde{\mathcal{D}}: C^{\infty}(M) \longrightarrow C^{\infty}(M)$	$\tilde{\mathcal{D}}: C^{\infty}(M) \longrightarrow C^{\infty}(M)$
cardona-qft-methods_ EQ0171	089	1.000 N|+0	`\mathcal{D}=\sum_{|\alpha| \leq k} a_{\alpha}(x) \frac{ \partial^{\alpha_{1}}}{\partial x_{1}^{\alpha_{1}}} \cdots \frac{\partial^{\alpha_{n}}}{\partial x_{n}^{ \alpha_{n}}} .`	$\mathcal{D}=\sum_{\|\alpha\| \leq k} a_{\alpha}(x) \frac{\partial^{\alpha_{1}}}{\partial x_{1}^{\alpha_{1}}} \cdots \frac{\partial^{\alpha_{n}}}{\partial x_{n}^{\alpha_{n}}} .$	$\mathcal{D}=\sum_{\|\alpha\| \leq k} a_{\alpha}(x) \frac{\partial^{\alpha_{1}}}{\partial x_{1}^{\alpha_{1}}} \cdots \frac{\partial^{\alpha_{n}}}{\partial x_{n}^{\alpha_{n}}}.$
cardona-qft-methods_ EQ0172	089	1.000 N|+0	`\tilde{\mathcal{D}}:=\sum_{|\alpha| \leq k} a_{\alpha}(x) \frac{\partial^{\alpha_{1}}}{\partial x_{1}^{\alpha_{1}}} \cdots \frac{\partial^{\alpha_{n}}}{\partial x_{n}^{ \alpha_{n}}} .`	$\tilde{\mathcal{D}}:=\sum_{\|\alpha\| \leq k} a_{\alpha}(x) \frac{\partial^{\alpha_{1}}}{\partial x_{1}^{\alpha_{1}}} \cdots \frac{\partial^{\alpha_{n}}}{\partial x_{n}^{\alpha_{n}}} .$	$\tilde{\mathcal{D}}:=\sum_{\|\alpha\| \leq k} a_{\alpha}(x) \frac{\partial^{\alpha_{1}}}{\partial x_{1}^{\alpha_{1}}} \cdots \frac{\partial^{\alpha_{n}}}{\partial x_{n}^{\alpha_{n}}}.$
cardona-qft-methods_ EQ0173 (1)	090	1.000 K|+0	`\operatorname{Ker}(\tilde{\mathcal{D}})= \operatorname{Ker}(\mathcal{D}) \otimes L^{2}(G), \quad \operatorname{Coker}(\tilde{\mathcal{D}})= \operatorname{Coker}(\mathcal{D}) \otimes L^{2}(G) .`	$\operatorname{Ker}(\tilde{\mathcal{D}})=\operatorname{Ker}(\mathcal{D}) \otimes L^{2}(G), \quad \operatorname{Coker}(\tilde{\mathcal{D}})=\operatorname{Coker}(\mathcal{D}) \otimes L^{2}(G) .$	$\operatorname{Ker}(\tilde{\mathcal{D}})=\operatorname{Ker}(\mathcal{D}) \otimes L^{2}(G), \quad \operatorname{Coker}(\tilde{\mathcal{D}})=\operatorname{Coker}(\mathcal{D}) \otimes L^{2}(G).$
cardona-qft-methods_ EQ0174 (2)	090	1.000 N|+1	`L^{2}(G)=\bigoplus_{V \in \operatorname{Irr}(G)}( \operatorname{dim} V) V .`	$L^{2}(G)=\bigoplus_{V \in \operatorname{Irr}(G)}(\operatorname{dim} V) V .$	$L^{2}(G)=\bigoplus_{V \in \operatorname{Irr}(G)}(\operatorname{dim} V) V.$
cardona-qft-methods_ EQ0176	090	1.000 K|+0	`\mathcal{O}(x):=\{g \cdot x \mid g \in G\} .`	$\mathcal{O}(x):=\{g \cdot x \mid g \in G\} .$	$\mathcal{O}(x):=\left\{g \cdot x \middle\| g \in G\right\}.$
cardona-qft-methods_ EQ0178	091	1.000 N|+0	`\mathcal{D}: C^{\infty}(M, E) \longrightarrow C^{ \infty}(M, F)`	$\mathcal{D}: C^{\infty}(M, E) \longrightarrow C^{\infty}(M, F)$	$\mathcal{D}: C^{\infty}(M, E) \longrightarrow C^{\infty}(M, F)$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0179 (5)	091	■ 1.000 ● N|+0	`\operatorname{Ker}(\mathcal{D})=\bigoplus_{V \in \operatorname{Irr}(G)} m_{V}^{+} V, \quad \operatorname{Coker}(\mathcal{D})=\bigoplus_{V \in \operatorname{Irr}(G)} m_{V}^{-} V .`	$\operatorname{Ker}(\mathcal{D})=\bigoplus_{V \in \operatorname{Irr}(G)} m_{V}^{+} V, \quad \operatorname{Coker}(\mathcal{D})=\bigoplus_{V \in \operatorname{Irr}(G)} m_{V}^{-} V .$	$\operatorname{Ker}(\mathcal{D}) = \bigoplus_{V \in \operatorname{Irr}(G)} m_{V}^{+} V, \qquad \operatorname{Coker}(\mathcal{D}) = \bigoplus_{V \in \operatorname{Irr}(G)} m_{V}^{-} V.$
cardona-qft-methods_ EQ0180 (6)	091	■ 1.000 ● N|+1	`\operatorname{Ind}_{G}(\mathcal{D}):=\bigoplus_{V \in \operatorname{Irr}(G)}\left(m_{V}^{+}-m_{V}^{-}\right) V \in \widehat{R}(G),`	$\operatorname{Ind}_{G}(\mathcal{D}):=\bigoplus_{V \in \operatorname{Irr}(G)}\left(m_{V}^{+}-m_{V}^{-}\right) V \in \widehat{R}(G),$	$\operatorname{Ind}_{G}(\mathcal{D}) := \bigoplus_{V \in \operatorname{Irr}(G)} (m_{V}^{+}-m_{V}^{-}) V \in \widehat{R}(G),$
cardona-qft-methods_ EQ0182 (8)	092	■ 1.000 ● K|+0	`\mathrm{t}-\operatorname{Ind}_{G}(\sigma):= \operatorname{Ind}_{G}(\mathcal{D}) .`	$\mathrm{t}-\operatorname{Ind}_{G}(\sigma):= \operatorname{Ind}_{G}(\mathcal{D}) .$	$\text{t-}\operatorname{Ind}_{G}(\sigma) := \operatorname{Ind}_{G}(\mathcal{D}).$
cardona-qft-methods_ EQ0183	092	■ 1.000 ● K|+0	`\operatorname{Ind}_{G}(\mathcal{D})=\mathrm{t}-\operatorname{Ind}_{G}(\sigma) \in \widehat{R}(G) .`	$\operatorname{Ind}_{G}(\mathcal{D})=\mathrm{t}-\operatorname{Ind}_{G}(\sigma) \in \widehat{R}(G) .$	$\operatorname{Ind}_{G}(\mathcal{D}) = \text{t-}\operatorname{Ind}_{G}(\sigma) \in \widehat{R}(G).$
cardona-qft-methods_ EQ0184	093	■ 1.000 ● K|+0	`c: V \rightarrow \operatorname{End}(W)`	$c: V \rightarrow \operatorname{End}(W)$	$c: V \rightarrow \operatorname{End}(W)$
cardona-qft-methods_ EQ0188 (3)	093	■ 1.000 ● N|+1	`c: \mathbb{R}^{3} \longrightarrow \operatorname{End}\left(\mathcal{C}^{2}\right), \quad c\left(x_{1}, x_{2}, x_{3}\right):=\frac{1}{i}\left(x_{1} \sigma_{1}+x_{2} \sigma_{2}+x_{2} \sigma_{3}\right)`	$c: \mathbb{R}^{3} \longrightarrow \operatorname{End}\left(\mathcal{C}^{2}\right), \quad c\left(x_{1}, x_{2}, x_{3}\right):=\frac{1}{i}\left(x_{1} \sigma_{1}+x_{2} \sigma_{2}+x_{2} \sigma_{3}\right)$	$c: \mathbb{R}^{3} \longrightarrow \operatorname{End}(\mathcal{C}^{2}), \quad c(x_{1}, x_{2}, x_{3}) := \frac{1}{i}\left(x_{1} \sigma_{1}+x_{2} \sigma_{2}+x_{2} \sigma_{3}\right)$
cardona-qft-methods_ EQ0189	093	■ 1.000 ● N|+0	`W=\Lambda^{*}\left(V^{\mathcal{C}}\right)`	$W=\Lambda^{*}\left(V^{\mathcal{C}}\right)$	$W = \Lambda^{*}(V^{\mathcal{C}})$
cardona-qft-methods_ EQ0190	093	■ 1.000 ● K|+0	`\varepsilon(v) \alpha:=v \wedge \alpha, \quad \alpha \in W .`	$\varepsilon(v) \alpha:=v \wedge \alpha, \quad \alpha \in W .$	$\varepsilon(v)\,\alpha := v \wedge \alpha, \qquad \alpha \in W.$
cardona-qft-methods_ EQ0191 (4)	093	■ 1.000 ● K|+0	`c(v):=\varepsilon_{v}-\iota_{v}: W \longrightarrow W .`	$c(v):=\varepsilon_{v}-\iota_{v}: W \longrightarrow W .$	$c(v) := \varepsilon_{v} - \iota_{v} : W \longrightarrow W.$
cardona-qft-methods_ EQ0192 (5)	094	■ 1.000 ● N|+0	`c: T^{*} M \longrightarrow \operatorname{End}(E),`	$c: T^{*} M \longrightarrow \operatorname{End}(E),$	$c: T^{*}M \longrightarrow \operatorname{End}(E),$
cardona-qft-methods_ EQ0193	094	■ 1.000 ● N|+0	`c: T_{x}^{*} M \rightarrow \operatorname{End}\left(E_{x}\right)`	$c: T_{x}^{*} M \rightarrow \operatorname{End}\left(E_{x}\right)$	$c: T_{x}^{*}M \rightarrow \operatorname{End}(E_{x})$
cardona-qft-methods_ EQ0194	094	■ 1.000 ● K|+0	`c(v) \omega=v \wedge \omega-\iota_{v} \omega .`	$c(v) \omega=v \wedge \omega-\iota_{v} \omega .$	$c(v)\,\omega = v \wedge \omega - \iota_{v}\,\omega.$
cardona-qft-methods_ EQ0195 (6)	094	■ 1.000 ● K|+0	`\nabla_{u}(c(v) s)=c\left(\nabla_{u}^{L C} v\right) s+c(v) \nabla_{u} s,`	$\nabla_{u}(c(v) s)=c\left(\nabla_{u}^{L C} v\right) s+c(v) \nabla_{u} s,$	$\nabla_{u}\big(c(v)s\big) = c(\nabla_{u}^{LC} v)\, s + c(v)\, \nabla_{u} s,$
cardona-qft-methods_ EQ0196 (7)	095	■ 1.000 ● N|+0	`\mathcal{D}:=\sum_{j=1}^{n} c\left(e_{i}\right) \nabla_{e_{i}}: C^{\infty}(M, E) \longrightarrow C^{\infty}(M, E),`	$\mathcal{D}:=\sum_{j=1}^{n} c\left(e_{i}\right) \nabla_{e_{i}}: C^{\infty}(M, E) \longrightarrow C^{\infty}(M, E),$	$\mathcal{D} := \sum_{j=1}^{n} c(e_{i})\, \nabla_{e_{i}} : C^{\infty}(M, E) \longrightarrow C^{\infty}(M, E),$
cardona-qft-methods_ EQ0197 (8)	095	■ 1.000 ● N|+0	`\sigma_{L}(\mathcal{D})(x, \xi)=i c(\xi) .`	$\sigma_{L}(\mathcal{D})(x, \xi)=i c(\xi) .$	$\sigma_{L}(\mathcal{D})(x, \xi) = i\, c(\xi).$
cardona-qft-methods_ EQ0198	095	■ 1.000 ● N|+1	`\nabla_{\partial / \partial x_{i}}=\frac{\partial}{\partial x_{i}} .`	$\nabla_{\partial / \partial x_{i}}=\frac{\partial}{\partial x_{i}} .$	$\nabla_{\partial/\partial x_{i}} = \frac{\partial}{\partial x_{i}}.$
cardona-qft-methods_ EQ0200	095	■ 1.000 ● K|+0	`d^{*}: \Omega^{*}(M, E) \rightarrow \Omega^{*-1}(M, E)`	$d^{*}: \Omega^{*}(M, E) \rightarrow \Omega^{*-1}(M, E)$	$d^{*} : \Omega^{*}(M, E) \rightarrow \Omega^{*-1}(M, E)$
cardona-qft-methods_ EQ0203 (11)	096	■ 1.000 ● N|+0	`\operatorname{Ker}(\mathcal{D}) \simeq H^{*}(M) .`	$\operatorname{Ker}(\mathcal{D}) \simeq H^{*}(M) .$	$\operatorname{Ker}(\mathcal{D}) \simeq H^{*}(M).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>cardona-qft-methods_ EQ0204 (12)</td><td>096</td><td>1.000 K|+0</td><td>E=E^{+} \oplus E^{-},</td><td>$E=E^{+} \oplus E^{-},$</td><td>$E = E^{+} \oplus E^{-},$</td></tr>
<tr><td>cardona-qft-methods_ EQ0205</td><td>096</td><td>1.000 K|+0</td><td>c(v): E^{ \pm} \longrightarrow E^{\mp} .</td><td>$c(v): E^{\pm} \longrightarrow E^{\mp}.$</td><td>$c(v): E^{\pm} \longrightarrow E^{\mp}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0206</td><td>096</td><td>1.000 K|+0</td><td>\nabla_{v}: C^{\infty}\left(M, E^{ \pm}\right) \longrightarrow C^{\infty}\left(M, E^{ \pm}\right) .</td><td>$\nabla_{v}: C^{\infty}\left(M, E^{\pm}\right) \longrightarrow C^{\infty}\left(M, E^{\pm}\right).$</td><td>$\nabla_{v}: C^{\infty}(M, E^{\pm}) \longrightarrow C^{\infty}(M, E^{\pm}).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0207</td><td>096</td><td>1.000 K|+0</td><td>\mathcal{D}: C^{\infty}\left(M, E^{ \pm}\right) \longrightarrow C^{\infty}\left(M, E^{\mp}\right) .</td><td>$\mathcal{D}: C^{\infty}\left(M, E^{\pm}\right) \longrightarrow C^{\infty}\left(M, E^{\mp}\right).$</td><td>$\mathcal{D}: C^{\infty}(M, E^{\pm}) \longrightarrow C^{\infty}(M, E^{\mp}).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0208 (13)</td><td>097</td><td>1.000 K|+0</td><td>\mathcal{D}=\left(\begin{array}{cc} 0 & \mathcal{D}^{-} \\ \mathcal{D}^{+} & 0 \end{array}\right) .</td><td>$\mathcal{D}=\left(\begin{array}{cc} 0 & \mathcal{D}^{-} \\ \mathcal{D}^{+} & 0 \end{array}\right).$</td><td>$\mathcal{D} = \left(\begin{array}{cc} 0 & \mathcal{D}^{-} \\ \mathcal{D}^{+} & 0 \end{array}\right).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0210</td><td>097</td><td>1.000 N|+1</td><td>\operatorname{Ind}\left(\mathcal{D}^{+}\right)=\sum_{j=0}^{n}(-1)^{n} \operatorname{dim} H^{j}(M)</td><td>$\operatorname{Ind}\left(\mathcal{D}^{+}\right)=\sum_{j=0}^{n}(-1)^{n} \operatorname{dim} H^{j}(M)$</td><td>$\operatorname{Ind}(\mathcal{D}^{+}) = \sum_{j=0}^{n}(-1)^{n} \dim H^{j}(M).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0211</td><td>097</td><td>1.000 N|+0</td><td>\Gamma \omega:=i^{p(p-1)+l} * \omega, \quad \omega \in \Lambda^{p}(M) \otimes \mathcal{C} .</td><td>$\Gamma \omega:=i^{p(p-1)+l} * \omega, \quad \omega \in \Lambda^{p}(M) \otimes \mathcal{C}.$</td><td>$\Gamma \omega := i^{p(p-1)+l} * \omega, \qquad \omega \in \Lambda^{p}(M) \otimes \mathcal{C}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0212</td><td>098</td><td>1.000 K|+0</td><td>\operatorname{Ind}_{G}\left(\mathcal{D}^{+}\right) \in R(G) .</td><td>$\operatorname{Ind}_{G}\left(\mathcal{D}^{+}\right) \in R(G).$</td><td>$\operatorname{Ind}_{G}(\mathcal{D}^{+}) \in R(G).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0213 (1)</td><td>098</td><td>1.000 N|+0</td><td>\mathbf{v}: M \rightarrow \mathfrak{g}=\operatorname{Lie} G .</td><td>$\mathbf{v}: M \rightarrow \mathfrak{g}=\operatorname{Lie} G.$</td><td>$\mathbf{v}: M \rightarrow \mathfrak{g} = \operatorname{Lie} G.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0215</td><td>099</td><td>1.000 K|+0</td><td>G=S^{1}=\{z \in \mathcal{C}:|z|=1\}</td><td>$G=S^{1}=\{z \in \mathcal{C}:|z|=1\}$</td><td>$G = S^{1} = \{z \in \mathcal{C} : |z| = 1\}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0216 (3)</td><td>099</td><td>1.000 N|-1</td><td>v(x) \neq 0, \quad \text { for all } \quad x \notin K .</td><td>$v(x) \neq 0, \quad \text{ for all } \quad x \notin K.$</td><td>$v(x) \neq 0, \qquad \text{for all} \quad x \notin K.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0218</td><td>100</td><td>1.000 N|+0</td><td>c(\xi+v): \pi^{*} E^{+} \rightarrow \pi^{*} E^{-} .</td><td>$c(\xi+v): \pi^{*} E^{+} \rightarrow \pi^{*} E^{-}.$</td><td>$c(\xi+v): \pi^{*} E^{+} \rightarrow \pi^{*} E^{-}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0219</td><td>100</td><td>1.000 N|+0</td><td>\mathrm{t}-\operatorname{Ind}_{G}(c(\xi+v)) \in \widehat{R}(G) .</td><td>$\mathrm{t}-\operatorname{Ind}_{G}(c(\xi+v)) \in \widehat{R}(G).$</td><td>$\text{t-}\operatorname{Ind}_{G}\left(c(\xi+v)\right) \in \widehat{R}(G).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0220 (5)</td><td>100</td><td>1.000 N|+0</td><td>\mathrm{t}-\operatorname{Ind}_{G}(E, \mathbf{v}):= \mathrm{t}-\operatorname{Ind}_{G}(c(\xi+v)) .</td><td>$\mathrm{t}-\operatorname{Ind}_{G}(E, \mathbf{v}):=\mathrm{t}-\operatorname{Ind}_{G}(c(\xi+v)).$</td><td>$\text{t-}\operatorname{Ind}_{G}(E, \mathbf{v}) := \text{t-}\operatorname{Ind}_{G}\left(c(\xi+v)\right).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0221 (6)</td><td>101</td><td>1.000 K|+0</td><td>\mathcal{D}_{f v}=\mathcal{D}+i c(f v),</td><td>$\mathcal{D}_{f v}=\mathcal{D}+i c(f v),$</td><td>$\mathcal{D}_{fv} = \mathcal{D} + ic(fv),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0222 (7)</td><td>101</td><td>1.000 N|+0</td><td>\operatorname{Ker} \mathcal{D}_{f v}^{+}=\sum_{V \in \operatorname{Irr}(G)} m_{V}^{+} \cdot V, \quad \operatorname{Coker} \mathcal{D}_{f v}^{+}=\sum_{V \in \operatorname{Irr}(G)} m_{V}^{-} \cdot V,</td><td>$\operatorname{Ker} \mathcal{D}_{f v}^{+}=\sum_{V \in \operatorname{Irr}(G)} m_{V}^{+} \cdot V, \quad \operatorname{Coker} \mathcal{D}_{f v}^{+}=\sum_{V \in \operatorname{Irr}(G)} m_{V}^{-} \cdot V,$</td><td>$\operatorname{Ker} \mathcal{D}_{fv}^{+} = \sum_{V \in \operatorname{Irr}(G)} m_{V}^{+} \cdot V, \quad \operatorname{Coker} \mathcal{D}_{fv}^{+} = \sum_{V \in \operatorname{Irr}(G)} m_{V}^{-} \cdot V,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0223 (8)</td><td>101</td><td>1.000 N|+0</td><td>\operatorname{Ind}_{G}\left(\mathcal{D}_{f v}^{+}\right)=\sum_{V \in \operatorname{Irr}(G)} \left(m_{V}^{+}-m_{V}^{-}\right) \cdot V \in \widehat{R}(G) .</td><td>$\operatorname{Ind}_{G}\left(\mathcal{D}_{f v}^{+}\right)=\sum_{V \in \operatorname{Irr}(G)}\left(m_{V}^{+}-m_{V}^{-}\right) \cdot V \in \widehat{R}(G).$</td><td>$\operatorname{Ind}_{G}(\mathcal{D}_{fv}^{+}) = \sum_{V \in \operatorname{Irr}(G)} (m_{V}^{+}-m_{V}^{-}) \cdot V \in \widehat{R}(G).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0224 (9)</td><td>101</td><td>1.000 K|+0</td><td>\operatorname{Ind}_{G}(\mathcal{D}, \mathbf{v}):= \operatorname{Ind}_{G}\left(\mathcal{D}_{f v}^{+}\right),</td><td>$\operatorname{Ind}_{G}(\mathcal{D}, \mathbf{v}):=\operatorname{Ind}_{G}\left(\mathcal{D}_{f v}^{+}\right),$</td><td>$\operatorname{Ind}_{G}(\mathcal{D}, \mathbf{v}) := \operatorname{Ind}_{G}(\mathcal{D}_{fv}^{+}),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0225</td><td>102</td><td>1.000 N|+0</td><td>E^{ \pm}=M \times \mathbb{R}</td><td>$E^{\pm}=M \times \mathbb{R}$</td><td>$E^{\pm} = M \times \mathbb{R}$</td></tr>
</table>

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0226	102	■ 1.000 ● N|+0	`c(\xi):=\frac{1}{i}\left(\xi_{1} \sigma_{1}+\xi_{2} \sigma_{2}\right), \quad \xi=\left(\xi_{1}, \xi_{2} \right) \in T^{*} M \simeq \mathbb{R}^{2},`	$c(\xi):=\frac{1}{i}\left(\xi_{1} \sigma_{1}+\xi_{2} \sigma_{2}\right), \quad \xi=\left(\xi_{1}, \xi_{2} \right) \in T^{*} M \simeq \mathbb{R}^{2},$	$c(\xi) \;:=\; \frac{1}{i}\left(\xi_1\sigma_1+\xi_2\sigma_2\right), \qquad \xi=(\xi_1,\xi_2)\in T^*M\simeq\mathbb{R}^2,$
cardona-qft-methods_ EQ0228 (10)	102	■ 1.000 ● K|+0	`\operatorname{Ind}(\mathcal{D}, \mathbf{v})=\mathrm{t}-\operatorname{Ind}(E, \mathbf{v}) .`	$\operatorname{Ind}(\mathcal{D}, \mathbf{v})=\mathrm{t}-\operatorname{Ind}(E, \mathbf{v}) .$	$\operatorname{Ind}(\mathcal{D},\mathbf{v}) \;=\; \text{t-Ind}(E,\mathbf{v}).$
cardona-qft-methods_ EQ0229 (11)	102	■ 1.000 ● K|+0	`\operatorname{Ind}_{G}(\mathcal{D}, \mathbf{v})= \operatorname{Ind}_{G}(\mathcal{D}) .`	$\operatorname{Ind}_{G}(\mathcal{D}, \mathbf{v})= \operatorname{Ind}_{G}(\mathcal{D}) .$	$\operatorname{Ind}_G(\mathcal{D},\mathbf{v}) \;=\; \operatorname{Ind}_G(\mathcal{D}).$
cardona-qft-methods_ EQ0230	103	■ 1.000 ● N|+0	`M \backslash \Sigma=M_{1} \sqcup M_{2} .`	$M \backslash \Sigma=M_{1} \sqcup M_{2} .$	$M\backslash\Sigma \;=\; M_1\sqcup M_2.$
cardona-qft-methods_ EQ0231 (12)	103	■ 1.000 ● K|+0	`\operatorname{Ind}_{G}(\mathcal{D}, \mathbf{v})= \operatorname{Ind}_{G}\left(\mathcal{D}_{1}, \mathbf{v}_{1}\right)+\operatorname{Ind}_{G}\left( \mathcal{D}_{2}, \mathbf{v}_{2}\right) .`	$\operatorname{Ind}_{G}(\mathcal{D}, \mathbf{v})= \operatorname{Ind}_{G}\left(\mathcal{D}_{1}, \mathbf{v}_{1}\right)+\operatorname{Ind}_{G}\left(\mathcal{D}_{2}, \mathbf{v}_{2}\right) .$	$\operatorname{Ind}_G(\mathcal{D},\mathbf{v}) \;=\; \operatorname{Ind}_G(\mathcal{D}_1,\mathbf{v}_1) \;+\; \operatorname{Ind}_G(\mathcal{D}_2,\mathbf{v}_2).$
cardona-qft-methods_ EQ0232	109	■ 1.000 ● K|+0	`\chi(X)=s_{0}-s_{1}+s_{2}-\cdots`	$\chi(X)=s_{0}-s_{1}+s_{2}-\cdots$	$\chi(X)=s_0-s_1+s_2-\cdots$
cardona-qft-methods_ EQ0235	112	■ 1.000 ● K|+0	`\left[\mathbb{P}^{n-1}\right]=1+\mathbb{L}+\cdots+ \mathbb{L}^{n}=\frac{\mathbb{L}^{n}-1}{\mathbb{L}-1}`	$\left[\mathbb{P}^{n-1}\right]=1+\mathbb{L}+\cdots+ \mathbb{L}^{n}=\frac{\mathbb{L}^{n}-1}{\mathbb{L}-1}$	$[\mathbb{P}^{n-1}]=1+\mathbb{L}+\cdots+\mathbb{L}^n=\frac{\mathbb{L}^n-1}{\mathbb{L}-1}$
cardona-qft-methods_ EQ0239 (4)	114	■ 1.000 ● N|+1	`\int c(T X) \cap[X]=\chi(X)`	$\int c(T X) \cap[X]=\chi(X)$	$\int c(TX)\cap[X]=\chi(X) \quad .$
cardona-qft-methods_ EQ0240	115	■ 1.000 ● N|+0	`f_{*}\left(\mathbb{1}_{Z}\right)(p)=\chi\left(f^{-1}(p) \cap Z\right)`	$f_{*}\left(\mathbb{1}_{Z}\right)(p)=\chi\left(f^{-1}(p) \cap Z\right)$	$f_*(\mathbb{1}_Z)(p)=\chi(f^{-1}(p)\cap Z)$
cardona-qft-methods_ EQ0242 (5)	116	■ 1.000 ● N|+0	`\int c_{\mathrm{SM}}(X)=\kappa_{*} c_{\mathrm{SM}}\left( \mathbb{1}_{X}\right)=c_{\mathrm{SM}}\left(\kappa_{*} \mathbb{1}_{X}\right)=\chi(X) c_{\mathrm{SM}}\left( \mathbb{1}_{\mathrm{pt}}\right)=\chi(X)`	$\int c_{\mathrm{SM}}(X)=\kappa_{*} c_{\mathrm{SM}}\left(\mathbb{1}_{X}\right)=c_{\mathrm{SM}}\left(\kappa_{*} \mathbb{1}_{X}\right)=\chi(X) c_{\mathrm{SM}}\left(\mathbb{1}_{\mathrm{pt}}\right)=\chi(X)$	$\int c_{\mathrm{SM}}(X)=\kappa_*c_{\mathrm{SM}}(\mathbb{1}_X)=c_{\mathrm{SM}}(\kappa_*\mathbb{1}_X)=\chi(X)c_{\mathrm{SM}}(\mathbb{1}_{\mathrm{pt}})=\chi(X)$
cardona-qft-methods_ EQ0243	116	■ 1.000 ● K|+0	`\begin{aligned} c_{\mathrm{SM}}(X) & =c_{\mathrm{SM}} \left(\mathbb{1}_{X}\right)=c_{\mathrm{SM}}\left(\mathbb{1}_{Z}+\mathbb{1}_{X \backslash Z}\right) \\ & =c_{\mathrm{SM}}\left(\mathbb{1}_{Z}\right)+c_{ \mathrm{SM}}\left(\mathbb{1}_{X \backslash Z}\right)=c_{ \mathrm{SM}}(Z)+c_{\mathrm{SM}}(X \backslash Z) \end{aligned}`	$\begin{aligned} c_{\mathrm{SM}}(X) & =c_{\mathrm{SM}} \left(\mathbb{1}_{X}\right)=c_{\mathrm{SM}}\left(\mathbb{1}_{Z}+\mathbb{1}_{X \backslash Z}\right) \\ & =c_{\mathrm{SM}}\left(\mathbb{1}_{Z}\right)+c_{ \mathrm{SM}}\left(\mathbb{1}_{X \backslash Z}\right)=c_{ \mathrm{SM}}(Z)+c_{\mathrm{SM}}(X \backslash Z) \end{aligned}$	$\begin{aligned} c_{\mathrm{SM}}(X) &= c_{\mathrm{SM}}(\mathbb{1}_X)=c_{\mathrm{SM}}(\mathbb{1}_Z+\mathbb{1}_{X\smallsetminus Z}) \\ &= c_{\mathrm{SM}}(\mathbb{1}_Z)+c_{\mathrm{SM}}(\mathbb{1}_{X\smallsetminus Z})=c_{\mathrm{SM}}(Z)+c_{\mathrm{SM}}(X\smallsetminus Z) \end{aligned}$
cardona-qft-methods_ EQ0244	116	■ 1.000 ● N|+0	`c_{\mathrm{SM}}\left(\mathbb{P}^{1}\right)=c\left(T \mathbb{P}^{1}\right) \cap\left[\mathbb{P}^{1}\right]= \left(1+c_{1}\left(T \mathbb{P}^{1}\right)\right) \cap \left[\mathbb{P}^{1}\right]=\left[\mathbb{P}^{1}\right]+ \chi\left(\mathbb{P}^{1}\right)\left[\mathbb{P}^{0} \right]=\left[\mathbb{P}^{1}\right]+2\left[\mathbb{P}^{0} \right] .`	$c_{\mathrm{SM}}\left(\mathbb{P}^{1}\right)=c\left(T \mathbb{P}^{1}\right) \cap\left[\mathbb{P}^{1}\right]= \left(1+c_{1}\left(T \mathbb{P}^{1}\right)\right) \cap \left[\mathbb{P}^{1}\right]=\left[\mathbb{P}^{1}\right]+ \chi\left(\mathbb{P}^{1}\right)\left[\mathbb{P}^{0} \right]=\left[\mathbb{P}^{1}\right]+2\left[\mathbb{P}^{0} \right] .$	$c_{\mathrm{SM}}(\mathbb{P}^1)=c(T\mathbb{P}^1)\cap[\mathbb{P}^1]=(1+c_1(T\mathbb{P}^1))\cap[\mathbb{P}^1]=[\mathbb{P}^1]+\chi(\mathbb{P}^1)[\mathbb{P}^0]=[\mathbb{P}^1]+2[\mathbb{P}^0].$
cardona-qft-methods_ EQ0245 (6)	116	■ 1.000 ● N|+1	`c_{\mathrm{SM}}\left(\mathbb{P}^{n}\right)= \left[\mathbb{P}^{n}\right]+\binom{n+1}{1}\left[ \mathbb{P}^{n-1}\right]+\binom{n+1}{2}\left[ \mathbb{P}^{n-2}\right]+\cdots+\binom{n+1}{n}\left[ \mathbb{P}^{0}\right] .`	$c_{\mathrm{SM}}\left(\mathbb{P}^{n}\right)= \left[\mathbb{P}^{n}\right]+\binom{n+1}{1}\left[\mathbb{P}^{n-1}\right]+\binom{n+1}{2}\left[\mathbb{P}^{n-2}\right]+\cdots+\binom{n+1}{n}\left[\mathbb{P}^{0}\right] .$	$c_{\mathrm{SM}}(\mathbb{P}^n)=[\mathbb{P}^n]+\binom{n+1}{1}[\mathbb{P}^{n-1}]+\binom{n+1}{2}[\mathbb{P}^{n-2}]+\cdots+\binom{n+1}{n}[\mathbb{P}^0].$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0246	116	1.000 N|+0	`c_{\mathrm{SM}}\left(\mathbb{P}^{n}\right)=(1+H)^{n+1} \cap\left[\mathbb{P}^{n}\right]`	$c_{\mathrm{SM}}\left(\mathbb{P}^{n}\right)=(1+H)^{n+1} \cap\left[\mathbb{P}^{n}\right]$	$c_{\mathrm{SM}}(\mathbb{P}^{n})=(1+H)^{n+1} \cap[\mathbb{P}^{n}]$
cardona-qft-methods_EQ0247	117	1.000 N|+0	`c_{\mathrm{SM}}(C)=c_{\mathrm{SM}}\left(\mathbb{1}_{C}\right)=c(T C) \cap[C]=\left(1+c_{1}(T C)\right) \cap[C]=2\left[\mathbb{P}^{1}\right]+2\left[ \mathbb{P}^{0}\right] .`	$c_{\mathrm{SM}}(C)=c_{\mathrm{SM}}\left(\mathbb{1}_{C}\right)=c(T C) \cap[C]=\left(1+c_{1}(T C)\right) \cap[C]=2\left[\mathbb{P}^{1}\right]+2\left[\mathbb{P}^{0}\right] .$	$c_{\mathrm{SM}}(C)=c_{\mathrm{SM}}(\mathbb{1}_{C})=c(TC) \cap[C]=(1+c_{1}(TC)) \cap[C]=2[\mathbb{P}^{1}]+2[\mathbb{P}^{0}].$
cardona-qft-methods_EQ0248	117	1.000 N|+0	`c_{\mathrm{SM}}\left(L_{1}\right)+c_{\mathrm{SM}}\left(L_{2}\right)-c_{\mathrm{SM}}\left(L_{1} \cap L_{2}\right)=2 c_{\mathrm{SM}}\left(\mathbb{P}^{1}\right)-c_{\mathrm{SM}}\left(\mathbb{P}^{0}\right)=2\left[\mathbb{P}^{1}\right]+3\left[\mathbb{P}^{0}\right] .`	$c_{\mathrm{SM}}\left(L_{1}\right)+c_{\mathrm{SM}}\left(L_{2}\right)-c_{\mathrm{SM}}\left(L_{1} \cap L_{2}\right)=2 c_{\mathrm{SM}}\left(\mathbb{P}^{1}\right)-c_{\mathrm{SM}}\left(\mathbb{P}^{0}\right)=2\left[\mathbb{P}^{1}\right]+3\left[\mathbb{P}^{0}\right] .$	$c_{\mathrm{SM}}(L_{1})+c_{\mathrm{SM}}(L_{2})-c_{\mathrm{SM}}(L_{1} \cap L_{2})=2 c_{\mathrm{SM}}(\mathbb{P}^{1})-c_{\mathrm{SM}}(\mathbb{P}^{0})=2[\mathbb{P}^{1}]+3[\mathbb{P}^{0}].$
cardona-qft-methods_EQ0249	118	1.000 N|+0	`S(\phi)=\int L(\phi) d^{D} x`	$S(\phi)=\int L(\phi) d^{D} x$	$S(\phi)=\int L(\phi) d^{D} x$
cardona-qft-methods_EQ0251	118	1.000 N|+1	`G(\phi)=\frac{1}{N!} \int_{\sum_{i} p_{i}=0} \hat{\phi} \left(p_{1}\right) \cdots \hat{\phi}\left(p_{N}\right) U \left(G\left(p_{1}, \ldots, p_{N}\right)\right) d p_{1} \cdots d p_{N}`	$G(\phi)=\frac{1}{N!} \int_{\sum_{i} p_{i}=0} \hat{\phi} \left(p_{1}\right) \cdots \hat{\phi}\left(p_{N}\right) U \left(G\left(p_{1}, \ldots, p_{N}\right)\right) d p_{1} \cdots d p_{N}$	$G(\phi)=\frac{1}{N!} \int_{\sum_{i} p_{i}=0} \hat{\phi}(p_{1}) \cdots \hat{\phi}(p_{N}) U(G(p_{1}, \ldots, p_{N})) d p_{1} \cdots d p_{N} \quad .$
cardona-qft-methods_EQ0252	118	1.000 N|+1	`U\left(G\left(p_{1}, \ldots, p_{N}\right)\right)=\int I_{G}\left(k_{1}, \ldots, k_{\ell}, p_{1}, \ldots, p_{N}\right) d^{D} k_{1} \cdots d^{D} k_{\ell}`	$U\left(G\left(p_{1}, \ldots, p_{N}\right)\right)=\int I_{G}\left(k_{1}, \ldots, k_{\ell}, p_{1}, \ldots, p_{N}\right) d^{D} k_{1} \cdots d^{D} k_{\ell}$	$U(G(p_{1}, \ldots, p_{N}))=\int I_{G}(k_{1}, \ldots, k_{\ell}, p_{1}, \ldots, p_{N}) d^{D} k_{1} \cdots d^{D} k_{\ell} \quad ,$
cardona-qft-methods_EQ0253 (2)	119	1.000 N|+0	`U(G)=\frac{\Gamma(n-D \ell / 2)}{(4 \pi)^{\ell D / 2}} \int_{\sigma_{n}} \frac{P_{G}(\underline{t}, p)^{-n+D \ell / 2} \omega_{n}}{\Psi_{G}(\underline{t})^{-n+D( \ell+1) / 2}} .`	$U(G)=\frac{\Gamma(n-D \ell / 2)}{(4 \pi)^{\ell D / 2}} \int_{\sigma_{n}} \frac{P_{G}(\underline{t}, p)^{-n+D \ell / 2} \omega_{n}}{\Psi_{G}(\underline{t})^{-n+D(\ell+1) / 2}} .$	$U(G)=\frac{\Gamma(n-D \ell / 2)}{(4 \pi)^{\ell D / 2}} \int_{\sigma_{n}} \frac{P_{G}(\underline{t}, p)^{-n+D \ell / 2} \omega_{n}}{\Psi_{G}(\underline{t})^{-n+D(\ell+1) / 2}} \quad .$
cardona-qft-methods_EQ0255	119	1.000 W|+0	`U(G)=\frac{\Gamma(2-D / 2)}{(4 \pi)^{D / 2}} \int_{ \sigma_{2}} \frac{\left(p^{2} t_{1} t_{2}\right)^{-2+D / 2} \omega_{2}}{\left(t_{1}+t_{2}\right)^{-2+D}}`	$U(G)=\frac{\Gamma(2-D / 2)}{(4 \pi)^{D / 2}} \int_{\sigma_{2}} \frac{\left(p^{2} t_{1} t_{2}\right)^{-2+D / 2} \omega_{2}}{\left(t_{1}+t_{2}\right)^{-2+D}}$	$U(G)=\frac{\Gamma(2-D / 2)}{(4 \pi)^{D / 2}} \int_{\sigma_{2}} \frac{(p^{2} t_{1} t_{2})^{-2+D / 2} \omega_{2}}{(t_{1}+t_{2})^{-2+D}}$
cardona-qft-methods_EQ0256	120	1.000 N|+0	`\int_{\text {cycle } \sigma} \frac{\text { diff. form }}{ \Psi_{G}(t)^{M}}`	$\int_{\text {cycle } \sigma} \frac{\text { diff. form }}{\Psi_{G}(t)^{M}}$	$\int_{\text {cycle } \sigma} \frac{\text { diff. form }}{\Psi_{G}(t)^{M}}$
cardona-qft-methods_EQ0257	120	1.000 N|+0	`\zeta\left(n_{1}, \ldots, n_{r}\right)=\sum_{0<k_{1}< \cdots<k_{r}} \frac{1}{k_{1}^{n_{1}} \cdots k_{r}^{n_{r}}}`	$\zeta\left(n_{1}, \ldots, n_{r}\right)=\sum_{0<k_{1}<\cdots<k_{r}} \frac{1}{k_{1}^{n_{1}} \cdots k_{r}^{n_{r}}}$	$\zeta(n_{1}, \ldots, n_{r})=\sum_{0<k_{1}<\cdots<k_{r}} \frac{1}{k_{1}^{n_{1}} \cdots k_{r}^{n_{r}}}$
cardona-qft-methods_EQ0258	121	1.000 N|+0	`\Psi_{G}\left(t_{1}, \ldots, t_{n}\right):=\sum_{T} \prod_{e \notin T} t_{e}`	$\Psi_{G}\left(t_{1}, \ldots, t_{n}\right):=\sum_{T} \prod_{e \notin T} t_{e}$	$\Psi_{G}(t_{1}, \ldots, t_{n}):=\sum_{T} \prod_{e \notin T} t_{e}$
cardona-qft-methods_EQ0259	122	1.000 K|+0	`\Psi_{G}(\underline{t})=t_{2} t_{3}+t_{1} t_{3}+t_{1} t_{2}=t_{1} t_{2} t_{3}\left(\frac{1}{t_{1}}+ \frac{1}{t_{2}}+\frac{1}{t_{3}}\right)`	$\Psi_{G}(\underline{t})=t_{2} t_{3}+t_{1} t_{3}+t_{1} t_{2}=t_{1} t_{2} t_{3}\left(\frac{1}{t_{1}}+\frac{1}{t_{2}}+\frac{1}{t_{3}}\right)$	$\Psi_{G}(\underline{t})=t_{2} t_{3}+t_{1} t_{3}+t_{1} t_{2}=t_{1} t_{2} t_{3}\left(\frac{1}{t_{1}}+\frac{1}{t_{2}}+\frac{1}{t_{3}}\right)$
cardona-qft-methods_EQ0260	122	1.000 N|+0	`\Psi_{G}(\underline{t})=t_{1} \cdots t_{n}\left( \frac{1}{t_{1}}+\cdots+\frac{1}{t_{n}}\right)`	$\Psi_{G}(\underline{t})=t_{1} \cdots t_{n}\left(\frac{1}{t_{1}}+\cdots+\frac{1}{t_{n}}\right)$	$\Psi_{G}(\underline{t})=t_{1} \cdots t_{n}\left(\frac{1}{t_{1}}+\cdots+\frac{1}{t_{n}}\right)$

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Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0266	127	1.000 N|+1	`\Psi_{G}\left(t_{1}, \ldots, t_{n_{1}}, u_{1}, \ldots, u_{n_{2}}\right) \neq 0`	$\Psi_{G}\left(t_{1}, \ldots, t_{n_{1}}, u_{1}, \ldots, u_{n_{2}}\right) \neq 0$	$\Psi_{G}(t_{1}, \ldots, t_{n_{1}}, u_{1}, \ldots, u_{n_{2}}) \neq 0 \quad ,$
cardona-qft-methods_ EQ0269	129	1.000 K|+0	`c_{\mathrm{SM}}\left(\mathbb{1}_{\hat{X}}\right)=a_{0} \left[\mathbb{P}^{0}\right]+a_{1}\left[\mathbb{P}^{1} \right]+\cdots+a_{n}\left[\mathbb{P}^{n}\right] .`	$c_{\mathrm{SM}}\left(\mathbb{1}_{\hat{X}}\right)=a_{0}\left[\mathbb{P}^{0}\right]+a_{1}\left[\mathbb{P}^{1}\right]+\cdots+a_{n}\left[\mathbb{P}^{n}\right] .$	$c_{\mathrm{SM}}(\mathbb{1}_{\hat{X}})=a_{0}[\mathbb{P}^{0}]+a_{1}[\mathbb{P}^{1}]+\cdots+a_{n}[\mathbb{P}^{n}] \quad .$
cardona-qft-methods_ EQ0270	130	1.000 N|+0	`\begin{aligned} c_{\mathrm{SM}}\left(\mathbb{1}_{ \mathbb{A}^{n}}\right) & =c_{\mathrm{SM}}\left(\mathbb{1}_{\mathbb{P}^{n}}-\mathbb{1}_{\mathbb{P}^{n-1}} \right) \\ & =\left((1+H)^{n+1}-(1+H)^{n} H\right) \cap\left[\mathbb{P}^{n}\right]=(1+H)^{n} \cap\left[\mathbb{P}^{n}\right] \\ & =\sum_{i=0}^{n}\binom{n}{i} \left[\mathbb{P}^{i}\right] \rightsquigarrow F_{ \mathbb{A}^{n}}(T)=\sum_{i=0}^{n}\binom{n}{i} T^{i} . \end{aligned}`	$\begin{aligned} c_{\mathrm{SM}}\left(\mathbb{1}_{\mathbb{A}^{n}}\right) & =c_{\mathrm{SM}}\left(\mathbb{1}_{\mathbb{P}^{n}}-\mathbb{1}_{\mathbb{P}^{n-1}}\right) \\ & =\left((1+H)^{n+1}-(1+H)^{n} H\right) \cap\left[\mathbb{P}^{n}\right]=(1+H)^{n} \cap\left[\mathbb{P}^{n}\right] \\ & =\sum_{i=0}^{n}\binom{n}{i}\left[\mathbb{P}^{i}\right] \rightsquigarrow F_{\mathbb{A}^{n}}(T)=\sum_{i=0}^{n}\binom{n}{i} T^{i} . \end{aligned}$	$\begin{aligned} c_{\mathrm{SM}}(\mathbb{1}_{\mathbb{A}^{n}}) & =c_{\mathrm{SM}}(\mathbb{1}_{\mathbb{P}^{n}}-\mathbb{1}_{\mathbb{P}^{n-1}}) \\ & =((1+H)^{n+1}-(1+H)^{n} H) \cap[\mathbb{P}^{n}]=(1+H)^{n} \cap[\mathbb{P}^{n}] \\ & =\sum_{i=0}^{n}\binom{n}{i}[\mathbb{P}^{i}] \rightsquigarrow F_{\mathbb{A}^{n}}(T)=\sum_{i=0}^{n}\binom{n}{i} T^{i} \quad . \end{aligned}$
cardona-qft-methods_ EQ0271	130	1.000 W|+0	`\begin{aligned} C_{G}(T) & =F_{\mathbb{A}^{n} \backslash \mathbb{A}^{n-1}}(T)=(1+T)^{n}-(1+T)^{n-1} \\ & =(1+T-1)(1+T)^{n-1}=T(1+T)^{n-1} \end{aligned}`	$\begin{aligned} C_{G}(T) & =F_{\mathbb{A}^{n} \backslash \mathbb{A}^{n-1}}(T)=(1+T)^{n}-(1+T)^{n-1} \\ & =(1+T-1)(1+T)^{n-1}=T(1+T)^{n-1} \end{aligned}$	$\begin{aligned} C_{G}(T) & =F_{\mathbb{A}^{n} \smallsetminus \mathbb{A}^{n-1}}(T)=(1+T)^{n}-(1+T)^{n-1} \\ & =(1+T-1)(1+T)^{n-1}=T(1+T)^{n-1} \end{aligned}$
cardona-qft-methods_ EQ0273	132	1.000 N|+0	`T_{G}(x, y)=T_{G / e}(x, y)+T_{G \backslash e}(x, y)`	$T_{G}(x, y)=T_{G / e}(x, y)+T_{G \backslash e}(x, y)$	$T_{G}(x, y)=T_{G / e}(x, y)+T_{G \smallsetminus e}(x, y)$
cardona-qft-methods_ EQ0274	133	1.000 N|+0	`\gamma^{b_{0}(G)} \alpha^{\# V(G)-b_{0}(G)} \beta^{b_{1}(G)} T_{G}\left(\frac{\gamma x}{\alpha}, \frac{y}{\beta}\right)`	$\gamma^{b_{0}(G)} \alpha^{\# V(G)-b_{0}(G)} \beta^{b_{1}(G)} T_{G}\left(\frac{\gamma x}{\alpha}, \frac{y}{\beta}\right)$	$\gamma^{b_{0}(G)} \alpha^{\# V(G)-b_{0}(G)} \beta^{b_{1}(G)} T_{G}\left(\frac{\gamma x}{\alpha}, \frac{y}{\beta}\right)$
cardona-qft-methods_ EQ0275	134	1.000 N|+0	`T_{G_{1} \cup G_{2}}(x, y)=T_{G_{1}}(x, y) \cdot T_{G_{2}}(x, y)`	$T_{G_{1} \cup G_{2}}(x, y)=T_{G_{1}}(x, y) \cdot T_{G_{2}}(x, y)$	$T_{G_{1} \cup G_{2}}(x, y)=T_{G_{1}}(x, y) \cdot T_{G_{2}}(x, y)$
cardona-qft-methods_ EQ0276	134	1.000 W|+0	`\begin{aligned} T_{G_{2 e}}(x, y) & =T_{G}(x, y)+y T_{G / e}(x, y) \\ & =T_{G \backslash e}(x, y)+(y+1) T_{G / e}(x, y) \end{aligned}`	$\begin{aligned} T_{G_{2 e}}(x, y) & =T_{G}(x, y)+y T_{G / e}(x, y) \\ & =T_{G \backslash e}(x, y)+(y+1) T_{G / e}(x, y) \end{aligned}$	$\begin{aligned} T_{G_{2 e}}(x, y) & =T_{G}(x, y)+y T_{G / e}(x, y) \\ & =T_{G \smallsetminus e}(x, y)+(y+1) T_{G / e}(x, y) \end{aligned}$
cardona-qft-methods_ EQ0277	134	1.000 N|+1	`\sum_{m \geq 0} T_{G_{m e}}(x, y) \frac{s^{m}}{m!}=e^{s} \left(T_{G \backslash e}(x, y)+\frac{e^{(y-1) s}-1}{y-1} T_{G / e}(x, y)\right)`	$\sum_{m \geq 0} T_{G_{m e}}(x, y) \frac{s^{m}}{m!}=e^{s}\left(T_{G \backslash e}(x, y)+\frac{e^{(y-1) s}-1}{y-1} T_{G / e}(x, y)\right)$	$\sum_{m \geq 0} T_{G_{m e}}(x, y) \frac{s^{m}}{m!}=e^{s}\left(T_{G \smallsetminus e}(x, y)+\frac{e^{(y-1) s}-1}{y-1} T_{G / e}(x, y)\right) \quad ,$
cardona-qft-methods_ EQ0280	136	1.000 K|+0	`\chi\left(Y_{G_{m e}}\right)=(-1)^{m-1}\left(\chi \left(Y_{G}\right)-\chi\left(Y_{G / e}\right)\right),`	$\chi\left(Y_{G_{m e}}\right)=(-1)^{m-1}\left(\chi\left(Y_{G}\right)-\chi\left(Y_{G / e}\right)\right),$	$\chi(Y_{G_{m e}})=(-1)^{m-1}(\chi(Y_{G})-\chi(Y_{G / e})) \quad ,$
cardona-qft-methods_ EQ0281 (1)	137	1.000 N|+1	`\Psi_{G}=t_{e} \Psi_{G \backslash e}+\Psi_{G / e}`	$\Psi_{G}=t_{e} \Psi_{G \backslash e}+\Psi_{G / e}$	$\Psi_{G}=t_{e} \Psi_{G \smallsetminus e}+\Psi_{G / e} \quad .$
cardona-qft-methods_ EQ0285	137	1.000 N|+1	`C_{G}(T)=C_{G \backslash e, G / e}(T)+(T-1) C_{G \backslash e}(T)`	$C_{G}(T)=C_{G \backslash e, G / e}(T)+(T-1) C_{G \backslash e}(T)$	$C_{G}(T)=C_{G \smallsetminus e, G / e}(T)+(T-1) C_{G \smallsetminus e}(T) \quad .$
cardona-qft-methods_ EQ0288	138	1.000 K|+0	`C_{G}(T)=(2 T-1) T(1+T)^{2}-T(T-1)(1+T)^{2}+T(1+T)=T(1+T) \left(1+T+T^{2}\right)`	$C_{G}(T)=(2 T-1) T(1+T)^{2}-T(T-1)(1+T)^{2}+T(1+T)=T(1+T)\left(1+T+T^{2}\right)$	$C_{G}(T)=(2 T-1) T(1+T)^{2}-T(T-1)(1+T)^{2}+T(1+T)=T(1+T)(1+T+T^{2})$
cardona-qft-methods_ EQ0291	139	1.000 N|+0	`\Psi_{G}=t_{e} \Psi_{G \backslash e}+\Psi_{G / e}`	$\Psi_{G}=t_{e} \Psi_{G \backslash e}+\Psi_{G / e}$	$\Psi_{G}=t_{e} \Psi_{G \smallsetminus e}+\Psi_{G / e}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

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cardona-qft-methods_ E00293	142	■ 1.000 ● N|+0	`\mathbb{U}(G) \equiv-\mathbb{U}(G \backslash e) \quad \bmod (\mathbb{L})`	$\mathbb{U}(G) \equiv-\mathbb{U}(G \backslash e) \quad \bmod (\mathbb{L})$	$\mathbb{U}(G) \equiv -\mathbb{U}(G \setminus e) \mod (\mathbb{L})$
cardona-qft-methods_ E00294	142	■ 1.000 ● N|+0	`\mathbb{U}(G)=(\mathbb{L}-1) \mathbb{U}(G \backslash e) \equiv-\mathbb{U}(G \backslash e) \quad \bmod ( \mathbb{L}) ;`	$\mathbb{U}(G)=(\mathbb{L}-1) \mathbb{U}(G \backslash e) \equiv-\mathbb{U}(G \backslash e) \quad \bmod (\mathbb{L}) ;$	$\mathbb{U}(G) = (\mathbb{L}-1)\mathbb{U}(G \setminus e) \equiv -\mathbb{U}(G \setminus e) \mod (\mathbb{L});$
cardona-qft-methods_ E00295	142	■ 1.000 ● N|+0	`\mathbb{U}(G)=\mathbb{L} \mathbb{U}(G \backslash e) \equiv 0 \quad \bmod (\mathbb{L})`	$\mathbb{U}(G)=\mathbb{L} \mathbb{U}(G \backslash e) \equiv 0 \quad \bmod (\mathbb{L})$	$\mathbb{U}(G) = \mathbb{L}\mathbb{U}(G \setminus e) \equiv 0 \mod (\mathbb{L})$
cardona-qft-methods_ E00297 (1)	150	■ 1.000 ● K|+0	`\mathbf{A}_{\mu}(x) \rightarrow \mathbf{A}_{\mu}^{\prime}(x)=g^{-1}(x)\left[\mathbf{A}_{\mu}(x)+\partial_{\mu}\right] g(x),`	$\mathbf{A}_{\mu}(x) \rightarrow \mathbf{A}_{\mu}^{\prime}(x)=g^{-1}(x)\left[\mathbf{A}_{\mu}(x)+\partial_{\mu}\right] g(x),$	$\mathbf{A}_{\mu}(x) \rightarrow \mathbf{A}'_{\mu}(x) = g^{-1}(x)[\mathbf{A}_{\mu}(x) + \partial_{\mu}]g(x),$
cardona-qft-methods_ E00298 (2)	150	■ 1.000 ● K|+0	`\mathcal{C}\left(\mathbf{A}^{\prime}\right)=\mathcal{C}(\mathbf{A})+d \Omega,`	$\mathcal{C}\left(\mathbf{A}^{\prime}\right)=\mathcal{C}(\mathbf{A})+d \Omega,$	$\mathcal{C}(\mathbf{A}') = \mathcal{C}(\mathbf{A}) + d\Omega,$
cardona-qft-methods_ E00299 (3)	150	■ 1.000 ● N|+1	`L(q, \dot{q}) d t \rightarrow L\left(q^{\prime}, \dot{q}^{\prime}\right) d t^{\prime}=L(q, \dot{q}) d t+d \Omega(q, t)`	$L(q, \dot{q}) d t \rightarrow L\left(q^{\prime}, \dot{q}^{\prime}\right) d t^{\prime}=L(q, \dot{q}) d t+d \Omega(q, t)$	$L(q,\dot{q})dt \rightarrow L(q',\dot{q}')dt' = L(q,\dot{q})dt + d\Omega(q,t).$
cardona-qft-methods_ E00300 (4)	150	■ 1.000 ● N|+1	`\int \mathbf{A}_{\mu}(x) j^{\mu}(x) d^{n} x`	$\int \mathbf{A}_{\mu}(x) j^{\mu}(x) d^{n} x$	$\int \mathbf{A}_{\mu}(x) j^{\mu}(x) d^{n}x.$
cardona-qft-methods_ E00304 (8)	152	■ 1.000 ● N|+1	`\mathbf{A}(x) \xrightarrow{g} \mathbf{A}^{\prime}(x)=g^{-1}(x) \mathbf{A}(x) g(x)+g^{-1}(x) d g(x) .`	$\mathbf{A}(x) \xrightarrow{g} \mathbf{A}^{\prime}(x)=g^{-1}(x) \mathbf{A}(x) g(x)+g^{-1}(x) d g(x) .$	$\mathbf{A}(x) \xrightarrow{g} \mathbf{A}'(x) = g^{-1}(x)\mathbf{A}(x)g(x) + g^{-1}(x)dg(x).$
cardona-qft-methods_ E00305 (9)	152	■ 1.000 ● N|+0	`\mathbf{F} \xrightarrow{g} \mathbf{F}^{\prime}=g \mathbf{F} g^{-1} .`	$\mathbf{F} \xrightarrow{g} \mathbf{F}^{\prime}=g \mathbf{F} g^{-1} .$	$\mathbf{F} \xrightarrow{g} \mathbf{F}' = g\mathbf{F}g^{-1}.$
cardona-qft-methods_ E00306 (10)	152	■ 1.000 ● K|+0	`\operatorname{Tr}\left(\mathbf{F}^{k}\right),`	$\operatorname{Tr}\left(\mathbf{F}^{k}\right),$	$Tr(\mathbf{F}^{k}),$
cardona-qft-methods_ E00307 (11)	154	■ 1.000 ● N|+0	`\delta_{\text {gauge }} P_{2 k}=d\left(\delta_{\text {gauge }} \mathcal{C}_{2 k-1}\right),`	$\delta_{\text {gauge }} P_{2 k}=d\left(\delta_{\text {gauge }} \mathcal{C}_{2 k-1}\right),$	$\delta_{gauge}P_{2k} = d(\delta_{gauge}\mathcal{C}_{2k-1}),$
cardona-qft-methods_ E00309 (13)	154	■ 1.000 ● N|+2	`\delta_{\text {gauge }} \mathcal{C}_{2 k-1}=d \Omega`	$\delta_{\text {gauge }} \mathcal{C}_{2 k-1}=d \Omega$	$\delta_{gauge}\mathcal{C}_{2k-1} = d\Omega.$ ■
cardona-qft-methods_ E00310 (14)	154	■ 1.000 ● N|+1	`\delta_{\text {gauge }} I[A]=\int_{M^{2 n-1}} \delta_{\text {gauge }} \mathcal{C}_{2 n-1}=\int_{M^{2 n-1}} d \Omega`	$\delta_{\text {gauge }} I[A]=\int_{M^{2 n-1}} \delta_{\text {gauge }} \mathcal{C}_{2 n-1}=\int_{M^{2 n-1}} d \Omega$	$\delta_{gauge}I[A] = \int_{M^{2n-1}} \delta_{gauge}\mathcal{C}_{2n-1} = \int_{M^{2n-1}} d\Omega,$
cardona-qft-methods_ E00311 (15)	155	■ 1.000 ● N|+0	`x^{\mu} \rightarrow x^{\prime \mu}=x^{\mu}+\xi^{\mu}(x) .`	$x^{\mu} \rightarrow x^{\prime \mu}=x^{\mu}+\xi^{\mu}(x) .$	$x^{\mu} \rightarrow x'^{\mu} = x^{\mu} + \xi^{\mu}(x).$
cardona-qft-methods_ E00312 (16)	155	■ 1.000 ● N|+0	`v^{\prime \mu}\left(x^{\prime}\right)=L_{\nu}^{\mu}(x) v^{\nu}(x), \text { where } L_{\nu}^{\mu}(x)=\frac{ \partial x^{\prime \mu}}{\partial x^{\nu}}=\delta_{\nu}^{ \mu}+\frac{\partial \xi^{\mu}}{\partial x^{\nu}} .`	$v^{\prime \mu}\left(x^{\prime}\right)=L_{\nu}^{\mu}(x) v^{\nu}(x), \text { where } L_{\nu}^{\mu}(x)=\frac{ \partial x^{\prime \mu}}{\partial x^{\nu}}=\delta_{\nu}^{ \mu}+\frac{\partial \xi^{\mu}}{\partial x^{\nu}} .$	$v'^{\mu}(x') = L^{\mu}_{\nu}(x)v^{\nu}(x),\ \text{where } L^{\mu}_{\nu}(x) = \frac{\partial x'^{\mu}}{\partial x^{\nu}} = \delta^{\mu}_{\nu} + \frac{\partial \xi^{\mu}}{\partial x^{\nu}}.$
cardona-qft-methods_ E00313 (17)	155	■ 1.000 ● W|+0	`\begin{aligned} v^{\prime \mu}(x) & =L_{\nu}^{\mu}(x) v^{ \nu}(x)-\xi^{\lambda}(x) \partial_{\lambda} v^{\mu}(x) \\ & =v^{\mu}(x)+\frac{\partial \xi^{\mu}}{\partial x^{\nu}} v^{\nu}(x)-\xi^{\lambda}(x) \partial_{\lambda} v^{\mu}(x) \end{aligned}`	$\begin{aligned} v^{\prime \mu}(x) & =L_{\nu}^{\mu}(x) v^{ \nu}(x)-\xi^{\lambda}(x) \partial_{\lambda} v^{\mu}(x) \\ & =v^{\mu}(x)+\frac{\partial \xi^{\mu}}{\partial x^{\nu}} v^{\nu}(x)-\xi^{\lambda}(x) \partial_{\lambda} v^{\mu}(x) \end{aligned}$	$\begin{aligned} v'^{\mu}(x) &= L^{\mu}_{\nu}(x)v^{\nu}(x) - \xi^{\lambda}(x)\partial_{\lambda}v^{\mu}(x) \\ &= v^{\mu}(x) + \frac{\partial \xi^{\mu}}{\partial x^{\nu}} v^{\nu}(x) - \xi^{\lambda}(x)\partial_{\lambda}v^{\mu}(x) \end{aligned}$
cardona-qft-methods_ E00317 (3)	159	■ 1.000 ● N|+0	`g_{\mu \nu}(x) \equiv \eta_{a b} e_{\mu}^{a}(x) e_{ \nu}^{b}(x),`	$g_{\mu \nu}(x) \equiv \eta_{a b} e_{\mu}^{a}(x) e_{ \nu}^{b}(x),$	$g_{\mu\nu}(x) \equiv \eta_{ab}\, e^{a}_{\mu}(x)\, e^{b}_{\nu}(x)\,,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

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cardona-qft-methods_ EQ0318 (4)	160	1.000 N|+0	`e_{\mu}^{a}(x) \longrightarrow e_{\mu}^{\prime a}(x)= \Lambda_{b}^{a}(x) e_{\mu}^{b}(x),`	$e_{\mu}^{a}(x) \longrightarrow e_{\mu}^{\prime a}(x)= \Lambda_{b}^{a}(x) e_{\mu}^{b}(x),$	$e_{\mu}^{a}(x) \longrightarrow e_{\mu}^{\prime a}(x)= \Lambda_{b}^{a}(x) e_{\mu}^{b}(x),$
cardona-qft-methods_ EQ0319 (5)	160	1.000 K|+0	`\Lambda_{c}^{a}(x) \Lambda_{d}^{b}(x) \eta_{a b}=\eta_{c d},`	$\Lambda_{c}^{a}(x) \Lambda_{d}^{b}(x) \eta_{a b}=\eta_{c d},$	$\Lambda_{c}^{a}(x) \Lambda_{d}^{b}(x) \eta_{a b}=\eta_{c d},$
cardona-qft-methods_ EQ0322 (8)	161	1.000 N|+0	`\omega_{b \mu}^{a}(x) \xrightarrow{\Lambda} \omega_{b \mu}^{\prime a}(x)=\Lambda_{c}^{a}(x) \Lambda_{b}^{d}(x) \omega_{d \mu}^{c}(x)+\Lambda_{c}^{a}(x) \partial_{\mu} \Lambda_{b}^{c}(x) .`	$\omega_{b \mu}^{a}(x) \xrightarrow{\Lambda} \omega_{b \mu}^{\prime a}(x)=\Lambda_{c}^{a}(x) \Lambda_{b}^{d}(x) \omega_{d \mu}^{c}(x)+\Lambda_{c}^{a}(x) \partial_{\mu} \Lambda_{b}^{c}(x) .$	$\omega^{a}{}_{b\mu}(x) \xrightarrow{\Lambda} \omega^{\prime a}{}_{b\mu}(x)=\Lambda_{c}^{a}(x) \Lambda_{b}^{d}(x) \omega^{c}{}_{d\mu}(x)+\Lambda_{c}^{a}(x) \partial_{\mu} \Lambda_{b}^{c}(x).$
cardona-qft-methods_ EQ0327 (14)	162	1.000 K|+0	`d\left(d \alpha_{p}\right)=d^{2} \alpha_{p}=0 .`	$d\left(d \alpha_{p}\right)=d^{2} \alpha_{p}=0 .$	$d(d\alpha_{p})=d^{2} \alpha_{p}=0.$
cardona-qft-methods_ EQ0328 (15)	162	1.000 W|+0	`\begin{aligned} D^{2} \phi^{a} & =D\left[d \phi^{a}+ \omega_{b}^{a} \phi^{b}\right] \\ & =d\left[d \phi^{a}+ \omega_{b}^{a} \phi^{b}\right]+\omega_{b}^{a}\left[d \phi^{b}+\omega_{c}^{b} \phi^{c}\right] \\ & =\left[d \omega_{b}^{a}+\omega_{c}^{a} \wedge \omega_{b}^{c} \right] \phi^{b} . \end{aligned}`	$\begin{aligned} D^{2} \phi^{a} & =D\left[d \phi^{a}+ \omega_{b}^{a} \phi^{b}\right] \\ & =d\left[d \phi^{a}+ \omega_{b}^{a} \phi^{b}\right]+\omega_{b}^{a}\left[d \phi^{b}+\omega_{c}^{b} \phi^{c}\right] \\ & =\left[d \omega_{b}^{a}+\omega_{c}^{a} \wedge \omega_{b}^{c} \right] \phi^{b} . \end{aligned}$	$\begin{aligned} D^{2} \phi^{a} & =D[d \phi^{a}+ \omega^{a}{}_{b} \phi^{b}] \\ & =d[d \phi^{a}+ \omega^{a}{}_{b} \phi^{b}]+\omega^{a}{}_{b}[d \phi^{b}+\omega^{b}{}_{c} \phi^{c}] \\ & =[d \omega^{a}{}_{b}+\omega^{a}{}_{c} \wedge \omega^{c}{}_{b}] \phi^{b} . \end{aligned}$
cardona-qft-methods_ EQ0331 (18)	163	1.000 N|+1	`T^{a}=d e^{a}+\omega_{b}^{a} \wedge e^{b}`	$T^{a}=d e^{a}+\omega_{b}^{a} \wedge e^{b}$	$T^{a}=d e^{a}+\omega^{a}{}_{b} \wedge e^{b},$
cardona-qft-methods_ EQ0340 (8)	166	1.000 N|+0	`\mathbf{P}_{D}=\mathbf{P}_{4 k_{1}} \mathbf{P}_{4 k_{2}} \cdots \mathbf{P}_{4 k_{r}},`	$\mathbf{P}_{D}=\mathbf{P}_{4 k_{1}} \mathbf{P}_{4 k_{2}} \cdots \mathbf{P}_{4 k_{r}},$	$\mathbf{P}_{D}=\mathbf{P}_{4 k_{1}} \mathbf{P}_{4 k_{2}} \cdots \mathbf{P}_{4 k_{r}},$
cardona-qft-methods_ EQ0341 (9)	166	1.000 N|+0	`I_{D}[e, \omega]=\kappa \int_{M} \sum_{p=0}^{[D / 2]} \alpha_{p} L_{p}^{D}`	$I_{D}[e, \omega]=\kappa \int_{M} \sum_{p=0}^{[D / 2]} \alpha_{p} L_{p}^{D}$	$I_{D}[e, \omega]=\kappa \int_{M} \sum_{p=0}^{[D / 2]} \alpha_{p} L_{p}^{D}$
cardona-qft-methods_ EQ0342 (10)	166	1.000 N|+2	`L_{p}^{D}=\epsilon_{a_{1} \cdots a_{D}} R^{a_{1} a_{2}} \cdots R^{a_{2 p-1} a_{2 p}} e^{a_{2 p+1}} \cdots e^{a_{D}}`	$L_{p}^{D}=\epsilon_{a_{1} \cdots a_{D}} R^{a_{1} a_{2}} \cdots R^{a_{2 p-1} a_{2 p}} e^{a_{2 p+1}} \cdots e^{a_{D}}$	$L_{p}^{D}=\epsilon_{a_{1} \cdots a_{D}} R^{a_{1} a_{2}} \cdots R^{a_{2 p-1} a_{2 p}} e^{a_{2 p+1}} \cdots e^{a_{D}}.$ ■
cardona-qft-methods_ EQ0344 (12)	167	1.000 C|+5	`\begin{aligned} I_{4} & =\kappa \int_{M} \epsilon_{a b c d}\left[\alpha_{2} R^{a b} R^{c d}+\alpha_{1} R^{a b} e^{c} e^{d}+\alpha_{0} e^{a} e^{b} e^{c} e^{d}\right] \\ & =-\kappa \int_{M} \sqrt{|g|}\left[\alpha_{2} \left(R^{\alpha \beta \gamma \delta} R_{\alpha \beta \gamma \delta}-4 R^{\alpha \beta} R_{\alpha \beta}+R^{2} \right)+2 \alpha_{1} R+24 \alpha_{0}\right] d^{4} x \\ & =-\kappa \alpha_{2} \cdot \mathbf{E}_{4}-2 \alpha_{1} \int_{M} \sqrt{|g|} R d^{4} x-24 \kappa \alpha_{0} \cdot V_{4} \end{aligned}`	$\begin{aligned} I_{4} & =\kappa \int_{M} \epsilon_{a b c d}\left[\alpha_{2} R^{a b} R^{c d}+\alpha_{1} R^{a b} e^{c} e^{d}+\alpha_{0} e^{a} e^{b} e^{c} e^{d}\right] \\ & =-\kappa \int_{M} \sqrt{\vert g\vert}\left[\alpha_{2} \left(R^{\alpha \beta \gamma \delta} R_{\alpha \beta \gamma \delta}-4 R^{\alpha \beta} R_{\alpha \beta}+R^{2} \right)+2 \alpha_{1} R+24 \alpha_{0}\right] d^{4} x \\ & =-\kappa \alpha_{2} \cdot \mathbf{E}_{4}-2 \alpha_{1} \int_{M} \sqrt{\vert g\vert} R d^{4} x-24 \kappa \alpha_{0} \cdot V_{4} \end{aligned}$	$\begin{aligned} I_{4} & =\kappa \int_{M} \epsilon_{a b c d}\left[\alpha_{2} R^{a b} R^{c d}+\alpha_{1} R^{a b} e^{c} e^{d}+\alpha_{0} e^{a} e^{b} e^{c} e^{d}\right] \\ & =-\kappa \int_{M} \sqrt{\vert g\vert}\left[\alpha_{2} \left(R^{\alpha \beta \gamma \delta} R_{\alpha \beta \gamma \delta}-4 R^{\alpha \beta} R_{\alpha \beta}+R^{2} \right)+2 \alpha_{1} R+24 \alpha_{0}\right] d^{4} x \\ & =-\kappa \alpha_{2} \cdot \mathbf{E}_{4}-2 \alpha_{1} \int_{M} \sqrt{\vert g\vert} R d^{4} x-24 \kappa \alpha_{0} \cdot V_{4}. \end{aligned} \qquad (12)$
cardona-qft-methods_ EQ0345 (13)	167	1.000 N|+0	`\epsilon_{a b c d e} R^{a b} R^{c d} e^{e}=\sqrt{|g|} \left[R^{\alpha \beta \gamma \delta} R_{\alpha \beta \gamma \delta}-4 R^{\alpha \beta} R_{\alpha \beta}+R^{2} \right] d^{5} x .`	$\epsilon_{a b c d e} R^{a b} R^{c d} e^{e}=\sqrt{\vert g\vert} \left[R^{\alpha \beta \gamma \delta} R_{\alpha \beta \gamma \delta}-4 R^{\alpha \beta} R_{\alpha \beta}+R^{2} \right] d^{5} x .$	$\epsilon_{a b c d e} R^{a b} R^{c d} e^{e}=\sqrt{\vert g\vert} \left[R^{\alpha \beta \gamma \delta} R_{\alpha \beta \gamma \delta}-4 R^{\alpha \beta} R_{\alpha \beta}+R^{2} \right] d^{5} x.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0346 (14)	168	■ 1.000 ● N|+1	`\delta I_{D}=\int\left[\delta e^{a} \mathcal{E}_{a}+ \delta \omega^{a b} \mathcal{E}_{a b}\right]=0`	$\delta I_{D}=\int\left[\delta e^{a} \mathcal{E}_{a}+\delta \omega^{a b} \mathcal{E}_{a b}\right]=0$	$\delta I_D=\int[\delta e^a\mathcal{E}_a+\delta\omega^{ab}\mathcal{E}_{ab}]=0,$
cardona-qft-methods_ EQ0347 (15)	168	■ 1.000 ● N|+1	`\mathcal{E}_{a}=\sum_{p=0}^{\left[\frac{D-1}{2}\right]} \alpha_{p}(D-2 p) \mathcal{E}_{a}^{(p)}=0`	$\mathcal{E}_{a}=\sum_{p=0}^{\left[\frac{D-1}{2}\right]} \alpha_{p}(D-2 p) \mathcal{E}_{a}^{(p)}=0$	$\mathcal{E}_a=\sum_{p=0}^{[\frac{D-1}{2}]}\alpha_p(D-2p)\mathcal{E}_a^{(p)}=0,$
cardona-qft-methods_ EQ0348 (16)	168	■ 1.000 ● N|+1	`\mathcal{E}_{a b}=\sum_{p=1}^{\left[\frac{D-1}{2}\right]} \alpha_{p} p(D-2 p) \mathcal{E}_{a b}^{(p)}=0`	$\mathcal{E}_{a b}=\sum_{p=1}^{\left[\frac{D-1}{2}\right]} \alpha_{p} p(D-2 p) \mathcal{E}_{a b}^{(p)}=0$	$\mathcal{E}_{ab}=\sum_{p=1}^{[\frac{D-1}{2}]}\alpha_p p(D-2p)\mathcal{E}_{ab}^{(p)}=0,$
cardona-qft-methods_ EQ0350 (19)	168	■ 1.000 ● N|+0	`T^{a}=d e^{a}+\omega^{a}{ }_{b} e^{b}=0,`	$T^{a}=d e^{a}+\omega^{a}{ }_{b} e^{b}=0,$	$T^a=de^a+\omega^a{}_be^b=0,$
cardona-qft-methods_ EQ0352	169	■ 1.000 ● N|+1	`I_{D}[g]=\int_{M} d^{D} x \sqrt{g}\left[\alpha_{0}^{\prime}+\alpha_{1}^{\prime} R+\alpha_{2}^{\prime}\left(R^{\alpha \beta \gamma \delta} R_{\alpha \beta \gamma \delta}-4 R^{\alpha \beta} R_{\alpha \beta}+R^{2}\right)+\cdots\right]`	$I_{D}[g]=\int_{M} d^{D} x \sqrt{g}\left[\alpha_{0}^{\prime}+\alpha_{1}^{\prime} R+\alpha_{2}^{\prime}\left(R^{\alpha \beta \gamma \delta} R_{\alpha \beta \gamma \delta}-4 R^{\alpha \beta} R_{\alpha \beta}+R^{2}\right)+\cdots\right]$	$I_D[g]=\int_M d^Dx\sqrt{g}\,[\alpha_0'+\alpha_1'R+\alpha_2'(R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}-4R^{\alpha\beta}R_{\alpha\beta}+R^2)+\cdots].$
cardona-qft-methods_ EQ0353 (21)	170	■ 1.000 ● N|+0	`\zeta_{0}=e^{a} T_{a},`	$\zeta_{0}=e^{a} T_{a},$	$\zeta_0=e^aT_a,$
cardona-qft-methods_ EQ0361 (1)	173	■ 1.000 ● N|+0	`L_{3}=\epsilon_{a b c} R^{a b} e^{c} .`	$L_{3}=\epsilon_{a b c} R^{a b} e^{c} .$	$L_3=\epsilon_{abc}R^{ab}e^c.$
cardona-qft-methods_ EQ0365 (4)	174	■ 1.000 ● K|+0	`\delta \omega^{a b}=0,`	$\delta \omega^{a b}=0,$	$\delta\omega^{ab}=0,$
cardona-qft-methods_ EQ0366 (5)	174	■ 1.000 ● K|+0	`\delta L_{3}=d\left[\epsilon_{a b c} R^{a b} \lambda^{c}\right],`	$\delta L_{3}=d\left[\epsilon_{a b c} R^{a b} \lambda^{c}\right],$	$\delta L_3=d[\epsilon_{abc}R^{ab}\lambda^c],$
cardona-qft-methods_ EQ0367 (6)	175	■ 1.000 ● N|+1	`L_{3}^{A d S}=\epsilon_{a b c}\left(R^{a b} e^{c} \pm \frac{1}{3 l^{2}} e^{a} e^{b} e^{c}\right)`	$L_{3}^{A d S}=\epsilon_{a b c}\left(R^{a b} e^{c} \pm \frac{1}{3 l^{2}} e^{a} e^{b} e^{c}\right)$	$L_3^{AdS}=\epsilon_{abc}(R^{ab}e^c\pm\frac{1}{3l^2}e^ae^be^c),$
cardona-qft-methods_ EQ0369 (9)	175	■ 1.000 ● N|+0	`\begin{aligned} \delta\left[\begin{array}{cc} \omega^{a b} & e^{a} l^{-l} \\ -e^{b} l^{-l} & 0 \end{array} \right]= & d\left[\begin{array}{cc} \lambda^{a b} & \lambda^{a} l^{-l} \\ -\lambda^{b} l^{-l} & 0 \end{array} \right] \\ & +\left[\begin{array}{cc} \omega_{c}^{a} & e^{a} l^{-l} \\ -e_{c} l^{-l} & 0 \end{array}\right] \left[\begin{array}{cc} \lambda^{c b} & \lambda^{c} l^{-1} \\ \pm \lambda^{b} l^{-1} & 0 \end{array}\right] \\ & -\left[\begin{array}{cc} \lambda^{a c} & \lambda^{a} l^{-1} \\ -\lambda^{c} l^{-1} & 0 \end{array}\right] \left[\begin{array}{cc} \omega_{c}^{b} & e_{c} l^{-1} \\ \pm e^{b} l^{-1} & 0 \end{array}\right] . \end{aligned}`	$\begin{aligned} \delta\left[\begin{array}{cc} \omega^{a b} & e^{a} l^{-l} \\ -e^{b} l^{-l} & 0 \end{array} \right]= & d\left[\begin{array}{cc} \lambda^{a b} & \lambda^{a} l^{-l} \\ -\lambda^{b} l^{-l} & 0 \end{array} \right] \\ & +\left[\begin{array}{cc} \omega_{c}^{a} & e^{a} l^{-l} \\ -e_{c} l^{-l} & 0 \end{array}\right] \left[\begin{array}{cc} \lambda^{c b} & \lambda^{c} l^{-1} \\ \pm \lambda^{b} l^{-1} & 0 \end{array}\right] \\ & -\left[\begin{array}{cc} \lambda^{a c} & \lambda^{a} l^{-1} \\ -\lambda^{c} l^{-1} & 0 \end{array}\right] \left[\begin{array}{cc} \omega_{c}^{b} & e_{c} l^{-1} \\ \pm e^{b} l^{-1} & 0 \end{array}\right] . \end{aligned}$	$\begin{aligned} \delta\begin{bmatrix} \omega^{ab} & e^al^{-l} \\ -e^bl^{-l} & 0 \end{bmatrix} = & d\begin{bmatrix} \lambda^{ab} & \lambda^al^{-l} \\ -\lambda^bl^{-l} & 0 \end{bmatrix} \\ & +\begin{bmatrix} \omega^a_c & e^al^{-l} \\ -e_cl^{-l} & 0 \end{bmatrix}\begin{bmatrix} \lambda^{cb} & \lambda^cl^{-1} \\ \pm\lambda^bl^{-1} & 0 \end{bmatrix} \\ & -\begin{bmatrix} \lambda^{ac} & \lambda^al^{-1} \\ -\lambda^cl^{-1} & 0 \end{bmatrix}\begin{bmatrix} \omega_c{}^b & e_cl^{-1} \\ \pm e^bl^{-1} & 0 \end{bmatrix}. \end{aligned}$
cardona-qft-methods_ EQ0370 (11)	175	■ 1.000 ● N|+0	`\delta W^{A B}=d \Lambda^{A B}+W_{C}^{A} \Lambda^{C B}-\Lambda^{A C} W_{C}^{B},`	$\delta W^{A B}=d \Lambda^{A B}+W_{C}^{A} \Lambda^{C B}-\Lambda^{A C} W_{C}^{B},$	$\delta W^{AB}=d\Lambda^{AB}+W^A_C\Lambda^{CB}-\Lambda^{AC}W^B_C,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0371 (10)	175	■1.000 ● N|+0	`\begin{aligned} W^{A B} & =\left[\begin{array}{cc} \omega^{a b} & e^{a} l^{-1} \\ -e^{b} l^{-1} & 0 \end{array}\right] \\ \Lambda^{A B} & =\left[ \begin{array}{cc} \lambda^{a b} & \lambda^{a} l^{-1} \\ -\lambda^{b} l^{-1} & 0 \end{array}\right], \end{aligned}`	$\begin{aligned} W^{A B} & =\left[\begin{array}{cc} \omega^{a b} & e^{a} l^{-1} \\ -e^{b} l^{-1} & 0 \end{array}\right] \\ \Lambda^{A B} & =\left[\begin{array}{cc} \lambda^{a b} & \lambda^{a} l^{-1} \\ -\lambda^{b} l^{-1} & 0 \end{array}\right], \end{aligned}$	$W^{AB} = \left[\begin{array}{cc} \omega^{ab} & e^{a} l^{-1} \\ -e^{b} l^{-1} & 0 \end{array}\right]$ $\Lambda^{AB} = \left[\begin{array}{cc} \lambda^{ab} & \lambda^{a} l^{-1} \\ -\lambda^{b} l^{-1} & 0 \end{array}\right],$
cardona-qft-methods_ EQ0373 (13)	176	■1.000 ● K|+0	`D_{W} \Upsilon^{A B}=d \Upsilon^{A B}+W_{C}^{A} \Upsilon^{C B}+W_{C}^{B} \Upsilon^{A C}=0 .`	$D_{W} \Upsilon^{A B}=d \Upsilon^{A B}+W_{C}^{A} \Upsilon^{C B}+W_{C}^{B} \Upsilon^{A C}=0 .$	$D_{W} \Upsilon^{AB}=d \Upsilon^{AB}+W_{C}^{A} \Upsilon^{CB}+W_{C}^{B} \Upsilon^{AC}=0.$
cardona-qft-methods_ EQ0376 (17)	177	■1.000 ● K|+0	`L_{3}^{E H}=\epsilon_{a b c} R^{a b} e^{c},`	$L_{3}^{E H}=\epsilon_{a b c} R^{a b} e^{c},$	$L_{3}^{EH}=\epsilon_{abc} R^{ab} e^{c},$
cardona-qft-methods_ EQ0381 (22)	178	■1.000 ● K|+0	`L_{2 n+1}=\epsilon_{a_{1} \cdots a_{2 n+1}} R^{a_{1} a_{2}} \cdots R^{a_{2 n-1} a_{2 n}} e^{a_{2 n+1}},`	$L_{2 n+1}=\epsilon_{a_{1} \cdots a_{2 n+1}} R^{a_{1} a_{2}} \cdots R^{a_{2 n-1} a_{2 n}} e^{a_{2 n+1}},$	$L_{2n+1}=\epsilon_{a_{1} \cdots a_{2n+1}} R^{a_{1} a_{2}} \ldots R^{a_{2n-1} a_{2n}} e^{a_{2n+1}},$
cardona-qft-methods_ EQ0382	178	■1.000 ● K|+0	`F^{A B}=d W^{A B}+W_{C}^{A} W^{C B},`	$F^{A B}=d W^{A B}+W_{C}^{A} W^{C B},$	$F^{AB}=d W^{AB}+W_{C}^{A} W^{CB},$
cardona-qft-methods_ EQ0383 (23)	178	■1.000 ● N|+0	`F^{A B}=\left[\begin{array}{cc} R^{a b} \pm l^{-2} e^{a} e^{b} & l^{-1} T^{a} \\ -l^{-1} T^{b} & 0 \end{array}\right] .`	$F^{A B}=\left[\begin{array}{cc} R^{a b} \pm l^{-2} e^{a} e^{b} & l^{-1} T^{a} \\ -l^{-1} T^{b} & 0 \end{array}\right] .$	$F^{AB}=\left[\begin{array}{cc} R^{ab} \pm l^{-2} e^{a} e^{b} & l^{-1} T^{a} \\ -l^{-1} T^{b} & 0 \end{array}\right].$
cardona-qft-methods_ EQ0384 (24)	178	■1.000 ● K|+0	`\mathbf{E}_{4}=\epsilon_{A B C D} F^{A B} F^{C D} .`	$\mathbf{E}_{4}=\epsilon_{A B C D} F^{A B} F^{C D} .$	$\mathbf{E}_{4}=\epsilon_{ABCD} F^{AB} F^{CD}.$
cardona-qft-methods_ EQ0385 (25)	179	■1.000 ● W|+2	`\begin{aligned} \mathbf{E}_{4} & =4 \epsilon_{a b c} \left(R^{a b} \pm l^{-2} e^{a} e^{b}\right) l^{-1} T^{a} \\ & =\frac{4}{l} d\left[\epsilon_{a b c}\left(R^{a b} \pm \frac{1}{3 l^{2}} e^{a} e^{b}\right) e^{c}\right] \end{aligned}`	$\begin{aligned} \mathbf{E}_{4} & =4 \epsilon_{a b c} \left(R^{a b} \pm l^{-2} e^{a} e^{b}\right) l^{-1} T^{a} \\ & =\frac{4}{l} d\left[\epsilon_{a b c}\left(R^{a b} \pm \frac{1}{3 l^{2}} e^{a} e^{b}\right) e^{c}\right] \end{aligned}$	$\begin{aligned} \mathbf{E}_{4} & =4 \epsilon_{abc}(R^{ab} \pm l^{-2} e^{a} e^{b}) l^{-1} T^{a} \\ & =\frac{4}{l} d\left[\epsilon_{abc}\left(R^{ab} \pm \frac{1}{3 l^{2}} e^{a} e^{b}\right) e^{c}\right], \end{aligned}$
cardona-qft-methods_ EQ0387	179	■1.000 ● K|+0	`\delta\left(d L_{3}^{A d S}\right)=0 .`	$\delta\left(d L_{3}^{A d S}\right)=0 .$	$\delta\left(d L_{3}^{AdS}\right)=0.$
cardona-qft-methods_ EQ0390 (28)	180	■1.000 ● N|+1	`L_{2 n-1}^{(A) d S}=\sum_{p=0}^{n-1} \bar{\alpha}_{p} L_{p}^{2 n-1}`	$L_{2 n-1}^{(A) d S}=\sum_{p=0}^{n-1} \bar{\alpha}_{p} L_{p}^{2 n-1}$	$L_{2n-1}^{(A)dS}=\sum_{p=0}^{n-1} \bar{\alpha}_{p} L_{p}^{2n-1},$
cardona-qft-methods_ EQ0391 (29)	180	■1.000 ● N|+0	`\bar{\alpha}_{p}=\kappa \cdot \frac{( \pm 1)^{p+1} l^{2 p-D}}{(D-2 p)}\binom{n-1}{p}, p=1,2, \ldots, n-1= \frac{D-1}{2},`	$\bar{\alpha}_{p}=\kappa \cdot \frac{(\pm 1)^{p+1} l^{2 p-D}}{(D-2 p)}\binom{n-1}{p}, p=1,2, \ldots, n-1= \frac{D-1}{2},$	$\bar{\alpha}_{p}=\kappa \cdot \frac{(\pm 1)^{p+1} l^{2p-D}}{(D-2p)}\binom{n-1}{p},\ p=1,2,...,n-1=\frac{D-1}{2},$
cardona-qft-methods_ EQ0393 (31)	180	■1.000 ● N|+0	`L_{5}^{(A) d S}=\frac{\kappa}{l} \epsilon_{a b c d e} \left[e^{a} R^{b c} R^{d e} \pm \frac{2}{3 l^{2}} e^{a} e^{b} e^{c} R^{d e}+\frac{1}{5 l^{4}} e^{a} e^{b} e^{c} e^{d} e^{e}\right] .`	$L_{5}^{(A) d S}=\frac{\kappa}{l} \epsilon_{a b c d e} \left[e^{a} R^{b c} R^{d e} \pm \frac{2}{3 l^{2}} e^{a} e^{b} e^{c} R^{d e}+\frac{1}{5 l^{4}} e^{a} e^{b} e^{c} e^{d} e^{e}\right] .$	$L_{5}^{(A)dS}=\frac{\kappa}{l} \epsilon_{abcde}\left[e^{a} R^{bc} R^{de} \pm \frac{2}{3 l^{2}} e^{a} e^{b} e^{c} R^{de}+\frac{1}{5 l^{4}} e^{a} e^{b} e^{c} e^{d} e^{e}\right].$
cardona-qft-methods_ EQ0394	180	■1.000 ● K|+0	`\tilde{e}^{a}=l^{-1} e^{a},`	$\tilde{e}^{a}=l^{-1} e^{a},$	$\tilde{e}^{a}=l^{-1} e^{a},$
cardona-qft-methods_ EQ0396	181	■1.000 ● S|-1	`L_{3}^{(A) d S}=\epsilon_{a b c}\left(R^{a b} \pm \frac{e^{a} e^{b}}{3 l^{2}}\right) e^{c}`	$L_{3}^{(A) d S}=\epsilon_{a b c}\left(R^{a b} \pm \frac{e^{a} e^{b}}{3 l^{2}}\right) e^{c}$	$L_{3}^{(A)dS}=\epsilon_{abc}(R^{ab} \pm \frac{e^{a} e^{b}}{3 l^{2}}) e^{c}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0400 (33)	183	1.000 N|+0	\kappa\left[\int_{\Omega} \mathbf{E}_{2 n}-\int_{\Omega^{\prime}} \mathbf{E}_{2 n}\right]=2 n \pi \hbar .	$\kappa\left[\int_{\Omega} \mathbf{E}_{2 n}-\int_{\Omega^{\prime}} \mathbf{E}_{2 n}\right]=2 n \pi \hbar .$	$\kappa\left[\int_{\Omega} \mathbf{E}_{2 n}-\int_{\Omega^{\prime}} \mathbf{E}_{2 n}\right]=2 n \pi \hbar .$
cardona-qft-methods_ EQ0401 (34)	183	1.000 K|+0	\kappa\left[\int_{\Omega} \mathbf{E}_{2 n}+\int_{\widetilde{\Omega}^{\prime}} \mathbf{E}_{2 n}\right]=2 n \pi \hbar .	$\kappa\left[\int_{\Omega} \mathbf{E}_{2 n}+\int_{\widetilde{\Omega}^{\prime}} \mathbf{E}_{2 n}\right]=2 n \pi \hbar .$	$\kappa\left[\int_{\Omega} \mathbf{E}_{2 n}+\int_{\widetilde{\Omega}^{\prime}} \mathbf{E}_{2 n}\right]=2 n \pi \hbar .$
cardona-qft-methods_ EQ0402	183	1.000 K|+0	\kappa=n h,	$\kappa=n h,$	$\kappa=n h,$
cardona-qft-methods_ EQ0403 (35)	183	1.000 N|+0	\epsilon_{a b_{1} b_{2} \cdots b_{2 n}}\left[R^{b_{1} b_{2}}+\frac{1}{l^{2}} e^{b_{1}} e^{b_{2}}\right] \cdots \left[R^{b_{2 n-1} b_{2 n}}+\frac{1}{l^{2}} e^{b_{2 n-1}} e^{b_{2 n}}\right]=0	$\epsilon_{a b_{1} b_{2} \cdots b_{2 n}}\left[R^{b_{1} b_{2}}+\frac{1}{l^{2}} e^{b_{1}} e^{b_{2}}\right] \cdots \left[R^{b_{2 n-1} b_{2 n}}+\frac{1}{l^{2}} e^{b_{2 n-1}} e^{b_{2 n}}\right]=0$	$\epsilon_{a b_{1} b_{2} \cdots b_{2 n}}\left[R^{b_{1} b_{2}}+\frac{1}{l^{2}} e^{b_{1}} e^{b_{2}}\right] \cdots \left[R^{b_{2 n-1} b_{2 n}}+\frac{1}{l^{2}} e^{b_{2 n-1}} e^{b_{2 n}}\right]=0$
cardona-qft-methods_ EQ0404 (36)	184	1.000 N|+1	\epsilon_{a b b_{2} \cdots b_{2 n}}\left[R^{b_{2} b_{3}}+\frac{1}{l^{2}} e^{b_{2}} e^{b_{3}}\right] \cdots \left[R^{b_{2 n-2} b_{2 n-1}}+\frac{1}{l^{2}} e^{b_{2 n-2}} e^{b_{2 n-1}}\right] T^{b_{2 n}}=0	$\epsilon_{a b b_{2} \cdots b_{2 n}}\left[R^{b_{2} b_{3}}+\frac{1}{l^{2}} e^{b_{2}} e^{b_{3}}\right] \cdots \left[R^{b_{2 n-2} b_{2 n-1}}+\frac{1}{l^{2}} e^{b_{2 n-2}} e^{b_{2 n-1}}\right] T^{b_{2 n}}=0$	$\epsilon_{a b b_{2} \cdots b_{2 n}}\left[R^{b_{2} b_{3}}+\frac{1}{l^{2}} e^{b_{2}} e^{b_{3}}\right] \cdots \left[R^{b_{2 n-2} b_{2 n-1}}+\frac{1}{l^{2}} e^{b_{2 n-2}} e^{b_{2 n-1}}\right] T^{b_{2 n}}=0$
cardona-qft-methods_ EQ0406	185	1.000 N|-1	\operatorname{Pfaff}[M]=\epsilon_{a_{1} b_{1} a_{2} b_{2} \cdots a_{n} b_{n}} M^{a_{1} b_{1}} \cdots M^{a_{n} b_{n}}=\sqrt{\operatorname{det}[M]} .	$\operatorname{Pfaff}[M]=\epsilon_{a_{1} b_{1} a_{2} b_{2} \cdots a_{n} b_{n}} M^{a_{1} b_{1}} \cdots M^{a_{n} b_{n}}=\sqrt{\operatorname{det}[M]} .$	$\operatorname{Pfaff}[M]=\epsilon_{a_{1} b_{1} a_{2} b_{2} \cdots a_{n} b_{n}} M^{a_{1} b_{1}} \cdots M^{a_{n} b_{n}}=\sqrt{det[M]} .$
cardona-qft-methods_ EQ0407 (38)	186	1.000 N|+1	\delta I[\phi]=\int_{M} \delta \phi^{r} \frac{\delta L}{\delta \phi^{r}}+\int_{\partial M} \Theta(\phi, \delta \phi) .	$\delta I[\phi]=\int_{M} \delta \phi^{r} \frac{\delta L}{\delta \phi^{r}}+\int_{\partial M} \Theta(\phi, \delta \phi) .$	$\delta I[\phi]=\int_{M} \delta \phi^{r} \frac{\delta L}{\delta \phi^{r}}+\int_{\partial M} \Theta(\phi, \delta \phi) .$
cardona-qft-methods_ EQ0410 (40)	187	1.000 N|+1	I_{2 n-1}=\int_{M} L_{2 n-1}^{(A) d S}+\int_{\partial M} B_{2 n}	$I_{2 n-1}=\int_{M} L_{2 n-1}^{(A) d S}+\int_{\partial M} B_{2 n}$	$I_{2 n-1}=\int_{M} L_{2 n-1}^{(A) d S}+\int_{\partial M} B_{2 n},$
cardona-qft-methods_ EQ0416 (1)	190	1.000 N|+1	I_{\mathrm{Int}}=\int_{\Gamma} A_{\mu} j^{\mu} d^{4} x	$I_{\mathrm{Int}}=\int_{\Gamma} A_{\mu} j^{\mu} d^{4} x$	$I_{\mathrm{Int}}=\int_{\Gamma} A_{\mu} j^{\mu} d^{4} x\,,$
cardona-qft-methods_ EQ0417 (2)	191	1.000 N|+1	I_{\mathrm{Int}}=q \int_{\Gamma} A_{\mu}(z) d z^{\mu}=q \int_{\Gamma} A	$I_{\mathrm{Int}}=q \int_{\Gamma} A_{\mu}(z) d z^{\mu}=q \int_{\Gamma} A$	$I_{\mathrm{Int}}=q \int_{\Gamma} A_{\mu}(z) d z^{\mu}=q \int_{\Gamma} A\,,$
cardona-qft-methods_ EQ0418 (3)	192	1.000 N|+1	I[\mathbf{A} ; \mathbf{j}]=\int\left\langle\mathbf{j}_{2 p} \wedge \mathfrak{C}_{2 p+1}(\mathbf{A})\right\rangle,	$I[\mathbf{A} ; \mathbf{j}]=\int\left\langle\mathbf{j}_{2 p} \wedge \mathfrak{C}_{2 p+1}(\mathbf{A})\right\rangle,$	$I[\mathbf{A} ; \mathbf{j}]=\int\langle\mathbf{j}_{2 p} \wedge \mathfrak{C}_{2 p+1}(\mathbf{A})\rangle,$
cardona-qft-methods_ EQ0419 (4)	192	1.000 N|+1	\mathbf{j}_{2 p}=q j^{a_{1} a_{2} \cdots a_{s}} \mathbf{K}_{a_{1}} \mathbf{K}_{a_{2}} \cdots \mathbf{K}_{a_{s}} \delta(\Gamma) d x^{\alpha_{1}} \wedge d x^{\alpha_{2}} \wedge \cdots d x^{\alpha_{D-2 p-1}},	$\mathbf{j}_{2 p}=q j^{a_{1} a_{2} \cdots a_{s}} \mathbf{K}_{a_{1}} \mathbf{K}_{a_{2}} \cdots \mathbf{K}_{a_{s}} \delta(\Gamma) d x^{\alpha_{1}} \wedge d x^{\alpha_{2}} \wedge \cdots d x^{\alpha_{D-2 p-1}},$	$\mathbf{j}_{2 p}=q j^{a_{1} a_{2} \cdots a_{s}} \mathbf{K}_{a_{1}} \mathbf{K}_{a_{2}} \cdots \mathbf{K}_{a_{s}} \delta(\Gamma) d x^{\alpha_{1}} \wedge d x^{\alpha_{2}} \wedge \cdots d x^{\alpha_{D-2 p-1}},$
cardona-qft-methods_ EQ0420 (5)	192	1.000 K|+0	D \mathbf{j}_{2 p}=d \mathbf{j}_{2 p}+\left[\mathbf{A}, \mathbf{j}_{2 p}\right]=0,	$D \mathbf{j}_{2 p}=d \mathbf{j}_{2 p}+\left[\mathbf{A}, \mathbf{j}_{2 p}\right]=0,$	$D \mathbf{j}_{2 p}=d \mathbf{j}_{2 p}+\left[\mathbf{A}, \mathbf{j}_{2 p}\right]=0,$
cardona-qft-methods_ EQ0421 (6)	193	1.000 N|+0	I[A]=\frac{1}{2} \int_{M}(\operatorname{Tr}[F \wedge * F]-\Theta(x) \operatorname{Tr}[F \wedge F]),	$I[A]=\frac{1}{2} \int_{M}(\operatorname{Tr}[F \wedge * F]-\Theta(x) \operatorname{Tr}[F \wedge F]),$	$I[A]=\frac{1}{2} \int_{M}(Tr[F \wedge * F]-\Theta(x) Tr[F \wedge F])\,,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0423	202	1.000 N|+1	S_{E H}\left(g_{\mu \nu}\right)=\frac{1}{16 \pi G} \int_{M} R \sqrt{g} d^{4} x	$S_{EH}\left(g_{\mu\nu}\right)=\frac{1}{16\pi G}\int_{M} R\sqrt{g}d^{4}x$	$S_{EH}(g_{\mu\nu})=\frac{1}{16\pi G}\int_{M} R\sqrt{g}\, d^{4}x,$
cardona-qft-methods_EQ0425	203	1.000 N|+0	\int_{M}\left(\frac{1}{2}\left|D_{\mu} \mathbf{H} \right|^{2}-\mu_{0}^{2}|\mathbf{H}|^{2}-\xi_{0} R| \mathbf{H}|^{2}+\lambda_{0}|\mathbf{H}|^{4}\right) \sqrt{g} d^{4} x .	$\int_{M}\left(\frac{1}{2}\left\lvert D_{\mu}\mathbf{H}\right\rvert^{2}-\mu_{0}^{2}\lvert\mathbf{H}\rvert^{2}-\xi_{0}R\lvert\mathbf{H}\rvert^{2}+\lambda_{0}\lvert\mathbf{H}\rvert^{4}\right)\sqrt{g}d^{4}x.$	$\int_{M}\left(\frac{1}{2}\lvert D_{\mu}\mathbf{H}\rvert^{2}-\mu_{0}^{2}\lvert\mathbf{H}\rvert^{2}-\xi_{0}\,R\,\lvert\mathbf{H}\rvert^{2}+\lambda_{0}\lvert\mathbf{H}\rvert^{4}\right)\sqrt{g}\,d^{4}x.$
cardona-qft-methods_EQ0426	204	1.000 K|+0	[\gamma, a]=0, \forall a \in \mathcal{A}, \quad \text { and } \quad D \gamma=-\gamma D .	$[\gamma, a]=0, \forall a \in \mathcal{A}, \quad \text{ and } \quad D\gamma=-\gamma D.$	$[\gamma, a]=0,\ \forall a \in \mathcal{A}, \quad \text{and} \quad D\gamma=-\gamma D.$
cardona-qft-methods_EQ0428	205	1.000 N|+0	\left[[D, a], b^{0}\right]=0 \quad \forall a, b \in \mathcal{A} .	$\left[[D, a], b^{0}\right]=0 \quad \forall a, b \in \mathcal{A}.$	$[[D, a], b^{0}]=0 \qquad \forall\, a, b \in \mathcal{A}.$
cardona-qft-methods_EQ0430	207	1.000 N|+0	\mathcal{A}_{L R}=\mathbb{C} \oplus \mathbb{H}_{L} \oplus \mathbb{H}_{R} \oplus M_{3}(\mathbb{C}) .	$\mathcal{A}_{LR}=\mathbb{C}\oplus\mathbb{H}_{L}\oplus\mathbb{H}_{R}\oplus M_{3}(\mathbb{C}).$	$\mathcal{A}_{LR}=\mathbb{C}\oplus\mathbb{H}_{L}\oplus\mathbb{H}_{R}\oplus M_{3}(\mathbb{C}).$
cardona-qft-methods_EQ0431	208	1.000 N|+0	\operatorname{Ad}(u) \xi=u \xi u^{*} \quad \forall \xi \in \mathcal{M} .	$\operatorname{Ad}(u)\xi=u\xi u^{*} \quad \forall\xi\in\mathcal{M}.$	$\operatorname{Ad}(u)\xi=u\xi u^{*} \qquad \forall\xi\in\mathcal{M}.$
cardona-qft-methods_EQ0433	208	1.000 N|+0	\mathcal{M}^{0}=\{\bar{\xi} ; \xi \in \mathcal{M}\}, \quad a \bar{\xi} b=\overline{b^{*} \xi a^{*}}	$\mathcal{M}^{0}=\{\bar{\xi};\xi\in\mathcal{M}\}, \quad a\bar{\xi}b=\overline{b^{*}\xi a^{*}}$	$\mathcal{M}^{0}=\{\bar{\xi}\ ;\ \xi\in\mathcal{M}\}\,, \quad a\,\bar{\xi}\,b=\overline{b^{*}\xi\, a^{*}}$
cardona-qft-methods_EQ0440	210	1.000 K|+0	\mathrm{SU}\left(\mathcal{A}_{F}\right) \sim \mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3),	$\mathrm{SU}\left(\mathcal{A}_{F}\right)\sim\mathrm{U}(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3),$	$\mathrm{SU}(\mathcal{A}_{F})\sim\mathrm{U}(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3),$
cardona-qft-methods_EQ0442	211	1.000 N|+0	\mathbb{C}_{F} \subset \mathcal{A}_{F}, \quad \mathbb{C}_{F}=\{(\lambda, \lambda, 0), \lambda \in \mathbb{C}\} .	$\mathbb{C}_{F}\subset\mathcal{A}_{F}, \quad \mathbb{C}_{F}=\{(\lambda,\lambda,0),\lambda\in\mathbb{C}\}.$	$\mathbb{C}_{F}\subset\mathcal{A}_{F}\,, \quad \mathbb{C}_{F}=\{(\lambda,\lambda,0)\,,\lambda\in\mathbb{C}\}.$
cardona-qft-methods_EQ0446	211	1.000 K|+0	T: E_{R}=\uparrow_{R} \otimes \mathbf{1}^{0} \rightarrow J_{F} E_{R} .	$T: E_{R}=\uparrow_{R}\otimes\mathbf{1}^{0}\rightarrow J_{F}E_{R}.$	$T: E_{R}=\uparrow_{R}\otimes\mathbf{1}^{0}\rightarrow J_{F}\,E_{R}.$
cardona-qft-methods_EQ0447	211	1.000 N|+0	Y_{(\downarrow 3)}^{\prime}=W_{1} Y_{(\downarrow 3)} W_{3}^{*}, \quad Y_{(\uparrow 3)}^{\prime}=W_{2} Y_{(\uparrow 3)} W_{3}^{*}	$Y_{(\downarrow 3)}^{\prime}=W_{1}Y_{(\downarrow 3)}W_{3}^{*}, \quad Y_{(\uparrow 3)}^{\prime}=W_{2}Y_{(\uparrow 3)}W_{3}^{*}$	$Y_{(\downarrow 3)}^{\prime}=W_{1}\,Y_{(\downarrow 3)}\,W_{3}^{*}, \quad Y_{(\uparrow 3)}^{\prime}=W_{2}\,Y_{(\uparrow 3)}\,W_{3}^{*}$
cardona-qft-methods_EQ0450	212	1.000 W|+0	\begin{gathered} Y_{(\uparrow 3)}=\delta_{(\uparrow 3)} \quad Y_{(\downarrow 3)}=U_{C K M} \delta_{(\downarrow 3)} U_{C K M}^{*} \\\\ Y_{(\uparrow 1)}=U_{P M N S}^{*} \delta_{(\uparrow 1)} U_{P M N S} \quad Y_{(\downarrow 1)}=\delta_{(\downarrow 1)} \end{gathered}	$\begin{gathered} Y_{(\uparrow 3)}=\delta_{(\uparrow 3)} \quad Y_{(\downarrow 3)}=U_{CKM}\delta_{(\downarrow 3)}U_{CKM}^{*} \\ Y_{(\uparrow 1)}=U_{PMNS}^{*}\delta_{(\uparrow 1)}U_{PMNS} \quad Y_{(\downarrow 1)}=\delta_{(\downarrow 1)} \end{gathered}$	$\begin{gathered} Y_{(\uparrow 3)}=\delta_{(\uparrow 3)} \qquad Y_{(\downarrow 3)}=U_{CKM}\,\delta_{(\downarrow 3)}\,U_{CKM}^{*} \\ Y_{(\uparrow 1)}=U_{PMNS}^{*}\,\delta_{(\uparrow 1)}\,U_{PMNS} \qquad Y_{(\downarrow 1)}=\delta_{(\downarrow 1)} \end{gathered}$
cardona-qft-methods_EQ0451	212	1.000 W|+2	U=\left(\begin{array}{ccc} c_{1} & -s_{1} c_{3} & -s_{1} s_{3} \\\\ s_{1} c_{2} & c_{1} c_{2} c_{3}-s_{2} s_{3} e_{\delta} & c_{1} c_{2} s_{3}+s_{2} c_{3} e_{\delta} \\\\ s_{1} s_{2} & c_{1} s_{2} c_{3}+c_{2} s_{3} e_{\delta} & c_{1} s_{2} s_{3}-c_{2} c_{3} e_{\delta} \end{array}\right)	$U=\left(\begin{array}{ccc} c_{1} & -s_{1}c_{3} & -s_{1}s_{3} \\ s_{1}c_{2} & c_{1}c_{2}c_{3}-s_{2}s_{3}e_{\delta} & c_{1}c_{2}s_{3}+s_{2}c_{3}e_{\delta} \\ s_{1}s_{2} & c_{1}s_{2}c_{3}+c_{2}s_{3}e_{\delta} & c_{1}s_{2}s_{3}-c_{2}c_{3}e_{\delta} \end{array}\right)$	$U=\left(\begin{array}{ccc} c_{1} & -s_{1}c_{3} & -s_{1}s_{3} \\ s_{1}c_{2} & c_{1}c_{2}c_{3}-s_{2}s_{3}e_{\delta} & c_{1}c_{2}s_{3}+s_{2}c_{3}e_{\delta} \\ s_{1}s_{2} & c_{1}s_{2}c_{3}+c_{2}s_{3}e_{\delta} & c_{1}s_{2}s_{3}-c_{2}c_{3}e_{\delta} \end{array}\right)$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0453	214	■1.000 ● W|+0	\begin{gathered} \mathcal{A}=C^{\infty}(M) \otimes \mathcal{A}_{F}=C^{\infty}\left(M, \mathcal{A}_{F}\right) \\ \mathcal{H}=L^{2}(M, S) \otimes \mathcal{H}_{F}=L^{2} \left(M, S \otimes \mathcal{H}_{F}\right) \\ D=\not \partial_{M} \otimes 1+\gamma_{5} \otimes D_{F} \end{gathered}	$\begin{gathered} \mathcal{A}=C^{\infty}(M) \otimes \mathcal{A}_{F}=C^{\infty}\left(M, \mathcal{A}_{F}\right) \\ \mathcal{H}=L^{2}(M, S) \otimes \mathcal{H}_{F}=L^{2} \left(M, S \otimes \mathcal{H}_{F}\right) \\ D=\not \partial_{M} \otimes 1+\gamma_{5} \otimes D_{F} \end{gathered}$	$\mathcal{A}=C^{\infty}(M)\otimes\mathcal{A}_F=C^{\infty}(M,\mathcal{A}_F)$ $\mathcal{H}=L^2(M,S)\otimes\mathcal{H}_F=L^2(M,S\otimes\mathcal{H}_F)$ $D=\not\partial_M\otimes 1+\gamma_5\otimes D_F$
cardona-qft-methods_EQ0455	215	■1.000 ● N|+0	\xi b=b^{0} \xi, \quad \xi \in \mathcal{H}, \quad b \in \mathcal{A}	$\xi b=b^{0} \xi, \quad \xi \in \mathcal{H}, \quad b \in \mathcal{A}$	$\xi\, b=b^0\,\xi, \quad \xi\in\mathcal{H}, \quad b\in\mathcal{A}$
cardona-qft-methods_EQ0456	215	■1.000 ● N|+0	\xi \in \mathcal{H} \rightarrow \operatorname{Ad}(u) \xi=u \xi u^{*} \quad \xi \in \mathcal{H}	$\xi \in \mathcal{H} \rightarrow \operatorname{Ad}(u) \xi=u \xi u^{*} \quad \xi \in \mathcal{H}$	$\xi\in\mathcal{H}\rightarrow\mathrm{Ad}(u)\,\xi=u\,\xi\,u^* \quad \xi\in\mathcal{H}$
cardona-qft-methods_EQ0457	215	■1.000 ● K|+0	D \rightarrow D_{A}=D+A+\varepsilon^{\prime} J A J^{-1}	$D \rightarrow D_{A}=D+A+\varepsilon^{\prime} J A J^{-1}$	$D\rightarrow D_A=D+A+\varepsilon'\,J\,A\,J^{-1}$
cardona-qft-methods_EQ0458	215	■1.000 ● N|+0	A=\sum a_{j}\left[D, b_{j}\right], \quad a_{j}, b_{j} \in \mathcal{A} .	$A=\sum a_{j}\left[D, b_{j}\right], \quad a_{j}, b_{j} \in \mathcal{A} .$	$A=\sum a_j[D,b_j], \quad a_j,b_j\in\mathcal{A}.$
cardona-qft-methods_EQ0459	216	■1.000 ● K|+0	\mathrm{SU}\left(\mathcal{A}_{F}\right)=\left\{u \in \mathrm{U}\left(\mathcal{A}_{F}\right) \mid \operatorname{det}(\mathrm{u})=1\right\},	$\mathrm{SU}\left(\mathcal{A}_{F}\right)=\left\{u \in \mathrm{U}\left(\mathcal{A}_{F}\right) \mid \operatorname{det}(\mathrm{u})=1\right\},$	$\mathrm{SU}(\mathcal{A}_F)=\{u\in\mathrm{U}(\mathcal{A}_F)\mid\det(\mathrm{u})=1\},$
cardona-qft-methods_EQ0460	216	■1.000 ● K|+0	\mathrm{SU}\left(\mathcal{A}_{F}\right) \sim \mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3),	$\mathrm{SU}\left(\mathcal{A}_{F}\right) \sim \mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3),$	$\mathrm{SU}(\mathcal{A}_F)\sim\mathrm{U}(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3),$
cardona-qft-methods_EQ0462	216	■1.000 ● N|+0	\left(\begin{array}{cccc} \frac{4}{3} \Lambda+V & 0 & 0 & 0 \\ 0 & -\frac{2}{3} \Lambda+V & 0 & 0 \\ 0 & 0 & Q_{11}+\frac{1}{3} \Lambda+V & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22}+\frac{1}{3} \Lambda+V \end{array}\right)	$\left(\begin{array}{cccc} \frac{4}{3} \Lambda+V & 0 & 0 & 0 \\ 0 & -\frac{2}{3} \Lambda+V & 0 & 0 \\ 0 & 0 & Q_{11}+\frac{1}{3} \Lambda+V & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22}+\frac{1}{3} \Lambda+V \end{array}\right)$	$\left(\begin{array}{cccc} \frac{4}{3}\Lambda+V & 0 & 0 & 0 \\ 0 & -\frac{2}{3}\Lambda+V & 0 & 0 \\ 0 & 0 & Q_{11}+\frac{1}{3}\Lambda+V & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22}+\frac{1}{3}\Lambda+V \end{array}\right)$
cardona-qft-methods_EQ0463	217	■1.000 ● K|+0	\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & -2 \Lambda & 0 & 0 \\ 0 & 0 & Q_{11}-\Lambda & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22}-\Lambda \end{array}\right)	$\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & -2 \Lambda & 0 & 0 \\ 0 & 0 & Q_{11}-\Lambda & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22}-\Lambda \end{array}\right)$	$\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & -2\Lambda & 0 & 0 \\ 0 & 0 & Q_{11}-\Lambda & Q_{12} \\ 0 & 0 & Q_{21} & Q_{22}-\Lambda \end{array}\right)$
cardona-qft-methods_EQ0464	217	■1.000 ● N|+0	\left.\sum_{i} a_{i}\left[\gamma_{5} \otimes D_{F}, a_{i}^{\prime}\right](x)\right|_{\mathcal{H}_{f}}= \gamma_{5} \otimes\left(A_{q}^{(0,1)}+A_{\ell}^{(0,1)}\right)	$\left.\sum_{i} a_{i}\left[\gamma_{5} \otimes D_{F}, a_{i}^{\prime}\right](x)\right\|_{\mathcal{H}_{f}}= \gamma_{5} \otimes\left(A_{q}^{(0,1)}+A_{\ell}^{(0,1)}\right)$	$\sum_i a_i[\gamma_5\otimes D_F,a_i'](x)\|_{\mathcal{H}_f}=\gamma_5\otimes(A_q^{(0,1)}+A_\ell^{(0,1)})$
cardona-qft-methods_EQ0466	217	■1.000 ● N|+0	q=\left(\begin{array}{cc} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{array}\right) .	$q=\left(\begin{array}{cc} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{array}\right) .$	$q=\left(\begin{array}{cc} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{array}\right).$
cardona-qft-methods_EQ0468	217	■1.000 ● N|+1	-E=\frac{1}{4} s \otimes \mathrm{id}+\sum_{\mu<\nu} \gamma^{\mu} \gamma^{\nu} \otimes \mathbb{F}_{\mu \nu}-i \gamma_{5} \gamma^{\mu} \otimes \mathbb{M}\left(D_{\mu} \varphi\right)+1_{4} \otimes\left(D^{0,1}\right)^{2}	$-E=\frac{1}{4} s \otimes \mathrm{id}+\sum_{\mu<\nu} \gamma^{\mu} \gamma^{\nu} \otimes \mathbb{F}_{\mu \nu}-i \gamma_{5} \gamma^{\mu} \otimes \mathbb{M}\left(D_{\mu} \varphi\right)+1_{4} \otimes\left(D^{0,1}\right)^{2}$	$-E=\frac{1}{4}\,s\otimes\mathrm{id}+\sum_{\mu<\nu}\gamma^\mu\gamma^\nu\otimes\mathbb{F}_{\mu\nu}-i\,\gamma_5\,\gamma^\mu\otimes\mathbb{M}(D_\mu\varphi)+1_4\otimes(D^{0,1})^2,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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<tr><td>cardona-qft-methods_EQ0469</td><td>218</td><td>1.000 K|+0</td><td><code>D_{\mu} \varphi=\partial_{\mu} \varphi+\frac{i}{2} g_{2} W_{\mu}^{\alpha} \varphi \sigma^{\alpha}-\frac{i}{2} g_{1} B_{\mu} \varphi</code></td><td>$D_{\mu} \varphi=\partial_{\mu} \varphi+\frac{i}{2} g_{2} W_{\mu}^{\alpha} \varphi \sigma^{\alpha}-\frac{i}{2} g_{1} B_{\mu} \varphi$</td><td>$D_\mu\varphi = \partial_\mu\varphi + \frac{i}{2}g_2 W_\mu^\alpha\varphi\,\sigma^\alpha - \frac{i}{2}g_1 B_\mu\,\varphi$</td></tr>
<tr><td>cardona-qft-methods_EQ0470</td><td>218</td><td>1.000 N|-1</td><td><code>\operatorname{Tr}(f(D / \Lambda)),</code></td><td>$\operatorname{Tr}(f(D / \Lambda)),$</td><td>$\mathrm{Tr}(f(D/\Lambda)),$</td></tr>
<tr><td>cardona-qft-methods_EQ0472</td><td>218</td><td>1.000 N|-1</td><td><code>\operatorname{Tr}\left(e^{-t \Delta}\right) \sim \sum a_{\alpha} t^{\alpha} \quad(t \rightarrow 0)</code></td><td>$\operatorname{Tr}\left(e^{-t \Delta}\right) \sim \sum a_{\alpha} t^{\alpha} \quad(t \rightarrow 0)$</td><td>$\mathrm{Tr}(e^{-t\Delta}) \sim \sum a_\alpha\, t^\alpha \qquad (t \to 0)$</td></tr>
<tr><td>cardona-qft-methods_EQ0473</td><td>218</td><td>1.000 N|-1</td><td><code>\zeta_{D}(s)=\operatorname{Tr}\left(\Delta^{-s / 2} \right) .</code></td><td>$\zeta_{D}(s)=\operatorname{Tr}\left(\Delta^{-s / 2} \right) .$</td><td>$\zeta_D(s) = \mathrm{Tr}(\Delta^{-s/2}).$</td></tr>
<tr><td>cardona-qft-methods_EQ0474</td><td>218</td><td>1.000 K|+0</td><td><code>\operatorname{Res}_{s=-2 \alpha} \zeta_{D}(s)=\frac{2 a_{\alpha}}{\Gamma(-\alpha)} .</code></td><td>$\operatorname{Res}_{s=-2 \alpha} \zeta_{D}(s)=\frac{2 a_{\alpha}}{\Gamma(-\alpha)} .$</td><td>$\mathrm{Res}_{s=-2\alpha}\,\zeta_D(s) = \frac{2\,a_\alpha}{\Gamma(-\alpha)}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0475</td><td>218</td><td>1.000 N|+0</td><td><code>|D|^{-s}=\Delta^{-s / 2}=\frac{1}{\Gamma\left(\frac{s}{2} \right)} \int_{0}^{\infty} e^{-t \Delta} t^{s / 2-1} d t</code></td><td>$|D|^{-s}=\Delta^{-s / 2}=\frac{1}{\Gamma\left(\frac{s}{2} \right)} \int_{0}^{\infty} e^{-t \Delta} t^{s / 2-1} d t$</td><td>$|D|^{-s} = \Delta^{-s/2} = \frac{1}{\Gamma\left(\frac{s}{2}\right)} \int_0^\infty e^{-t\Delta}\, t^{s/2-1}\, dt$</td></tr>
<tr><td>cardona-qft-methods_EQ0476</td><td>219</td><td>1.000 N|+0</td><td><code>\frac{1}{\Gamma\left(\frac{s}{2}\right)} \sim \frac{s}{2} \quad \text { as } \quad s \rightarrow 0</code></td><td>$\frac{1}{\Gamma\left(\frac{s}{2}\right)} \sim \frac{s}{2} \quad \text { as } \quad s \rightarrow 0$</td><td>$\frac{1}{\Gamma\left(\frac{s}{2}\right)} \sim \frac{s}{2} \qquad \text{as} \quad s \to 0.$</td></tr>
<tr><td>cardona-qft-methods_EQ0477</td><td>219</td><td>1.000 N|-1</td><td><code>\int_{0}^{\infty} \operatorname{Tr}\left(e^{-t \Delta} \right) t^{s / 2-1} d t</code></td><td>$\int_{0}^{\infty} \operatorname{Tr}\left(e^{-t \Delta} \right) t^{s / 2-1} d t$</td><td>$\int_0^\infty \mathrm{Tr}(e^{-t\Delta})\, t^{s/2-1}\, dt$</td></tr>
<tr><td>cardona-qft-methods_EQ0479</td><td>219</td><td>1.000 N|+0</td><td><code>\frac{1}{2}\left\langle J \tilde{\xi}, D_{A} \tilde{\xi} \right\rangle</code></td><td>$\frac{1}{2}\left\langle J \tilde{\xi}, D_{A} \tilde{\xi} \right\rangle$</td><td>$\frac{1}{2}\langle J\tilde{\xi}, D_A\tilde{\xi}\rangle$</td></tr>
<tr><td>cardona-qft-methods_EQ0480</td><td>219</td><td>1.000 N|+0</td><td><code>\mathcal{H}_{c l}^{+}=\left\{\xi \in \mathcal{H}_{c l} \mid \gamma \xi=\xi\right\}</code></td><td>$\mathcal{H}_{c l}^{+}=\left\{\xi \in \mathcal{H}_{c l} \mid \gamma \xi=\xi\right\}$</td><td>$\mathcal{H}_{cl}^+ = \{\xi \in \mathcal{H}_{cl} \,|\, \gamma\xi = \xi\}$</td></tr>
<tr><td>cardona-qft-methods_EQ0482</td><td>220</td><td>1.000 N|+0</td><td><code>\frac{1}{2}\left\langle J \tilde{\xi}, D_{A} \tilde{\xi} \right\rangle</code></td><td>$\frac{1}{2}\left\langle J \tilde{\xi}, D_{A} \tilde{\xi} \right\rangle$</td><td>$\frac{1}{2}\langle J\tilde{\xi}, D_A\tilde{\xi}\rangle$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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● W|+0</td><td>\begin{aligned} S & =\frac{1}{\pi^{2}}\left(48 f_{4} \Lambda^{4}-f_{2} \Lambda^{2} \mathfrak{c}+ \frac{f_{0}}{4} \mathfrak{d}\right) \int \sqrt{g} d^{4} x \\ & +\frac{96 f_{2} \Lambda^{2}-f_{0} \mathfrak{c}}{24 \pi^{2}} \int R \sqrt{g} d^{4} x \\ & +\frac{f_{0}}{10 \pi^{2}} \int\left(\frac{11}{6} R^{*} R^{*}-3 C_{\mu \nu \rho \sigma} C^{\mu \nu \rho \sigma}\right) \sqrt{g} d^{4} x \\ & +\frac{\left(-2 \mathfrak{a} f_{2} \Lambda^{2}+\mathfrak{e} f_{0} \right)}{\pi^{2}} \int|\varphi|^{2} \sqrt{g} d^{4} x \\ & +\frac{f_{0} \mathfrak{a}}{2 \pi^{2}} \int\left|D_{\mu} \varphi\right|^{2} \sqrt{g} d^{4} x \\ & -\frac{f_{0} \mathfrak{a}}{12 \pi^{2}} \int R|\varphi|^{2} \sqrt{g} d^{4} x \\ & +\frac{f_{0} \mathfrak{b}}{2 \pi^{2}} \int|\varphi|^{4} \sqrt{g} d^{4} x \\ & +\frac{f_{0}}{2 \pi^{2}} \int\left(g_{3}^{2} G_{\mu \nu}^{i} G^{\mu \nu i}+g_{2}^{2} F_{\mu \nu}^{\alpha} F^{\mu \nu \alpha}+ \frac{5}{3} g_{1}^{2} B_{\mu \nu} B^{\mu \nu}\right) \sqrt{g} d^{4} x, \end{aligned}</td><td>$$\begin{aligned} S & =\frac{1}{\pi^{2}}\left(48 f_{4} \Lambda^{4}-f_{2} \Lambda^{2} \mathfrak{c}+ \frac{f_{0}}{4} \mathfrak{d}\right) \int \sqrt{g} d^{4} x \\ & +\frac{96 f_{2} \Lambda^{2}-f_{0} \mathfrak{c}}{24 \pi^{2}} \int R \sqrt{g} d^{4} x \\ & +\frac{f_{0}}{10 \pi^{2}} \int\left(\frac{11}{6} R^{*} R^{*}-3 C_{\mu \nu \rho \sigma} C^{\mu \nu \rho \sigma}\right) \sqrt{g} d^{4} x \\ & +\frac{\left(-2 \mathfrak{a} f_{2} \Lambda^{2}+\mathfrak{e} f_{0} \right)}{\pi^{2}} \int|\varphi|^{2} \sqrt{g} d^{4} x \\ & +\frac{f_{0} \mathfrak{a}}{2 \pi^{2}} \int\left|D_{\mu} \varphi\right|^{2} \sqrt{g} d^{4} x \\ & -\frac{f_{0} \mathfrak{a}}{12 \pi^{2}} \int R|\varphi|^{2} \sqrt{g} d^{4} x \\ & +\frac{f_{0} \mathfrak{b}}{2 \pi^{2}} \int|\varphi|^{4} \sqrt{g} d^{4} x \\ & +\frac{f_{0}}{2 \pi^{2}} \int\left(g_{3}^{2} G_{\mu \nu}^{i} G^{\mu \nu i}+g_{2}^{2} F_{\mu \nu}^{\alpha} F^{\mu \nu \alpha}+ \frac{5}{3} g_{1}^{2} B_{\mu \nu} B^{\mu \nu}\right) \sqrt{g} d^{4} x, \end{aligned}$$</td><td>$$\begin{aligned} S = & \frac{1}{\pi^{2}}(48\, f_{4}\, \Lambda^{4}-f_{2}\, \Lambda^{2}\, \mathfrak{c}+ \frac{f_{0}}{4}\, \mathfrak{d}) \int \sqrt{g}\, d^{4} x \\ & +\frac{96\, f_{2}\, \Lambda^{2}-f_{0}\, \mathfrak{c}}{24 \pi^{2}} \int R \sqrt{g}\, d^{4} x \\ & +\frac{f_{0}}{10\, \pi^{2}} \int (\frac{11}{6}\, R^{*} R^{*}-3\, C_{\mu \nu \rho \sigma}\, C^{\mu \nu \rho \sigma}) \sqrt{g}\, d^{4} x \\ & +\frac{(-2\, \mathfrak{a}\, f_{2}\, \Lambda^{2}\,+\,\mathfrak{e}\, f_{0})}{\pi^{2}} \int |\varphi|^{2} \sqrt{g}\, d^{4} x \\ & +\frac{f_{0} \mathfrak{a}}{2\, \pi^{2}} \int |D_{\mu} \varphi|^{2} \sqrt{g}\, d^{4} x \\ & -\frac{f_{0} \mathfrak{a}}{12\, \pi^{2}} \int R\, |\varphi|^{2} \sqrt{g}\, d^{4} x \\ & +\frac{f_{0} \mathfrak{b}}{2\, \pi^{2}} \int |\varphi|^{4} \sqrt{g}\, d^{4} x \\ & +\frac{f_{0}}{2\, \pi^{2}} \int (g_{3}^{2}\, G_{\mu \nu}^{i}\, G^{\mu \nu i}+g_{2}^{2}\, F_{\mu \nu}^{\alpha}\, F^{\mu \nu \alpha}+ \frac{5}{3}\, g_{1}^{2}\, B_{\mu \nu}\, B^{\mu \nu}) \sqrt{g}\, d^{4} x, \end{aligned}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0484</td><td>220</td><td>■ 1.000
● N|+1</td><td>f_{k}=\int_{0}^{\infty} f(v) v^{k-1} d v</td><td>$$f_{k}=\int_{0}^{\infty} f(v) v^{k-1} d v$$</td><td>$$f_{k}=\int_{0}^{\infty} f(v) v^{k-1} dv.$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0486</td><td>221</td><td>■ 1.000
● N|-1</td><td>\operatorname{Tr} e^{-t P} \sim \sum_{n \geq 0} t^{ \frac{n-m}{2}} \int_{M} a_{n}(x, P) d v(x)</td><td>$$\operatorname{Tr} e^{-t P} \sim \sum_{n \geq 0} t^{ \frac{n-m}{2}} \int_{M} a_{n}(x, P) d v(x)$$</td><td>$$\operatorname{Tr} e^{-t P} \sim \sum_{n \geq 0} t^{\frac{n-m}{2}} \int_{M} a_{n}(x, P)\, dv(x)$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0487</td><td>221</td><td>■ 1.000
● W|+1</td><td>\begin{gathered} \nabla_{\mu}=\partial_{\mu}+\omega_{ \mu}^{\prime}, \quad \omega_{\mu}^{\prime}=\frac{1}{2} g_{\mu \nu}\left(A^{\nu}+\Gamma^{\nu} \cdot \mathrm{id} \right) \\ E=B-g^{\mu \nu}\left(\partial_{\mu} \omega_{ \nu}^{\prime}+\omega_{\mu}^{\prime} \omega_{\nu}^{ \prime}-\Gamma_{\mu \nu}^{\rho} \omega_{\rho}^{\prime} \right) \\ \Omega_{\mu \nu}=\partial_{\mu} \omega_{ \nu}^{\prime}-\partial_{\nu} \omega_{\mu}^{\prime}+\left[\omega_{\mu}^{\prime}, \omega_{\nu}^{\prime}\right] \end{gathered}</td><td>$$\begin{gathered} \nabla_{\mu}=\partial_{\mu}+\omega_{ \mu}^{\prime}, \quad \omega_{\mu}^{\prime}=\frac{1}{2} g_{\mu \nu}\left(A^{\nu}+\Gamma^{\nu} \cdot \mathrm{id} \right) \\ E=B-g^{\mu \nu}\left(\partial_{\mu} \omega_{ \nu}^{\prime}+\omega_{\mu}^{\prime} \omega_{\nu}^{ \prime}-\Gamma_{\mu \nu}^{\rho} \omega_{\rho}^{\prime} \right) \\ \Omega_{\mu \nu}=\partial_{\mu} \omega_{ \nu}^{\prime}-\partial_{\nu} \omega_{\mu}^{\prime}+\left[\omega_{\mu}^{\prime}, \omega_{\nu}^{\prime}\right] \end{gathered}$$</td><td>$$\begin{gathered} \nabla_{\mu}\,=\,\partial_{\mu}\,+\,\omega_{\mu}^{\prime}, \quad \omega_{\mu}^{\prime}=\frac{1}{2}\, g_{\mu \nu}(A^{\nu}+\Gamma^{\nu} \cdot \mathrm{id}) \\ E=B-g^{\mu \nu}(\partial_{\mu}\, \omega_{\nu}^{\prime}+\omega_{\mu}^{\prime}\, \omega_{\nu}^{\prime}-\Gamma_{\mu \nu}^{\rho}\, \omega_{\rho}^{\prime}) \\ \Omega_{\mu \nu}=\partial_{\mu}\, \omega_{\nu}^{\prime}-\partial_{\nu}\, \omega_{\mu}^{\prime}+[\omega_{\mu}^{\prime},\, \omega_{\nu}^{\prime}]. \end{gathered}$$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>cardona-qft-methods_ EQ0493</td><td>222</td><td>1.000 K|+0</td><td>\frac{g^{2} f_{0}}{2 \pi^{2}}=\frac{1}{4} .</td><td>$$\frac{g^{2} f_{0}}{2 \pi^{2}}=\frac{1}{4} .$$</td><td>$$\frac{g^{2} f_{0}}{2 \pi^{2}}=\frac{1}{4} .$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0495</td><td>225</td><td>1.000 N|+0</td><td>\partial_{t} x_{i}(t)=\beta_{x_{i}}(x(t)),</td><td>$$\partial_{t} x_{i}(t)=\beta_{x_{i}}(x(t)),$$</td><td>$$\partial_{t} x_{i}(t)=\beta_{x_{i}}(x(t)),$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0496 (1)</td><td>225</td><td>1.000 N|+0</td><td>\beta_{1}=\frac{41}{96 \pi^{2}} g_{1}^{3}, \quad \beta_{2}=-\frac{19}{96 \pi^{2}} g_{2}^{3}, \quad \beta_{3}=-\frac{7}{16 \pi^{2}} g_{3}^{3} .</td><td>$$\beta_{1}=\frac{41}{96 \pi^{2}} g_{1}^{3}, \quad \beta_{2}=-\frac{19}{96 \pi^{2}} g_{2}^{3}, \quad \beta_{3}=-\frac{7}{16 \pi^{2}} g_{3}^{3} .$$</td><td>$$\beta_{1}=\frac{41}{96 \pi^{2}} g_{1}^{3}, \qquad \beta_{2}=-\frac{19}{96 \pi^{2}} g_{2}^{3}, \qquad \beta_{3}=-\frac{7}{16 \pi^{2}} g_{3}^{3} .$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0497 (2)</td><td>225</td><td>1.000 N|+0</td><td>g_{1}(0)=0.3575, \quad g_{2}(0)=0.6514, \quad g_{3}(0)=1.221</td><td>$$g_{1}(0)=0.3575, \quad g_{2}(0)=0.6514, \quad g_{3}(0)=1.221$$</td><td>$$g_{1}(0)=0.3575, \qquad g_{2}(0)=0.6514, \qquad g_{3}(0)=1.221$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0499</td><td>226</td><td>1.000 N|+0</td><td>\frac{\sqrt{\mathfrak{a} f_{0}}}{\pi}=\frac{2 M_{W}}{g}</td><td>$$\frac{\sqrt{\mathfrak{a} f_{0}}}{\pi}=\frac{2 M_{W}}{g}$$</td><td>$$\frac{\sqrt{\mathfrak{a} f_{0}}}{\pi}=\frac{2 M_{W}}{g}$$</td></tr>
<tr><td>cardona-qft-methods_ EQ0503 (4)</td><td>227</td><td>1.000 N|+0</td><td>\begin{gathered} \delta_{(\uparrow 1)}=\frac{1}{246} \left(\begin{array}{ccc} 12.2 \times 10^{-9} & 0 & 0 \\ 0 & 170 \times 10^{-6} & 0 \\ 0 & 0 & 15.5 \times 10^{-3} \end{array}\right) \\ Y_{\nu}=U_{P M N S}^{\dagger} \delta_{(\uparrow 1)} U_{P M N S} \end{gathered}</td><td>$$\begin{gathered} \delta_{(\uparrow 1)}=\frac{1}{246}\left(\begin{array}{ccc} 12.2 \times 10^{-9} & 0 & 0 \\ 0 & 170 \times 10^{-6} & 0 \\ 0 & 0 & 15.5 \times 10^{-3} \end{array}\right) \\ Y_{\nu}=U_{P M N S}^{\dagger} \delta_{(\uparrow 1)} U_{P M N S} \end{gathered}$$</td><td>$$\begin{gathered} \delta_{(\uparrow 1)}=\frac{1}{246}\left(\begin{array}{ccc} 12.2 \times 10^{-9} & 0 & 0 \\ 0 & 170 \times 10^{-6} & 0 \\ 0 & 0 & 15.5 \times 10^{-3} \end{array}\right) \\ Y_{\nu}=U_{P M N S}^{\dagger} \delta_{(\uparrow 1)} U_{P M N S} \end{gathered}$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0504 (3)	227	1.000 C|+7	\begin{gathered} \beta_{y}=\frac{1}{16 \pi^{2}}\left(\frac{9}{2} y^{3}-8 g_{3}^{2} y-\frac{9}{4} g_{2}^{2} y-\frac{17}{12} g_{1}^{2} y\right) . \\ \beta_{\lambda}= \frac{1}{16 \pi^{2}}\left(24 \lambda^{2}+12 \lambda y^{2}-9 \lambda\left(g_{2}^{2}+\frac{1}{3} g_{1}^{2} \right)-6 y^{4}+\frac{9}{8} g_{2}^{4}+\frac{3}{8} g_{1}^{4}+\frac{3}{4} g_{2}^{2} g_{1}^{2}\right), \end{gathered}	$\begin{gathered} \beta_{y}=\frac{1}{16 \pi^{2}}\left(\frac{9}{2} y^{3}-8 g_{3}^{2} y-\frac{9}{4} g_{2}^{2} y-\frac{17}{12} g_{1}^{2} y\right) . \\ \beta_{\lambda}=\frac{1}{16 \pi^{2}}\left(24 \lambda^{2}+12 \lambda y^{2}-9 \lambda\left(g_{2}^{2}+\frac{1}{3} g_{1}^{2}\right)-6 y^{4}+\frac{9}{8} g_{2}^{4}+\frac{3}{8} g_{1}^{4}+\frac{3}{4} g_{2}^{2} g_{1}^{2}\right), \end{gathered}$	$\beta_{y}=\frac{1}{16 \pi^{2}}\left(\frac{9}{2} y^{3}-8 g_{3}^{2} y-\frac{9}{4} g_{2}^{2} y-\frac{17}{12} g_{1}^{2} y\right) . \quad (3)$ $\beta_{\lambda}=\frac{1}{16 \pi^{2}}\left(24 \lambda^{2}+12 \lambda y^{2}-9 \lambda(g_{2}^{2}+\frac{1}{3} g_{1}^{2})-6 y^{4}+\frac{9}{8} g_{2}^{4}+\frac{3}{8} g_{1}^{4}+\frac{3}{4} g_{2}^{2} g_{1}^{2}\right),$
cardona-qft-methods_EQ0505	228	1.000 N|+0	\frac{f_{0}}{2 \pi^{2}} \int b|\varphi|^{4} \sqrt{g} d^{4} x=\frac{\pi^{2}}{2 f_{0}} \frac{\mathfrak{b}}{\mathfrak{a}^{2}} \int|\mathbf{H}|^{4} \sqrt{g} d^{4} x	$\frac{f_{0}}{2 \pi^{2}} \int b\vert\varphi\vert^{4} \sqrt{g} d^{4} x=\frac{\pi^{2}}{2 f_{0}} \frac{\mathfrak{b}}{\mathfrak{a}^{2}} \int\vert\mathbf{H}\vert^{4} \sqrt{g} d^{4} x$	$\frac{f_{0}}{2 \pi^{2}} \int b\,\vert\varphi\vert^{4} \sqrt{g}\, d^{4} x=\frac{\pi^{2}}{2 f_{0}} \frac{\mathfrak{b}}{\mathfrak{a}^{2}} \int \vert\mathbf{H}\vert^{4} \sqrt{g}\, d^{4} x$
cardona-qft-methods_EQ0506	228	1.000 K|+0	\tilde{\lambda}(\Lambda)=g_{3}^{2} \frac{\mathfrak{b}}{\mathfrak{a}^{2}} .	$\tilde{\lambda}(\Lambda)=g_{3}^{2} \frac{\mathfrak{b}}{\mathfrak{a}^{2}} .$	$\tilde{\lambda}(\Lambda)=g_{3}^{2} \frac{\mathfrak{b}}{\mathfrak{a}^{2}} .$
cardona-qft-methods_EQ0507	228	1.000 N|+0	\frac{d \lambda}{d t}=\lambda \gamma+\frac{1}{8 \pi^{2}} \left(12 \lambda^{2}+B\right)	$\frac{d \lambda}{d t}=\lambda \gamma+\frac{1}{8 \pi^{2}} \left(12 \lambda^{2}+B\right)$	$\frac{d \lambda}{d t}=\lambda \gamma+\frac{1}{8 \pi^{2}}(12 \lambda^{2}+B)$
cardona-qft-methods_EQ0508	228	1.000 N|+0	\gamma=\frac{1}{16 \pi^{2}}\left(12 y_{t}^{2}-9 g_{2}^{2}-3 g_{1}^{2}\right) \quad B=\frac{3}{16}\left(3 g_{2}^{4}+2 g_{1}^{2} g_{2}^{2}+g_{1}^{4}\right)-3 y_{t}^{4}	$\gamma=\frac{1}{16 \pi^{2}}\left(12 y_{t}^{2}-9 g_{2}^{2}-3 g_{1}^{2}\right) \quad B=\frac{3}{16}\left(3 g_{2}^{4}+2 g_{1}^{2} g_{2}^{2}+g_{1}^{4}\right)-3 y_{t}^{4}$	$\gamma=\frac{1}{16 \pi^{2}}(12 y_{t}^{2}-9 g_{2}^{2}-3 g_{1}^{2}) \quad B=\frac{3}{16}(3 g_{2}^{4}+2 g_{1}^{2} g_{2}^{2}+g_{1}^{4})-3 y_{t}^{4}$
cardona-qft-methods_EQ0509	228	1.000 N|+0	m_{H}^{2}=8 \lambda \frac{M^{2}}{g^{2}}, \quad m_{H}= \sqrt{2 \lambda} \frac{2 M}{g} .	$m_{H}^{2}=8 \lambda \frac{M^{2}}{g^{2}}, \quad m_{H}= \sqrt{2 \lambda} \frac{2 M}{g} .$	$m_{H}^{2}=8 \lambda \frac{M^{2}}{g^{2}}, \quad m_{H}= \sqrt{2 \lambda} \frac{2 M}{g} .$
cardona-qft-methods_EQ0510	229	1.000 W|+1	\int\left(\frac{1}{2 \eta} C_{\mu \nu \rho \sigma} C^{ \mu \nu \rho \sigma}-\frac{\omega}{3 \eta} R^{2}+\frac{ \theta}{\eta} R^{*} R^{*}\right) \sqrt{g} d^{4} x	$\int\left(\frac{1}{2 \eta} C_{\mu \nu \rho \sigma} C^{\mu \nu \rho \sigma}-\frac{\omega}{3 \eta} R^{2}+\frac{\theta}{\eta} R^{*} R^{*}\right) \sqrt{g} d^{4} x$	$\int\left(\frac{1}{2 \eta} C_{\mu \nu \rho \sigma} C^{\mu \nu \rho \sigma}-\frac{\omega}{3 \eta} R^{2}+\frac{\theta}{\eta} R^{*} R^{*}\right) \sqrt{g}\, d^{4} x$
cardona-qft-methods_EQ0511	229	1.000 W|+0	\begin{aligned} & \beta_{\eta}=-\frac{1}{(4 \pi)^{2}} \frac{133}{10} \eta^{2} \\ & \beta_{\omega}=-\frac{1}{(4 \pi)^{2}} \frac{25+1098 \omega+200 \omega^{2}}{60} \eta \\ & \beta_{\theta}=\frac{1}{(4 \pi)^{2}} \frac{7(56-171 \theta)}{90} \eta . \end{aligned}	$\begin{aligned} & \beta_{\eta}=-\frac{1}{(4 \pi)^{2}} \frac{133}{10} \eta^{2} \\ & \beta_{\omega}=-\frac{1}{(4 \pi)^{2}} \frac{25+1098 \omega+200 \omega^{2}}{60} \eta \\ & \beta_{\theta}=\frac{1}{(4 \pi)^{2}} \frac{7(56-171 \theta)}{90} \eta . \end{aligned}$	$\begin{aligned} & \beta_{\eta}=-\frac{1}{(4 \pi)^{2}} \frac{133}{10} \eta^{2} \\ & \beta_{\omega}=-\frac{1}{(4 \pi)^{2}} \frac{25+1098\, \omega+200\, \omega^{2}}{60} \eta \\ & \beta_{\theta}=\frac{1}{(4 \pi)^{2}} \frac{7(56-171\, \theta)}{90} \eta . \end{aligned}$
cardona-qft-methods_EQ0516 (2)	236	1.000 N|+0	\Delta x \Delta p \geq \frac{\hbar}{2},	$\Delta x \Delta p \geq \frac{\hbar}{2},$	$\Delta x \Delta p \geq \frac{\hbar}{2},$
cardona-qft-methods_EQ0517 (3)	236	1.000 N|+0	x y-y x=i \theta,	$x y-y x=i \theta,$	$x\, y-y\, x=i \theta,$
cardona-qft-methods_EQ0518 (4)	236	1.000 N|+0	\Delta x \Delta y \geq \frac{\theta}{2},	$\Delta x \Delta y \geq \frac{\theta}{2},$	$\Delta x\, \Delta y \geq \frac{\theta}{2},$

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Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>cardona-qft-methods_ EQ0521 (1)</td><td>239</td><td>■1.000 ● N|+0</td><td><code>U(g)=\bigoplus_{N} U^{(N)}(g),</code></td><td>$U(g)=\bigoplus_{N} U^{(N)}(g),$</td><td>$U(g)=\bigoplus_{N} U^{(N)}(g),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0522 (2)</td><td>239</td><td>■1.000 ● N|+0</td><td><code>U^{(N)}(g)\left|\rho_{1}\right\rangle \otimes\left|\rho_{2}\right\rangle \otimes \cdots \otimes\left|\rho_{N}\right\rangle=U^{(1)}(g)\left|\rho_{1}\right\rangle \otimes U^{(1)}(g)\left|\rho_{2}\right\rangle \otimes \cdots \otimes U^{(1)}(g)\left|\rho_{N}\right\rangle .</code></td><td>$U^{(N)}(g)\left|\rho_{1}\right\rangle \otimes\left|\rho_{2}\right\rangle \otimes \cdots \otimes\left|\rho_{N}\right\rangle=U^{(1)}(g)\left|\rho_{1}\right\rangle \otimes U^{(1)}(g)\left|\rho_{2}\right\rangle \otimes \cdots \otimes U^{(1)}(g)\left|\rho_{N}\right\rangle .$</td><td>$U^{(N)}(g)|\rho_1\rangle\otimes|\rho_2\rangle\otimes\cdots\otimes|\rho_N\rangle = U^{(1)}(g)|\rho_1\rangle\otimes U^{(1)}(g)|\rho_2\rangle\otimes\cdots\otimes U^{(1)}(g)|\rho_N\rangle.$</td></tr>
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<tr><td>cardona-qft-methods_ EQ0524 (4)</td><td>240</td><td>■1.000 ● N|+0</td><td><code>\alpha^{*} \beta^{*}:=\sum_{g \in G}(\alpha * \beta)(g) g,</code></td><td>$\alpha^{*} \beta^{*}:=\sum_{g \in G}(\alpha * \beta)(g) g,$</td><td>$\alpha^{*}\, \beta^{*}:=\sum_{g \in G}(\alpha * \beta)(g)\, g,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0525 (5)</td><td>240</td><td>■1.000 ● N|+0</td><td><code>(\alpha * \beta)(g)=\sum_{h \in G} \alpha\left(g h^{-1}\right) \beta(h) .</code></td><td>$(\alpha * \beta)(g)=\sum_{h \in G} \alpha\left(g h^{-1}\right) \beta(h) .$</td><td>$(\alpha * \beta)(g)=\sum_{h \in G} \alpha(g h^{-1}) \beta(h).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0526 (6)</td><td>240</td><td>■1.000 ● N|+0</td><td><code>\alpha^{*}=\int d \mu(g) \alpha(g) g .</code></td><td>$\alpha^{*}=\int d \mu(g) \alpha(g) g .$</td><td>$\alpha^{*}=\int d \mu(g) \alpha(g) g.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0527 (7)</td><td>240</td><td>■1.000 ● N|+2</td><td><code>\begin{aligned} U^{(N)}\left(\alpha^{*}\right) & =\int d \mu(g) \alpha(g) U^{(N)}(g) \\ U\left(\alpha^{*}\right) & =\int d \mu(g) \alpha(g) U(g) \end{aligned}</code></td><td>$\begin{aligned} U^{(N)}\left(\alpha^{*}\right) & =\int d \mu(g) \alpha(g) U^{(N)}(g) \\ U\left(\alpha^{*}\right) & =\int d \mu(g) \alpha(g) U(g) \end{aligned}$</td><td>$\begin{aligned} U^{(N)}(\alpha^{*}) & =\int d \mu(g) \alpha(g) U^{(N)}(g), \\ U(\alpha^{*}) & =\int d \mu(g) \alpha(g) U(g). \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0528 (8)</td><td>241</td><td>■1.000 ● N|+0</td><td><code>U^{(1)}(g)|\rho\rangle=|\sigma\rangle D_{\sigma \rho}(g),</code></td><td>$U^{(1)}(g)|\rho\rangle=|\sigma\rangle D_{\sigma \rho}(g),$</td><td>$U^{(1)}(g)|\rho\rangle=|\sigma\rangle D_{\sigma \rho}(g),$</td></tr>
<tr><td>cardona-qft-methods_ EQ0529 (9)</td><td>241</td><td>■1.000 ● K|+0</td><td><code>\Delta(g)=g \otimes g .</code></td><td>$\Delta(g)=g \otimes g .$</td><td>$\Delta(g)=g \otimes g.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0531</td><td>241</td><td>■1.000 ● N|+0</td><td><code>\begin{aligned} & U^{(3)}(g)=\left(U^{(2)} \otimes U^{(1)}\right) \Delta(g), \\ & U^{(4)}(g)=\left(U^{(3)} \otimes U^{(1)}\right) \Delta(g), \end{aligned}</code></td><td>$\begin{aligned} & U^{(3)}(g)=\left(U^{(2)} \otimes U^{(1)}\right) \Delta(g), \\ & U^{(4)}(g)=\left(U^{(3)} \otimes U^{(1)}\right) \Delta(g), \end{aligned}$</td><td>$\begin{aligned} & U^{(3)}(g)=\left(U^{(2)} \otimes U^{(1)}\right) \Delta(g), \\ & U^{(4)}(g)=\left(U^{(3)} \otimes U^{(1)}\right) \Delta(g), \end{aligned}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0532 (11)</td><td>241</td><td>■1.000 ● N|+0</td><td><code>U^{(N)}(g)=U^{(1)}(g) \otimes U^{(1)}(g) \otimes \cdots \otimes U^{(1)}(g) .</code></td><td>$U^{(N)}(g)=U^{(1)}(g) \otimes U^{(1)}(g) \otimes \cdots \otimes U^{(1)}(g) .$</td><td>$U^{(N)}(g)=U^{(1)}(g) \otimes U^{(1)}(g) \otimes \cdots \otimes U^{(1)}(g).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0533 (12)</td><td>241</td><td>■1.000 ● K|+0</td><td><code>(\Delta \otimes \mathrm{Id}) \Delta=(\mathrm{Id} \otimes \Delta) \Delta .</code></td><td>$(\Delta \otimes \mathrm{Id}) \Delta=(\mathrm{Id} \otimes \Delta) \Delta .$</td><td>$(\Delta \otimes \mathrm{Id}) \Delta=(\mathrm{Id} \otimes \Delta) \Delta.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0534 (13)</td><td>241</td><td>■1.000 ● W|+0</td><td><code>\mathbb{C} G \rightarrow \underbrace{\mathbb{C} G \otimes \mathbb{C} G \otimes \cdots \mathbb{C} G}_{N \text { factors }} .</code></td><td>$\mathbb{C} G \rightarrow \underbrace{\mathbb{C} G \otimes \mathbb{C} G \otimes \cdots \mathbb{C} G}_{N \text { factors }} .$</td><td>$\mathbb{C} G \rightarrow \underbrace{\mathbb{C} G \otimes \mathbb{C} G \otimes \cdots \mathbb{C} G}_{N \text { factors }}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0535 (14)</td><td>241</td><td>■1.000 ● W|+2</td><td><code>\mathbb{C} G \rightarrow \underbrace{\mathbb{C} G \otimes \mathbb{C} G \otimes \cdots \otimes \mathbb{C} G}_{(N+1) \text { factors }}</code></td><td>$\mathbb{C} G \rightarrow \underbrace{\mathbb{C} G \otimes \mathbb{C} G \otimes \cdots \otimes \mathbb{C} G}_{(N+1) \text { factors }}$</td><td>$\mathbb{C} G \rightarrow \underbrace{\mathbb{C} G \otimes \mathbb{C} G \otimes \cdots \otimes \mathbb{C} G}_{(N+1) \text { factors }},$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0536 (15)	242	1.000 N|+0	`\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta},`	$\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta},$	$\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta},$
cardona-qft-methods_ EQ0537	242	1.000 N|+0	`F_{\theta}=e^{\frac{1}{2} P_{\mu} \theta^{\mu \nu} \otimes P_{\nu}},`	$F_{\theta}=e^{\frac{1}{2} P_{\mu} \theta^{\mu \nu} \otimes P_{\nu}},$	$F_{\theta}=e^{\frac{1}{2} P_{\mu} \theta^{\mu \nu} \otimes P_{\nu}},$
cardona-qft-methods_ EQ0538	242	1.000 N|+0	`\epsilon: G \rightarrow \mathbb{C}, \quad \epsilon(g)=1 .`	$\epsilon: G \rightarrow \mathbb{C}, \quad \epsilon(g)=1 .$	$\epsilon: G \rightarrow \mathbb{C}, \quad \epsilon(g)=1 .$
cardona-qft-methods_ EQ0539	242	1.000 K|+0	`\epsilon: \int d \mu(g) \alpha(g) g \rightarrow \int d \mu(g) \alpha(g)`	$\epsilon: \int d \mu(g) \alpha(g) g \rightarrow \int d \mu(g) \alpha(g)$	$\epsilon: \int d \mu(g) \alpha(g) g \rightarrow \int d \mu(g) \alpha(g)$
cardona-qft-methods_ EQ0540	243	1.000 N|+1	`S: \hat{\alpha}=\int d \mu(g) \alpha(g) g \rightarrow S(\hat{\alpha})=\int d \mu(g) \alpha(g) S(g)`	$S: \hat{\alpha}=\int d \mu(g) \alpha(g) g \rightarrow S(\hat{\alpha})=\int d \mu(g) \alpha(g) S(g)$	$S: \hat{\alpha}=\int d \mu(g) \alpha(g) g \rightarrow S(\hat{\alpha})=\int d \mu(g) \alpha(g) S(g).$
cardona-qft-methods_ EQ0541	243	1.000 N|+1	`m(g \otimes h)=g h, \quad ; g, h \in G .`	$m(g \otimes h)=g h, \quad ; g, h \in G .$	$m(g \otimes h)=g h, \quad ; g, h \in G .$
cardona-qft-methods_ EQ0544	243	1.000 K|+0	`\left(m_{13} \otimes m_{24}\right)\left(a_{1} \otimes a_{2} \otimes a_{3} \otimes a_{4}\right)=a_{1} a_{3} \otimes a_{2} a_{4} .`	$\left(m_{13} \otimes m_{24}\right)\left(a_{1} \otimes a_{2} \otimes a_{3} \otimes a_{4}\right)=a_{1} a_{3} \otimes a_{2} a_{4} .$	$(m_{13} \otimes m_{24})(a_{1} \otimes a_{2} \otimes a_{3} \otimes a_{4})=a_{1} a_{3} \otimes a_{2} a_{4} .$
cardona-qft-methods_ EQ0545	243	1.000 N|+1	`\left(\int d \mu(g) \alpha(g) g\right)^{*}=\int d \mu(g) \bar{\alpha}(g) g^{-1}`	$\left(\int d \mu(g) \alpha(g) g\right)^{*}=\int d \mu(g) \bar{\alpha}(g) g^{-1}$	$\left(\int d \mu(g) \alpha(g) g\right)^{*}=\int d \mu(g) \bar{\alpha}(g) g^{-1}.$
cardona-qft-methods_ EQ0546 (16)	244	1.000 K|+0	`\Delta: H \rightarrow H \otimes H`	$\Delta: H \rightarrow H \otimes H$	$\Delta: H \rightarrow H \otimes H$
cardona-qft-methods_ EQ0550	244	1.000 N|+0	`m(\alpha \otimes \beta)=\alpha \beta .`	$m(\alpha \otimes \beta)=\alpha \beta .$	$m(\alpha \otimes \beta)=\alpha \beta .$
cardona-qft-methods_ EQ0551	244	1.000 N|+1	`\left(m_{13} \otimes m_{24}\right)\left(\alpha_{1} \otimes \alpha_{2} \otimes \alpha_{3} \otimes \alpha_{4}\right)=\alpha_{1} \alpha_{3} \otimes \alpha_{2} \alpha_{4} .`	$\left(m_{13} \otimes m_{24}\right)\left(\alpha_{1} \otimes \alpha_{2} \otimes \alpha_{3} \otimes \alpha_{4}\right)=\alpha_{1} \alpha_{3} \otimes \alpha_{2} \alpha_{4} .$	$(m_{13} \otimes m_{24})(\alpha_{1} \otimes \alpha_{2} \otimes \alpha_{3} \otimes \alpha_{4})=\alpha_{1} \alpha_{3} \otimes \alpha_{2} \alpha_{4} .$
cardona-qft-methods_ EQ0552 (19)	244	1.000 N|+0	`\Delta m(\alpha \otimes \beta)=\left(m_{13} \otimes m_{24}\right)(\Delta \otimes \Delta)(\alpha \otimes \beta) .`	$\Delta m(\alpha \otimes \beta)=\left(m_{13} \otimes m_{24}\right)(\Delta \otimes \Delta)(\alpha \otimes \beta) .$	$\Delta m(\alpha \otimes \beta)=(m_{13} \otimes m_{24})(\Delta \otimes \Delta)(\alpha \otimes \beta) .$
cardona-qft-methods_ EQ0553	245	1.000 N|-2	`(F \otimes I d)(\Delta \otimes I d) F=(I d \otimes F)(I d \otimes \Delta) F, \quad(\epsilon \otimes I d) F=(I d \otimes \epsilon) F=1 .`	$(F \otimes I d)(\Delta \otimes I d) F=(I d \otimes F)(I d \otimes \Delta) F, \quad(\epsilon \otimes I d) F=(I d \otimes \epsilon) F=1 .$	$(F \otimes Id)(\Delta \otimes Id) F=(Id \otimes F)(Id \otimes \Delta) F, \quad(\epsilon \otimes Id) F=(Id \otimes \epsilon) F=1 .$
cardona-qft-methods_ EQ0554	245	1.000 N|+0	`\Delta(g)=g \otimes g, \quad g \in G \subset G \mathbb{C} G,`	$\Delta(g)=g \otimes g, \quad g \in G \subset G \mathbb{C} G,$	$\Delta(g)=g \otimes g, \quad g \in G \subset G \mathbb{C} G,$
cardona-qft-methods_ EQ0555	245	1.000 N|+0	`\xi \otimes \eta \rightarrow(\rho \otimes \rho) \Delta(g) \xi \otimes \eta=\rho(g) \xi \otimes \rho(g) \eta .`	$\xi \otimes \eta \rightarrow(\rho \otimes \rho) \Delta(g) \xi \otimes \eta=\rho(g) \xi \otimes \rho(g) \eta .$	$\xi \otimes \eta \rightarrow(\rho \otimes \rho) \Delta(g) \xi \otimes \eta=\rho(g) \xi \otimes \rho(g) \eta .$
cardona-qft-methods_ EQ0556	245	1.000 N|+0	`\tau(\xi \otimes \eta)=\eta \otimes \xi .`	$\tau(\xi \otimes \eta)=\eta \otimes \xi .$	$\tau(\xi \otimes \eta)=\eta \otimes \xi .$
cardona-qft-methods_ EQ0557	246	1.000 N|+0	`\tau_{i, i+1}=\underbrace{1 \otimes \cdots \otimes 1}_{(i-1) \text { factors }} \otimes \tau \underbrace{\otimes 1 \otimes \cdots \otimes 1}_{(N-i-1) \text { factors }} .`	$\tau_{i, i+1}=\underbrace{1 \otimes \cdots \otimes 1}_{(i-1) \text { factors }} \otimes \tau \underbrace{\otimes 1 \otimes \cdots \otimes 1}_{(N-i-1) \text { factors }} .$	$\tau_{i, i+1}=\underbrace{1 \otimes \cdots \otimes 1}_{(i-1) \text { factors }} \otimes \tau \underbrace{\otimes 1 \otimes \cdots \otimes 1}_{(N-i-1) \text { factors }} .$
cardona-qft-methods_ EQ0559	246	1.000 N|+1	`\Delta^{\mathrm{op}}(\eta)=\eta^{(2)} \otimes \eta^{(2)}`	$\Delta^{\mathrm{op}}(\eta)=\eta^{(2)} \otimes \eta^{(2)}$	$\Delta^{\mathrm{op}}(\eta)=\eta^{(2)} \otimes \eta^{(2)}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0560	247	■ 1.000 ● N|+0	`R_{i, i+1}=\underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(i-1) \text { factors }} \otimes R \otimes \underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(N-i-1) \text { factors }}`	$R_{i, i+1}=\underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(i-1) \text { factors }} \otimes R \otimes \underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(N-i-1) \text { factors }}$	$R_{i,i+1}=\underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(i-1) \text { factors }} \otimes R \otimes \underbrace{1 \otimes 1 \otimes \cdots \otimes 1}_{(N-i-1) \text { factors }}$
cardona-qft-methods_ EQ0561	247	■ 1.000 ● K|+0	`\mathcal{R}_{i, i+1}=\tau_{i, i+1}(\rho \otimes \rho \otimes \cdots \otimes \rho) R_{i, i+1}`	$\mathcal{R}_{i, i+1}=\tau_{i, i+1}(\rho \otimes \rho \otimes \cdots \otimes \rho) R_{i, i+1}$	$\mathcal{R}_{i,i+1}=\tau_{i,i+1}(\rho \otimes \rho \otimes \cdots \otimes \rho) R_{i,i+1}$
cardona-qft-methods_ EQ0562	247	■ 1.000 ● N|+0	`R_{i+1, i+2} R_{i, i+2} R_{i, i+1}=R_{i, i+1} R_{i, i+2} R_{i+1, i+2} .`	$R_{i+1, i+2} R_{i, i+2} R_{i, i+1}=R_{i, i+1} R_{i, i+2} R_{i+1, i+2} .$	$R_{i+1,i+2} R_{i,i+2} R_{i,i+1}=R_{i,i+1} R_{i,i+2} R_{i+1,i+2}.$
cardona-qft-methods_ EQ0563	247	■ 1.000 ● N|+0	`\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta}`	$\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta}$	$\Delta_{\theta}(g)=F_{\theta}^{-1}(g \otimes g) F_{\theta}$
cardona-qft-methods_ EQ0564	247	■ 1.000 ● K|+0	`\tau_{\theta}=(\rho \otimes \rho) F_{\theta}^{-1} \tau(\rho \otimes \rho) F_{\theta}`	$\tau_{\theta}=(\rho \otimes \rho) F_{\theta}^{-1} \tau(\rho \otimes \rho) F_{\theta}$	$\tau_{\theta}=(\rho \otimes \rho) F_{\theta}^{-1} \tau(\rho \otimes \rho) F_{\theta}$
cardona-qft-methods_ EQ0565	247	■ 1.000 ● K|+0	`\tau_{\theta}=\tau(\rho \otimes \rho) F_{\theta}^{2},`	$\tau_{\theta}=\tau(\rho \otimes \rho) F_{\theta}^{2},$	$\tau_{\theta}=\tau(\rho \otimes \rho) F_{\theta}^{2},$
cardona-qft-methods_ EQ0566	247	■ 1.000 ● K|+0	`R_{\theta}=F_{\theta}^{2} .`	$R_{\theta}=F_{\theta}^{2} .$	$R_{\theta}=F_{\theta}^{2}.$
cardona-qft-methods_ EQ0567 (1)	248	■ 1.000 ● K|+0	`R(t)=0 \quad t<0`	$R(t)=0 \quad t<0$	$R(t)=0 \quad t<0$
cardona-qft-methods_ EQ0569 (3)	248	■ 1.000 ● K|+0	`\Im \omega>0 .`	$\Im \omega>0 .$	$\Im \omega>0.$
cardona-qft-methods_ EQ0570 (4)	248	■ 1.000 ● W|-2	`\begin{gathered} {[\rho(x), \eta(y)]=0 \quad \text { if }} \\ \left(x^{0}-y^{0}\right)^{2}-(\vec{x}-\vec{y})^{2}<0 \end{gathered}`	$\begin{gathered} {[\rho(x), \eta(y)]=0 \quad \text { if }} \\ \left(x^{0}-y^{0}\right)^{2}-(\vec{x}-\vec{y})^{2}<0 \end{gathered}$	$\begin{gathered} [\rho(x), \eta(y)]=0 \quad \text{if} \\ (x^{0}-y^{0})^{2}-(\vec{x}-\vec{y})^{2}<0 \end{gathered}$
cardona-qft-methods_ EQ0571 (6)	249	■ 1.000 ● N|+0	`[\varphi(x), \chi(y)]_{-}=0 \quad x \sim y`	$[\varphi(x), \chi(y)]_{-}=0 \quad x \sim y$	$[\varphi(x), \chi(y)]_{-}=0 \quad x \sim y$
cardona-qft-methods_ EQ0572 (7)	249	■ 1.000 ● N|+0	`\left.\left[\psi_{\alpha}^{(1)}(x), \psi_{\alpha}^{(2)}(y)\right)\right]_{+}=0 \quad x \sim y`	$\left.\left[\psi_{\alpha}^{(1)}(x), \psi_{\alpha}^{(2)}(y)\right)\right]_{+}=0 \quad x \sim y$	$[\psi_{\alpha}^{(1)}(x), \psi_{\alpha}^{(2)}(y))]_{+}=0 \quad x \sim y$
cardona-qft-methods_ EQ0575 (10)	249	■ 1.000 ● N|+0	`g \rightarrow\left[U_{1} \otimes U_{2}\right](g \otimes g)`	$g \rightarrow\left[U_{1} \otimes U_{2}\right](g \otimes g)$	$g \rightarrow [U_{1} \otimes U_{2}](g \otimes g)$
cardona-qft-methods_ EQ0576 (11)	250	■ 1.000 ● N|+0	`\begin{aligned} \Delta: G & \rightarrow G \otimes G, \\ g & \rightarrow \Delta(g):=g \otimes g, \end{aligned}`	$\begin{aligned} \Delta: G & \rightarrow G \otimes G, \\ g & \rightarrow \Delta(g):=g \otimes g, \end{aligned}$	$\begin{aligned} \Delta: G & \rightarrow G \otimes G, \\ g & \rightarrow \Delta(g):=g \otimes g, \end{aligned}$
cardona-qft-methods_ EQ0577 (12)	250	■ 1.000 ● N|+1	`F_{\theta}=e^{\frac{i}{2} \partial_{\mu} \otimes \theta^{\mu \nu} \partial_{\nu}}=\text { "Twist element" }`	$F_{\theta}=e^{\frac{i}{2} \partial_{\mu} \otimes \theta^{\mu \nu} \partial_{\nu}}=\text { "Twist element" }$	$F_{\theta}=e^{\frac{i}{2} \partial_{\mu} \otimes \theta^{\mu \nu} \partial_{\nu}}=\text{“}\mathit{Twist\ element}\text{”}$
cardona-qft-methods_ EQ0578 (13)	250	■ 1.000 ● N|+0	`f \star g=m_{0}\left[F_{\theta} f \otimes g\right],`	$f \star g=m_{0}\left[F_{\theta} f \otimes g\right],$	$f \star g=m_{0}[F_{\theta} f \otimes g],$
cardona-qft-methods_ EQ0579 (14)	250	■ 1.000 ● N|+0	`m_{0}(\alpha \otimes \beta)(x)=\alpha(x) \beta(x)`	$m_{0}(\alpha \otimes \beta)(x)=\alpha(x) \beta(x)$	$m_{0}(\alpha \otimes \beta)(x)=\alpha(x) \beta(x)$
cardona-qft-methods_ EQ0581	250	■ 1.000 ● N|+0	`\Lambda: \alpha \rightarrow \Lambda^{*} \alpha, \quad\left(\Lambda^{*} \alpha\right)(x)=\alpha\left[\Lambda^{-1}(x)\right] .`	$\Lambda: \alpha \rightarrow \Lambda^{*} \alpha, \quad\left(\Lambda^{*} \alpha\right)(x)=\alpha\left[\Lambda^{-1}(x)\right] .$	$\Lambda: \alpha \rightarrow \Lambda^{*} \alpha, \quad (\Lambda^{*} \alpha)(x)=\alpha[\Lambda^{-1}(x)].$
cardona-qft-methods_ EQ0582 (15)	250	■ 1.000 ● K|+0	`\Delta_{\theta}(\Lambda)=F_{\theta}^{-1}(\Lambda \otimes \lambda) F_{\theta} .`	$\Delta_{\theta}(\Lambda)=F_{\theta}^{-1}(\Lambda \otimes \lambda) F_{\theta} .$	$\Delta_{\theta}(\Lambda)=F_{\theta}^{-1}(\Lambda \otimes \lambda) F_{\theta}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0583 (16)	250	1.000 N|+0	`\phi \otimes_{S, A} \chi \equiv \frac{1}{2}(\phi \otimes \chi \pm \chi \otimes \phi)`	$\phi \otimes_{S, A} \chi \equiv \frac{1}{2}(\phi \otimes \chi \pm \chi \otimes \phi)$	$\phi \otimes_{S,A} \chi \equiv \frac{1}{2}(\phi \otimes \chi \pm \chi \otimes \phi)$
cardona-qft-methods_EQ0584 (17)	251	1.000 N|+0	`\tau_{0}(\phi \otimes \chi)=\chi \otimes \phi`	$\tau_{0}(\phi \otimes \chi)=\chi \otimes \phi$	$\tau_0(\phi \otimes \chi) = \chi \otimes \phi$
cardona-qft-methods_EQ0585 (18)	251	1.000 K|+0	`\tau_{0} \Delta_{0}(\Lambda)=\Delta_{0}(\Lambda) \tau_{0}`	$\tau_{0} \Delta_{0}(\Lambda)=\Delta_{0}(\Lambda) \tau_{0}$	$\tau_0\Delta_0(\Lambda) = \Delta_0(\Lambda)\tau_0$
cardona-qft-methods_EQ0586 (19)	251	1.000 N|+0	`\left(\frac{1 \pm \tau_{0}}{2}\right)(\phi \otimes \chi) .`	$\left(\frac{1 \pm \tau_{0}}{2}\right)(\phi \otimes \chi) .$	$\left(\frac{1 \pm \tau_0}{2}\right)(\phi \otimes \chi).$
cardona-qft-methods_EQ0587 (20)	251	1.000 N|+0	`\tau_{0} F_{\theta}=F_{\theta}^{-1} \tau_{0},`	$\tau_{0} F_{\theta}=F_{\theta}^{-1} \tau_{0},$	$\tau_0 F_\theta = F_\theta^{-1}\tau_0,$
cardona-qft-methods_EQ0589 (22)	251	1.000 N|+0	`\tau_{\theta} \equiv F_{\theta}^{-1} \tau_{0} F_{\theta}, \quad \tau_{\theta}^{2}=1 \otimes 1,`	$\tau_{\theta} \equiv F_{\theta}^{-1} \tau_{0} F_{\theta}, \quad \tau_{\theta}^{2}=1 \otimes 1,$	$\tau_\theta \equiv F_\theta^{-1}\tau_0 F_\theta, \qquad \tau_\theta^2 = 1 \otimes 1,$
cardona-qft-methods_EQ0590 (23)	251	1.000 K|+0	`\Delta_{\theta}(\Lambda)=F_{\theta}^{-1} \Lambda \otimes \Lambda F_{\theta} .`	$\Delta_{\theta}(\Lambda)=F_{\theta}^{-1} \Lambda \otimes \Lambda F_{\theta} .$	$\Delta_\theta(\Lambda) = F_\theta^{-1}\Lambda \otimes \Lambda F_\theta.$
cardona-qft-methods_EQ0593 (2)	252	1.000 K|+0	`U:(a, \Lambda) \rightarrow U(a, \Lambda)`	$U:(a, \Lambda) \rightarrow U(a, \Lambda)$	$U : (a, \Lambda) \rightarrow U(a, \Lambda)$
cardona-qft-methods_EQ0594 (3)	252	1.000 K|+0	`U(a, \Lambda) \varphi(x) U(a, \Lambda)^{-1}=\varphi((a, \Delta) x)`	$U(a, \Lambda) \varphi(x) U(a, \Lambda)^{-1}=\varphi((a, \Delta) x)$	$U(a, \Lambda)\varphi(x)U(a, \Lambda)^{-1} = \varphi((a, \Delta)x)$
cardona-qft-methods_EQ0595 (4)	252	1.000 K|+0	`\operatorname{Ad} U(a, \Lambda) \varphi=U(a, \Lambda) \varphi U(a, \Lambda)^{-1}`	$\operatorname{Ad} U(a, \Lambda) \varphi=U(a, \Lambda) \varphi U(a, \Lambda)^{-1}$	$\mathrm{Ad}U(a, \Lambda)\varphi = U(a, \Lambda)\varphi U(a, \Lambda)^{-1}$
cardona-qft-methods_EQ0596 (5)	252	1.000 K|+0	`(a, \Delta): \alpha \rightarrow(a, \Lambda) \triangleright \alpha, \quad[(a, \Lambda) \alpha](x) \quad \alpha \in S\left(M^{d+1}\right)`	$(a, \Delta): \alpha \rightarrow(a, \Lambda) \triangleright \alpha, \quad[(a, \Lambda) \alpha](x) \quad \alpha \in S\left(M^{d+1}\right)$	$(a, \Delta) : \alpha \rightarrow (a, \Lambda) \triangleright \alpha, \quad [(a, \Lambda)\alpha](x) \quad \alpha \in S(M^{d+1})$
cardona-qft-methods_EQ0597 (6)	252	1.000 K|+0	`\Delta_{0}((a, \Lambda))=(a, \Lambda) \otimes(a, \Lambda)`	$\Delta_{0}((a, \Lambda))=(a, \Lambda) \otimes(a, \Lambda)$	$\Delta_0((a, \Lambda)) = (a, \Lambda) \otimes (a, \Lambda)$
cardona-qft-methods_EQ0599 (8)	252	1.000 K|+0	`\varphi\left(x_{1}\right) \varphi\left(x_{2}\right) \ldots \varphi\left(x_{N}\right) .`	$\varphi\left(x_{1}\right) \varphi\left(x_{2}\right) \ldots \varphi\left(x_{N}\right) .$	$\varphi(x_1)\varphi(x_2)\ldots\varphi(x_N).$
cardona-qft-methods_EQ0600 (9)	253	1.000 K|+0	`(\varphi \otimes \ldots \otimes \varphi)\left(x_{1}, x_{2}, \ldots, x_{N}\right)=\varphi\left(x_{1}\right) \varphi\left(x_{2}\right) \ldots \varphi\left(x_{N}\right)`	$(\varphi \otimes \ldots \otimes \varphi)\left(x_{1}, x_{2}, \ldots, x_{N}\right)=\varphi\left(x_{1}\right) \varphi\left(x_{2}\right) \ldots \varphi\left(x_{N}\right)$	$(\varphi \otimes \ldots \otimes \varphi)(x_1, x_2, \ldots, x_N) = \varphi(x_1)\varphi(x_2)\ldots\varphi(x_N)$
cardona-qft-methods_EQ0601 (10)	253	1.000 S|-1	`(\underbrace{\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \ldots \mathbb{I} \otimes \Delta_{0}}_{N-1}(\underbrace{\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \ldots \mathbb{I} \otimes \Delta_{0}}_{N-2}) \ldots \Delta_{0}`	$(\underbrace{\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \ldots \mathbb{I} \otimes \Delta_{0}}_{N-1}(\underbrace{\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \ldots \mathbb{I} \otimes \Delta_{0}}_{N-2}) \ldots \Delta_{0}$	$(\underbrace{\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \ldots \mathbb{I} \otimes \Delta_0}_{N-1}(\underbrace{\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \ldots \mathbb{I} \otimes \Delta_0}_{N-2}) \ldots \Delta_0$
cardona-qft-methods_EQ0602 (11)	253	1.000 K|+0	`\Delta_{0}^{N}(a, \Lambda)=(a, \Lambda) \otimes(a, \Lambda) \otimes \cdots \otimes(a, \Lambda) .`	$\Delta_{0}^{N}(a, \Lambda)=(a, \Lambda) \otimes(a, \Lambda) \otimes \cdots \otimes(a, \Lambda) .$	$\Delta_0^N(a, \Lambda) = (a, \Lambda) \otimes (a, \Lambda) \otimes \cdots \otimes (a, \Lambda).$
cardona-qft-methods_EQ0603 (12)	253	1.000 K|+0	`U(a, \Lambda)\left(\varphi^{\otimes N}\left((a, \Lambda)^{-1} x_{1}, \ldots,(a, \Lambda)^{-1} x_{N}\right)\right) U(a, \Lambda)^{-1}=\varphi^{\otimes N}\left(x_{1}, x_{2}, \ldots x_{N}\right)`	$U(a, \Lambda)\left(\varphi^{\otimes N}\left((a, \Lambda)^{-1} x_{1}, \ldots,(a, \Lambda)^{-1} x_{N}\right)\right) U(a, \Lambda)^{-1}=\varphi^{\otimes N}\left(x_{1}, x_{2}, \ldots x_{N}\right)$	$U(a, \Lambda)(\varphi^{\otimes N}((a, \Lambda)^{-1}x_1, \ldots, (a, \Lambda)^{-1}x_N))U(a, \Lambda)^{-1} = \varphi^{\otimes N}(x_1, x_2, \ldots x_N)$
cardona-qft-methods_EQ0604 (13)	253	1.000 K|+0	`\varphi=\int d \mu(p)\left(c_{p}^{\dagger} e_{p}+c_{p} e_{-p}\right)=\varphi^{(-)}+\varphi^{(+)}`	$\varphi=\int d \mu(p)\left(c_{p}^{\dagger} e_{p}+c_{p} e_{-p}\right)=\varphi^{(-)}+\varphi^{(+)}$	$\varphi = \int d\mu(p)(c_p^\dagger e_p + c_p e_{-p}) = \varphi^{(-)} + \varphi^{(+)}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>cardona-qft-methods_EQ0607 (15)</td><td>253</td><td>1.000 K|+0</td><td>\left[P_{\mu}, c_{p}^{\dagger}\right]=P_{\mu} c_{p}^{\dagger}, \quad P_{\mu}|0\rangle=0</td><td>$\left[P_{\mu}, c_{p}^{\dagger}\right]=P_{\mu} c_{p}^{\dagger}, \quad P_{\mu}|0\rangle=0$</td><td>$[P_\mu, c_p^\dagger] = P_\mu c_p^\dagger, \quad P_\mu|0\rangle = 0$</td></tr>
<tr><td>cardona-qft-methods_EQ0608 (16)</td><td>254</td><td>1.000 N|+0</td><td>\mathcal{P}_{\mu} e_{p}=-p_{\mu} e_{p}</td><td>$\mathcal{P}_{\mu} e_{p}=-p_{\mu} e_{p}$</td><td>$\mathcal{P}_\mu e_p = -p_\mu e_p$</td></tr>
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<tr><td>cardona-qft-methods_EQ0610 (18)</td><td>254</td><td>1.000 K|+0</td><td>\left(\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \ldots \mathbb{I} \otimes \Delta_{0}\right) \ldots \Delta_{0}\left(\mathcal{P}_{\mu}\right) e_{p_{1}} \otimes e_{p_{2}} \otimes \ldots e_{p_{N}}=-\sum_{i} p_{i j} e_{p_{1}} \otimes e_{p_{2}} \otimes \ldots e_{p_{N}} .</td><td>$\left(\mathbb{I} \otimes \mathbb{I} \otimes \mathbb{I} \ldots \mathbb{I} \otimes \Delta_{0}\right) \ldots \Delta_{0}\left(\mathcal{P}_{\mu}\right) e_{p_{1}} \otimes e_{p_{2}} \otimes \ldots e_{p_{N}}=-\sum_{i} p_{i j} e_{p_{1}} \otimes e_{p_{2}} \otimes \ldots e_{p_{N}} .$</td><td>$(\mathbb{I}\otimes\mathbb{I}\otimes\mathbb{I}\ldots\mathbb{I}\otimes\Delta_0)\ldots\Delta_0(\mathcal{P}_\mu)e_{p_1}\otimes e_{p_2}\otimes\ldots e_{p_N} = -\sum_i p_{ij}e_{p_1}\otimes e_{p_2}\otimes\ldots e_{p_N}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0614 (22)</td><td>254</td><td>1.000 K|+0</td><td>d \mu\left(\Lambda^{-1} p_{i}\right)=d \mu\left(p_{i} \right)</td><td>$d \mu\left(\Lambda^{-1} p_{i}\right)=d \mu\left(p_{i} \right)$</td><td>$d\mu(\Lambda^{-1}p_i) = d\mu(p_i)$</td></tr>
<tr><td>cardona-qft-methods_EQ0616 (24)</td><td>255</td><td>1.000 N|+0</td><td>\left(\alpha^{*} f\right)(p)=f(\alpha p)</td><td>$\left(\alpha^{*} f\right)(p)=f(\alpha p)$</td><td>$(\alpha^* f)(p) = f(\alpha p)$</td></tr>
<tr><td>cardona-qft-methods_EQ0619 (27)</td><td>256</td><td>1.000 K|+0</td><td>U(a, \Lambda) \varphi^{2}\left((a, \Lambda)^{-1} x\right) U(a, \Lambda)^{-1}=\varphi^{2}(x) .</td><td>$U(a, \Lambda) \varphi^{2}\left((a, \Lambda)^{-1} x\right) U(a, \Lambda)^{-1}=\varphi^{2}(x) .$</td><td>$U(a,\Lambda)\varphi^2((a,\Lambda)^{-1}x)U(a,\Lambda)^{-1} = \varphi^2(x).$</td></tr>
<tr><td>cardona-qft-methods_EQ0620 (28)</td><td>256</td><td>1.000 N|+0</td><td>U(a, \Delta) \Psi\left((a, \Delta)^{-1} x\right) U(a, \Delta)^{-1}=\Psi</td><td>$U(a, \Delta) \Psi\left((a, \Delta)^{-1} x\right) U(a, \Delta)^{-1}=\Psi$</td><td>$U(a,\Delta)\Psi((a,\Delta)^{-1}x)U(a,\Delta)^{-1} = \Psi$</td></tr>
<tr><td>cardona-qft-methods_EQ0622 (30)</td><td>257</td><td>1.000 K|+0</td><td>m_{\theta}(\gamma \otimes \delta)=\gamma(x) \delta(x), \quad \gamma, \delta \in \mathcal{A}_{0}\left(M^{(d+1)} \right) .</td><td>$m_{\theta}(\gamma \otimes \delta)=\gamma(x) \delta(x), \quad \gamma, \delta \in \mathcal{A}_{0}\left(M^{(d+1)} \right) .$</td><td>$m_\theta(\gamma\otimes\delta) = \gamma(x)\delta(x), \quad \gamma,\delta\in\mathcal{A}_0(M^{(d+1)}).$</td></tr>
<tr><td>cardona-qft-methods_EQ0625 (33)</td><td>257</td><td>1.000 K|+0</td><td>U_{\theta}(a, \Lambda) \varphi_{\theta}\left((a, \Lambda)^{-1} x\right) U_{\theta}(a, \Lambda)^{-1}= \varphi_{\theta}(x)</td><td>$U_{\theta}(a, \Lambda) \varphi_{\theta}\left((a, \Lambda)^{-1} x\right) U_{\theta}(a, \Lambda)^{-1}= \varphi_{\theta}(x)$</td><td>$U_\theta(a,\Lambda)\varphi_\theta((a,\Lambda)^{-1}x)U_\theta(a,\Lambda)^{-1} = \varphi_\theta(x)$</td></tr>
<tr><td>cardona-qft-methods_EQ0626 (34)</td><td>257</td><td>1.000 N|-1</td><td>\varphi=\int d \mu(p)\left[a_{p}^{\dagger} e_{p}+a_{p} e_{-p}\right]=\varphi_{\theta}^{(-)}+\varphi_{ \theta}^{(+)}, \quad d \mu(p)=\frac{d^{d} p}{2\left|p_{0} \right|} .</td><td>$\varphi=\int d \mu(p)\left[a_{p}^{\dagger} e_{p}+a_{p} e_{-p}\right]=\varphi_{\theta}^{(-)}+\varphi_{ \theta}^{(+)}, \quad d \mu(p)=\frac{d^{d} p}{2\left|p_{0} \right|} .$</td><td>$\varphi = \int d\mu(p)[a_p^\dagger e_p + a_p e_{-p}] = \varphi_\theta^{(-)} + \varphi_\theta^{(+)}, \quad d\mu(p) = \frac{d^dp}{2|p_0|}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0627 (35)</td><td>257</td><td>1.000 K|+0</td><td>a_{p}|0\rangle=0, \forall p .</td><td>$a_{p}|0\rangle=0, \forall p .$</td><td>$a_p|0\rangle = 0, \forall p.$</td></tr>
<tr><td>cardona-qft-methods_EQ0628 (36)</td><td>257</td><td>1.000 K|+0</td><td>U_{\theta}(a, \Lambda) a_{p}^{\dagger} U_{\theta}(a, \Lambda)^{-1}=a_{\Lambda p}^{\dagger}, \quad U_{ \theta}(a, \Lambda) a_{p} U_{\theta}(a, \Lambda)^{-1}=a_{ \Delta p}</td><td>$U_{\theta}(a, \Lambda) a_{p}^{\dagger} U_{\theta}(a, \Lambda)^{-1}=a_{\Lambda p}^{\dagger}, \quad U_{ \theta}(a, \Lambda) a_{p} U_{\theta}(a, \Lambda)^{-1}=a_{ \Delta p}$</td><td>$U_\theta(a,\Lambda)a_p^\dagger U_\theta(a,\Lambda)^{-1} = a_{\Lambda p}^\dagger, \quad U_\theta(a,\Lambda)a_p U_\theta(a,\Lambda)^{-1} = a_{\Delta p}$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0629 (37)	257	■1.000 ● K|+0	U_{\theta}(a, \Lambda)|0\rangle=a_{\Lambda p}^{\dagger}|0 \rangle .	$U_\theta(a,\Lambda)\vert 0\rangle = a^\dagger_{\Lambda p}\vert 0\rangle.$	$U_\theta(a,\Lambda)\vert 0\rangle = a^\dagger_{\Lambda p}\vert 0\rangle.$
cardona-qft-methods_ EQ0630 (38)	258	■1.000 ● N|+0	\int \prod_{i} d \mu\left(p_{i}\right) a_{p_{1}}^{\dagger} a_{p_{2}}^{\dagger}|0\rangle e_{p_{1}} \otimes e_{p_{2}}	$\int \prod_{i} d \mu\left(p_{i}\right) a_{p_{1}}^{\dagger} a_{p_{2}}^{\dagger}\vert 0\rangle e_{p_{1}} \otimes e_{p_{2}}$	$\int \prod_{i} d\mu(p_{i}) a_{p_{1}}^{\dagger} a_{p_{2}}^{\dagger}\vert 0\rangle e_{p_{1}} \otimes e_{p_{2}}$
cardona-qft-methods_ EQ0631 (39)	258	■1.000 ● K|+0	\Delta_{\theta}\left(P_{\mu}\right) e_{p_{1}} \otimes e_{p_{2}}=\left(\mathbb{I} \otimes P_{\mu}+P_{\mu} \otimes \mathbb{I}\right) e_{p_{1}} \otimes e_{p_{2}}=-\left(\sum_{i} p_{i \mu} e_{p_{1}} \otimes e_{p_{2}}\right) .	$\Delta_{\theta}\left(P_{\mu}\right) e_{p_{1}} \otimes e_{p_{2}}=\left(\mathbb{I} \otimes P_{\mu}+P_{\mu} \otimes \mathbb{I}\right) e_{p_{1}} \otimes e_{p_{2}}=-\left(\sum_{i} p_{i \mu} e_{p_{1}} \otimes e_{p_{2}}\right) .$	$\Delta_\theta(P_\mu) e_{p_1} \otimes e_{p_2} = (\mathbb{I} \otimes P_\mu + P_\mu \otimes \mathbb{I}) e_{p_1} \otimes e_{p_2} = -(\sum_i p_{i\mu} e_{p_1} \otimes e_{p_2}).$
cardona-qft-methods_ EQ0632 (40)	258	■1.000 ● N|+1	\left[P_{\mu}^{\theta}, a_{p}^{\dagger}\right]=P_{\mu} a_{p}^{\dagger}	$\left[P_{\mu}^{\theta}, a_{p}^{\dagger}\right]=P_{\mu} a_{p}^{\dagger}$	$[P_{\mu}^{\theta}, a_{p}^{\dagger}]=P_{\mu} a_{p}^{\dagger},$
cardona-qft-methods_ EQ0633 (41)	258	■1.000 ● K|+0	\Delta_{\theta}(\Lambda) \triangleright e_{p_{1}} \otimes e_{p_{2}}=\mathcal{F}_{\theta}^{-1}(\Lambda \otimes \Lambda) \mathcal{F} e_{p_{1}} \otimes e_{p_{2}} .	$\Delta_{\theta}(\Lambda) \triangleright e_{p_{1}} \otimes e_{p_{2}}=\mathcal{F}_{\theta}^{-1}(\Lambda \otimes \Lambda) \mathcal{F} e_{p_{1}} \otimes e_{p_{2}} .$	$\Delta_{\theta}(\Lambda) \triangleright e_{p_{1}} \otimes e_{p_{2}}=\mathcal{F}_{\theta}^{-1}(\Lambda \otimes \Lambda) \mathcal{F} e_{p_{1}} \otimes e_{p_{2}} .$
cardona-qft-methods_ EQ0635 (42)	258	■1.000 ● N|+0	a_{p}^{\dagger}=c_{p}^{\dagger} e^{\frac{i}{2} p \wedge P}	$a_{p}^{\dagger}=c_{p}^{\dagger} e^{\frac{i}{2} p \wedge P}$	$a_{p}^{\dagger}=c_{p}^{\dagger} e^{\frac{i}{2} p \wedge P}$
cardona-qft-methods_ EQ0636 (43)	258	■1.000 ● K|+0	U_{\theta}(a, \Lambda)=U_{0}(a, \Lambda)=U(a, \Lambda) .	$U_{\theta}(a, \Lambda)=U_{0}(a, \Lambda)=U(a, \Lambda) .$	$U_{\theta}(a, \Lambda)=U_{0}(a, \Lambda)=U(a, \Lambda) .$
cardona-qft-methods_ EQ0637 (44)	258	■1.000 ● N|+0	a_{p}=e^{-\frac{i}{2} p \wedge P} c_{p}=c_{p} e^{-\frac{i}{2} p \wedge P}	$a_{p}=e^{-\frac{i}{2} p \wedge P} c_{p}=c_{p} e^{-\frac{i}{2} p \wedge P}$	$a_{p}=e^{-\frac{i}{2} p \wedge P} c_{p}=c_{p} e^{-\frac{i}{2} p \wedge P},$
cardona-qft-methods_ EQ0639 (46)	259	■1.000 ● W|+0	\begin{gathered} U(a, \Lambda) \varphi_{\theta}\left(x_{1}\right) \varphi_{\theta}\left(x_{2}\right) \ldots \varphi_{\theta}\left(x_{N}\right) U(a, \Lambda)^{-1}|0\rangle= \\\\ \varphi_{\theta}\left((a, \Lambda)\left(x_{1}\right)\right) \varphi_{\theta}\left((a, \Lambda)\left(x_{2}\right)\right) \ldots \varphi_{\theta}\left((a, \Lambda)\left(x_{N}\right)\right)|0\rangle \end{gathered}	$\begin{gathered} U(a, \Lambda) \varphi_{\theta}\left(x_{1}\right) \varphi_{\theta}\left(x_{2}\right) \ldots \varphi_{\theta}\left(x_{N}\right) U(a, \Lambda)^{-1}\vert 0\rangle= \\ \varphi_{\theta}\left((a, \Lambda)\left(x_{1}\right)\right) \varphi_{\theta}\left((a, \Lambda)\left(x_{2}\right)\right) \ldots \varphi_{\theta}\left((a, \Lambda)\left(x_{N}\right)\right)\vert 0\rangle \end{gathered}$	$\begin{gathered} U(a,\Lambda)\varphi_\theta(x_1)\varphi_\theta(x_2)\ldots\varphi_\theta(x_N)U(a,\Lambda)^{-1}\vert 0\rangle = \\ \varphi_\theta((a,\Lambda)(x_1))\varphi_\theta((a,\Lambda)(x_2))\ldots\varphi_\theta((a,\Lambda)(x_N))\vert 0\rangle \end{gathered}$
cardona-qft-methods_ EQ0640 (47)	259	■1.000 ● N|+0	\begin{gathered} \tau_{\theta}=\mathcal{F}_{\theta}^{-1} \tau_{0} \mathcal{F}, \\\\ \tau_{0} \alpha \otimes \beta:= \beta \otimes \alpha, \end{gathered}	$\begin{gathered} \tau_{\theta}=\mathcal{F}_{\theta}^{-1} \tau_{0} \mathcal{F}, \\ \tau_{0} \alpha \otimes \beta:= \beta \otimes \alpha, \end{gathered}$	$\begin{gathered} \tau_{\theta}=\mathcal{F}_{\theta}^{-1} \tau_{0} \mathcal{F}, \\ \tau_{0} \alpha \otimes \beta:= \beta \otimes \alpha, \end{gathered}$
cardona-qft-methods_ EQ0641 (49)	259	■1.000 ● K|+0	\tau_{0}^{2}=\mathbb{I} \quad \tau_{\theta}^{2}= \mathbb{I} .	$\tau_{0}^{2}=\mathbb{I} \quad \tau_{\theta}^{2}= \mathbb{I} .$	$\tau_{0}^{2}=\mathbb{I} \quad \tau_{\theta}^{2}= \mathbb{I} .$
cardona-qft-methods_ EQ0642 (50)	259	■1.000 ● N|+0	e_{p_{1}} \otimes S_{\theta} e_{p_{2}}=\frac{\mathbb{I} \pm \tau_{\theta}}{2} e_{p_{1}} \otimes e_{p_{2}}	$e_{p_{1}} \otimes S_{\theta} e_{p_{2}}=\frac{\mathbb{I} \pm \tau_{\theta}}{2} e_{p_{1}} \otimes e_{p_{2}}$	$e_{p_{1}} \otimes S_{\theta} e_{p_{2}}=\frac{\mathbb{I} \pm \tau_{\theta}}{2} e_{p_{1}} \otimes e_{p_{2}}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>cardona-qft-methods_EQ0645 (52)</td><td>260</td><td>1.000 K|+0</td><td><code>=\int \prod_{i}^{2} d \mu\left(p_{i}\right)\left(\exp ^{i p_{1} \wedge p_{2}} a_{p_{2}}^{\dagger} a_{p_{1}}^{\dagger}\right)|0\rangle e^{i p_{1} \wedge p_{2}} e_{\Lambda p_{1}} \otimes S_{\theta} e_{\Lambda p_{2}} .</code></td><td>$=\int \prod_{i}^{2} d \mu\left(p_{i}\right)\left(\exp ^{i p_{1} \wedge p_{2}} a_{p_{2}}^{\dagger} a_{p_{1}}^{\dagger}\right)|0\rangle e^{i p_{1} \wedge p_{2}} e_{\Lambda p_{1}} \otimes S_{\theta} e_{\Lambda p_{2}} .$</td><td>$=\int \prod_{i}^{2} d\mu(p_{i})(\exp^{i p_{1} \wedge p_{2}} a_{p_{2}}^{\dagger} a_{p_{1}}^{\dagger})|0\rangle e^{i p_{1} \wedge p_{2}} e_{\Lambda p_{1}} \otimes S_{\theta} e_{\Lambda p_{2}}.$</td></tr>
<tr><td>cardona-qft-methods_EQ0647 (54)</td><td>260</td><td>1.000 N|+0</td><td><code>\tau_{\theta}^{i j}=\mathcal{F}_{\theta}^{-1} \tau_{0}^{i j} \mathcal{F}=\mathcal{F}_{\theta}^{-2} \tau_{0}^{i j},</code></td><td>$\tau_{\theta}^{i j}=\mathcal{F}_{\theta}^{-1} \tau_{0}^{i j} \mathcal{F}=\mathcal{F}_{\theta}^{-2} \tau_{0}^{i j},$</td><td>$\tau_{\theta}^{ij}=\mathcal{F}_{\theta}^{-1} \tau_{0}^{ij} \mathcal{F}=\mathcal{F}_{\theta}^{-2} \tau_{0}^{ij},$</td></tr>
<tr><td>cardona-qft-methods_EQ0648 (55)</td><td>260</td><td>1.000 K|+0</td><td><code>\left(\tau_{0}^{i j}\right)^{2}=\mathbb{I}</code></td><td>$\left(\tau_{0}^{i j}\right)^{2}=\mathbb{I}$</td><td>$(\tau_{0}^{ij})^{2}=\mathbb{I}$</td></tr>
<tr><td>cardona-qft-methods_EQ0649 (56)</td><td>260</td><td>1.000 N|+0</td><td><code>\tau_{\theta}^{i, i+1} \tau_{\theta}^{i+1, i+2} \tau_{\theta}^{i, i+1}=\tau_{\theta}^{i+1, i+2} \tau_{\theta}^{i, i+1} \tau_{\theta}^{i+1, i+2}</code></td><td>$\tau_{\theta}^{i, i+1} \tau_{\theta}^{i+1, i+2} \tau_{\theta}^{i, i+1}=\tau_{\theta}^{i+1, i+2} \tau_{\theta}^{i, i+1} \tau_{\theta}^{i+1, i+2}$</td><td>$\tau_{\theta}^{i,i+1} \tau_{\theta}^{i+1,i+2} \tau_{\theta}^{i,i+1}=\tau_{\theta}^{i+1,i+2} \tau_{\theta}^{i,i+1} \tau_{\theta}^{i+1,i+2}$</td></tr>
<tr><td>cardona-qft-methods_EQ0651 (58)</td><td>260</td><td>1.000 N|+1</td><td><code>\varphi_{\theta}^{*}=\varphi_{0}^{*} e-\frac{i}{2} \overleftarrow{\partial} \wedge P</code></td><td>$\varphi_{\theta}^{*}=\varphi_{0}^{*} e-\frac{i}{2} \overleftarrow{\partial} \wedge P$</td><td>$\varphi_{\theta}^{*}=\varphi_{0}^{*} e-\frac{i}{2} \overleftarrow{\partial} \wedge P.$</td></tr>
<tr><td>cardona-qft-methods_EQ0653 (3)</td><td>261</td><td>1.000 K|+0</td><td><code>\tau_{\theta}\left|x_{1}, x_{2}\right\rangle=-\left|x_{1}, x_{2}\right\rangle .</code></td><td>$\tau_{\theta}\left|x_{1}, x_{2}\right\rangle=-\left|x_{1}, x_{2}\right\rangle .$</td><td>$\tau_{\theta}|x_{1}, x_{2}\rangle=-|x_{1}, x_{2}\rangle.$</td></tr>
<tr><td>cardona-qft-methods_EQ0656 (6)</td><td>262</td><td>1.000 N|+0</td><td><code>\left\langle\left[0 \mid \rho_{\theta}(x), \eta_{\theta}(y)\right] \mid 0\right\rangle=0 .</code></td><td>$\left\langle\left[0 \mid \rho_{\theta}(x), \eta_{\theta}(y)\right] \mid 0\right\rangle=0 .$</td><td>$\langle[0|\rho_{\theta}(x), \eta_{\theta}(y)]|0\rangle=0.$</td></tr>
<tr><td>cardona-qft-methods_EQ0657 (7)</td><td>262</td><td>1.000 W|+0</td><td><code>\omega_{\beta}\left(\rho_{\theta}(x) \eta_{\theta}(y)\right)=\frac{\operatorname{Tr}\left[e^{-\beta H} \rho_{\theta}(x) \eta_{\theta}(y)\right]}{\operatorname{Tr} e^{-\beta H}},</code></td><td>$\omega_{\beta}\left(\rho_{\theta}(x) \eta_{\theta}(y)\right)=\frac{\operatorname{Tr}\left[e^{-\beta H} \rho_{\theta}(x) \eta_{\theta}(y)\right]}{\operatorname{Tr} e^{-\beta H}},$</td><td>$\omega_{\beta}(\rho_{\theta}(x) \eta_{\theta}(y))=\frac{Tr\left[e^{-\beta H} \rho_{\theta}(x) \eta_{\theta}(y)\right]}{Tr e^{-\beta H}},$</td></tr>
<tr><td>cardona-qft-methods_EQ0658 (8)</td><td>262</td><td>1.000 K|+0</td><td><code>d s^{2}=d t^{2}-a(t)^{2}\left[d r^{2}+r^{2} d \Omega^{2}\right]</code></td><td>$d s^{2}=d t^{2}-a(t)^{2}\left[d r^{2}+r^{2} d \Omega^{2}\right]$</td><td>$ds^{2}=dt^{2}-a(t)^{2}\left[dr^{2}+r^{2} d\Omega^{2}\right]$</td></tr>
<tr><td>cardona-qft-methods_EQ0662 (13)</td><td>263</td><td>1.000 W|+2</td><td><code>U(\Lambda) S^{(2)} U^{-1}(\Lambda)=\frac{-i^{2}}{2!} \int d_{x}^{4} d^{4} y\left(\theta\left(x_{0}-y_{0}\right)\left[H_{I}(x), H_{I}(y)\right]+H_{I}(y) H_{I}(x)\right)</code></td><td>$U(\Lambda) S^{(2)} U^{-1}(\Lambda)=\frac{-i^{2}}{2!} \int d_{x}^{4} d^{4} y\left(\theta\left(x_{0}-y_{0}\right)\left[H_{I}(x), H_{I}(y)\right]+H_{I}(y) H_{I}(x)\right)$</td><td>$U(\Lambda) S^{(2)} U^{-1}(\Lambda)=\frac{-i^{2}}{2!} \int d_{x}^{4} d^{4} y\,(\theta(x_{0}-y_{0})[H_{I}(x), H_{I}(y)]+H_{I}(y) H_{I}(x))$</td></tr>
<tr><td>cardona-qft-methods_EQ0663 (14)</td><td>263</td><td>1.000 K|+0</td><td><code>\theta\left(x_{0}-y_{0}\right)\left[H_{I}(\Lambda x), H_{I}(\Lambda y)\right]=\theta\left((\Lambda x)_{0}-(\Lambda y)_{0}\right)\left[H_{I}(\Lambda x), H_{I}(\Lambda y)\right]</code></td><td>$\theta\left(x_{0}-y_{0}\right)\left[H_{I}(\Lambda x), H_{I}(\Lambda y)\right]=\theta\left((\Lambda x)_{0}-(\Lambda y)_{0}\right)\left[H_{I}(\Lambda x), H_{I}(\Lambda y)\right]$</td><td>$\theta(x_{0}-y_{0})[H_{I}(\Lambda x), H_{I}(\Lambda y)]=\theta((\Lambda x)_{0}-(\Lambda y)_{0})[H_{I}(\Lambda x), H_{I}(\Lambda y)]$</td></tr>
<tr><td>cardona-qft-methods_EQ0665 (16)</td><td>263</td><td>1.000 N|+0</td><td><code>\left[H_{I}(\Lambda x), H_{I}(\Lambda y)\right]=0, \quad x \sim y,</code></td><td>$\left[H_{I}(\Lambda x), H_{I}(\Lambda y)\right]=0, \quad x \sim y,$</td><td>$[H_{I}(\Lambda x), H_{I}(\Lambda y)]=0, \quad x \sim y,$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0666 (17)	263	1.000 K|+0	`\theta_{0 i}\left(\vec{P}_{i n c}\right)_{i}`	$\theta_{0 i}\left(\vec{P}_{i n c}\right)_{i}$	$\theta_{0 i}\left(\vec{P}_{i n c}\right)_{i}$
cardona-qft-methods_ EQ0667 (18)	263	1.000 K|+0	`\theta_{0 i}\left(\vec{P}_{i n c}\right)_{i}=0`	$\theta_{0 i}\left(\vec{P}_{i n c}\right)_{i}=0$	$\theta_{0 i}\left(\vec{P}_{i n c}\right)_{i}=0$
cardona-qft-methods_ EQ0668 (19)	264	1.000 K|+0	`\frac{\Delta T(\widehat{n})}{T}=\sum_{l m} a_{l m} Y_{l m}(\widehat{n}) .`	$\frac{\Delta T(\widehat{n})}{T}=\sum_{l m} a_{l m} Y_{l m}(\widehat{n}) .$	$\frac{\Delta T(\widehat{n})}{T}=\sum_{l m} a_{l m} Y_{l m}(\widehat{n}) .$
cardona-qft-methods_ EQ0669 (20)	264	1.000 K|+0	`a_{l m}=4 \pi(-i)^{l} \int \frac{d^{3} k}{(2 \pi)^{3}} \Phi(k) \Delta_{l}^{T}(k) Y_{l m}^{*}(\widehat{k})`	$a_{l m}=4 \pi(-i)^{l} \int \frac{d^{3} k}{(2 \pi)^{3}} \Phi(k) \Delta_{l}^{T}(k) Y_{l m}^{*}(\widehat{k})$	$a_{l m}=4 \pi(-i)^{l} \int \frac{d^{3} k}{(2 \pi)^{3}} \Phi(k) \Delta_{l}^{T}(k) Y_{l m}^{*}(\widehat{k})$
cardona-qft-methods_ EQ0671 (21)	264	1.000 K|+0	`P_{\Phi}=\left.\frac{8 \pi G H^{2}}{9 \varepsilon k^{3}} \right|_{a H=k}`	$P_{\Phi}=\left.\frac{8 \pi G H^{2}}{9 \varepsilon k^{3}} \right\|_{a H=k}$	$P_{\Phi}=\frac{8 \pi G H^{2}}{9 \varepsilon k^{3}}\|_{a H=k}$
cardona-qft-methods_ EQ0672 (22)	264	1.000 N|+1	`\varphi(\alpha, x)=\left(\frac{\alpha}{3}\right)^{\frac{3}{2}} \int d^{3} x^{\prime} \varphi_{\theta}(x) \exp ^{-\alpha\left(x^{\prime}-x\right)^{2}}`	$\varphi(\alpha, x)=\left(\frac{\alpha}{3}\right)^{\frac{3}{2}} \int d^{3} x^{\prime} \varphi_{\theta}(x) \exp ^{-\alpha\left(x^{\prime}-x\right)^{2}}$	$\varphi(\alpha, x)=\left(\frac{\alpha}{3}\right)^{\frac{3}{2}} \int d^{3} x^{\prime} \varphi_{\theta}(x) \exp ^{-\alpha\left(x^{\prime}-x\right)^{2}},$
cardona-qft-methods_ EQ0673 (23)	264	1.000 K|+0	`\Delta \varphi_{\theta}\left(\alpha, x_{1}\right) \Delta \varphi_{\theta}\left(\alpha, x_{2}\right) \geq\langle 0| \varphi_{\theta}\left(\alpha, x_{1}\right), \varphi_{\theta}\left(\alpha, x_{2}\right)|0\rangle`	$\Delta \varphi_{\theta}\left(\alpha, x_{1}\right) \Delta \varphi_{\theta}\left(\alpha, x_{2}\right) \geq\langle 0\| \varphi_{\theta}\left(\alpha, x_{1}\right), \varphi_{\theta}\left(\alpha, x_{2}\right)\|0\rangle$	$\Delta \varphi_{\theta}(\alpha, x_{1}) \Delta \varphi_{\theta}(\alpha, x_{2}) \geq\langle 0\| \varphi_{\theta}(\alpha, x_{1}), \varphi_{\theta}(\alpha, x_{2})\|0\rangle$
cardona-qft-methods_ EQ0675 (1)	268	1.000 K|+0	`\vec{S}=\frac{1}{2} \vec{\sigma},`	$\vec{S}=\frac{1}{2} \vec{\sigma},$	$\vec{S}=\frac{1}{2} \vec{\sigma},$
cardona-qft-methods_ EQ0676 (2)	268	1.000 K|+0	`\sigma^{1}=\left(\begin{array}{cc} 0 & +1 \\ +1 & 0 \end{array}\right), \quad \sigma^{2}=\left(\begin{array}{cc} 0 & -i \\ +i & 0 \end{array}\right), \quad \sigma^{3}=\left(\begin{array}{cc} +1 & 0 \\ 0 & -1 \end{array}\right) .`	$\sigma^{1}=\left(\begin{array}{cc} 0 & +1 \\ +1 & 0 \end{array}\right), \quad \sigma^{2}=\left(\begin{array}{cc} 0 & -i \\ +i & 0 \end{array}\right), \quad \sigma^{3}=\left(\begin{array}{cc} +1 & 0 \\ 0 & -1 \end{array}\right) .$	$\sigma^{1}=\left(\begin{array}{cc} 0 & +1 \\ +1 & 0 \end{array}\right), \quad \sigma^{2}=\left(\begin{array}{cc} 0 & -i \\ +i & 0 \end{array}\right), \quad \sigma^{3}=\left(\begin{array}{cc} +1 & 0 \\ 0 & -1 \end{array}\right) .$
cardona-qft-methods_ EQ0677	268	1.000 N|+1	`\left[\sigma^{i}, \sigma^{j}\right]=2 i \epsilon^{i j k} \sigma^{k}`	$\left[\sigma^{i}, \sigma^{j}\right]=2 i \epsilon^{i j k} \sigma^{k}$	$[\sigma^{i}, \sigma^{j}]=2 i \epsilon^{i j k} \sigma^{k},$
cardona-qft-methods_ EQ0679 (4)	269	1.000 N|+1	`s=0, \frac{1}{2}, 1, \frac{3}{2}, \ldots,`	$s=0, \frac{1}{2}, 1, \frac{3}{2}, \ldots,$	$s=0, \frac{1}{2}, 1, \frac{3}{2}, \ldots,$
cardona-qft-methods_ EQ0680 (5)	269	1.000 K|+0	`\vec{S} \cdot \mid \text { state }\rangle=0,`	$\vec{S} \cdot \mid \text { state }\rangle=0,$	$\vec{S} \cdot \|\text{state}\rangle=0,$
cardona-qft-methods_ EQ0682 (6)	269	1.000 N|+1	`S^{ \pm}=S^{1} \pm i S^{2}`	$S^{ \pm}=S^{1} \pm i S^{2}$	$S^{ \pm}=S^{1} \pm i S^{2} .$
cardona-qft-methods_ EQ0683	269	1.000 N|+1	`\left[S^{3}, S^{ \pm}\right]= \pm S^{ \pm}, \quad \left[S^{+}, S^{-}\right]=2 S^{3}`	$\left[S^{3}, S^{ \pm}\right]= \pm S^{ \pm}, \quad \left[S^{+}, S^{-}\right]=2 S^{3}$	$[S^{3}, S^{ \pm}]= \pm S^{ \pm}, \qquad [S^{+}, S^{-}]=2 S^{3} .$
cardona-qft-methods_ EQ0685	269	1.000 K|+0	`|\uparrow\rangle=\binom{1}{0}, \quad|\downarrow\rangle=\binom{0}{1} .`	$\|\uparrow\rangle=\binom{1}{0}, \quad\|\downarrow\rangle=\binom{0}{1} .$	$\|\uparrow\rangle=\binom{1}{0}, \qquad\|\downarrow\rangle=\binom{0}{1} .$
cardona-qft-methods_ EQ0686	269	1.000 N|+0	`S^{3}|\uparrow\rangle=+\frac{1}{2}|\uparrow\rangle, \quad S^{3}|\downarrow\rangle=-\frac{1}{2}|\downarrow\rangle,`	$S^{3}\|\uparrow\rangle=+\frac{1}{2}\|\uparrow\rangle, \quad S^{3}\|\downarrow\rangle=-\frac{1}{2}\|\downarrow\rangle,$	$S^{3}\|\uparrow\rangle=+\frac{1}{2}\|\uparrow\rangle, \qquad S^{3}\|\downarrow\rangle=-\frac{1}{2}\|\downarrow\rangle,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0688 (7)	270	■1.000 ● N|+0	`\mathcal{H}=\mathbb{C}^{2} \otimes \cdots \otimes \mathbb{C}^{2}=\left(\mathbb{C}^{2}\right)^{\otimes L}`	$\mathcal{H}=\mathbb{C}^{2} \otimes \cdots \otimes \mathbb{C}^{2}=\left(\mathbb{C}^{2}\right)^{\otimes L}$	$\mathcal{H}=\mathbb{C}^{2} \otimes \cdots \otimes \mathbb{C}^{2}=(\mathbb{C}^{2})^{\otimes L}$
cardona-qft-methods_ EQ0689	270	■1.000 ● N|-1	`|\uparrow \uparrow \cdots \downarrow \uparrow\rangle= \binom{1}{0} \otimes\binom{1}{0} \otimes \cdots \otimes \binom{0}{1} \otimes\binom{1}{0} .`	$\vert\uparrow \uparrow \cdots \downarrow \uparrow\rangle= \binom{1}{0} \otimes\binom{1}{0} \otimes \cdots \otimes \binom{0}{1} \otimes\binom{1}{0} .$	$\vert\uparrow \uparrow \cdots \downarrow \uparrow\rangle= \binom{1}{0} \otimes\binom{1}{0} \otimes \cdots \otimes \binom{0}{1} \otimes\binom{1}{0} .$
cardona-qft-methods_ EQ0690	270	■1.000 ● N|+0	`\frac{1}{2} \otimes \frac{1}{2}=1 \oplus 0 .`	$\frac{1}{2} \otimes \frac{1}{2}=1 \oplus 0 .$	$\frac{1}{2} \otimes \frac{1}{2}=1 \oplus 0 .$
cardona-qft-methods_ EQ0691 (8)	270	■1.000 ● N|+0	`\left\{|\uparrow \uparrow\rangle, \frac{1}{\sqrt{2}} \left(|\uparrow \downarrow\rangle+|\downarrow \uparrow \rangle,|\downarrow \downarrow\rangle, \frac{1}{ \sqrt{2}}(|\uparrow \downarrow\rangle-|\downarrow \uparrow\rangle\} .\right.\right.`	$\left\{\vert\uparrow \uparrow\rangle, \frac{1}{\sqrt{2}} \left(\vert\uparrow \downarrow\rangle+\vert\downarrow \uparrow \rangle,\vert\downarrow \downarrow\rangle, \frac{1}{ \sqrt{2}}(\vert\uparrow \downarrow\rangle-\vert\downarrow \uparrow\rangle\} .\right.\right.$	$\{\vert\uparrow\uparrow\rangle, \frac{1}{\sqrt{2}}(\vert\uparrow\downarrow\rangle+\vert\downarrow\uparrow\rangle), \vert\downarrow\downarrow\rangle, \frac{1}{\sqrt{2}}(\vert\uparrow\downarrow\rangle-\vert\downarrow\uparrow\rangle)\} .$
cardona-qft-methods_ EQ0692 (8)	270	■1.000 ● N|+1	`\hat{H}=4 \sum_{l=1}^{L}\left(\frac{1}{4}-\vec{S}_{l} \cdot \vec{S}_{l+1}\right),`	$\hat{H}=4 \sum_{l=1}^{L}\left(\frac{1}{4}-\vec{S}_{l} \cdot \vec{S}_{l+1}\right),$	$\hat{H}=4 \sum_{l=1}^{L}\left(\frac{1}{4}-\vec{S}_{l} \cdot \vec{S}_{l+1}\right),$
cardona-qft-methods_ EQ0693	270	■1.000 ● K|+0	`\vec{S}_{L+1}=\vec{S}_{1} .`	$\vec{S}_{L+1}=\vec{S}_{1} .$	$\vec{S}_{L+1}=\vec{S}_{1} .$
cardona-qft-methods_ EQ0694 (9)	271	■1.000 ● N|+0	`\hat{H}|\psi\rangle=E|\psi\rangle, \quad|\psi\rangle \in \left(\mathbb{C}^{2}\right)^{\otimes L},`	$\hat{H}\vert\psi\rangle=E\vert\psi\rangle, \quad\vert\psi\rangle \in \left(\mathbb{C}^{2}\right)^{\otimes L},$	$\hat{H}\vert\psi\rangle=E\vert\psi\rangle, \qquad \vert\psi\rangle \in (\mathbb{C}^{2})^{\otimes L},$
cardona-qft-methods_ EQ0696	271	■1.000 ● K|+0	`\mathbb{P}_{l, l+1}=\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right),`	$\mathbb{P}_{l, l+1}=\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right),$	$\mathbb{P}_{l, l+1}=\left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right),$
cardona-qft-methods_ EQ0700	272	■1.000 ● K|+0	`\hat{U}=\mathbb{P}_{1,2} \cdots \mathbb{P}_{L-1, L} .`	$\hat{U}=\mathbb{P}_{1,2} \cdots \mathbb{P}_{L-1, L} .$	$\hat{U}=\mathbb{P}_{1,2} \cdots \mathbb{P}_{L-1, L} .$
cardona-qft-methods_ EQ0701 (11)	272	■1.000 ● K|+0	`\hat{U}^{L}=\mathbb{I} .`	$\hat{U}^{L}=\mathbb{I} .$	$\hat{U}^{L}=\mathbb{I} .$
cardona-qft-methods_ EQ0702 (12)	272	■1.000 ● K|+0	`U=e^{2 \pi i n / L},`	$U=e^{2 \pi i n / L},$	$U=e^{2 \pi i n / L},$
cardona-qft-methods_ EQ0703 (13)	272	■1.000 ● N|+2	`\left(\frac{u_{k}+\frac{i}{2}}{u_{k}-\frac{i}{2}}\right)^{L}=\prod_{j \neq k}^{M} \frac{u_{k}-u_{j}+i}{u_{k}-u_{j}-i}, \quad k=1, \ldots, M`	$\left(\frac{u_{k}+\frac{i}{2}}{u_{k}-\frac{i}{2}}\right)^{L}=\prod_{j \neq k}^{M} \frac{u_{k}-u_{j}+i}{u_{k}-u_{j}-i}, \quad k=1, \ldots, M$	$\left(\frac{u_{k}+\frac{i}{2}}{u_{k}-\frac{i}{2}}\right)^{L}=\prod_{j \neq k}^{M} \frac{u_{k}-u_{j}+i}{u_{k}-u_{j}-i}, \qquad k=1, \ldots, M .$
cardona-qft-methods_ EQ0704	272	■1.000 ● N|+0	`E=2 \sum_{k=1}^{M} \frac{1}{u_{k}^{2}+\frac{1}{4}}, \quad U=\prod_{k=1}^{M} \frac{u_{k}+\frac{i}{2}}{u_{k}-\frac{i}{2}} .`	$E=2 \sum_{k=1}^{M} \frac{1}{u_{k}^{2}+\frac{1}{4}}, \quad U=\prod_{k=1}^{M} \frac{u_{k}+\frac{i}{2}}{u_{k}-\frac{i}{2}} .$	$E=2 \sum_{k=1}^{M} \frac{1}{u_{k}^{2}+\frac{1}{4}}, \qquad U=\prod_{k=1}^{M} \frac{u_{k}+\frac{i}{2}}{u_{k}-\frac{i}{2}} .$
cardona-qft-methods_ EQ0705 (1)	272	■1.000 ● K|+0	`|\psi\rangle=\sum_{1 \leq l_{1}<\cdots<l_{M} \leq L} \psi \left(l_{1}, \ldots, l_{M}\right) S_{l_{1}}^{-} \cdots S_{l_{M}}^{-}|0\rangle,`	$\vert\psi\rangle=\sum_{1 \leq l_{1}<\cdots<l_{M} \leq L} \psi \left(l_{1}, \ldots, l_{M}\right) S_{l_{1}}^{-} \cdots S_{l_{M}}^{-}\vert 0\rangle,$	$\vert\psi\rangle=\sum_{1 \leq l_{1}<\cdots<l_{M} \leq L} \psi(l_{1}, \ldots, l_{M}) S_{l_{1}}^{-} \cdots S_{l_{M}}^{-}\vert 0\rangle,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0706 (2)	273	■1.000 N|+1	`\psi\left(l_{1}, \ldots, l_{M}\right)=\sum_{\tau \in S_{n}} A(\tau) e^{i p_{\tau_{1}} l_{1}+\cdots+i p_{\tau_{M}} l_{M}}`	$\psi\left(l_{1}, \ldots, l_{M}\right)=\sum_{\tau \in S_{n}} A(\tau) e^{i p_{\tau_{1}} l_{1}+\cdots+i p_{\tau_{M}} l_{M}}$	$\psi(l_1,\ldots,l_M)=\sum_{\tau\in S_n} A(\tau)e^{ip_{\tau_1}l_1+\cdots+ip_{\tau_M}l_M}\,,$
cardona-qft-methods_EQ0707	273	■1.000 N|+0	`A(\tau)=\operatorname{sgn}(\tau) \prod_{j<k}\left(e^{i p_{k}+i p_{j}}-2 e^{i p_{k}}+1\right),`	$A(\tau)=\operatorname{sgn}(\tau) \prod_{j<k}\left(e^{i p_{k}+i p_{j}}-2 e^{i p_{k}}+1\right),$	$A(\tau)=\mathrm{sgn}(\tau)\prod_{j<k}(e^{ip_k+ip_j}-2e^{ip_k}+1)\,,$
cardona-qft-methods_EQ0708	273	■1.000 K|+0	`E=\sum_{k=1}^{M} 8 \sin ^{2}\left(\frac{p_{k}}{2}\right) .`	$E=\sum_{k=1}^{M} 8 \sin ^{2}\left(\frac{p_{k}}{2}\right) .$	$E=\sum_{k=1}^{M} 8\,\sin^2(\frac{p_k}{2})\,.$
cardona-qft-methods_EQ0710	273	■1.000 N|+0	`S\left(p_{k}, p_{j}\right)=-\frac{e^{i p_{k}+i p_{j}}-2 e^{i p_{k}}+1}{e^{i p_{k}+i p_{j}}-2 e^{i p_{j}}+1} .`	$S\left(p_{k}, p_{j}\right)=-\frac{e^{i p_{k}+i p_{j}}-2 e^{i p_{k}}+1}{e^{i p_{k}+i p_{j}}-2 e^{i p_{j}}+1} .$	$S(p_k,p_j)=-\frac{e^{ip_k+ip_j}-2e^{ip_k}+1}{e^{ip_k+ip_j}-2e^{ip_j}+1}\,.$
cardona-qft-methods_EQ0711	273	■1.000 N|+0	`e^{i p_{k}}=\frac{u_{k}+\frac{i}{2}}{u_{k}-\frac{i}{2}} \quad \Longleftrightarrow \quad \frac{1}{2} \cot \left( \frac{p_{k}}{2}\right)=u_{k},`	$e^{i p_{k}}=\frac{u_{k}+\frac{i}{2}}{u_{k}-\frac{i}{2}} \quad \Longleftrightarrow \quad \frac{1}{2} \cot \left(\frac{p_{k}}{2}\right)=u_{k},$	$e^{ip_k}=\frac{u_k+\frac{i}{2}}{u_k-\frac{i}{2}} \qquad \Longleftrightarrow \qquad \frac12\cot(\frac{p_k}{2})=u_k\,,$
cardona-qft-methods_EQ0713	274	■1.000 N|+2	`R_{a, b}\left(u-u^{\prime}\right) R_{a, c}(u) R_{b, c} \left(u^{\prime}\right)=R_{b, c}\left(u^{\prime}\right) R_{a, c}(u) R_{a, b}\left(u-u^{\prime}\right)`	$R_{a, b}\left(u-u^{\prime}\right) R_{a, c}(u) R_{b, c} \left(u^{\prime}\right)=R_{b, c}\left(u^{\prime}\right) R_{a, c}(u) R_{a, b}\left(u-u^{\prime}\right)$	$R_{a,b}(u-u')R_{a,c}(u)R_{b,c}(u')=R_{b,c}(u')R_{a,c}(u)R_{a,b}(u-u')$
cardona-qft-methods_EQ0714	274	■1.000 N|+0	`R_{a, b}\left(u-u^{\prime}\right)=`	$R_{a, b}\left(u-u^{\prime}\right)=$	$R_{a,b}(u-u')=$
cardona-qft-methods_EQ0715	275	■1.000 K|+0	`R_{a, b}\left(u-u^{\prime}\right) L_{a, l}(u) L_{b, l} \left(u^{\prime}\right)=L_{b, l}\left(u^{\prime}\right) L_{a, l}(u) R_{a, b}\left(u-u^{\prime}\right) .`	$R_{a, b}\left(u-u^{\prime}\right) L_{a, l}(u) L_{b, l} \left(u^{\prime}\right)=L_{b, l}\left(u^{\prime}\right) L_{a, l}(u) R_{a, b}\left(u-u^{\prime}\right) .$	$R_{a,b}(u-u')L_{a,l}(u)L_{b,l}(u')=L_{b,l}(u')L_{a,l}(u)R_{a,b}(u-u').$
cardona-qft-methods_EQ0719	276	■1.000 K|+0	`\hat{T}(u) \hat{T}\left(u^{\prime}\right)=\hat{T} \left(u^{\prime}\right) \hat{T}(u)`	$\hat{T}(u) \hat{T}\left(u^{\prime}\right)=\hat{T} \left(u^{\prime}\right) \hat{T}(u)$	$\hat{T}(u)\hat{T}(u')=\hat{T}(u')\hat{T}(u)$
cardona-qft-methods_EQ0720	276	■1.000 K|+0	`\left[\hat{T}(u), \hat{T}\left(u^{\prime}\right)\right]=0 .`	$\left[\hat{T}(u), \hat{T}\left(u^{\prime}\right)\right]=0 .$	$[\hat{T}(u),\hat{T}(u')]=0.$
cardona-qft-methods_EQ0721	276	■1.000 N|+0	`\frac{1}{i^{L}} \hat{T}\left(\frac{i}{2}\right)= \operatorname{tr}_{a}\left(P_{a, L} \cdots P_{a, L-1} \cdots P_{a, 1}\right)=P_{12} P_{23} \cdots P_{L-1, L}= \hat{U}=e^{i \hat{P}}`	$\frac{1}{i^{L}} \hat{T}\left(\frac{i}{2}\right)= \operatorname{tr}_{a}\left(P_{a, L} \cdots P_{a, L-1} \cdots P_{a, 1}\right)=P_{12} P_{23} \cdots P_{L-1, L}= \hat{U}=e^{i \hat{P}}$	$\frac{1}{i^L}\hat{T}(\frac{i}{2})=\mathrm{tr}_a(P_{a,L}\cdots P_{a,L-1}\cdots P_{a,1})=P_{12}P_{23}\cdots P_{L-1,L}=\hat{U}=e^{i\hat{P}},$
cardona-qft-methods_EQ0722	276	■1.000 N|+0	`\left.i \hat{T}(u)^{-1} \frac{\mathrm{~d}}{\mathrm{~d} u} \hat{T}(u)\right|_{u=\frac{i}{2}}=\sum_{l=1}^{L} P_{l-1, l} .`	$\left.i \hat{T}(u)^{-1} \frac{\mathrm{~d}}{\mathrm{~d} u} \hat{T}(u)\right\|_{u=\frac{i}{2}}=\sum_{l=1}^{L} P_{l-1, l} .$	$i\hat{T}(u)^{-1}\frac{\mathrm{d}}{\mathrm{d}u}\hat{T}(u)\Big\|_{u=\frac{i}{2}}=\sum_{l=1}^{L}P_{l-1,l}.$
cardona-qft-methods_EQ0723	276	■1.000 N|+0	`\hat{H}=2 \sum_{l=1}^{L}\left(1_{l, l+1}-P_{l, l+1} \right)=2 L-\left.2 i \frac{\mathrm{~d}}{\mathrm{~d} u} \log \hat{T}(u)\right|_{u=\frac{i}{2}} .`	$\hat{H}=2 \sum_{l=1}^{L}\left(1_{l, l+1}-P_{l, l+1} \right)=2 L-\left.2 i \frac{\mathrm{~d}}{\mathrm{~d} u} \log \hat{T}(u)\right\|_{u=\frac{i}{2}} .$	$\hat{H}=2\sum_{l=1}^{L}(1_{l,l+1}-P_{l,l+1})=2L-2i\,\frac{\mathrm{d}}{\mathrm{d}u}\log\hat{T}(u)\Big\|_{u=\frac{i}{2}}\,.$
cardona-qft-methods_EQ0724	276	■1.000 N|+0	`\log \hat{T}(u)=\hat{Q}_{1}+\left(u-\frac{i}{2}\right) \hat{Q}_{2}+\left(u-\frac{i}{2}\right)^{2} \hat{Q}_{3}+ \left(u-\frac{i}{2}\right)^{3} \hat{Q}_{4}+\ldots,`	$\log \hat{T}(u)=\hat{Q}_{1}+\left(u-\frac{i}{2}\right) \hat{Q}_{2}+\left(u-\frac{i}{2}\right)^{2} \hat{Q}_{3}+ \left(u-\frac{i}{2}\right)^{3} \hat{Q}_{4}+\ldots,$	$\log\hat{T}(u)=\hat{Q}_1+(u-\frac{i}{2})\hat{Q}_2+(u-\frac{i}{2})^2\hat{Q}_3+(u-\frac{i}{2})^3\hat{Q}_4+\ldots,$
cardona-qft-methods_EQ0725	276	■1.000 N|+0	`\hat{Q}_{1} \propto \hat{P}, \quad \hat{Q}_{2} \propto \hat{H}-2 L .`	$\hat{Q}_{1} \propto \hat{P}, \quad \hat{Q}_{2} \propto \hat{H}-2 L .$	$\hat{Q}_1\propto\hat{P}, \quad \hat{Q}_2\propto\hat{H}-2L.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
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<tr><td>cardona-qft-methods_ EQ0726</td><td>276</td><td>■ 1.000 ● N|+1</td><td>\left[\hat{Q}_{n}, \hat{Q}_{m}\right]=0</td><td>$\left[\hat{Q}_{n}, \hat{Q}_{m}\right]=0$</td><td>$[\hat{Q}_n,\hat{Q}_m]=0.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0727</td><td>277</td><td>■ 1.000 ● K|+0</td><td>\hat{T}(u)|\psi\rangle=T(u)|\psi\rangle .</td><td>$\hat{T}(u)|\psi\rangle=T(u)|\psi\rangle .$</td><td>$\hat{T}(u)|\psi\rangle=T(u)|\psi\rangle.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0729</td><td>277</td><td>■ 1.000 ● N|+1</td><td>|\psi\rangle=\hat{B}\left(u_{1}\right) \cdots \hat{B} \left(u_{M}\right)|0\rangle</td><td>$|\psi\rangle=\hat{B}\left(u_{1}\right) \cdots \hat{B}\left(u_{M}\right)|0\rangle$</td><td>$|\psi\rangle=\hat{B}(u_1)\cdots\hat{B}(u_M)|0\rangle$</td></tr>
<tr><td>cardona-qft-methods_ EQ0731</td><td>277</td><td>■ 1.000 ● N|+0</td><td>T(u)=\left(u+\frac{i}{2}\right)^{L} \prod_{j=1}^{M} \frac{u-u_{j}-i}{u-u_{j}}+\left(u-\frac{i}{2}\right)^{L} \prod_{j=1}^{M} \frac{u-u_{j}+i}{u-u_{j}} .</td><td>$T(u)=\left(u+\frac{i}{2}\right)^{L} \prod_{j=1}^{M} \frac{u-u_{j}-i}{u-u_{j}}+\left(u-\frac{i}{2}\right)^{L} \prod_{j=1}^{M} \frac{u-u_{j}+i}{u-u_{j}} .$</td><td>$T(u)=(u+\frac{i}{2})^L\prod_{j=1}^M\frac{u-u_j-i}{u-u_j}+(u-\frac{i}{2})^L\prod_{j=1}^M\frac{u-u_j+i}{u-u_j}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0732</td><td>278</td><td>■ 1.000 ● N|+1</td><td>Q(u)=\prod_{j=1}^{M}\left(u-u_{j}\right) .</td><td>$Q(u)=\prod_{j=1}^{M}\left(u-u_{j}\right) .$</td><td>$Q(u)=\prod_{j=1}^M(u-u_j).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0733</td><td>278</td><td>■ 1.000 ● K|+0</td><td>T(u) Q(u)=\left(u+\frac{i}{2}\right)^{L} Q(u-i)+\left(u-\frac{i}{2}\right)^{L} Q(u+i) .</td><td>$T(u) Q(u)=\left(u+\frac{i}{2}\right)^{L} Q(u-i)+\left(u-\frac{i}{2}\right)^{L} Q(u+i) .$</td><td>$T(u)Q(u)=(u+\frac{i}{2})^LQ(u-i)+(u-\frac{i}{2})^LQ(u+i).$</td></tr>
<tr><td>cardona-qft-methods_ EQ0737 (7)</td><td>281</td><td>■ 1.000 ● N|+0</td><td>\left[\mathcal{D}, P_{\mu}\right]=i P_{\mu} \quad, \quad \left[\mathcal{D}, K_{\mu}\right]=-i K_{\mu}</td><td>$\left[\mathcal{D}, P_{\mu}\right]=i P_{\mu} \quad, \quad \left[\mathcal{D}, K_{\mu}\right]=-i K_{\mu}$</td><td>$[\mathcal{D},P_\mu]=iP_\mu \quad , \quad [\mathcal{D},K_\mu]=-iK_\mu$</td></tr>
<tr><td>cardona-qft-methods_ EQ0738</td><td>281</td><td>■ 1.000 ● N|+2</td><td>L=\begin{array}{|c|c|} \hline S^{3} & S^{+} \\ \hline S^{-} & -S^{3} \\ \hline \end{array} .</td><td>$L=\begin{array}{|c|c|} \hline S^{3} & S^{+} \\ \hline S^{-} & -S^{3} \\ \hline \end{array} .$</td><td>$L=\begin{array}{|c|c|} \hline S^{3} & S^{+} \\ \hline S^{-} & -S^{3} \\ \hline \end{array}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0739 (9)</td><td>282</td><td>■ 1.000 ● N|+1</td><td>\begin{aligned} \left\{Q_{a \alpha}, \bar{Q}_{\dot{ \alpha}}^{b}\right\} & =\left(\sigma^{\mu}\right)_{ \alpha \dot{\alpha}} P_{\mu} \delta_{a}^{b} \\ {\left[\mathcal{D}, Q_{a \alpha}\right] } & =\frac{i}{2} Q_{a \alpha} \end{aligned}</td><td>$\begin{aligned} \left\{Q_{a \alpha}, \bar{Q}_{\dot{\alpha}}^{b}\right\} & =\left(\sigma^{\mu}\right)_{\alpha \dot{\alpha}} P_{\mu} \delta_{a}^{b} \\ {\left[\mathcal{D}, Q_{a \alpha}\right] } & =\frac{i}{2} Q_{a \alpha} \end{aligned}$</td><td>$\{Q_{a\alpha},\bar{Q}^b_{\dot\alpha}\}=(\sigma^\mu)_{\alpha\dot\alpha}P_\mu\delta^b_a$
$[\mathcal{D},Q_{a\alpha}]=\frac{i}{2}Q_{a\alpha}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0741</td><td>282</td><td>■ 1.000 ● N|-1</td><td>\mathcal{O} \equiv \mathcal{O}_{(x)}=\operatorname{Tr} \left(Z X \bar{Z}(D Y)\left(D^{5} Y\right) \psi \left(D^{3} \psi\right) \bar{\psi} F\left(D^{7} \bar{F} \right)_{\ldots}\right)_{(x)} .</td><td>$\mathcal{O} \equiv \mathcal{O}_{(x)}=\operatorname{Tr}\left(Z X \bar{Z}(D Y)\left(D^{5} Y\right) \psi\left(D^{3} \psi\right) \bar{\psi} F\left(D^{7} \bar{F}\right)_{\ldots}\right)_{(x)} .$</td><td>$\mathcal{O}\equiv\mathcal{O}_{(x)}=\mathrm{Tr}\Big(ZX\bar{Z}(DY)(D^5Y)\psi(D^3\psi)\bar{\psi}F(D^7\bar{F})...\Big)_{(x)}.$</td></tr>
<tr><td>cardona-qft-methods_ EQ0742</td><td>282</td><td>■ 1.000 ● K|+0</td><td>[\mathcal{D}, \mathcal{O}]=i \Delta \mathcal{O},</td><td>$[\mathcal{D}, \mathcal{O}]=i \Delta \mathcal{O},$</td><td>$[\mathcal{D},\mathcal{O}]=i\Delta\mathcal{O}\,,$</td></tr>
<tr><td>cardona-qft-methods_ EQ0743</td><td>282</td><td>■ 1.000 ● N|+0</td><td>\gamma=g_{Y M}^{2} \underbrace{\gamma_{1}}_{1-\text { loop }}+g_{Y M}^{4} \underbrace{\gamma_{2}}_{2-\text { loop }}+\ldots</td><td>$\gamma=g_{Y M}^{2} \underbrace{\gamma_{1}}_{1-\text { loop }}+g_{Y M}^{4} \underbrace{\gamma_{2}}_{2-\text { loop }}+\ldots$</td><td>$\gamma=g^2_{YM}\underbrace{\gamma_1}_{1-loop}+g^4_{YM}\underbrace{\gamma_2}_{2-loop}+...$</td></tr>
<tr><td>cardona-qft-methods_ EQ0746</td><td>283</td><td>■ 1.000 ● N|+0</td><td>\left\langle\left(\bar{Z}^{j}\right)(x)\left(Z^{j} \right)(y)\right\rangle=\frac{1}{(x-y)^{2 j}}</td><td>$\left\langle\left(\bar{Z}^{j}\right)(x)\left(Z^{j}\right)(y)\right\rangle=\frac{1}{(x-y)^{2 j}}$</td><td>$\langle(\bar{Z}^j)(x)(Z^j)(y)\rangle=\frac{1}{(x-y)^{2j}}$</td></tr>
<tr><td>cardona-qft-methods_ EQ0750</td><td>284</td><td>■ 1.000 ● W|+0</td><td>\mathfrak{p s u}(2,2 \mid 4)=\begin{array}{|c|c|} \hline \mathfrak{u}(2,2) & \text { supercharges } \\ \hline \text { supercharges } & \mathfrak{u}(4) \\ \hline \end{array}</td><td>$\mathfrak{p s u}(2,2 \mid 4)=\begin{array}{|c|c|} \hline \mathfrak{u}(2,2) & \text { supercharges } \\ \hline \text { supercharges } & \mathfrak{u}(4) \\ \hline \end{array}$</td><td>$\mathfrak{psu}(2,2|4)=\begin{array}{|c|c|} \hline \mathfrak{u}(2,2) & \text{supercharges} \\ \hline \text{supercharges} & \mathfrak{u}(4) \\ \hline \end{array}$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0752 (1)	284	■1.000 ● N|+1	S=\frac{R^{2}}{\alpha^{\prime}} \int d \tau \frac{d \sigma}{2 \pi}\left(\frac{1}{2} \partial_{a} Z^{M} \partial^{a} Z_{M}+\frac{1}{2} \partial_{a} Y^{N} \partial^{a} Y_{N}\right)+\text { fermions }	$S=\frac{R^{2}}{\alpha^{\prime}} \int d \tau \frac{d \sigma}{2 \pi}\left(\frac{1}{2} \partial_{a} Z^{M} \partial^{a} Z_{M}+\frac{1}{2} \partial_{a} Y^{N} \partial^{a} Y_{N}\right)+\text { fermions }$	$S=\frac{R^{2}}{\alpha'}\int d\tau\frac{d\sigma}{2\pi}\left(\frac{1}{2}\partial_a Z^M\partial^a Z_M+\frac{1}{2}\partial_a Y^N\partial^a Y_N\right)+\text{fermions},$
cardona-qft-methods_EQ0753	285	■1.000 ● K|+0	\sqrt{\lambda}=\frac{R^{2}}{\alpha^{\prime}} \quad, \quad 4 \pi g_{\mathrm{s}}=\frac{\lambda}{N} .	$\sqrt{\lambda}=\frac{R^{2}}{\alpha^{\prime}} \quad, \quad 4 \pi g_{\mathrm{s}}=\frac{\lambda}{N} .$	$\sqrt{\lambda}=\frac{R^2}{\alpha'}\quad,\quad 4\pi g_s=\frac{\lambda}{N}.$
cardona-qft-methods_EQ0754	285	■1.000 ● N|+0	S_{S Y M}=\int \frac{N}{\lambda} \frac{d^{4} x}{4 \pi^{2}} \operatorname{Tr}\{\ldots\} \longleftrightarrow S_{\sigma-\text { model }}=\sqrt{\lambda} \int d \tau \frac{d \sigma}{2 \pi}\{\ldots\},	$S_{S Y M}=\int \frac{N}{\lambda} \frac{d^{4} x}{4 \pi^{2}} \operatorname{Tr}\{\ldots\} \longleftrightarrow S_{\sigma-\text { model }}=\sqrt{\lambda} \int d \tau \frac{d \sigma}{2 \pi}\{\ldots\},$	$S_{SYM}=\int\frac{N}{\lambda}\frac{d^4x}{4\pi^2}Tr\{\ldots\}\longleftrightarrow S_{\sigma-model}=\sqrt{\lambda}\int d\tau\frac{d\sigma}{2\pi}\{\ldots\},$
cardona-qft-methods_EQ0755	285	■1.000 ● K|+0	Z_{4}+Z_{5}=\frac{R}{z}, \quad\left(Z_{0}, Z_{1}, Z_{2}, Z_{3}\right)=\frac{R}{z} x_{\mu}, \quad d s_{A d S_{5}}^{2}=\frac{R^{2}}{z^{2}}\left(d x^{\mu} d x_{\mu}+d z^{2}\right) .	$Z_{4}+Z_{5}=\frac{R}{z}, \quad\left(Z_{0}, Z_{1}, Z_{2}, Z_{3}\right)=\frac{R}{z} x_{\mu}, \quad d s_{A d S_{5}}^{2}=\frac{R^{2}}{z^{2}}\left(d x^{\mu} d x_{\mu}+d z^{2}\right) .$	$Z_4+Z_5=\frac{R}{z},\quad (Z_0,Z_1,Z_2,Z_3)=\frac{R}{z}x_\mu,\quad ds^2_{AdS_5}=\frac{R^2}{z^2}(dx^\mu dx_\mu+dz^2).$
cardona-qft-methods_EQ0756	286	■1.000 ● N|+1	Y_{1}+i Y_{2}=r_{1} e^{i \phi_{1}}, \quad Y_{3}+i Y_{4}=r_{2} e^{i \phi_{2}}, \quad Y_{5}+i Y_{6}=r_{3} e^{i \phi_{3}}	$Y_{1}+i Y_{2}=r_{1} e^{i \phi_{1}}, \quad Y_{3}+i Y_{4}=r_{2} e^{i \phi_{2}}, \quad Y_{5}+i Y_{6}=r_{3} e^{i \phi_{3}}$	$Y_1+iY_2=r_1e^{i\phi_1},\qquad Y_3+iY_4=r_2e^{i\phi_2},\qquad Y_5+iY_6=r_3e^{i\phi_3},$
cardona-qft-methods_EQ0757	286	■1.000 ● N|+1	Z_{0}-i Z_{5}=\rho_{3} e^{i t}, \quad Z_{1}+i Z_{2}=\rho_{1} e^{i \alpha_{1}}, \quad Z Y_{3}+i Z_{4}=\rho_{2} e^{i \alpha_{2}}	$Z_{0}-i Z_{5}=\rho_{3} e^{i t}, \quad Z_{1}+i Z_{2}=\rho_{1} e^{i \alpha_{1}}, \quad Z Y_{3}+i Z_{4}=\rho_{2} e^{i \alpha_{2}}$	$Z_0-iZ_5=\rho_3e^{it},\qquad Z_1+iZ_2=\rho_1e^{i\alpha_1},\qquad ZY_3+iZ_4=\rho_2e^{i\alpha_2}.$
cardona-qft-methods_EQ0760 (1)	292	■1.000 ● N|+1	S=-\frac{1}{4 \pi \alpha^{\prime}} \int \mathrm{d} \sigma \mathrm{~d} \tau \sqrt{\gamma} \gamma^{\alpha \beta} \eta_{M N} \partial_{\alpha} X^{M} \partial_{\beta} X^{N}	$S=-\frac{1}{4 \pi \alpha^{\prime}} \int \mathrm{d} \sigma \mathrm{~d} \tau \sqrt{\gamma} \gamma^{\alpha \beta} \eta_{M N} \partial_{\alpha} X^{M} \partial_{\beta} X^{N}$	$S=-\frac{1}{4\pi\alpha'}\int \mathrm{d}\sigma\mathrm{d}\tau\sqrt{\gamma}\gamma^{\alpha\beta}\eta_{MN}\partial_\alpha X^M\partial_\beta X^N,$
cardona-qft-methods_EQ0761 (2)	292	■1.000 ● N|+0	T=\frac{1}{2 \pi \alpha^{\prime}}=\frac{1}{2 \pi l_{s}^{2}} .	$T=\frac{1}{2 \pi \alpha^{\prime}}=\frac{1}{2 \pi l_{s}^{2}} .$	$T=\frac{1}{2\pi\alpha'}=\frac{1}{2\pi l_s^2}.$
cardona-qft-methods_EQ0762 (3)	293	■1.000 ● N|+1	\left(\frac{\partial^{2}}{\partial \sigma^{2}}-\frac{\partial^{2}}{\partial \tau^{2}}\right) X^{M}(\tau, \sigma)=0	$\left(\frac{\partial^{2}}{\partial \sigma^{2}}-\frac{\partial^{2}}{\partial \tau^{2}}\right) X^{M}(\tau, \sigma)=0$	$\left(\frac{\partial^2}{\partial\sigma^2}-\frac{\partial^2}{\partial\tau^2}\right)X^M(\tau,\sigma)=0.$
cardona-qft-methods_EQ0763 (4)	293	■1.000 ● N|+1	\frac{\partial}{\partial \sigma^{+}} \frac{\partial}{\partial \sigma^{-}} X^{M}(\tau, \sigma)=0	$\frac{\partial}{\partial \sigma^{+}} \frac{\partial}{\partial \sigma^{-}} X^{M}(\tau, \sigma)=0$	$\frac{\partial}{\partial\sigma^+}\frac{\partial}{\partial\sigma^-}X^M(\tau,\sigma)=0,$
cardona-qft-methods_EQ0764 (5)	293	■1.000 ● N|+0	X^{M}(\tau, \sigma)=X_{R}^{M}\left(\sigma^{-}\right)+X_{L}^{M}\left(\sigma^{+}\right) .	$X^{M}(\tau, \sigma)=X_{R}^{M}\left(\sigma^{-}\right)+X_{L}^{M}\left(\sigma^{+}\right) .$	$X^M(\tau,\sigma)=X_R^M(\sigma^-)+X_L^M(\sigma^+).$
cardona-qft-methods_EQ0769 (10)	294	■1.000 ● N|+0	M^{2}=\frac{2}{\alpha^{\prime}}\left(\sum_{n=1}^{\infty} \alpha_{n} \cdot \alpha_{-n}+\tilde{\alpha}_{n} \cdot \tilde{\alpha}_{-n}-2\right)	$M^{2}=\frac{2}{\alpha^{\prime}}\left(\sum_{n=1}^{\infty} \alpha_{n} \cdot \alpha_{-n}+\tilde{\alpha}_{n} \cdot \tilde{\alpha}_{-n}-2\right)$	$M^2=\frac{2}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\tilde{\alpha}_n\cdot\tilde{\alpha}_{-n}-2\right)$
cardona-qft-methods_EQ0770 (11)	294	■1.000 ● N|+1	\hat{L}_{0}=\frac{1}{2}\left(\sum_{n=1}^{\infty} \alpha_{n} \cdot \alpha_{-n}-\tilde{\alpha}_{n} \cdot \tilde{\alpha}_{-n}\right)	$\hat{L}_{0}=\frac{1}{2}\left(\sum_{n=1}^{\infty} \alpha_{n} \cdot \alpha_{-n}-\tilde{\alpha}_{n} \cdot \tilde{\alpha}_{-n}\right)$	$\hat{L}_0=\frac{1}{2}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}-\tilde{\alpha}_n\cdot\tilde{\alpha}_{-n}\right)$
cardona-qft-methods_EQ0771 (12)	294	■1.000 ● N|+0	\left|\xi_{M N}\right\rangle \equiv \xi_{M} \tilde{\xi}_{N} \alpha_{1}^{M} \tilde{\alpha}_{1}^{N}|0\rangle	$\left\vert\xi_{M N}\right\rangle \equiv \xi_{M} \tilde{\xi}_{N} \alpha_{1}^{M} \tilde{\alpha}_{1}^{N}\vert 0\rangle$	$\vert\xi_{MN}\rangle\equiv\xi_M\tilde{\xi}_N\alpha_1^M\tilde{\alpha}_1^N\vert 0\rangle$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0772 (13)	294	■ 1.000 ● N|+1	`\xi_{M N}=\xi_{M N}^{s}+\xi_{t} \eta_{M N}+\xi_{M N}^{a}`	$\xi_{M N}=\xi_{M N}^{s}+\xi_{t} \eta_{M N}+\xi_{M N}^{a}$	$\xi_{MN}=\xi^{s}_{MN}+\xi_t\,\eta_{MN}+\xi^{a}_{MN}\ ,$
cardona-qft-methods_ EQ0774 (15)	295	■ 1.000 ● N|+2	`S=-\frac{1}{4 \pi \alpha^{\prime}} \int \mathrm{d} \sigma \mathrm{~d} \tau \sqrt{\gamma} \gamma^{\alpha \beta} G_{M N}(X) \partial_{\alpha} X^{M} \partial_{\beta} X^{N}`	$S=-\frac{1}{4 \pi \alpha^{\prime}} \int \mathrm{d} \sigma \mathrm{~d} \tau \sqrt{\gamma} \gamma^{\alpha \beta} G_{M N}(X) \partial_{\alpha} X^{M} \partial_{\beta} X^{N}$	$S=-\frac{1}{4\pi\alpha'}\int \mathrm{d}\sigma\mathrm{d}\tau\sqrt{\gamma}\,\gamma^{\alpha\beta}G_{MN}(X)\partial_\alpha X^M\partial_\beta X^N\ ,$
cardona-qft-methods_ EQ0776 (17)	295	■ 1.000 ● C|+4	`\begin{aligned} S= & -\frac{1}{4 \pi \alpha^{\prime}} \int \mathrm{d} \sigma \mathrm{~d} \tau \sqrt{\gamma} \left[\left(\gamma^{\alpha \beta} g_{M N}(X)+\mathrm{i} \epsilon^{\alpha \beta} B_{M N}(X)\right) \partial_{\alpha} X^{M} \partial_{\beta} X^{N}\right. \\ & \left.+ \alpha^{\prime} \Phi R\right] \end{aligned}`	$\begin{aligned} S= & -\frac{1}{4 \pi \alpha^{\prime}} \int \mathrm{d} \sigma \mathrm{~d} \tau \sqrt{\gamma} \left[\left(\gamma^{\alpha \beta} g_{M N}(X)+\mathrm{i} \epsilon^{\alpha \beta} B_{M N}(X)\right) \partial_{\alpha} X^{M} \partial_{\beta} X^{N}\right. \\ & \left.+\alpha^{\prime} \Phi R\right] \end{aligned}$	$\begin{aligned} S= & -\frac{1}{4\pi\alpha'}\int \mathrm{d}\sigma\mathrm{d}\tau\sqrt{\gamma}\ \left[\left(\gamma^{\alpha\beta}g_{MN}(X)+\mathrm{i}\epsilon^{\alpha\beta}B_{MN}(X)\right)\partial_\alpha X^M\partial_\beta X^N\right. \\ & \left.+\alpha'\Phi R\right]\ . \end{aligned}$ (
cardona-qft-methods_ EQ0777 (18)	296	■ 1.000 ● K|+0	`\chi=\frac{1}{4 \pi} \int d^{2} \sigma(-\gamma)^{1 / 2} R .`	$\chi=\frac{1}{4 \pi} \int d^{2} \sigma(-\gamma)^{1 / 2} R .$	$\chi=\frac{1}{4\pi}\int d^2\sigma(-\gamma)^{1/2}R\ .$
cardona-qft-methods_ EQ0782 (23)	298	■ 1.000 ● N|+0	`M^{2}=\frac{1}{\alpha^{\prime}}\left(\sum_{n=1}^{\infty} \alpha_{n} \cdot \alpha_{-n}-1\right) .`	$M^{2}=\frac{1}{\alpha^{\prime}}\left(\sum_{n=1}^{\infty} \alpha_{n} \cdot \alpha_{-n}-1\right) .$	$M^2=\frac{1}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}-1\right)\ .$
cardona-qft-methods_ EQ0783 (24)	298	■ 1.000 ● N|+0	`\left|\xi_{M}\right\rangle \equiv \xi_{M} \alpha_{1}^{M}|0\rangle`	$\left\lvert\xi_{M}\right\rangle \equiv \xi_{M} \alpha_{1}^{M}\lvert 0\rangle$	$\lvert\xi_M\rangle\equiv\xi_M\alpha_1^M\lvert 0\rangle$
cardona-qft-methods_ EQ0784 (25)	298	■ 1.000 ● N|+0	`\int_{\partial M} d \tau A_{M} \partial_{\tau} X^{M}`	$\int_{\partial M} d \tau A_{M} \partial_{\tau} X^{M}$	$\int_{\partial M} d\tau A_M\partial_\tau X^M$
cardona-qft-methods_ EQ0785 (26)	298	■ 1.000 ● N|+1	`\mathrm{S}=-\frac{1}{4} \int \mathrm{~d}^{D} X e^{-\Phi} F_{M N} F^{M N}+O\left(\alpha^{\prime}\right)`	$\mathrm{S}=-\frac{1}{4} \int \mathrm{~d}^{D} X e^{-\Phi} F_{M N} F^{M N}+O\left(\alpha^{\prime}\right)$	$\mathrm{S}=-\frac{1}{4}\int \mathrm{d}^D X\,e^{-\Phi}F_{MN}F^{MN}+O(\alpha')\ ,$
cardona-qft-methods_ EQ0786 (27)	299	■ 1.000 ● N|+0	`S=\frac{1}{4 \pi} \int_{M} \mathrm{~d}^{2} \sigma \eta_{M N}\left[\frac{1}{\alpha^{\prime}} \partial X^{M} \bar{\partial} X^{N}+\Psi^{M} \bar{\partial} \Psi^{N}+\tilde{\Psi}^{M} \bar{\partial} \tilde{\Psi}^{N}\right]`	$S=\frac{1}{4 \pi} \int_{M} \mathrm{~d}^{2} \sigma \eta_{M N}\left[\frac{1}{\alpha^{\prime}} \partial X^{M} \bar{\partial} X^{N}+\Psi^{M} \bar{\partial} \Psi^{N}+\tilde{\Psi}^{M} \bar{\partial} \tilde{\Psi}^{N}\right]$	$S=\frac{1}{4\pi}\int_M \mathrm{d}^2\sigma\,\eta_{MN}\left[\frac{1}{\alpha'}\partial X^M\bar\partial X^N+\Psi^M\bar\partial\Psi^N+\tilde\Psi^M\bar\partial\tilde\Psi^N\right]$
cardona-qft-methods_ EQ0788 (29)	299	■ 1.000 ● N|+0	`\left\{\psi_{r}^{M}, \psi_{s}^{N}\right\}=\left\{\tilde{\psi}_{r}^{M}, \tilde{\psi}_{s}^{N}\right\}=\eta^{M N} \delta_{r+s} .`	$\left\{\psi_{r}^{M}, \psi_{s}^{N}\right\}=\left\{\tilde{\psi}_{r}^{M}, \tilde{\psi}_{s}^{N}\right\}=\eta^{M N} \delta_{r+s} .$	$\{\psi^M_r,\psi^N_s\}=\{\tilde\psi^M_r,\tilde\psi^N_s\}=\eta^{MN}\delta_{r+s}\ .$
cardona-qft-methods_ EQ0789 (30)	300	■ 1.000 ● N|+0	`M^{2}=\frac{1}{\alpha^{\prime}}\left(\sum_{n=1}^{\infty} \alpha_{n} \cdot \alpha_{-n}+\sum_{r} r \psi_{r} \cdot \psi_{-r}-a+\text { same with tilde }\right),`	$M^{2}=\frac{1}{\alpha^{\prime}}\left(\sum_{n=1}^{\infty} \alpha_{n} \cdot \alpha_{-n}+\sum_{r} r \psi_{r} \cdot \psi_{-r}-a+\text { same with tilde }\right),$	$M^2=\frac{1}{\alpha'}\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\sum_r r\psi_r\cdot\psi_{-r}-a+\text{same with tilde}\right)\ ,$
cardona-qft-methods_ EQ0790 (31)	300	■ 1.000 ● N|+0	`\tilde{L}_{0}=\left(\sum_{n=1}^{\infty} \alpha_{n} \cdot \alpha_{-n}+\sum_{r} r \psi_{r} \cdot \psi_{-r}-a-\text { same with tilde }\right) .`	$\tilde{L}_{0}=\left(\sum_{n=1}^{\infty} \alpha_{n} \cdot \alpha_{-n}+\sum_{r} r \psi_{r} \cdot \psi_{-r}-a-\text { same with tilde }\right) .$	$\tilde L_0=\left(\sum_{n=1}^{\infty}\alpha_n\cdot\alpha_{-n}+\sum_r r\psi_r\cdot\psi_{-r}-a-\text{same with tilde}\right)\ .$
cardona-qft-methods_ EQ0792 (33)	300	■ 1.000 ● N|+0	`R: \psi_{0}^{M}|0\rangle \rightarrow|s\rangle=\left|\pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}\right\rangle`	$R: \psi_{0}^{M}\lvert 0\rangle \rightarrow\lvert s\rangle=\left\lvert\pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}\right\rangle$	$R:\psi_0^M\lvert 0\rangle\rightarrow\lvert s\rangle=\lvert\pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2}\rangle$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0793 (34)	301	1.000 K|+0	`N S: \psi_{1 / 2}^{M}|0\rangle,`	$NS:\psi_{1/2}^{M}\|0\rangle,$	$NS:\psi_{1/2}^{M}\|0\rangle\ ,$
cardona-qft-methods_EQ0797 (38)	302	1.000 K|+0	`M^{2}=\frac{n^{2}}{R^{2}}+\frac{m^{2} R^{2}}{\alpha^{\prime 2}}+\frac{2}{\alpha^{\prime}}(N+\tilde{N}-2)`	$M^{2}=\frac{n^{2}}{R^{2}}+\frac{m^{2} R^{2}}{\alpha^{\prime 2}}+\frac{2}{\alpha^{\prime}}(N+\tilde{N}-2)$	$M^{2}=\frac{n^{2}}{R^{2}}+\frac{m^{2} R^{2}}{\alpha'^{2}}+\frac{2}{\alpha'}(N+\tilde{N}-2)$
cardona-qft-methods_EQ0798 (39)	304	1.000 N|+0	`S_{D B I}=-T_{p} \int \mathrm{~d}^{p+1} \xi e^{-\Phi} \sqrt{\operatorname{det}\left(G_{a b}+B_{a b}+2 \pi \alpha^{\prime} F_{a b}\right)}`	$S_{D B I}=-T_{p} \int \mathrm{~d}^{p+1} \xi e^{-\Phi} \sqrt{\operatorname{det}\left(G_{a b}+B_{a b}+2 \pi \alpha^{\prime} F_{a b}\right)}$	$S_{DBI}=-T_{p} \int \mathrm{d}^{p+1} \xi e^{-\Phi} \sqrt{\det(G_{ab}+B_{ab}+2 \pi \alpha' F_{ab})}$
cardona-qft-methods_EQ0799 (40)	304	1.000 K|+0	`S_{C S}=\mathrm{i} T_{p} \int_{p+1} e^{2 \pi \alpha^{\prime} F+B} \wedge C`	$S_{C S}=\mathrm{i} T_{p} \int_{p+1} e^{2 \pi \alpha^{\prime} F+B} \wedge C$	$S_{CS}=\mathrm{i} T_{p} \int_{p+1} e^{2 \pi \alpha' F+B} \wedge C$
cardona-qft-methods_EQ0800 (41)	304	1.000 N|+1	`G_{a b}(\xi)=\frac{\partial X^{M}}{\partial \xi^{a}} \frac{\partial X^{N}}{\partial \xi^{b}} G_{M N}(X)`	$G_{a b}(\xi)=\frac{\partial X^{M}}{\partial \xi^{a}} \frac{\partial X^{N}}{\partial \xi^{b}} G_{M N}(X)$	$G_{ab}(\xi)=\frac{\partial X^{M}}{\partial \xi^{a}} \frac{\partial X^{N}}{\partial \xi^{b}} G_{MN}(X)$
cardona-qft-methods_EQ0801 (1)	305	1.000 N|+0	`\mathcal{G}=\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) .`	$\mathcal{G}=\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) .$	$\mathcal{G}=\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\ .$
cardona-qft-methods_EQ0802 (2)	306	1.000 N|+0	`\mathcal{B}=\left(\begin{array}{ll} 0 & 0 \\ B & 0 \end{array}\right),`	$\mathcal{B}=\left(\begin{array}{ll} 0 & 0 \\ B & 0 \end{array}\right),$	$\mathcal{B}=\begin{pmatrix} 0 & 0 \\ B & 0 \end{pmatrix}\ ,$
cardona-qft-methods_EQ0805 (5)	306	1.000 N|+0	`\Lambda_{(\alpha \beta)}+\Lambda_{(\beta \gamma)}+\Lambda_{(\gamma \alpha)}=g_{(\alpha \beta \gamma)} \mathrm{d} g_{(\alpha \beta \gamma)}`	$\Lambda_{(\alpha \beta)}+\Lambda_{(\beta \gamma)}+\Lambda_{(\gamma \alpha)}=g_{(\alpha \beta \gamma)} \mathrm{d} g_{(\alpha \beta \gamma)}$	$\Lambda_{(\alpha \beta)}+\Lambda_{(\beta \gamma)}+\Lambda_{(\gamma \alpha)}=g_{(\alpha \beta \gamma)} \mathrm{d} g_{(\alpha \beta \gamma)}$
cardona-qft-methods_EQ0806 (6)	306	1.000 N|+1	`\mathcal{J}_{1} \equiv\left(\begin{array}{cc} I & 0 \\ 0 & -I^{t} \end{array}\right)`	$\mathcal{J}_{1} \equiv\left(\begin{array}{cc} I & 0 \\ 0 & -I^{t} \end{array}\right)$	$\mathcal{J}_{1} \equiv\begin{pmatrix} I & 0 \\ 0 & -I^{t} \end{pmatrix}\ ,$
cardona-qft-methods_EQ0807 (7)	307	1.000 N|+1	`\mathcal{J}_{2} \equiv\left(\begin{array}{cc} 0 & -J^{-1} \\ J & 0 \end{array}\right) .`	$\mathcal{J}_{2} \equiv\left(\begin{array}{cc} 0 & -J^{-1} \\ J & 0 \end{array}\right) .$	$\mathcal{J}_{2} \equiv\begin{pmatrix} 0 & -J^{-1} \\ J & 0 \end{pmatrix}\ .$
cardona-qft-methods_EQ0810 (10)	308	1.000 K|+0	`\mathcal{H}=\mathcal{G} \mathcal{J}_{1} \mathcal{J}_{2} .`	$\mathcal{H}=\mathcal{G} \mathcal{J}_{1} \mathcal{J}_{2} .$	$\mathcal{H}=\mathcal{G} \mathcal{J}_{1} \mathcal{J}_{2}\ .$
cardona-qft-methods_EQ0812 (12)	308	1.000 N|+0	`\eta_{+}^{1} \otimes \eta_{ \pm}^{2 \dagger}=\frac{1}{4} \sum_{k=0}^{6} \frac{1}{k!} \eta_{ \pm}^{2 \dagger} \gamma_{i_{1} \ldots i_{k}} \eta_{+}^{1} \gamma^{i_{k} \ldots i_{1}}`	$\eta_{+}^{1} \otimes \eta_{ \pm}^{2 \dagger}=\frac{1}{4} \sum_{k=0}^{6} \frac{1}{k!} \eta_{ \pm}^{2 \dagger} \gamma_{i_{1} \ldots i_{k}} \eta_{+}^{1} \gamma^{i_{k} \ldots i_{1}}$	$\eta_{+}^{1} \otimes \eta_{\pm}^{2 \dagger}=\frac{1}{4} \sum_{k=0}^{6} \frac{1}{k!} \eta_{\pm}^{2 \dagger} \gamma_{i_{1} \ldots i_{k}} \eta_{+}^{1} \gamma^{i_{k} \ldots i_{1}}$
cardona-qft-methods_EQ0813 (13)	308	1.000 W|+1	`\sum_{k} \frac{1}{k!} C_{i_{1} \ldots i_{k}}^{(k)} \gamma_{\alpha \beta}^{i_{i} \ldots i_{k}} \quad \longleftrightarrow \quad \sum_{k} \frac{1}{k!} C_{i_{1} \ldots i_{k}}^{(k)} \mathrm{d} x^{i_{i}} \wedge \ldots \wedge \mathrm{~d} x^{i_{k}}`	$\sum_{k} \frac{1}{k!} C_{i_{1} \ldots i_{k}}^{(k)} \gamma_{\alpha \beta}^{i_{i} \ldots i_{k}} \quad \longleftrightarrow \quad \sum_{k} \frac{1}{k!} C_{i_{1} \ldots i_{k}}^{(k)} \mathrm{d} x^{i_{i}} \wedge \ldots \wedge \mathrm{~d} x^{i_{k}}$	$\sum_{k} \frac{1}{k!} C_{i_{1} \ldots i_{k}}^{(k)} \gamma_{\alpha \beta}^{i_{i} \ldots i_{k}} \qquad \longleftrightarrow \qquad \sum_{k} \frac{1}{k!} C_{i_{1} \ldots i_{k}}^{(k)} \mathrm{d} x^{i_{i}} \wedge \ldots \wedge \mathrm{d} x^{i_{k}}\ .$
cardona-qft-methods_EQ0814 (14)	309	1.000 N|+0	`\Phi_{ \pm} \rightarrow e^{-B} \Phi_{ \pm} \equiv \Phi_{ \pm}^{D}`	$\Phi_{ \pm} \rightarrow e^{-B} \Phi_{ \pm} \equiv \Phi_{ \pm}^{D}$	$\Phi_{\pm} \rightarrow e^{-B} \Phi_{\pm} \equiv \Phi_{\pm}^{D}$
cardona-qft-methods_EQ0815 (15)	309	1.000 K|+0	`\Phi_{+}=e^{-\mathrm{i} J}, \quad \Phi_{-}=-\mathrm{i} \Omega,`	$\Phi_{+}=e^{-\mathrm{i} J}, \quad \Phi_{-}=-\mathrm{i} \Omega,$	$\Phi_{+}=e^{-\mathrm{i} J}\ , \qquad \Phi_{-}=-\mathrm{i} \Omega\ ,$
cardona-qft-methods_EQ0817 (17)	309	1.000 N|+0	`J \wedge \Omega=0, \quad J \wedge J \wedge J=\frac{3 \mathrm{i}}{4} \Omega \wedge \bar{\Omega},`	$J \wedge \Omega=0, \quad J \wedge J \wedge J=\frac{3 \mathrm{i}}{4} \Omega \wedge \bar{\Omega},$	$J \wedge \Omega=0\ , \qquad J \wedge J \wedge J=\frac{3 \mathrm{i}}{4} \Omega \wedge \bar{\Omega}\ ,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0821 (21)	311	■1.000 ● N|+0	`[X+\zeta, Y+\eta]_{H}=[X+\zeta, Y+\eta]_{C}+\iota_{X} \iota_{Y} H,`	$[X+\zeta, Y+\eta]_{H}=[X+\zeta, Y+\eta]_{C}+\iota_{X} \iota_{Y} H,$	$[X+\zeta, Y+\eta]_{H}=[X+\zeta, Y+\eta]_{C}+\iota_{X} \iota_{Y} H\ ,$
cardona-qft-methods_ EQ0822	311	■1.000 ● K|+0	`(\mathrm{d}-H \wedge) \Phi=0`	$(\mathrm{d}-H \wedge) \Phi=0$	$(\mathrm{d}-H \wedge) \Phi=0$
cardona-qft-methods_ EQ0825 (4)	313	■1.000 ● K|+0	`F^{(10)}=\mathrm{d} C-H \wedge C+m e^{B}=\hat{F}-H \wedge C`	$F^{(10)}=\mathrm{d} C-H \wedge C+m e^{B}=\hat{F}-H \wedge C$	$F^{(10)}=\mathrm{d} C-H \wedge C+m\, e^{B}=\hat{F}-H \wedge C$
cardona-qft-methods_ EQ0826 (5)	313	■1.000 ● N|+1	`F_{n}^{(10)}=(-1)^{\operatorname{Int}[n / 2]} \star F_{10-n}^{(10)},`	$F_{n}^{(10)}=(-1)^{\operatorname{Int}[n / 2]} \star F_{10-n}^{(10)},$	$F_{n}^{(10)}=(-1)^{\operatorname{Int}[n / 2]} \star F_{10-n}^{(10)}\ ,$
cardona-qft-methods_ EQ0827 (6)	314	■1.000 ● K|+0	`d H=0, \quad d F^{(10)}-H \wedge F^{(10)}=0 .`	$d H=0, \quad d F^{(10)}-H \wedge F^{(10)}=0 .$	$dH=0\ , \quad dF^{(10)}-H \wedge F^{(10)}=0\ .$
cardona-qft-methods_ EQ0828 (7)	314	■1.000 ● K|+0	`\tilde{\nabla}_{\mu} \epsilon+\frac{e^{-A}}{2}\left( \gamma_{\mu} \gamma_{5} \otimes \not \nabla A\right) \epsilon=0`	$\tilde{\nabla}_{\mu} \epsilon+\frac{e^{-A}}{2}\left(\gamma_{\mu} \gamma_{5} \otimes \not \nabla A\right) \epsilon=0$	$\tilde{\nabla}_{\mu} \epsilon+\frac{e^{-A}}{2}(\gamma_{\mu} \gamma_{5} \otimes \not\nabla A) \epsilon=0$
cardona-qft-methods_ EQ0829 (8)	314	■1.000 ● K|+0	`\left[\tilde{\nabla}_{\mu}, \tilde{\nabla}_{\nu}\right] \epsilon=-\frac{1}{2}\left(\nabla_{m} A\right)\left( \nabla^{m} A\right) \gamma_{\mu \nu} \epsilon`	$\left[\tilde{\nabla}_{\mu}, \tilde{\nabla}_{\nu}\right] \epsilon=-\frac{1}{2}\left(\nabla_{m} A\right)\left(\nabla^{m} A\right) \gamma_{\mu \nu} \epsilon$	$[\tilde{\nabla}_{\mu}, \tilde{\nabla}_{\nu}] \epsilon=-\frac{1}{2}(\nabla_{m} A)(\nabla^{m} A)\, \gamma_{\mu \nu}\, \epsilon$
cardona-qft-methods_ EQ0830 (9)	314	■1.000 ● K|+0	`\left[\tilde{\nabla}_{\mu}, \tilde{\nabla}_{\nu}\right] \epsilon=\frac{1}{4} \tilde{R}_{\mu \nu \lambda \rho} \gamma^{\lambda \rho} \epsilon=0`	$\left[\tilde{\nabla}_{\mu}, \tilde{\nabla}_{\nu}\right] \epsilon=\frac{1}{4} \tilde{R}_{\mu \nu \lambda \rho} \gamma^{\lambda \rho} \epsilon=0$	$[\tilde{\nabla}_{\mu}, \tilde{\nabla}_{\nu}] \epsilon=\frac{1}{4} \tilde{R}_{\mu \nu \lambda \rho} \gamma^{\lambda \rho} \epsilon=0$
cardona-qft-methods_ EQ0832 (11)	314	■1.000 ● N|+0	`\epsilon_{\mathrm{IIB}}^{A}=\xi_{+}^{A} \otimes \eta_{+}^{A}+\xi_{-}^{A} \otimes \eta_{-}^{A}, \quad A=1,2 .`	$\epsilon_{\mathrm{IIB}}^{A}=\xi_{+}^{A} \otimes \eta_{+}^{A}+\xi_{-}^{A} \otimes \eta_{-}^{A}, \quad A=1,2 .$	$\epsilon_{\mathrm{IIB}}^{A}=\xi_{+}^{A} \otimes \eta_{+}^{A}+\xi_{-}^{A} \otimes \eta_{-}^{A}\ , \qquad A=1,2\ .$
cardona-qft-methods_ EQ0833 (12)	314	■1.000 ● K|+0	`\nabla_{m} \eta_{ \pm}^{A}=0 .`	$\nabla_{m} \eta_{ \pm}^{A}=0 .$	$\nabla_{m} \eta_{\pm}^{A}=0\,.$
cardona-qft-methods_ EQ0836 (15)	317	■1.000 ● K|+0	`\left|\Phi_{+}\right|=\left|\Phi_{-}\right|=e^{A}`	$\left\lvert\Phi_{+}\right\rvert=\left\lvert\Phi_{-}\right\rvert=e^{A}$	$\lvert\Phi_{+}\rvert=\lvert\Phi_{-}\rvert=e^{A}$
cardona-qft-methods_ EQ0839 (18)	317	■1.000 ● K|+0	`\lambda\left(A_{n}\right)=(-1)^{\operatorname{Int}[n / 2]} A_{n} .`	$\lambda\left(A_{n}\right)=(-1)^{\operatorname{Int}[n / 2]} A_{n} .$	$\lambda(A_{n})=(-1)^{\operatorname{Int}[n / 2]} A_{n}\ .$
cardona-qft-methods_ EQ0840	328	■1.000 ● K|+0	`\{f, g\}=\Pi(d f, d g)`	$\{f, g\}=\Pi(d f, d g)$	$\{f, g\}=\Pi(df, dg)$
cardona-qft-methods_ EQ0841	328	■1.000 ● N|+0	`\{f, g h\}=g\{f, h\}+h\{f, g\}, \forall f, g, h \in \mathcal{C}^{\infty}(M) .`	$\{f, g h\}=g\{f, h\}+h\{f, g\}, \forall f, g, h \in \mathcal{C}^{\infty}(M) .$	$\{f, gh\}=g\{f, h\}+h\{f, g\}, \forall f, g, h \in \mathcal{C}^{\infty}(M).$
cardona-qft-methods_ EQ0843	329	■1.000 ● N|+1	`S(X, \eta):=\int_{\Sigma}\langle\eta, d X\rangle+ \frac{1}{2}\left\langle\eta,\left(\Pi^{\#} \circ X\right) \eta\right\rangle`	$S(X, \eta):=\int_{\Sigma}\langle\eta, d X\rangle+ \frac{1}{2}\left\langle\eta,\left(\Pi^{\#} \circ X\right) \eta\right\rangle$	$S(X, \eta):=\int_{\Sigma}\langle\eta,\, dX\rangle+\frac{1}{2}\langle\eta,\, (\Pi^{\#} \circ X)\eta\rangle,$
cardona-qft-methods_ EQ0844	329	■1.000 ● N|+0	`\mathrm{EL}=\{\text { Solutions of the Euler-Lagrange equations }\} \subset \Phi,`	$\mathrm{EL}=\{\text { Solutions of the Euler-Lagrange equations }\} \subset \Phi,$	$\mathrm{EL}=\{\text{Solutions of the Euler-Lagrange equations}\} \subset \Phi,$
cardona-qft-methods_ EQ0847	330	■1.000 ● K|+0	`\Phi_{\partial}:=\left\{\text { vector bundle morphisms between } T(\partial \Sigma) \text { and } T^{*} M\right\} .`	$\Phi_{\partial}:=\left\{\text { vector bundle morphisms between } T(\partial \Sigma) \text { and } T^{*} M\right\} .$	$\Phi_{\partial}:=\{\text{vector bundle morphisms between } T(\partial \Sigma) \text{ and } T^{*} M\}.$
cardona-qft-methods_ EQ0848	330	■1.000 ● N|+0	`L_{\Sigma}:=p(E L) .`	$L_{\Sigma}:=p(E L) .$	$L_{\Sigma}:=p(EL).$
cardona-qft-methods_ EQ0849	330	■1.000 ● N|+0	`C_{\Pi}:=\left\{(X, \eta) \mid d X=\pi^{\#}(X) \eta, X: \partial \Sigma \rightarrow M, \eta \in \Gamma\left(T^{*} I \otimes X^{*}\left(T^{*} M\right)\right)\right\} .`	$C_{\Pi}:=\left\{(X, \eta) \mid d X=\pi^{\#}(X) \eta, X: \partial \Sigma \rightarrow M, \eta \in \Gamma\left(T^{*} I \otimes X^{*}\left(T^{*} M\right)\right)\right\} .$	$C_{\Pi}:=\{(X, \eta) \vert dX=\pi^{\#}(X)\eta,\, X: \partial \Sigma \rightarrow M,\, \eta \in \Gamma(T^{*} I \otimes X^{*}(T^{*} M))\}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0850	330	1.000 N|+0	`[X, f Y]_{A}=f[X, Y]+\rho_{*}(X)(f) Y, \forall X, Y \in \Gamma(A), f \in \mathcal{C}^{\infty}(M) .`	$[X, f Y]_{A}=f[X, Y]+\rho_{*}(X)(f) Y, \forall X, Y \in \Gamma(A), f \in \mathcal{C}^{\infty}(M) .$	$[X, fY]_A = f[X,Y] + \rho_*(X)(f)Y, \forall X, Y \in \Gamma(A), f \in \mathcal{C}^\infty(M).$
cardona-qft-methods_ EQ0852	331	1.000 N|+0	`\delta_{A} f:=\rho^{*} d f,`	$\delta_{A} f:=\rho^{*} d f,$	$\delta_A f := \rho^* df,$
cardona-qft-methods_ EQ0853	331	1.000 N|+0	`\left\langle\delta_{A} \alpha, X \wedge Y\right\rangle:=-\left\langle\alpha,[X, Y]_{A}\right\rangle+\langle\delta\langle\alpha, X\rangle, Y\rangle-\langle\delta\langle\alpha, Y\rangle, X\rangle,`	$\left\langle\delta_{A} \alpha, X \wedge Y\right\rangle:=-\left\langle\alpha,[X, Y]_{A}\right\rangle+\langle\delta\langle\alpha, X\rangle, Y\rangle-\langle\delta\langle\alpha, Y\rangle, X\rangle,$	$\langle\delta_A\alpha, X \wedge Y\rangle := -\langle\alpha, [X,Y]_A\rangle + \langle\delta\langle\alpha, X\rangle, Y\rangle - \langle\delta\langle\alpha, Y\rangle, X\rangle,$
cardona-qft-methods_ EQ0854	331	1.000 N|+0	`\delta_{A} \varphi^{*}=\varphi^{*} \delta_{B} .`	$\delta_{A} \varphi^{*}=\varphi^{*} \delta_{B} .$	$\delta_A\varphi^* = \varphi^*\delta_B.$
cardona-qft-methods_ EQ0856	332	1.000 K|+0	`G r_{\mu}:=\left\{(a, b, c) \in G^{3} \mid c=\mu(a, b)\right\}`	$G r_{\mu}:=\left\{(a, b, c) \in G^{3} \mid c=\mu(a, b)\right\}$	$Gr_\mu := \{(a,b,c) \in G^3 \mid c = \mu(a,b)\}$
cardona-qft-methods_ EQ0857	333	1.000 W|+0	`\overline{\mathcal{G}} \times \overline{\mathcal{G}} \nrightarrow \mathcal{G} \text { and } \mathcal{G} \nrightarrow \overline{\mathcal{G}}`	$\overline{\mathcal{G}} \times \overline{\mathcal{G}} \nrightarrow \mathcal{G} \text { and } \mathcal{G} \nrightarrow \overline{\mathcal{G}}$	$\bar{\mathcal{G}} \times \bar{\mathcal{G}} \nrightarrow \mathcal{G}$ and $\mathcal{G} \nrightarrow \bar{\mathcal{G}}$
cardona-qft-methods_ EQ0858	333	1.000 W|+0	`\begin{aligned} T: \mathcal{G} \times \mathcal{G} & \rightarrow \mathcal{G} \times \mathcal{G} \\ (x, y) & \mapsto(y, x) \end{aligned}`	$\begin{aligned} T: \mathcal{G} \times \mathcal{G} & \rightarrow \mathcal{G} \times \mathcal{G} \\ (x, y) & \mapsto(y, x) \end{aligned}$	$\begin{aligned} T: \mathcal{G} \times \mathcal{G} & \rightarrow \mathcal{G} \times \mathcal{G} \\ (x, y) & \mapsto (y, x) \end{aligned}$
cardona-qft-methods_ EQ0859	334	1.000 W|+0	`T_{r e l}: \mathcal{G} \times \mathcal{G} \nrightarrow \mathcal{G} \times \mathcal{G} \text { and } \overline{T_{r e l}}: \overline{\mathcal{G}} \times \overline{\mathcal{G}} \nrightarrow \overline{\mathcal{G}} \times \overline{\mathcal{G}} .`	$T_{r e l}: \mathcal{G} \times \mathcal{G} \nrightarrow \mathcal{G} \times \mathcal{G} \text { and } \overline{T_{r e l}}: \overline{\mathcal{G}} \times \overline{\mathcal{G}} \nrightarrow \overline{\mathcal{G}} \times \overline{\mathcal{G}} .$	$T_{rel} : \mathcal{G} \times \mathcal{G} \nrightarrow \mathcal{G} \times \mathcal{G}$ and $\overline{T_{rel}} : \bar{\mathcal{G}} \times \bar{\mathcal{G}} \nrightarrow \bar{\mathcal{G}} \times \bar{\mathcal{G}}.$
cardona-qft-methods_ EQ0860	334	1.000 N|+0	`L_{3}:=I_{r e l} \circ L_{r e l}: \mathcal{G} \times \mathcal{G} \nrightarrow \mathcal{G} .`	$L_{3}:=I_{r e l} \circ L_{r e l}: \mathcal{G} \times \mathcal{G} \nrightarrow \mathcal{G} .$	$L_3 := I_{rel} \circ L_{rel} : \mathcal{G} \times \mathcal{G} \nrightarrow \mathcal{G}.$
cardona-qft-methods_ EQ0861	334	1.000 K|+0	`\overline{I_{r e l}} \circ L_{1}=\overline{L_{1}},`	$\overline{I_{r e l}} \circ L_{1}=\overline{L_{1}},$	$\overline{I_{rel}} \circ L_1 = \overline{L_1},$
cardona-qft-methods_ EQ0862	334	1.000 K|+0	`I\left(L_{1}\right)=\overline{L_{1}},`	$I\left(L_{1}\right)=\overline{L_{1}},$	$I(L_1) = \overline{L_1},$
cardona-qft-methods_ EQ0864	335	1.000 N|+0	`L_{2}:=L_{3} \circ\left(L_{1} \times I d\right): \mathcal{G} \nrightarrow \mathcal{G} .`	$L_{2}:=L_{3} \circ\left(L_{1} \times I d\right): \mathcal{G} \nrightarrow \mathcal{G} .$	$L_2 := L_3 \circ (L_1 \times Id) : \mathcal{G} \nrightarrow \mathcal{G}.$
cardona-qft-methods_ EQ0865	335	1.000 K|+0	`L_{2}=L_{3} \circ\left(I d \times L_{1}\right) .`	$L_{2}=L_{3} \circ\left(I d \times L_{1}\right) .$	$L_2 = L_3 \circ (Id \times L_1).$
cardona-qft-methods_ EQ0866	335	1.000 K|+0	`\begin{aligned} & L_{2} \circ L_{1}=L_{1} \\ & L_{2} \circ L_{2}=L_{2} \\ & L_{2} \circ L_{3}=L_{3} . \end{aligned}`	$\begin{aligned} & L_{2} \circ L_{1}=L_{1} \\ & L_{2} \circ L_{2}=L_{2} \\ & L_{2} \circ L_{3}=L_{3} . \end{aligned}$	$\begin{aligned} & L_2 \circ L_1 = L_1 \\ & L_2 \circ L_2 = L_2 \\ & L_2 \circ L_3 = L_3. \end{aligned}$
cardona-qft-methods_ EQ0867	335	1.000 K|+0	`\overline{I_{r e l}} \circ L_{2}=\overline{L_{2}} \circ \overline{I_{r e l}} \text { and } L_{2}^{\dagger}=L_{2} .`	$\overline{I_{r e l}} \circ L_{2}=\overline{L_{2}} \circ \overline{I_{r e l}} \text { and } L_{2}^{\dagger}=L_{2} .$	$\overline{I_{rel}} \circ L_2 = \overline{L_2} \circ \overline{I_{rel}}$ *and* $L_2^\dagger = L_2.$
cardona-qft-methods_ EQ0868	335	1.000 K|+0	`C^{*}=\mathcal{G}^{*} \circ L_{2}`	$C^{*}=\mathcal{G}^{*} \circ L_{2}$	$C^* = \mathcal{G}^* \circ L_2$
cardona-qft-methods_ EQ0869	336	1.000 N|+0	`T:=\left\{(c,[l]) \in C \times M: \exists l \in[l], g \in \mathcal{G} \mid(c, l, g) \in L_{3}\right\}`	$T:=\left\{(c,[l]) \in C \times M: \exists l \in[l], g \in \mathcal{G} \mid(c, l, g) \in L_{3}\right\}$	$T := \{(c, [l]) \in C \times M : \exists l \in [l], g \in \mathcal{G} \mid (c, l, g) \in L_3\}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0871	337	1.000 W|+0	`\begin{aligned} \mathcal{G} & =G . \\ L & =\mathcal{L} \times \mathcal{L} \times \mathcal{L} . \\ I & =\{\text { identity of } G\} . \end{aligned}`	$\begin{aligned} \mathcal{G} & =G . \\ L & =\mathcal{L} \times \mathcal{L} \times \mathcal{L} . \\ I & =\{\text { identity of } G\} . \end{aligned}$	$\mathcal{G}=G.$ $L=\mathcal{L}\times\mathcal{L}\times\mathcal{L}.$ $I=\{\text{identity of } G\}.$
cardona-qft-methods_EQ0872	338	1.000 N|+0	`P:=\left\{(x, \alpha, \beta) \mid x \in \underline{G_{(n)}},[\alpha]=[\beta]=x\right\},`	$P:=\left\{(x, \alpha, \beta) \mid x \in \underline{G_{(n)}},[\alpha]=[\beta]=x\right\},$	$P:=\{(x,\alpha,\beta)\mid x\in \underline{G_{(n)}},\,[\alpha]=[\beta]=x\},$
cardona-qft-methods_EQ0873	338	1.000 K|+0	`P^{\dagger} \circ P=G r\left(I d_{G}\right), P \circ P^{\dagger}=\left\{(g, h) \in G^{n} \mid[g]=[h]\right\} .`	$P^{\dagger} \circ P=G r\left(I d_{G}\right), P \circ P^{\dagger}=\left\{(g, h) \in G^{n} \mid[g]=[h]\right\} .$	$P^{\dagger}\circ P=Gr(Id_G),\, P\circ P^{\dagger}=\{(g,h)\in G^n \mid [g]=[h]\}.$
cardona-qft-methods_EQ0875	338	1.000 N|+0	`\begin{aligned} \phi:[0,1] & \rightarrow[0,1] \\ t & \mapsto 1-t \end{aligned}`	$\begin{aligned} \phi:[0,1] & \rightarrow[0,1] \\ t & \mapsto 1-t \end{aligned}$	$\phi:[0,1]\rightarrow[0,1]$ $t\mapsto 1-t$
cardona-qft-methods_EQ0876	338	1.000 N|+0	`\overline{L_{1}}:=\cup_{x_{0} \in M} T_{(\overline{X, \eta})}^{*} P M \cap L_{1},`	$\overline{L_{1}}:=\cup_{x_{0} \in M} T_{(\overline{X, \eta})}^{*} P M \cap L_{1},$	$\overline{L_1}:=\cup_{x_0\in M}T^{*}_{(\overline{X,\eta})}PM\cap L_1,$
cardona-qft-methods_EQ0877	339	1.000 K|+0	`m: H \otimes H \rightarrow H .`	$m: H \otimes H \rightarrow H .$	$m: H\otimes H\rightarrow H.$
cardona-qft-methods_EQ0878	339	1.000 K|+0	`\langle a, b\rangle_{H}:=\langle\bar{a}, b\rangle,`	$\langle a, b\rangle_{H}:=\langle\bar{a}, b\rangle,$	$\langle a,b\rangle_H:=\langle\bar{a},b\rangle,$
cardona-qft-methods_EQ0879	339	1.000 K|+0	`\langle m(a, b), c\rangle_{H}=\langle a, m(b, c) \rangle_{H}`	$\langle m(a, b), c\rangle_{H}=\langle a, m(b, c) \rangle_{H}$	$\langle m(a,b),c\rangle_H=\langle a,m(b,c)\rangle_H$
cardona-qft-methods_EQ0880	339	1.000 N|+0	`e:=\sum_{i} m\left(e_{i}, \overline{e_{i}}\right)`	$e:=\sum_{i} m\left(e_{i}, \overline{e_{i}}\right)$	$e:=\sum_i m(e_i,\bar{e_i})$
cardona-qft-methods_EQ0881	339	1.000 K|+0	`\begin{aligned} P: H & \rightarrow H \\ a & \mapsto m(a, e) \end{aligned}`	$\begin{aligned} P: H & \rightarrow H \\ a & \mapsto m(a, e) \end{aligned}$	$P:H\rightarrow H$ $a\mapsto m(a,e)$
cardona-qft-methods_EQ0882	342	1.000 N|+0	`H_{p}(L M) \otimes H_{q}(L M) \rightarrow H_{p+q-n}(L M),`	$H_{p}(L M) \otimes H_{q}(L M) \rightarrow H_{p+q-n}(L M),$	$H_p(LM)\otimes H_q(LM)\rightarrow H_{p+q-n}(LM),$
cardona-qft-methods_EQ0885	344	1.000 N|+0	`\tau_{\Delta}: M \times M \rightarrow(M \times M) / \nu_{\Delta}^{c}`	$\tau_{\Delta}: M \times M \rightarrow(M \times M) / \nu_{\Delta}^{c}$	$\tau_\Delta: M\times M\rightarrow (M\times M)/\nu^c_\Delta$
cardona-qft-methods_EQ0886	344	1.000 N|+0	`\tau_{\Delta}: M \times M \rightarrow M^{T M}`	$\tau_{\Delta}: M \times M \rightarrow M^{T M}$	$\tau_\Delta: M\times M\rightarrow M^{TM}$
cardona-qft-methods_EQ0891	345	1.000 N|+0	`L M^{-T M} \wedge L M^{-T M} \rightarrow L M^{-T M},`	$L M^{-T M} \wedge L M^{-T M} \rightarrow L M^{-T M},$	$LM^{-TM}\wedge LM^{-TM}\rightarrow LM^{-TM},$
cardona-qft-methods_EQ0892	346	1.000 N|+0	`H_{p}\left(M^{-T M}\right) \cong H_{p+n}(M)`	$H_{p}\left(M^{-T M}\right) \cong H_{p+n}(M)$	$H_p(M^{-TM})\cong H_{p+n}(M)$
cardona-qft-methods_EQ0893	346	1.000 N|+0	`M^{-T M}:=\Sigma^{-k} M^{\nu(e)}`	$M^{-T M}:=\Sigma^{-k} M^{\nu(e)}$	$M^{-TM}:=\Sigma^{-k}M^{\nu(e)}$
cardona-qft-methods_EQ0894	346	1.000 N|+0	`\Delta^{*}: M^{-T M} \wedge M^{-T M}=(M \times M)^{-2 T M} \rightarrow M^{T M-2 T M}=M^{-T M}`	$\Delta^{*}: M^{-T M} \wedge M^{-T M}=(M \times M)^{-2 T M} \rightarrow M^{T M-2 T M}=M^{-T M}$	$\Delta^*: M^{-TM}\wedge M^{-TM}=(M\times M)^{-2TM}\rightarrow M^{TM-2TM}=M^{-TM}$
cardona-qft-methods_EQ0901	347	1.000 N|+0	`H_{*+n}(L M) \cong H H^{*}\left(C^{*}(M), C^{*}(M)\right) .`	$H_{*+n}(L M) \cong H H^{*}\left(C^{*}(M), C^{*}(M)\right) .$	$H_{*+n}(LM)\cong HH^*(C^*(M),C^*(M)).$
cardona-qft-methods_EQ0902	348	1.000 N|+0	`G_{x}=\{g \in G \mid g \bar{x}=\bar{x}\}`	$G_{x}=\{g \in G \mid g \bar{x}=\bar{x}\}$	$G_x=\{g\in G\mid g\bar{x}=\bar{x}\}$
cardona-qft-methods_EQ0903	349	1.000 N|+0	`P_{G} M \times_{G} E G \simeq L\left(M \times_{G} E G\right) .`	$P_{G} M \times_{G} E G \simeq L\left(M \times_{G} E G\right) .$	$P_GM\times_G EG\simeq L(M\times_G EG).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_EQ0904	349	■1.000 ● N|+1	`H_{*}\left(\mathcal{L}\left(M \times_{G} E G\right) ; \mathbb{Q}\right) \cong H_{*}\left(P_{G} M ; \mathbb{Q} \right)^{G}`	$H_{*}\left(\mathcal{L}\left(M \times_{G} E G\right) ; \mathbb{Q}\right) \cong H_{*}\left(P_{G} M ; \mathbb{Q} \right)^{G}$	$H_*(\mathcal{L}(M \times_G EG); \mathbb{Q}) \cong H_*(P_G M; \mathbb{Q})^G$
cardona-qft-methods_EQ0905	349	■1.000 ● N|+0	`P_{G} X:=\bigsqcup_{g \in G} P_{g} X \times\{g\} \quad \text { with } \quad P_{g} X:=\{f:[0,1] \rightarrow X \mid f(0) g=f(1)\}`	$P_{G} X:=\bigsqcup_{g \in G} P_{g} X \times\{g\} \quad \text { with } \quad P_{g} X:=\{f:[0,1] \rightarrow X \mid f(0) g=f(1)\}$	$P_G X := \bigsqcup_{g\in G} P_g X \times \{g\} \quad \text{with} \quad P_g X := \{f : [0,1] \to X \mid f(0)g = f(1)\}$
cardona-qft-methods_EQ0907	350	■1.000 ● N|+0	`P_{g} M:{ }_{1} \times_{0} P_{h} M=\left\{(\phi, \psi) \in P_{g} M \times P_{h} M \mid \phi(1)=\psi(0)\right\}`	$P_{g} M:{ }_{1} \times_{0} P_{h} M=\left\{(\phi, \psi) \in P_{g} M \times P_{h} M \mid \phi(1)=\psi(0)\right\}$	$P_g M : {}_1\times_0 P_h M = \{(\phi, \psi) \in P_g M \times P_h M \mid \phi(1) = \psi(0)\}$
cardona-qft-methods_EQ0909	350	■1.000 ● K|+0	`P_{G} M^{-T M}:=\bigsqcup_{g \in G} P_{g} M^{-T M}`	$P_{G} M^{-T M}:=\bigsqcup_{g \in G} P_{g} M^{-T M}$	$P_G M^{-TM} := \bigsqcup_{g\in G} P_g M^{-TM}$
cardona-qft-methods_EQ0910	350	■1.000 ● K|+0	`\mu: P_{G} M^{-T M} \wedge P_{G} M^{-T M} \rightarrow P_{G} M^{-T M} .`	$\mu: P_{G} M^{-T M} \wedge P_{G} M^{-T M} \rightarrow P_{G} M^{-T M} .$	$\mu : P_G M^{-TM} \wedge P_G M^{-TM} \to P_G M^{-TM}.$
cardona-qft-methods_EQ0911	350	■1.000 ● N|+0	`H_{*}\left(P_{G} M^{-T M} ; \mathbb{Q}\right)^{G} \cong H_{*+n}\left(L\left(M \times_{G} E G\right) ; \mathbb{Q} \right)`	$H_{*}\left(P_{G} M^{-T M} ; \mathbb{Q}\right)^{G} \cong H_{*+n}\left(L\left(M \times_{G} E G\right) ; \mathbb{Q} \right)$	$H_*(P_G M^{-TM}; \mathbb{Q})^G \cong H_{*+n}(L(M \times_G EG); \mathbb{Q})$
cardona-qft-methods_EQ0912	351	■1.000 ● N|+0	`H H^{*}\left(C^{*}(M ; \mathbb{Q}) \# G, C^{*}(M ; \mathbb{Q}) \# G\right) \cong H_{*}\left(P_{G} M^{-T M} ; \mathbb{Q}\right)^{G}`	$H H^{*}\left(C^{*}(M ; \mathbb{Q}) \# G, C^{*}(M ; \mathbb{Q}) \# G\right) \cong H_{*}\left(P_{G} M^{-T M} ; \mathbb{Q}\right)^{G}$	$HH^*(C^*(M; \mathbb{Q})\#G, C^*(M; \mathbb{Q})\#G) \cong H_*(P_G M^{-TM}; \mathbb{Q})^G$
cardona-qft-methods_EQ0913	351	■1.000 ● N|+1	`C^{*}(M) \# G:=C^{*}(M) \otimes \mathbb{Q} G \quad( \alpha, g) \cdot(\beta, h):=\left(\alpha \cdot g^{*} \beta, g h\right) .`	$C^{*}(M) \# G:=C^{*}(M) \otimes \mathbb{Q} G \quad(\alpha, g) \cdot(\beta, h):=\left(\alpha \cdot g^{*} \beta, g h\right) .$	$C^*(M)\#G := C^*(M) \otimes \mathbb{Q}G \quad (\alpha, g) \cdot (\beta, h) := (\alpha \cdot g^*\beta, gh).$
cardona-qft-methods_EQ0914	351	■1.000 ● N|+0	`H H^{*}\left(C^{*}(M ; \mathbb{Q}) \# G, C^{*}(M ; \mathbb{Q}) \# G\right) \cong \operatorname{Ext}_{ \mathbb{Q} G}^{*}\left(\mathbb{Q}, C_{*}\left(P_{G} M^{-T M} ; \mathbb{Q}\right)\right) .`	$H H^{*}\left(C^{*}(M ; \mathbb{Q}) \# G, C^{*}(M ; \mathbb{Q}) \# G\right) \cong \operatorname{Ext}_{ \mathbb{Q} G}^{*}\left(\mathbb{Q}, C_{*}\left(P_{G} M^{-T M} ; \mathbb{Q}\right)\right) .$	$HH^*(C^*(M; \mathbb{Q})\#G, C^*(M; \mathbb{Q})\#G) \cong \operatorname{Ext}^*_{\mathbb{Q}G}(\mathbb{Q}, C_*(P_G M^{-TM}; \mathbb{Q})).$
cardona-qft-methods_EQ0918	351	■1.000 ● K|+0	`C^{-*}(M) \rightarrow C_{*}\left(M^{-T M}\right)`	$C^{-*}(M) \rightarrow C_{*}\left(M^{-T M}\right)$	$C^{-*}(M) \to C_*(M^{-TM})$
cardona-qft-methods_EQ0920	352	■1.000 ● N|+0	`H H^{*}\left(C^{*}(M ; \mathbb{Q}) \# G, C^{*}(M ; \mathbb{Q}) \# G\right) \cong \operatorname{Ext}_{ \mathbb{Q} G}^{*}\left(\mathbb{Q}, C_{*}\left(P_{G} M^{-T M} ; \mathbb{Q}\right)\right)`	$H H^{*}\left(C^{*}(M ; \mathbb{Q}) \# G, C^{*}(M ; \mathbb{Q}) \# G\right) \cong \operatorname{Ext}_{ \mathbb{Q} G}^{*}\left(\mathbb{Q}, C_{*}\left(P_{G} M^{-T M} ; \mathbb{Q}\right)\right)$	$HH^*(C^*(M; \mathbb{Q})\#G, C^*(M; \mathbb{Q})\#G) \cong \operatorname{Ext}^*_{\mathbb{Q}G}(\mathbb{Q}, C_*(P_G M^{-TM}; \mathbb{Q}))$
cardona-qft-methods_EQ0921	352	■1.000 ● N|+0	`H H^{*}\left(C^{*}(M ; \mathbb{Q}) \# G, C^{*}(M ; \mathbb{Q}) \# G\right) \cong H_{*}\left(P_{G} M^{-T M} ; \mathbb{Q}\right)^{G} .`	$H H^{*}\left(C^{*}(M ; \mathbb{Q}) \# G, C^{*}(M ; \mathbb{Q}) \# G\right) \cong H_{*}\left(P_{G} M^{-T M} ; \mathbb{Q}\right)^{G} .$	$HH^*(C^*(M; \mathbb{Q})\#G, C^*(M; \mathbb{Q})\#G) \cong H_*(P_G M^{-TM}; \mathbb{Q})^G.$
cardona-qft-methods_EQ0922	352	■1.000 ● K|+0	`H H^{*}\left(C^{*}(M) \# G, C^{*}(M) \# G\right)`	$H H^{*}\left(C^{*}(M) \# G, C^{*}(M) \# G\right)$	$HH^*(C^*(M)\#G, C^*(M)\#G)$
cardona-qft-methods_EQ0923	352	■1.000 ● N|+0	`\operatorname{Ext}_{\mathbb{Z} G}^{*}\left(\mathbb{Z}, C_{*}\left(P_{G} M^{-T M}\right)\right) \cong H^{*}\left( \left(C^{*}(E G) \otimes C_{*}\left(P_{G} M^{-T M}\right) \right)^{G}\right)`	$\operatorname{Ext}_{\mathbb{Z} G}^{*}\left(\mathbb{Z}, C_{*}\left(P_{G} M^{-T M}\right)\right) \cong H^{*}\left(\left(C^{*}(E G) \otimes C_{*}\left(P_{G} M^{-T M}\right) \right)^{G}\right)$	$\operatorname{Ext}^*_{\mathbb{Z}G}(\mathbb{Z}, C_*(P_G M^{-TM})) \cong H^*\left((C^*(EG) \otimes C_*(P_G M^{-TM}))^G\right)$
cardona-qft-methods_EQ0924	352	■1.000 ● N|-1	`P_{G} M \times_{G} E G_{n} \xrightarrow{e_{0}} M \times_{G} E G_{n}`	$P_{G} M \times_{G} E G_{n} \xrightarrow{e_{0}} M \times_{G} E G_{n}$	$P_G M \times_G EG_n \xrightarrow{e_0} M \times_G EG_n$
cardona-qft-methods_EQ0926	352	■1.000 ● N|+0	`\left(P_{G} M \times_{G} E G_{n}\right)^{-e_{0}^{*} T \left(M_{n}\right)} \leftarrow\left(P_{G} M \times_{G} E G_{n+1}\right)^{-e_{0}^{*} T\left(M_{n+1}\right)} \leftarrow \cdots`	$\left(P_{G} M \times_{G} E G_{n}\right)^{-e_{0}^{*} T \left(M_{n}\right)} \leftarrow\left(P_{G} M \times_{G} E G_{n+1}\right)^{-e_{0}^{*} T\left(M_{n+1}\right)} \leftarrow \cdots$	$(P_G M \times_G EG_n)^{-e_0^* T(M_n)} \leftarrow (P_G M \times_G EG_{n+1})^{-e_0^* T(M_{n+1})} \leftarrow \cdots$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0929	353	1.000 N|+0	`\left[G^{a d} / G\right] \cong \bigsqcup_{(g)}[* / C(g)]`	$\left[G^{a d} / G\right] \cong \bigsqcup_{(g)}[* / C(g)]$	$[G^{ad}/G] \cong \bigsqcup_{(g)}[*/C(g)]$
cardona-qft-methods_ EQ0930 (g)	353	1.000 K|+0	`\bigsqcup B C(g)_{n}^{-T B C(g)_{n}}`	$\bigsqcup B C(g)_{n}^{-T B C(g)_{n}}$	$\bigsqcup BC(g)_{n}^{-TBC(g)_{n}}$
cardona-qft-methods_ EQ0932	354	1.000 N|+0	`H_{*}\left(E G_{n} / C(g) \times E G_{n} / C(h)\right) \rightarrow H_{*-k_{n}}\left(E G_{n} /(C(g) \cap C(h))\right)`	$H_{*}\left(E G_{n} / C(g) \times E G_{n} / C(h)\right) \rightarrow H_{*-k_{n}}\left(E G_{n} /(C(g) \cap C(h))\right)$	$H_*(EG_n/C(g) \times EG_n/C(h)) \rightarrow H_{*-k_n}(EG_n/(C(g) \cap C(h)))$
cardona-qft-methods_ EQ0933	354	1.000 N|+0	`H_{*}\left(E G_{n} /(C(g) \cap C(h))\right) \rightarrow H_{*}\left(E G_{n} / C(g h)\right)`	$H_{*}\left(E G_{n} /(C(g) \cap C(h))\right) \rightarrow H_{*}\left(E G_{n} / C(g h)\right)$	$H_*(EG_n/(C(g) \cap C(h))) \rightarrow H_*(EG_n/C(gh))$
cardona-qft-methods_ EQ0934	354	1.000 N|+0	`H^{*}\left(E G_{n} /(C(g) \cap C(h))\right) \rightarrow H^{*}\left(E G_{n} / C(g h)\right)`	$H^{*}\left(E G_{n} /(C(g) \cap C(h))\right) \rightarrow H^{*}\left(E G_{n} / C(g h)\right)$	$H^*(EG_n/(C(g) \cap C(h))) \rightarrow H^*(EG_n/C(gh))$
cardona-qft-methods_ EQ0935	354	1.000 N|+0	`H^{*}(B(C(g) \cap C(h))) \rightarrow H^{*}(B C(g h)) .`	$H^{*}(B(C(g) \cap C(h))) \rightarrow H^{*}(B C(g h)) .$	$H^*(B(C(g) \cap C(h))) \rightarrow H^*(BC(gh)).$
cardona-qft-methods_ EQ0938	355	1.000 N|+0	`H_{*}^{p r o}\left(L B G^{-T B G}\right) \cong H^{*}(L B G)`	$H_{*}^{p r o}\left(L B G^{-T B G}\right) \cong H^{*}(L B G)$	$H_*^{pro}(LBG^{-TBG}) \cong H^*(LBG)$
cardona-qft-methods_ EQ0939	355	1.000 N|+0	`L B G \leftarrow \operatorname{Map}(\Theta, B G) \rightarrow L B G \times L B G`	$L B G \leftarrow \operatorname{Map}(\Theta, B G) \rightarrow L B G \times L B G$	$LBG \leftarrow Map(\Theta, BG) \rightarrow LBG \times LBG$
cardona-qft-methods_ EQ0940	355	1.000 N|+0	`H^{*}(L B G) \otimes H^{*}(L B G) \rightarrow H^{*}(L B G)`	$H^{*}(L B G) \otimes H^{*}(L B G) \rightarrow H^{*}(L B G)$	$H^*(LBG) \otimes H^*(LBG) \rightarrow H^*(LBG)$
cardona-qft-methods_ EQ0941 (1)	358	1.000 K|+0	`[X]-[Y]=[X \backslash Y],`	$[X]-[Y]=[X \backslash Y],$	$[X]-[Y]=[X \setminus Y],$
cardona-qft-methods_ EQ0944 (4)	358	1.000 N|+0	`\left[\mathbb{A}_{k}^{n}\right]=\mathbb{L}^{n} .`	$\left[\mathbb{A}_{k}^{n}\right]=\mathbb{L}^{n} .$	$[\mathbb{A}_k^n]=\mathbb{L}^n .$
cardona-qft-methods_ EQ0945 (5)	359	1.000 N|+0	`\mathbb{T}=\left[\mathbb{A}_{k}^{1}\right]-\left[ \mathbb{A}_{k}^{0}\right]=\mathbb{L}-1,`	$\mathbb{T}=\left[\mathbb{A}_{k}^{1}\right]-\left[\mathbb{A}_{k}^{0}\right]=\mathbb{L}-1,$	$\mathbb{T}=[\mathbb{A}_k^1]-[\mathbb{A}_k^0]=\mathbb{L}-1,$
cardona-qft-methods_ EQ0946 (6)	359	1.000 K|+0	`[X \cup Y]=[X]+[Y]-[X \cap Y]`	$[X \cup Y]=[X]+[Y]-[X \cap Y]$	$[X \cup Y]=[X]+[Y]-[X \cap Y]$
cardona-qft-methods_ EQ0948 (8)	359	1.000 N|+0	`\left[\mathbb{P}_{k}^{n}\right]=\frac{ \mathbb{L}^{n+1}-1}{\mathbb{L}-1},`	$\left[\mathbb{P}_{k}^{n}\right]=\frac{ \mathbb{L}^{n+1}-1}{\mathbb{L}-1},$	$[\mathbb{P}_k^n]=\frac{\mathbb{L}^{n+1}-1}{\mathbb{L}-1},$
cardona-qft-methods_ EQ0949 (9)	359	1.000 N|+0	`\left[\mathbb{P}_{k}^{n}\right]=1+\mathbb{L}+ \mathbb{L}^{2}+\cdots+\mathbb{L}^{n},`	$\left[\mathbb{P}_{k}^{n}\right]=1+\mathbb{L}+ \mathbb{L}^{2}+\cdots+\mathbb{L}^{n},$	$[\mathbb{P}_k^n]=1+\mathbb{L}+\mathbb{L}^2+\cdots+\mathbb{L}^n,$
cardona-qft-methods_ EQ0950 (1)	359	1.000 K|+0	`\Psi_{\Gamma}(t)=\sum_{T \subset \Gamma} \prod_{e \notin E(T)} t_{e},`	$\Psi_{\Gamma}(t)=\sum_{T \subset \Gamma} \prod_{e \notin E(T)} t_{e},$	$\Psi_\Gamma(t)=\sum_{T\subset\Gamma}\prod_{e\notin E(T)} t_e,$
cardona-qft-methods_ EQ0951 (2)	360	1.000 N|+0	`X_{\Gamma}=\left\{t=\left(t_{1}: t_{2}: \cdots: t_{n} \right) \in \mathbb{P}_{k}^{n-1} \mid \Psi_{\Gamma}(t)=0 \right\} .`	$X_{\Gamma}=\left\{t=\left(t_{1}: t_{2}: \cdots: t_{n} \right) \in \mathbb{P}_{k}^{n-1} \mid \Psi_{\Gamma}(t)=0 \right\} .$	$X_\Gamma=\{t=(t_1:t_2:\cdots:t_n)\in\mathbb{P}_k^{n-1} \mid \Psi_\Gamma(t)=0\}.$
cardona-qft-methods_ EQ0952 (4)	360	1.000 N|+0	`\begin{gathered} \mathcal{C}: \mathbb{P}_{k}^{n-1} \rightarrow \mathbb{P}_{k}^{n-1} \\ \left(t_{1}: t_{2}: \cdots: t_{n}\right) \mapsto\left(\frac{1}{t_{1}}: \frac{1}{t_{2}}: \cdots: \frac{1}{t_{n}}\right) \end{gathered}`	$\begin{gathered} \mathcal{C}: \mathbb{P}_{k}^{n-1} \rightarrow \mathbb{P}_{k}^{n-1} \\ \left(t_{1}: t_{2}: \cdots: t_{n}\right) \mapsto\left(\frac{1}{t_{1}}: \frac{1}{t_{2}}: \cdots: \frac{1}{t_{n}}\right) \end{gathered}$	$\begin{gathered} \mathcal{C}: \mathbb{P}_k^{n-1} \rightarrow \mathbb{P}_k^{n-1} \\ (t_1:t_2:\cdots:t_n) \mapsto \left(\frac{1}{t_1}:\frac{1}{t_2}:\cdots:\frac{1}{t_n}\right) \end{gathered}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0953 (5)	360	■1.000 ● N|+0	`\Sigma_{n}=\left\{\left(t_{1}: t_{2}: \cdots: t_{n} \right) \in \mathbb{P}_{k}^{n-1} \mid \prod_{i=1}^{n} t_{i}=0\right\} .`	$\Sigma_{n}=\left\{\left(t_{1}: t_{2}: \cdots: t_{n}\right) \in \mathbb{P}_{k}^{n-1} \mid \prod_{i=1}^{n} t_{i}=0\right\} .$	$\Sigma_{n}=\left\{(t_{1}: t_{2}: \cdots: t_{n}) \in \mathbb{P}_{k}^{n-1} \vert \prod_{i=1}^{n} t_{i}=0\right\} .$
cardona-qft-methods_ EQ0954 (6)	361	■1.000 ● K|+0	`\mathcal{C} \circ \pi_{1}=\pi_{2} .`	$\mathcal{C} \circ \pi_{1}=\pi_{2} .$	$\mathcal{C} \circ \pi_{1}=\pi_{2} .$
cardona-qft-methods_ EQ0955 (7)	361	■1.000 ● K|+0	`t_{1} t_{n+1}=t_{2} t_{n+2}=\cdots=t_{n} t_{2 n},`	$t_{1} t_{n+1}=t_{2} t_{n+2}=\cdots=t_{n} t_{2 n},$	$t_{1} t_{n+1}=t_{2} t_{n+2}=\cdots=t_{n} t_{2 n} ,$
cardona-qft-methods_ EQ0961 (1)	362	■1.000 ● K|+0	`\Psi_{\Gamma}(t)=1,`	$\Psi_{\Gamma}(t)=1,$	$\Psi_{\Gamma}(t)=1 ,$
cardona-qft-methods_ EQ0962 (2)	362	■1.000 ● K|+0	`X_{\Gamma}=\emptyset .`	$X_{\Gamma}=\emptyset .$	$X_{\Gamma}=\emptyset .$
cardona-qft-methods_ EQ0963 (3)	362	■1.000 ● K|+0	`\Psi_{\Gamma^{\vee}}(t)=t_{1} t_{2} \cdots t_{n},`	$\Psi_{\Gamma^{\vee}}(t)=t_{1} t_{2} \cdots t_{n},$	$\Psi_{\Gamma^{\vee}}(t)=t_{1} t_{2} \cdots t_{n} ,$
cardona-qft-methods_ EQ0964 (4)	362	■1.000 ● N|+0	`X_{\Gamma^{\vee}}=\left\{t \in \mathbb{P}_{k}^{n-1} \mid \Psi_{\Gamma^{\vee}}(t)=0\right\}=\Sigma_{n} .`	$X_{\Gamma^{\vee}}=\left\{t \in \mathbb{P}_{k}^{n-1} \mid \Psi_{\Gamma^{\vee}}(t)=0\right\}=\Sigma_{n} .$	$X_{\Gamma^{\vee}}=\{t \in \mathbb{P}_{k}^{n-1} \vert \Psi_{\Gamma^{\vee}}(t)=0\}=\Sigma_{n} .$
cardona-qft-methods_ EQ0966 (6)	362	■1.000 ● K|+0	`\left[X_{\Gamma}\right]=0 .`	$\left[X_{\Gamma}\right]=0 .$	$[X_{\Gamma}]=0 .$
cardona-qft-methods_ EQ0968 (8)	363	■1.000 ● K|+0	`\Psi_{\Gamma}(t)=t_{1}+t_{2}+\cdots+t_{n},`	$\Psi_{\Gamma}(t)=t_{1}+t_{2}+\cdots+t_{n},$	$\Psi_{\Gamma}(t)=t_{1}+t_{2}+\cdots+t_{n} ,$
cardona-qft-methods_ EQ0970 (10)	363	■1.000 ● N|+0	`[\mathcal{L}]=\left[X_{\Gamma}\right]=\left[ \mathbb{P}_{k}^{n-2}\right]=\frac{\mathbb{L}^{n-1}-1}{ \mathbb{L}-1}=\frac{(\mathbb{T}+1)^{n-1}-1}{\mathbb{T}} .`	$[\mathcal{L}]=\left[X_{\Gamma}\right]=\left[\mathbb{P}_{k}^{n-2}\right]=\frac{\mathbb{L}^{n-1}-1}{\mathbb{L}-1}=\frac{(\mathbb{T}+1)^{n-1}-1}{\mathbb{T}} .$	$[\mathcal{L}]=[X_{\Gamma}]=[\mathbb{P}_{k}^{n-2}]=\frac{\mathbb{L}^{n-1}-1}{\mathbb{L}-1}=\frac{(\mathbb{T}+1)^{n-1}-1}{\mathbb{T}} .$
cardona-qft-methods_ EQ0973 (13)	363	■1.000 ● N|+0	`\mathcal{H}: g(\mathbb{T}) \mapsto \frac{g( \mathbb{T})-g(-1)}{\mathbb{T}+1}`	$\mathcal{H}: g(\mathbb{T}) \mapsto \frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}$	$\mathcal{H}: g(\mathbb{T}) \mapsto \frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}$
cardona-qft-methods_ EQ0974 (14)	364	■1.000 ● W|+0	`\frac{g(\mathbb{T}-g(-1))}{\mathbb{T}+1}=\frac{( \mathbb{T}-1)^{n-1}-1}{\mathbb{T}}=\left[\mathbb{P}^{n-1} \right],`	$\frac{g(\mathbb{T}-g(-1))}{\mathbb{T}+1}=\frac{(\mathbb{T}-1)^{n-1}-1}{\mathbb{T}}=\left[\mathbb{P}^{n-1}\right],$	$\frac{g(\mathbb{T}-g(-1))}{\mathbb{T}+1}=\frac{(\mathbb{T}-1)^{n-1}-1}{\mathbb{T}}=[\mathbb{P}^{n-1}] ,$
cardona-qft-methods_ EQ0975 (15)	364	■1.000 ● W|+0	`\left[\mathcal{L} \cap \Sigma_{n}\right]= \frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}= \frac{(1+\mathbb{T})^{n-1}-1}{\mathbb{T}}-\frac{ \mathbb{T}^{n-1}-(-1)^{n-1}}{\mathbb{T}+1},`	$\left[\mathcal{L} \cap \Sigma_{n}\right]=\frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}=\frac{(1+\mathbb{T})^{n-1}-1}{\mathbb{T}}-\frac{\mathbb{T}^{n-1}-(-1)^{n-1}}{\mathbb{T}+1},$	$[\mathcal{L} \cap \Sigma_{n}]=\frac{g(\mathbb{T})-g(-1)}{\mathbb{T}+1}=\frac{(1+\mathbb{T})^{n-1}-1}{\mathbb{T}}-\frac{\mathbb{T}^{n-1}-(-1)^{n-1}}{\mathbb{T}+1} ,$
cardona-qft-methods_ EQ0976 (16)	364	■1.000 ● W|+0	`\left[X_{\Gamma^{\vee}} \backslash \Sigma_{n}\right]= \left[\mathcal{L} \backslash \Sigma_{n}\right]=[ \mathcal{L}]-\left[\mathcal{L} \cap \Sigma_{n}\right]= \frac{\mathbb{T}^{n-1}-(-1)^{n-1}}{\mathbb{T}+1} .`	$\left[X_{\Gamma^{\vee}} \backslash \Sigma_{n}\right]=\left[\mathcal{L} \backslash \Sigma_{n}\right]=[\mathcal{L}]-\left[\mathcal{L} \cap \Sigma_{n}\right]=\frac{\mathbb{T}^{n-1}-(-1)^{n-1}}{\mathbb{T}+1} .$	$[X_{\Gamma^{\vee}} \setminus \Sigma_{n}]=[\mathcal{L} \setminus \Sigma_{n}]=[\mathcal{L}]-[\mathcal{L} \cap \Sigma_{n}]=\frac{\mathbb{T}^{n-1}-(-1)^{n-1}}{\mathbb{T}+1} .$
cardona-qft-methods_ EQ0977 (17)	364	■1.000 ● K|+0	`\mathcal{I}=\left(t_{2} t_{3} \cdots t_{n}, t_{1} t_{3} \cdots t_{n}, \ldots, t_{1} t_{2} \cdots \hat{t_{i}} \cdots t_{n}, \ldots, t_{1} t_{2} \cdots t_{n-1}\right) .`	$\mathcal{I}=\left(t_{2} t_{3} \cdots t_{n}, t_{1} t_{3} \cdots t_{n}, \ldots, t_{1} t_{2} \cdots \hat{t_{i}} \cdots t_{n}, \ldots, t_{1} t_{2} \cdots t_{n-1}\right) .$	$\mathcal{I}=(t_{2} t_{3} \cdots t_{n}, t_{1} t_{3} \cdots t_{n}, \ldots, t_{1} t_{2} \cdots \hat{t_{i}} \cdots t_{n}, \ldots, t_{1} t_{2} \cdots t_{n-1}) .$
cardona-qft-methods_ EQ0980 (20)	365	■1.000 ● K|+0	`\left[X_{\Gamma^{\vee}}\right]=\left[X_{\Gamma^{\vee}} \backslash \Sigma\right]+\left[X_{\Gamma^{\vee}} \cap \Sigma\right]=\left[X_{\Gamma^{\vee}} \backslash \Sigma \right]+\left[\mathcal{S}_{n}\right],`	$\left[X_{\Gamma^{\vee}}\right]=\left[X_{\Gamma^{\vee}} \backslash \Sigma\right]+\left[X_{\Gamma^{\vee}} \cap \Sigma\right]=\left[X_{\Gamma^{\vee}} \backslash \Sigma\right]+\left[\mathcal{S}_{n}\right],$	$[X_{\Gamma^{\vee}}]=[X_{\Gamma^{\vee}} \setminus \Sigma]+[X_{\Gamma^{\vee}} \cap \Sigma]=[X_{\Gamma^{\vee}} \setminus \Sigma]+[\mathcal{S}_{n}] ,$
cardona-qft-methods_ EQ0981 (1)	366	■1.000 ● K|+0	`\Psi_{\Gamma}(t)=p(t) q(t)`	$\Psi_{\Gamma}(t)=p(t) q(t)$	$\Psi_{\Gamma}(t)=p(t) q(t)$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ0982 (2)	367	1.000 N|+0	`P=\left\{t_{i} \mid \text { such that } t_{i} \text { appears in } p(t)\right\}`	$P=\left\{t_{i} \mid \text { such that } t_{i} \text { appears in } p(t)\right\}$	$P=\{t_i \| \textit{ such that } t_i \textit{ appears in } p(t)\}$
cardona-qft-methods_ EQ0984	369	1.000 K|+0	`X Y-Y X-[X, Y], \quad X, Y \in \tilde{m} g .`	$X Y-Y X-[X, Y], \quad X, Y \in \tilde{m} g .$	$XY-YX-[X,Y], \quad X,Y \in \tilde{m}g.$
cardona-qft-methods_ EQ0985	369	1.000 N|+0	`g f(x)=f\left(g^{-1} x\right), \quad f \in C(M) .`	$g f(x)=f\left(g^{-1} x\right), \quad f \in C(M) .$	$gf(x)=f(g^{-1}x), \quad f \in C(M).$
cardona-qft-methods_ EQ0986	369	1.000 N|+0	`e_{\xi}: g \in G \mapsto\langle J \xi, g \nu\rangle \in \mathbb{C} .`	$e_{\xi}: g \in G \mapsto\langle J \xi, g \nu\rangle \in \mathbb{C} .$	$e_\xi : g \in G \mapsto \langle J\xi, g\nu\rangle \in \mathbb{C}.$
cardona-qft-methods_ EQ0990	371	1.000 N|+0	`\omega(A)=\langle\nu, A \nu\rangle, \quad A \in \mathcal{A} .`	$\omega(A)=\langle\nu, A \nu\rangle, \quad A \in \mathcal{A} .$	$\omega(A)=\langle\nu, A\nu\rangle, \quad A \in \mathcal{A}.$
cardona-qft-methods_ EQ0991	372	1.000 K|+0	`\langle[A],[B]\rangle=\omega\left(A^{\dagger} B\right), \quad A, B \in \mathcal{A},`	$\langle[A],[B]\rangle=\omega\left(A^{\dagger} B\right), \quad A, B \in \mathcal{A},$	$\langle[A],[B]\rangle=\omega(A^\dagger B), \quad A,B \in \mathcal{A},$
cardona-qft-methods_ EQ0993	373	1.000 N|+0	`X \in \tilde{m} g \mapsto 1 \otimes X+X \otimes 1 \in \mathcal{T}(\tilde{m} g) \otimes \mathcal{T}(\tilde{m} g)`	$X \in \tilde{m} g \mapsto 1 \otimes X+X \otimes 1 \in \mathcal{T}(\tilde{m} g) \otimes \mathcal{T}(\tilde{m} g)$	$X \in \tilde{m}g \mapsto 1\otimes X + X \otimes 1 \in \mathcal{T}(\tilde{m}g)\otimes\mathcal{T}(\tilde{m}g)$
cardona-qft-methods_ EQ0994	373	1.000 N|-1	`\Delta \mathcal{I} \subseteq \mathcal{I} \otimes \mathcal{T}(\tilde{m} g)+\mathcal{T}(\tilde{m} g) \otimes \mathcal{I} .`	$\Delta \mathcal{I} \subseteq \mathcal{I} \otimes \mathcal{T}(\tilde{m} g)+\mathcal{T}(\tilde{m} g) \otimes \mathcal{I} .$	$\Delta\mathcal{I} \subseteq \mathcal{I}\otimes\mathcal{T}(\tilde{m}g)+\mathcal{T}(\tilde{m}g)\otimes\mathcal{I}.$
cardona-qft-methods_ EQ0996	374	1.000 K|+0	`a b(E)=a \otimes b(\Delta E), \quad a, b \in \mathcal{E}^{\dagger}, E \in \mathcal{E} .`	$a b(E)=a \otimes b(\Delta E), \quad a, b \in \mathcal{E}^{\dagger}, E \in \mathcal{E} .$	$ab(E)=a\otimes b(\Delta E), \quad a,b \in \mathcal{E}^\dagger,\ E \in \mathcal{E}.$
cardona-qft-methods_ EQ0997 (1)	374	1.000 K|+0	`E a(F)=a\left(\bar{E}^{\dagger} F\right), \quad E, F \in \mathcal{E}, a \in \mathcal{E}^{\dagger} .`	$E a(F)=a\left(\bar{E}^{\dagger} F\right), \quad E, F \in \mathcal{E}, a \in \mathcal{E}^{\dagger} .$	$Ea(F)=a(\overline{E}^\dagger F), \quad E,F \in \mathcal{E},\ a \in \mathcal{E}^\dagger.$
cardona-qft-methods_ EQ0998	374	1.000 N|+0	`\left(\forall a, b \in \mathcal{E}^{\dagger}\right) \delta(a b)=\delta(a) b+a \delta(b) .`	$\left(\forall a, b \in \mathcal{E}^{\dagger}\right) \delta(a b)=\delta(a) b+a \delta(b) .$	$(\forall a,b \in \mathcal{E}^\dagger)\ \delta(ab)=\delta(a)b+a\delta(b).$
cardona-qft-methods_ EQ0999	374	1.000 N|+0	`X(a b)=X((a \otimes b) \circ \Delta)=(a \otimes b) \circ \Delta \circ X^{\dagger}(\cdot) .`	$X(a b)=X((a \otimes b) \circ \Delta)=(a \otimes b) \circ \Delta \circ X^{\dagger}(\cdot) .$	$X(ab)=X((a\otimes b)\circ\Delta)=(a\otimes b)\circ\Delta\circ X^\dagger(\cdot).$
cardona-qft-methods_ EQ1002	375	1.000 N|+0	`a \in \mathcal{E}^{\dagger} \mapsto f_{a} \in \mathbb{C} \llbracket x_{1}, \ldots x_{n} \rrbracket, \quad f_{a}(x)=\sum_{\alpha \in \mathbb{N}^{n}} \frac{x^{\alpha}}{\alpha!} a\left(X^{\alpha}\right) .`	$a \in \mathcal{E}^{\dagger} \mapsto f_{a} \in \mathbb{C} \llbracket x_{1}, \ldots x_{n} \rrbracket, \quad f_{a}(x)=\sum_{\alpha \in \mathbb{N}^{n}} \frac{x^{\alpha}}{\alpha!} a\left(X^{\alpha}\right) .$	$a \in \mathcal{E}^\dagger \mapsto f_a \in \mathbb{C}\llbracket x_1,\ldots x_n\rrbracket, \quad f_a(x)=\sum_{\alpha\in\mathbb{N}^n}\frac{x^\alpha}{\alpha!}a(X^\alpha).$
cardona-qft-methods_ EQ1004	375	1.000 N|+0	`a b(1)=a \otimes b(\Delta 1)=a(1) b(1) .`	$a b(1)=a \otimes b(\Delta 1)=a(1) b(1) .$	$ab(1)=a\otimes b(\Delta 1)=a(1)b(1).$
cardona-qft-methods_ EQ1005	375	1.000 N|+0	`a^{\dagger}(E+f F)=\overline{a(E)}+f \overline{a(F)}, \quad E, F \in \mathcal{E}_{\mathbb{R}} .`	$a^{\dagger}(E+f F)=\overline{a(E)}+f \overline{a(F)}, \quad E, F \in \mathcal{E}_{\mathbb{R}} .$	$a^\dagger(E+\fint F)=\overline{a(E)}+\fint\ \overline{a(F)}, \quad E,F\in\mathcal{E}_\mathbb{R}.$
cardona-qft-methods_ EQ1006	375	1.000 N|+0	`\chi_{0}\left(a^{\dagger}\right)=a^{\dagger}(1)=\overline{a(1)}=\overline{\chi_{0}(a)} .`	$\chi_{0}\left(a^{\dagger}\right)=a^{\dagger}(1)=\overline{a(1)}=\overline{\chi_{0}(a)} .$	$\chi_0(a^\dagger)=a^\dagger(1)=\overline{a(1)}=\overline{\chi_0(a)}.$
cardona-qft-methods_ EQ1007	376	1.000 N|+0	`\xi \in \mathcal{F} \mapsto e_{\xi} \in \mathcal{E}^{\dagger}, \quad e_{\xi}(E)=\langle J \xi, E \nu\rangle,`	$\xi \in \mathcal{F} \mapsto e_{\xi} \in \mathcal{E}^{\dagger}, \quad e_{\xi}(E)=\langle J \xi, E \nu\rangle,$	$\xi\in\mathcal{F}\mapsto e_\xi\in\mathcal{E}^\dagger, \quad e_\xi(E)=\langle J\xi, E\nu\rangle,$
cardona-qft-methods_ EQ1008	376	1.000 N|+0	`M=\left\{: \chi \in \mathcal{A}^{\dagger} \mid \chi(a b)=\chi(a) \chi(b), \chi\left(a^{\dagger}\right)=\overline{\chi(a)}:\right\} .`	$M=\left\{: \chi \in \mathcal{A}^{\dagger} \mid \chi(a b)=\chi(a) \chi(b), \chi\left(a^{\dagger}\right)=\overline{\chi(a)}:\right\} .$	$M=\left\{:\chi\in\mathcal{A}^\dagger\ \middle\|\ \chi(ab)=\chi(a)\chi(b),\ \chi(a^\dagger)=\overline{\chi(a)}:\right\}.$
cardona-qft-methods_ EQ1009	376	1.000 N|+0	`E e_{\xi}(F)=\left\langle J \xi, \bar{E}^{\dagger} F \nu\right\rangle=\langle\bar{E} J \xi, F \nu\rangle=\langle J E \xi, F \nu\rangle=e_{E \xi}(F) .`	$E e_{\xi}(F)=\left\langle J \xi, \bar{E}^{\dagger} F \nu\right\rangle=\langle\bar{E} J \xi, F \nu\rangle=\langle J E \xi, F \nu\rangle=e_{E \xi}(F) .$	$Ee_\xi(F)=\langle J\xi,\overline{E}^\dagger F\nu\rangle=\langle\overline{E}J\xi,F\nu\rangle=\langle JE\xi,F\nu\rangle=e_{E\xi}(F).$
cardona-qft-methods_ EQ1010	376	1.000 N|+0	`\omega \in \mathcal{E}^{\dagger}, \quad \omega(E)=e_{\nu}(E)=\langle\nu, E \nu\rangle`	$\omega \in \mathcal{E}^{\dagger}, \quad \omega(E)=e_{\nu}(E)=\langle\nu, E \nu\rangle$	$\omega\in\mathcal{E}^\dagger, \quad \omega(E)=e_\nu(E)=\langle\nu,E\nu\rangle$

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Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_ EQ1011	377	■ 1.000 ● N|+0	`\tilde{m} g=\mathbb{R} X, \quad \mathcal{F}=\mathcal{H}= \mathbb{C}^{2}, \quad X=\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right), \quad \nu=\binom{1}{0} .`	$\tilde{m} g=\mathbb{R} X, \quad \mathcal{F}=\mathcal{H}=\mathbb{C}^{2}, \quad X=\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right), \quad \nu=\binom{1}{0} .$	$\tilde{m} g=\mathbb{R} X, \quad \mathcal{F}=\mathcal{H}=\mathbb{C}^{2}, \quad X=\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right), \quad \nu=\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right) .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

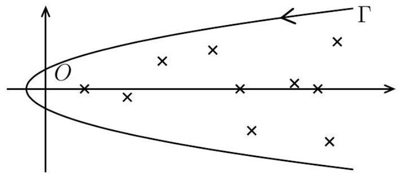
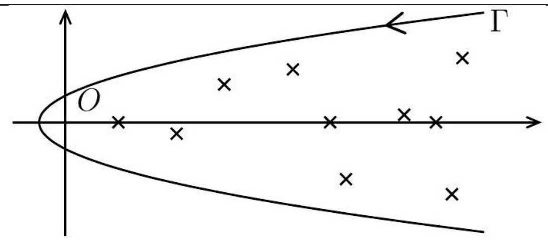
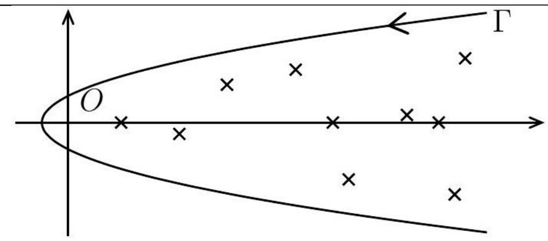
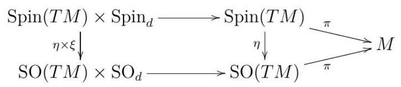
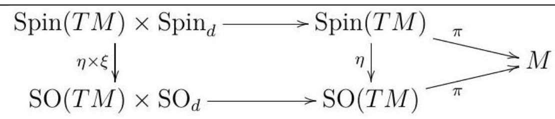
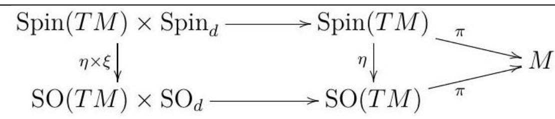
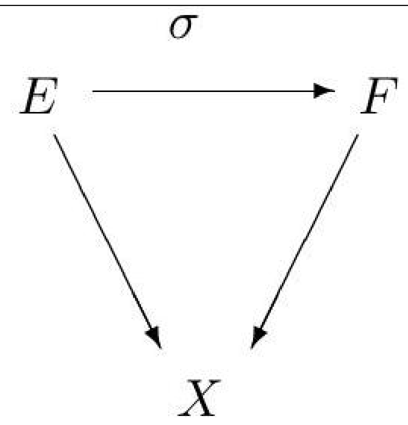
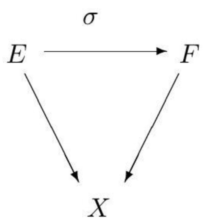
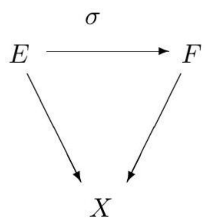
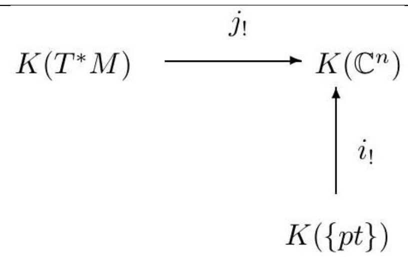
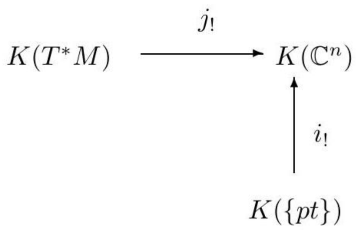
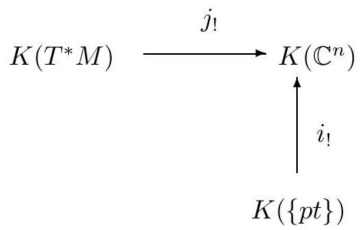
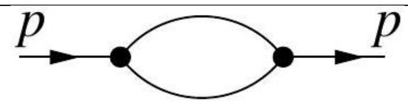
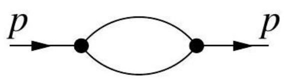
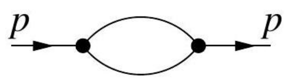
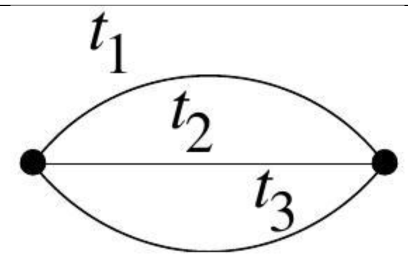
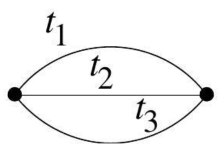
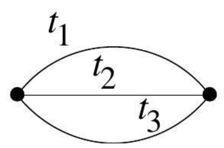
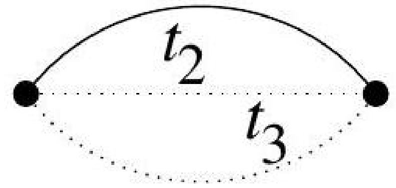
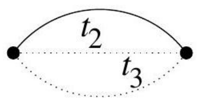
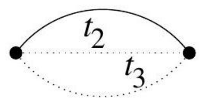
Tables

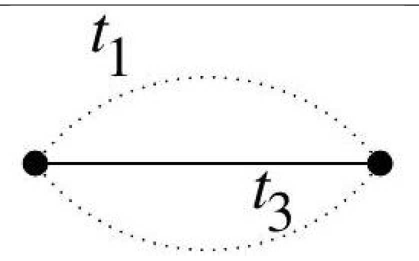
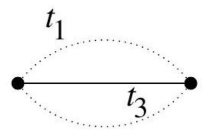
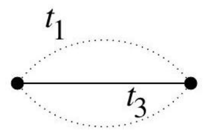
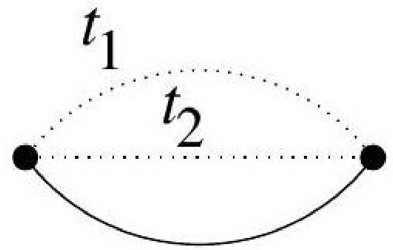
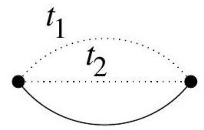
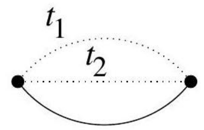
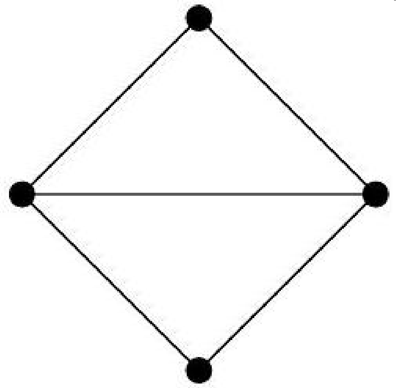
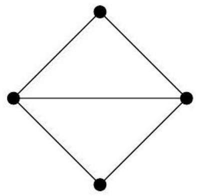
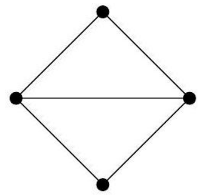
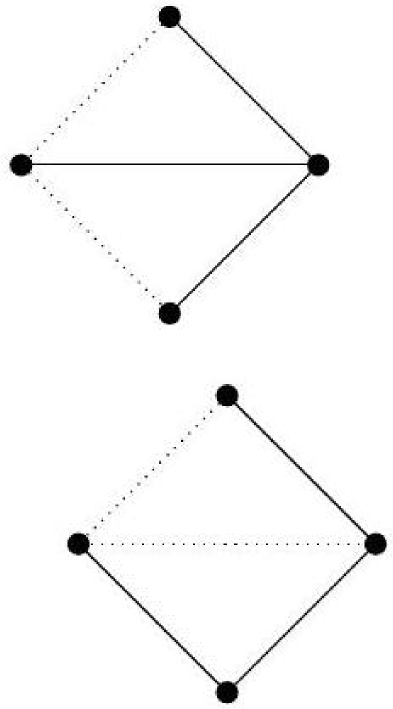
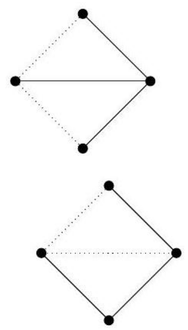
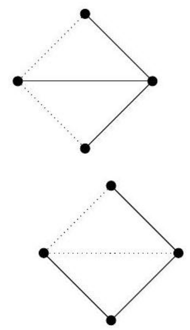
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
cardona-qft-methods_TAB_001	043	—	(no LaTeX source; 600 × 188 px region)	—	$[\pi(a), \pi(b)^\circ] = 0, \qquad [[\mathcal{D}, \pi(a)], \pi(b)^\circ] = 0, \forall a, b \in \mathcal{A}$ $a)^\circ = J\pi(a^*)J^{-1}$ is a representation of the opposite algel l-summable (or has metric dimension d) if the singular val ike $\mu_n(\mathcal{D}^{-1}) = \mathcal{O}(n^{-1/d})$. egular if $\mathcal{A}$ and $[\mathcal{D}, \mathcal{A}]$ are in the domain of δ^n for all $n \in$
cardona-qft-methods_TAB_002	181	—	(no LaTeX source; 1137 × 397 px region)	—	topological invariants, $\begin{array}{l\|l\|l} D=3 \text{ CS Lagrangians} & \text{Characteristic classes} & \text{Groups} \\ \hline L_3^{(A)dS} = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{3l^2})e^c & \mathbf{E}_4 = \epsilon_{abc}(R^{ab} \pm \frac{e^a e^b}{l^2})T^c & SO(4),\ SO(3,1) \text{ or } SO(2,2) \\ L_3^{Lor} = \omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a & \mathbf{P}_4^{Lor} = R^a{}_b R^b{}_a & SO(2,1) \\ L_3^{Tor} = e^a T_a & \mathbf{N}_4 = T^a T_a - e^a e^b R_{ab} & SO(2,1) \\ L_3^{U(1)} = AdA & \mathbf{P}_4^{U(1)} = FF & U(1) \\ L_3^{SU(N)} = Tr[\mathbf{A}d\mathbf{A} + \frac{2}{3}\mathbf{AAA}] & \mathbf{P}_4^{SU(4)} = Tr[\mathbf{FF}] & SU(N) \end{array}$ **Table 1:** Three-dimensional gravitational CS Lagrangians, their related
cardona-qft-methods_TAB_003	182	—	(no LaTeX source; 1066 × 244 px region)	—	two Lorentz-Pontryagin, and one torsional CS forms, $\begin{array}{l\|l} D=7 \text{ CS Lagrangians} & \text{Characteristic classes} \\ \hline L_7^{Lor} = \omega(d\omega)^3 + \frac{8}{5}\omega^3(d\omega)^2 + \cdots + \frac{4}{7}\omega^7 & \mathbf{P}_8^{Lor} = R^a{}_b R^b{}_c R^c{}_d R^d{}_a \\ L_7^{I} = (\omega^a{}_b d\omega^b{}_a + \frac{2}{3}\omega^a{}_b\omega^b{}_c\omega^c{}_a)R^a{}_b R^b{}_a & (\mathbf{P}_4^{Lor})^2 = (R^a{}_b R^b{}_a)^2 \\ L_7^{II} = (e^a T_a)(R^a{}_b R^b{}_a) & \mathbf{N}_4\mathbf{P}_4^{Lor} = (T^a T_a - e^a e^b R_{ab})R^c{}_d R^d{}_c \end{array}$ **Table 2:** Seven-dimensional gravitational CS Lagrangians, their related
cardona-qft-methods_TAB_004	205	—	(no LaTeX source; 625 × 198 px region)	—	$[[D, a], b^0] = 0 \qquad \forall\, a, b \in \mathcal{A}.$ table of signs listed above is modeled on the case of ordinar vhere it reflects the properties of the mod eight periodicity gy and real K-theory. However, imposing that real structu
cardona-qft-methods_TAB_005	281	—	(no LaTeX source; 158 × 191 px region)	—	glues the $\mathfrak{su}(2,$ ra, expressed i $\begin{array}{\|c\|c\|c\|} L & P & Q \\ K & \bar{L} & \bar{S} \\ S & \bar{Q} & R \end{array}$.
cardona-qft-methods_TAB_006	281	—	(no LaTeX source; 163 × 235 px region)	—	
cardona-qft-methods_TAB_007	286	—	(no LaTeX source; 158 × 189 px region)	—	$\begin{array}{\|c\|c\|} \begin{matrix} k & k \\ k & k \end{matrix} & \bar{L} \end{array}$. ierator one has t that there are *ey prediction o*
cardona-qft-methods_TAB_008	287	—	(no LaTeX source; 294 × 280 px region)	—	e organised in a 8×8 matri $\begin{array}{\|c\|c\|} \mathfrak{su}(2\|2) & \text{create } 8_B + 8_F \text{ magnons} \\ \text{destroy } 8+8 \text{ magnons} & \overline{\mathfrak{su}(2\|2)} \end{array}$

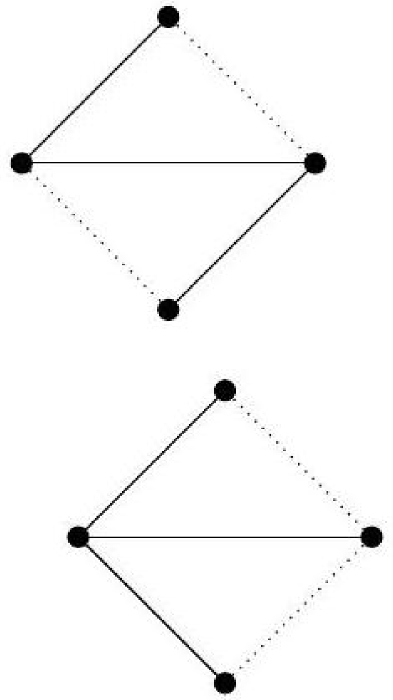
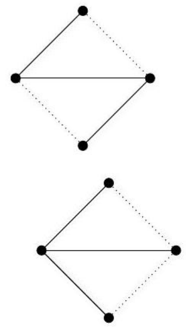
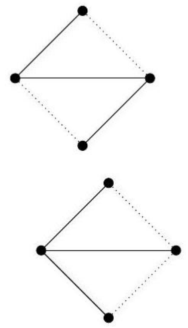
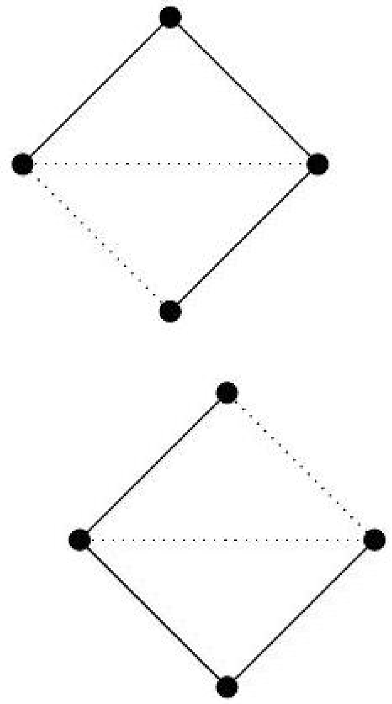
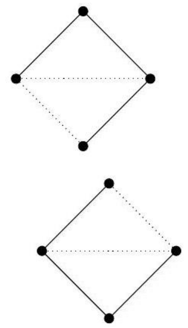
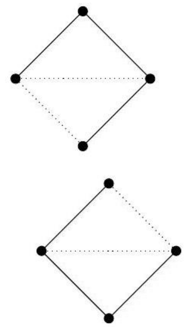
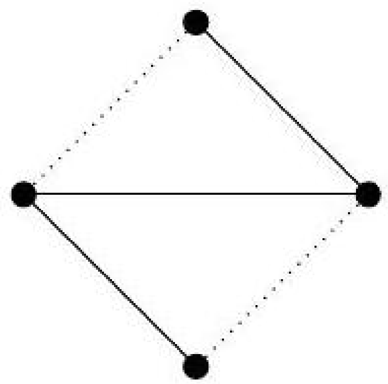
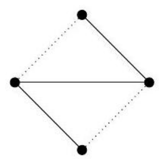
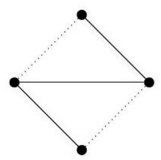
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

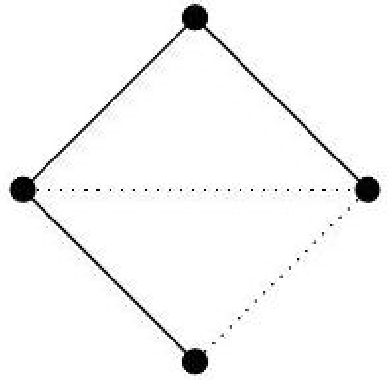
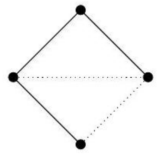
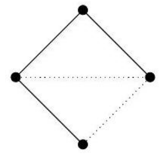
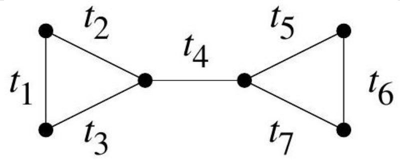
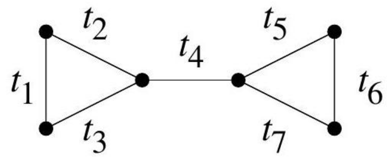
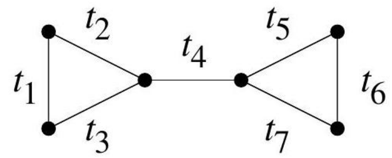
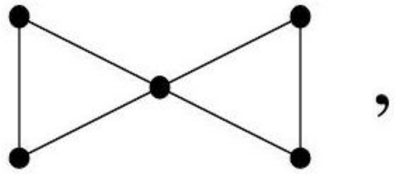
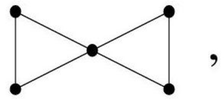
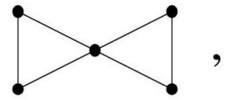
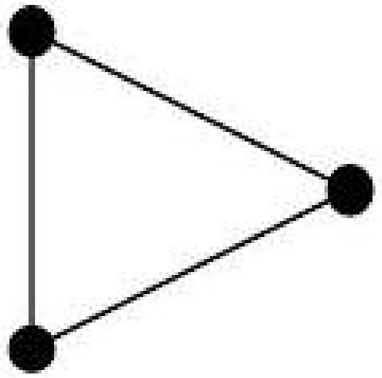
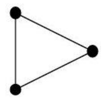
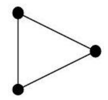
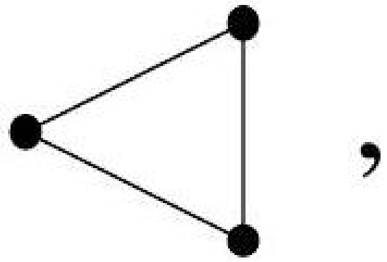
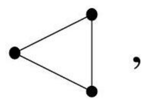
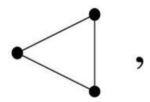
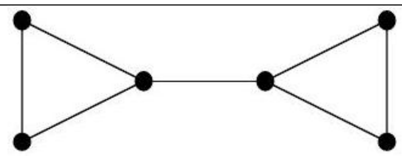
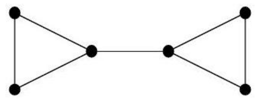
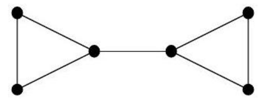
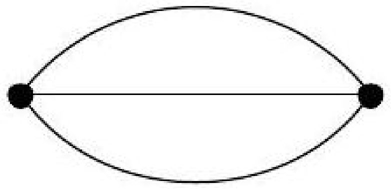
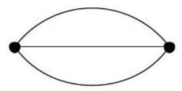
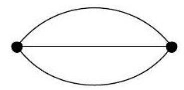
Image regions — rendered against scan

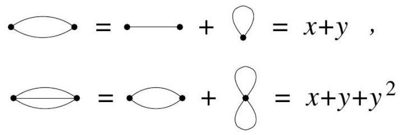
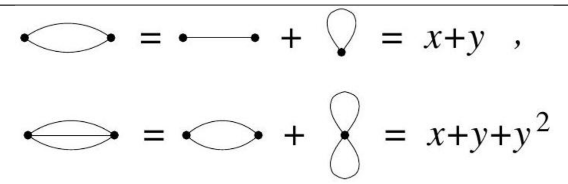
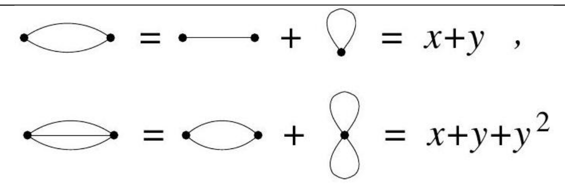
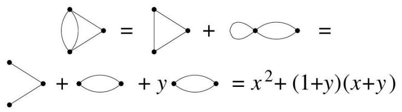
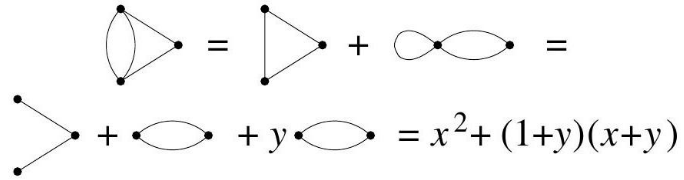
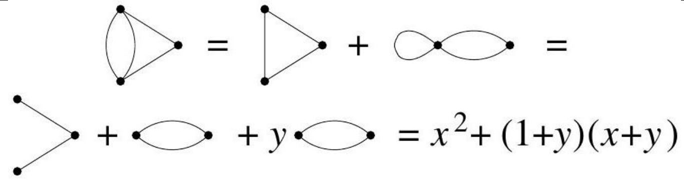
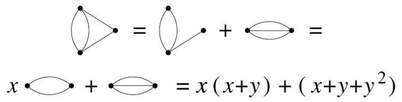
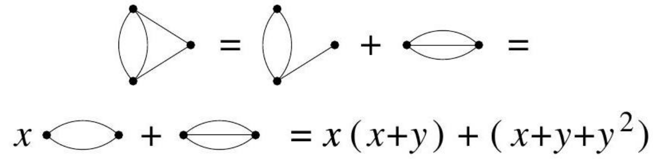
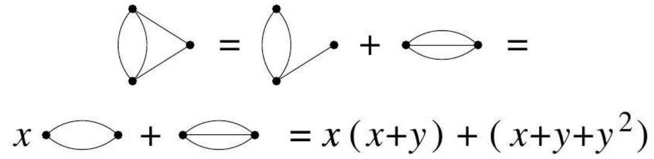
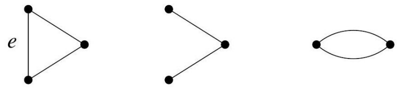
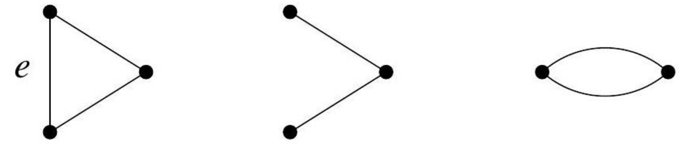
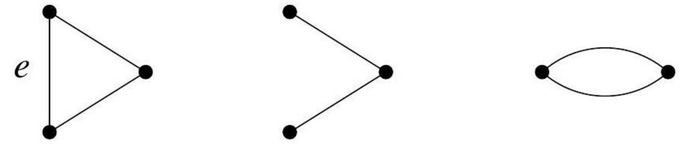
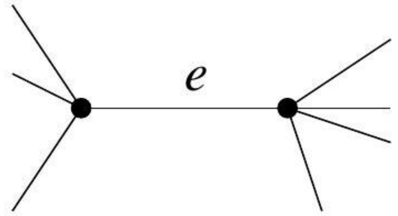
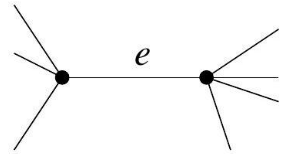
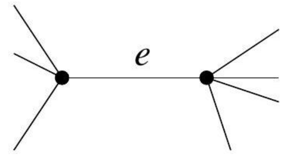
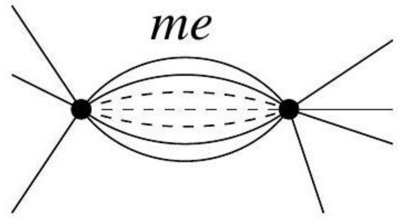
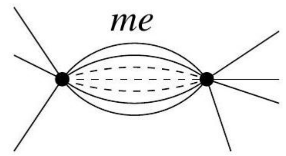
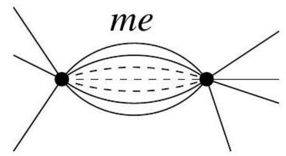
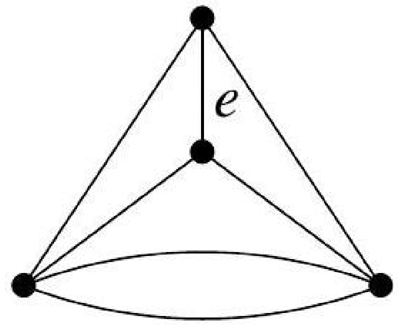
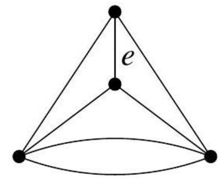
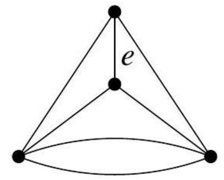
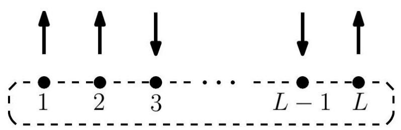
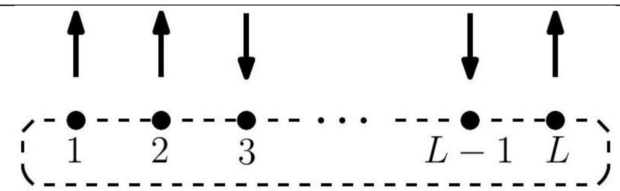
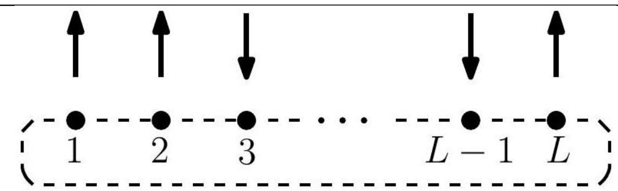
Two comparable cells per row. **Rendered**: the region's own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the `tex.zip` image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix's confidence, as the equation rows do. Setting a region's LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with `pdfdrill vision`; verify any LLM result against the real ink with inkdrill.

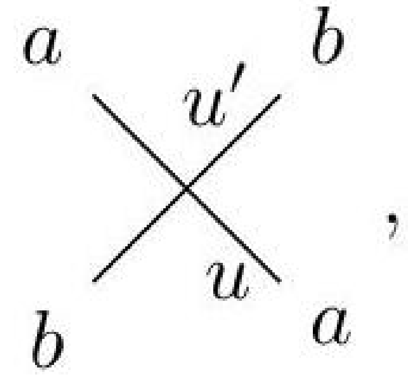
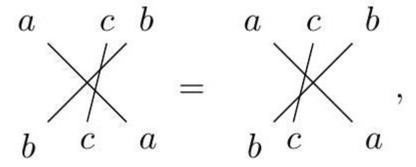
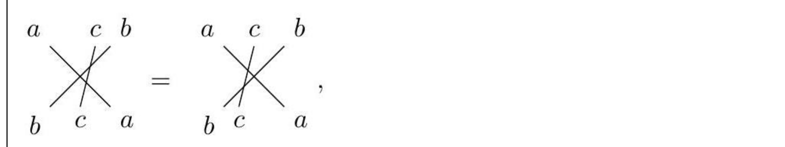
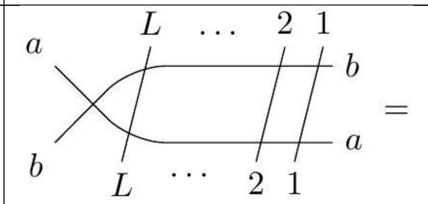
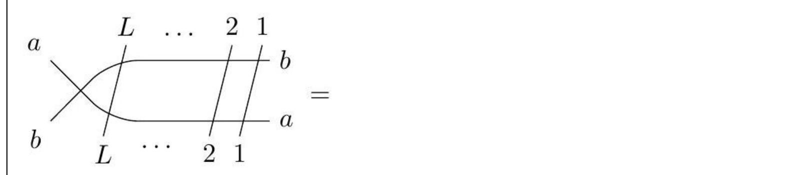
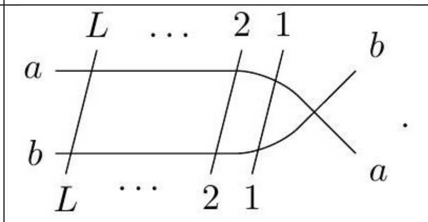
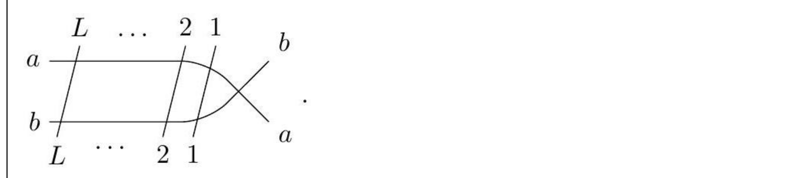
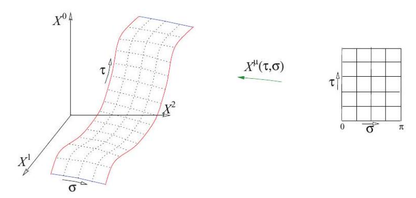
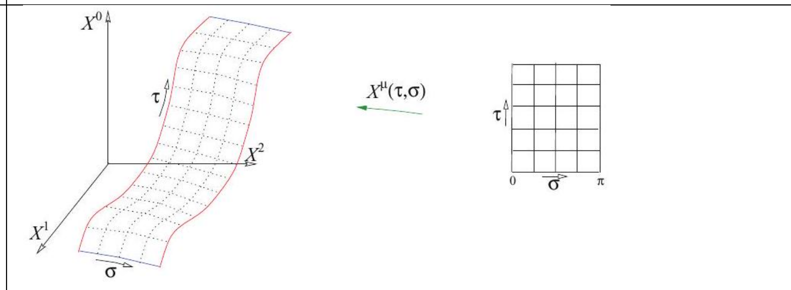
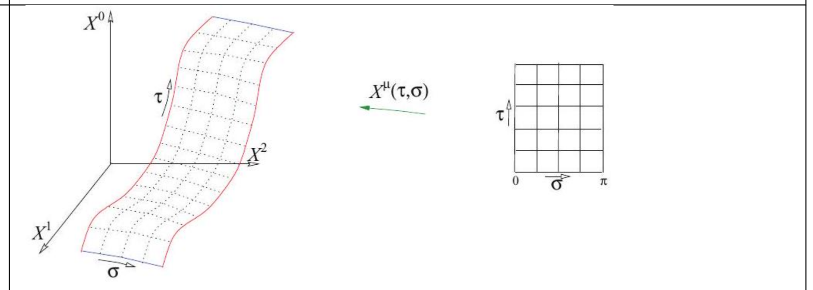
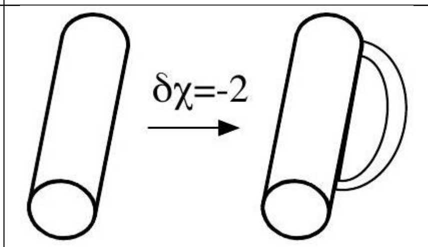
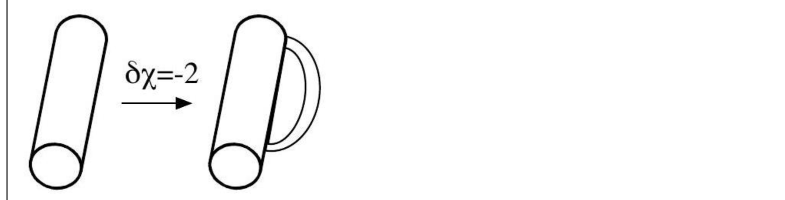
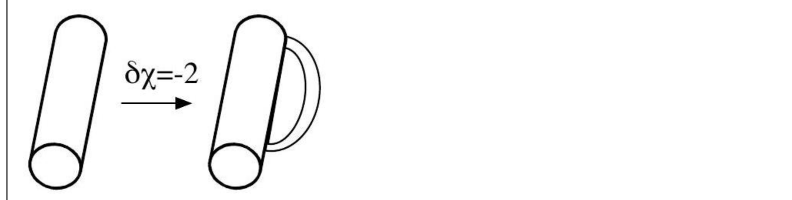
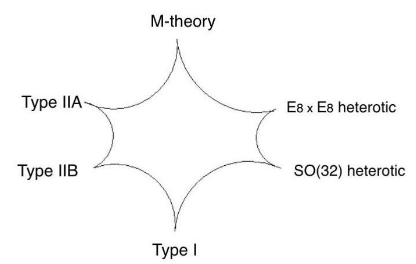
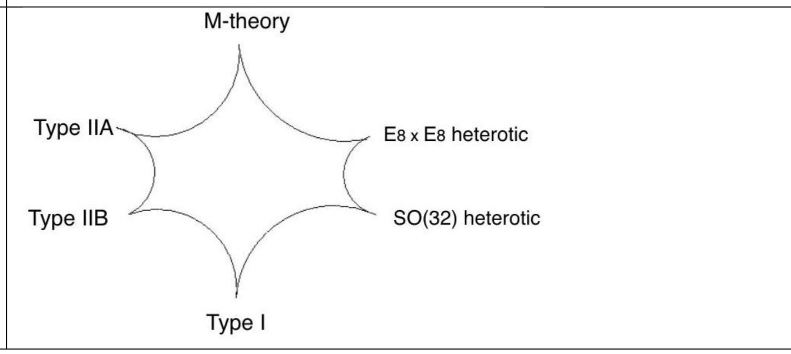
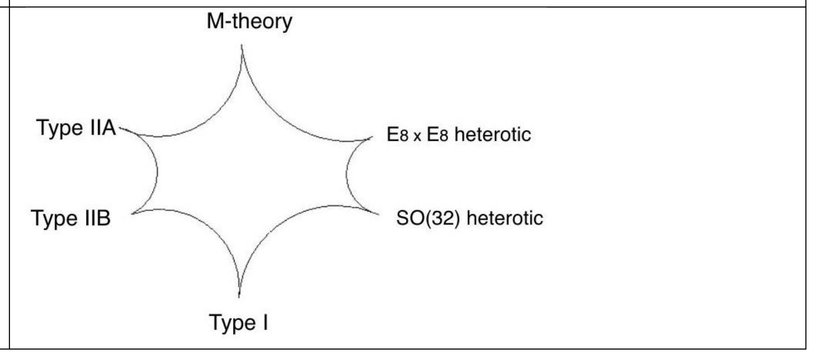
Identifier	Page	Class	Source	Author source	Rendered	Scan
cardona-qft-methods_PIC_0001	023	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-023_303_699_1682_342.jpg			
cardona-qft-methods_DIA_0001	034	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-034_145_712_1101_333.jpg			
cardona-qft-methods_DIA_0002	084	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-084_276_260_1585_505.jpg			
cardona-qft-methods_DIA_0003	086	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-086_294_460_1735_404.jpg			
cardona-qft-methods_DIA_0004	119	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-119_90_363_1455_526.jpg			
cardona-qft-methods_DIA_0005	122	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_181_280_312_568.jpg			
cardona-qft-methods_DIA_0006	122	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_120_253_623_212.jpg			

cardona-qft-methods_DIA_0007	122	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_163_257_581_584.jpg			
cardona-qft-methods_DIA_0008	122	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_166_255_581_949.jpg			
cardona-qft-methods_DIA_0009	122	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_250_251_1222_582.jpg			
cardona-qft-methods_DIA_0010	122	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_425_232_1579_198.jpg			

cardona-qft-methods_DIA_001[illegible]	122	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_423_235_1581_460.jpg			
cardona-qft-methods_DIA_001[illegible]	122	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_421_234_1583_722.jpg			
cardona-qft-methods_DIA_001[illegible]	122	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_200_200_1581_986.jpg			

cardona-qft-methods_DIA_0[illegible]	122	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-122_198_200_1806_1018.jpg			
cardona-qft-methods_DIA_0[illegible]	126	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-126_202_496_1818_460.jpg			
cardona-qft-methods_DIA_0[illegible]	127	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-127_127_280_460_197.jpg			
cardona-qft-methods_DIA_0[illegible]	127	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-127_124_124_462_522.jpg			
cardona-qft-methods_DIA_0[illegible]	127	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-127_124_181_462_673.jpg			
cardona-qft-methods_DIA_0[illegible]	127	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-127_124_324_462_898.jpg			
cardona-qft-methods_DIA_0[illegible]	132	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-132_110_225_253_597.jpg			

cardona-qft-methods_DIA_0[illegible]	133	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-133_230_701_995_370.jpg			
cardona-qft-methods_DIA_0[illegible]	133	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-133_234_875_1328_276.jpg			
cardona-qft-methods_DIA_0[illegible]	133	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-133_207_839_1664_294.jpg			
cardona-qft-methods_DIA_0[illegible]	135	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-135_191_876_1466_142.jpg			
cardona-qft-methods_DIA_0[illegible]	136	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-136_198_358_775_248.jpg			
cardona-qft-methods_DIA_0[illegible]	136	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-136_198_356_775_811.jpg			
cardona-qft-methods_DIA_0[illegible]	138	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-138_230_276_591_570.jpg			
cardona-qft-methods_DIA_0[illegible]	270	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-270_243_783_250_317.jpg			

cardona-qft-methods_DIA_0029	274	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-274_159_166_1108_754.jpg			
cardona-qft-methods_DIA_0030	274	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-274_163_395_1416_526.jpg			
cardona-qft-methods_DIA_0031	275	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-275_200_404_1696_319.jpg			
cardona-qft-methods_DIA_0032	275	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-275_202_374_1696_754.jpg			
cardona-qft-methods_DIA_0033	292	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-292_361_765_1261_326.jpg			
cardona-qft-methods_DIA_0034	296	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-296_239_393_1319_535.jpg			
cardona-qft-methods_DIA_0035	305	— *(dup)*	06a07804-7dd2-4dff-b89b-10ec98715e3f-305_432_678_251_370.jpg			

cardona-qft-methods_DIA_0036	344	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-344_193_652_251_382.jpg	$\begin{tikzcd} H_p(M)\otimes H_q(M) \arrow[r, dashed, "\star"] \arrow[d, "P.D\otimes P.D"] & H_{p+q-n}(M) \\ H^{n-p}(M)\otimes H^{n-q}(M) \arrow[r, "\sqcup"] & H^{2n-p-q}(M) \arrow[u, "P.D^{-1}"'] \end{tikzcd}$	$\begin{tikzcd} H_p(M)\otimes H_q(M) \arrow[r, dashed, "\star"] \arrow[d, "P.D\otimes P.D"] & H_{p+q-n}(M) \\ H^{n-p}(M)\otimes H^{n-q}(M) \arrow[r, "\sqcup"] & H^{2n-p-q}(M) \arrow[u, "P.D^{-1}"'] \end{tikzcd}$	$\begin{tikzcd} H_p(M)\otimes H_q(M) \arrow[r, dashed, "\star"] \arrow[d, "P.D\otimes P.D"] & H_{p+q-n}(M) \\ H^{n-p}(M)\otimes H^{n-q}(M) \arrow[r, "\sqcup"] & H^{2n-p-q}(M) \arrow[u, "P.D^{-1}"'] \end{tikzcd}$
cardona-qft-methods_DIA_0037	373	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-373_170_248_269_584.jpg	$\begin{tikzcd} \tilde{\mathcal{E}} \arrow[r, dashed] \arrow[d, dashed] & \mathcal{C}^*(\tilde{\mathcal{F}}) \arrow[d] \\ \mathcal{E} \arrow[r] & \mathcal{L}^\dagger(\mathcal{F}), \end{tikzcd}$	$\begin{tikzcd} \tilde{\mathcal{E}} \arrow[r, dashed] \arrow[d, dashed] & \mathcal{C}^*(\tilde{\mathcal{F}}) \arrow[d] \\ \mathcal{E} \arrow[r] & \mathcal{L}^\dagger(\mathcal{F}), \end{tikzcd}$	$\begin{tikzcd} \tilde{\mathcal{E}} \arrow[r, dashed] \arrow[d, dashed] & \mathcal{C}^*(\tilde{\mathcal{F}}) \arrow[d] \\ \mathcal{E} \arrow[r] & \mathcal{L}^\dagger(\mathcal{F}), \end{tikzcd}$
cardona-qft-methods_DIA_0038	373	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-373_191_354_1310_535.jpg	$\begin{tikzcd} \mathcal{E} \arrow[r, "\Delta"] \arrow[d, "\Delta"] & \mathcal{E}\otimes\mathcal{E} \arrow[d, "\mathrm{Id}\otimes\Delta"] \\ \mathcal{E}\otimes\mathcal{E} \arrow[r, "\Delta\otimes\mathrm{Id}"] & \mathcal{E}\otimes\mathcal{E}\otimes\mathcal{E}. \end{tikzcd}$	$\begin{tikzcd} \mathcal{E} \arrow[r, "\Delta"] \arrow[d, "\Delta"] & \mathcal{E}\otimes\mathcal{E} \arrow[d, "\mathrm{Id}\otimes\Delta"] \\ \mathcal{E}\otimes\mathcal{E} \arrow[r, "\Delta\otimes\mathrm{Id}"] & \mathcal{E}\otimes\mathcal{E}\otimes\mathcal{E}. \end{tikzcd}$	$\begin{tikzcd} \mathcal{E} \arrow[r, "\Delta"] \arrow[d, "\Delta"] & \mathcal{E}\otimes\mathcal{E} \arrow[d, "\mathrm{Id}\otimes\Delta"] \\ \mathcal{E}\otimes\mathcal{E} \arrow[r, "\Delta\otimes\mathrm{Id}"] & \mathcal{E}\otimes\mathcal{E}\otimes\mathcal{E}. \end{tikzcd}$
cardona-qft-methods_DIA_0039	374	— (dup)	06a07804-7dd2-4dff-b89b-10ec98715e3f-374_181_244_937_586.jpg	$\begin{tikzcd} & \mathcal{E}\otimes\mathcal{E} \arrow[dd, "\tau"] \\ \mathcal{E} \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & \\ & \mathcal{E}\otimes\mathcal{E}, \end{tikzcd}$	$\begin{tikzcd} & \mathcal{E}\otimes\mathcal{E} \arrow[dd, "\tau"] \\ \mathcal{E} \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & \\ & \mathcal{E}\otimes\mathcal{E}, \end{tikzcd}$	$\begin{tikzcd} & \mathcal{E}\otimes\mathcal{E} \arrow[dd, "\tau"] \\ \mathcal{E} \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & \\ & \mathcal{E}\otimes\mathcal{E}, \end{tikzcd}$